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BIGGER LEANER STRONGER



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THE SIMPLE SCIENCE OF BUILDING THE ULTIMATE MALE BODY

MICHAEL MATTHEWS

FOURTH EDITION

BIGGER LEANER STRONGER

The Simple Science of Building the Ultimate Male Body

Fourth Edition

Also by Michael Matthews

The Year One Challenge for Men

The Shredded Chef

Beyond Bigger Leaner Stronger

Muscle for Life

Thinner Leaner Stronger

The Year One Challenge for Women

The Little Black Book of Workout Motivation

Fitness Science Explained

Bigger Leaner Stronger
The Simple Science of Building the Ultimate Male Body
Fourth Edition

By Michael Matthews (*www.mikematthews.co*)

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Specific results mentioned in this book should be considered extraordinary, and there are no "typical" results. Because individuals differ, results will differ.

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Free Bonus Material (Workouts, Meal Plans, and More!)

Giving up on our long-term goals for immediate gratification, my friends, is procrastination.

—DAN ARIELY

Thank you for reading *Bigger Leaner Stronger*. I hope you find this book insightful, inspiring, and practical, and I hope it helps you build the fit, strong, and healthy body you really desire.

I want to make sure that you get as much value from this book as possible, so I've put together a number of additional free resources to help you, including:

- A savable, shareable, printable reference guide with all of this book's key takeaways, checklists, and action items.
- Links to form demonstration videos for all *Bigger Leaner Stronger* exercises.
- An entire year's worth of *Bigger Leaner Stronger* workouts neatly laid out and provided in several formats, including PDF, Excel, and Google Sheets. If you'd prefer the workouts in a digital or hard-copy book, check out *The Year One Challenge for Men* (www.biggerleanerstronger.com/challenge).

- Twenty *Bigger Leaner Stronger* meal plans that make losing fat and gaining lean muscle as simple as possible.
- My favorite tools for getting and staying motivated and on track inside and outside of the gym.
- My all-time favorite fitness books.
- And more.

To get instant access to all of those free bonuses (plus a few additional surprise gifts), go here now:

www.biggerleanerstronger.com/bonus

Remember too that if you have any questions or run into any difficulties, you can shoot me an email at mikem@legionsupplements.com, and I'll do my best to help!

To all the people who have supported me and my work and made this fourth edition possible.

This is for you.

And to be honest, I hope that after reading it, you'll say it's the best fitness book for men ever written. If nothing else, it's the absolute best fitness book for men I'm currently capable of writing.

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Foreword

Mark Rippetoe

Mike and I have been doing podcasts together for a while now, and we've gotten to be buddies despite the fact that he's too skinny and has too many abs.

I keep telling him he'd make a great lifter at 275 pounds, but he plays golf, and you know how those people can be.

Nonetheless, I have a great deal of respect for anyone who insists on telling the truth in this ridiculous industry, the home of charlatans, frauds, liars, and fools all over the world. Mike and I agree about far more than we disagree, and we learn from each other.

More importantly, Mike is his own guy—he works for himself, thinks for himself, and writes for himself. And you are the beneficiary. He sells no useless quasi-pharmaceuticals, no silly devices, no "breakthrough" programming, and no bullshit. He gets no money from sponsors, and he is owned by no one.

Mike may be mistaken (as may I), but he doesn't lie to you. He tells you the truth, and you should listen.

If you want to use strength training for aesthetics, Mike is your source. He understands the importance of getting strong. He knows that muscle determines the shape and appearance of the human body, and that muscles grow when they are stressed by heavy loads. He knows that hard work is the only effective use of your training time. He understands the science of nutrition, and he can explain it to you—and even to me.

I'm an old hardheaded strength coach, but I have enough sense to know that most people are primarily interested in their appearance. So read this book, and read mine too, and come away with what you need to know. The rest will be up to you.

Mark Rippetoe

Author of *Starting Strength: Basic Barbell Training* and Practical *Programming for Strength Training*

PART 1

What's In This For You?



“My mental focus, energy, strength, and self confidence have all gone up!”

LUKE M.



"I lost 16 pounds in 2 months without losing any muscle or strength!"

JUSTIN C.



"I am much more confident, self-assured, and disciplined!"

PAUL K.



“I've gotten the results I dreamed about my whole life, sounds cheesy, but it's the truth!”

ROMAN T.



"The program has been incredibly easy and the results came much quicker than expected."

HUNTER L.



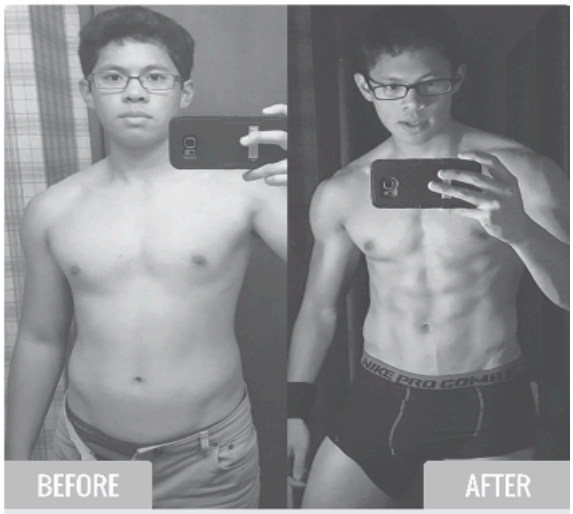
"I lost 24 lbs in 90 days!"

EMMETT S.



"Last summer I finally snapped and committed to making a change. In 12 months of Bigger Leaner Stronger, I went from 20% body fat down to 8-9%."

JOSH B.



"This journey has been nothing short of amazing and life changing."

DARICK C.



"I've gained 20 lbs of muscle and am able to stay lean while continuing to make gains at the gym."

GILSON C.



"I lost 26 pounds and 11% bodyfat in 3 months on the BLS program!"

SAM R.



"I finally feel relief and comfort in my diet compared to the worry of the unknown."

BEAU L.



"I'm completely off all medication, my cholesterol is the lowest ever, depression is gone, no alcohol, cigs or dope."

GREGORY M.



“The program is just plain easy and fun to follow, enjoy the journey!”

ANDREW M.



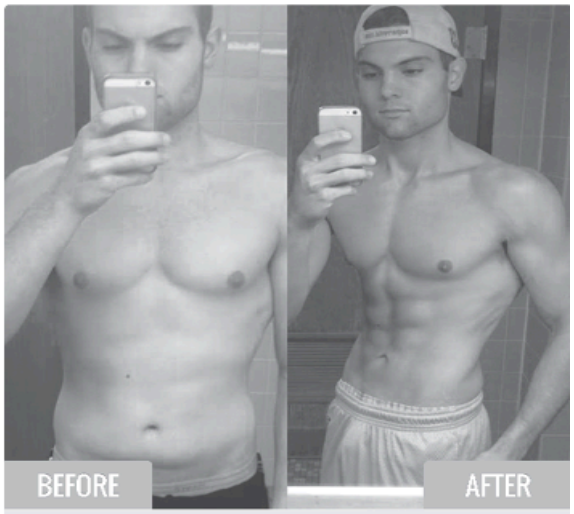
"I have gained almost 25 lbs of lean muscle and significantly increased my strength across the board."

STUART M.



"The program was so easy. I started out with a cut and seeing the scale and my waist change so rapidly was incredibly motivating."

ERIC B.



"Bigger Leaner Stronger changed my outlook on life and fitness."

CARTER G.



"At age 54 I have lost almost 20 pounds and 9% body fat."

JAY P.



"I am stronger, healthier, and look better than any other time in my life!"

GARY L.



"I lost 29 pounds while cutting!"

DYLAN M.



"I've lost nearly 25 pounds, and I can now dumbbell press 75 lb dumbbells in each hand."

STEPHEN S.



"I truly thought I could never get a defined mid-section and all it took was 3 months of the Bigger Leaner Stronger program."

IRA C.



"Challenging yourself to improve becomes its own game with a reward every day!"

ROB N.



"Once you understand the program, all it takes is keeping with your macros and showing up to get the lifts done."

ALEX R.



"I gained 25 lbs and all of my big lifts more than doubled in weight."

TYLER M.



"I definitely have more energy and I'm more productive!"

ARSENIO A.



"I've lost about 13 pounds overall and have built significant muscle!"

MATT H.



"I've never had so much energy before and I have more confidence!"

JOE T.



"I'm energetic, positive, and enjoy life so much more."

JONATHAN M.

It Works

Ordinary People, Extraordinary Results. Will You Be Next?

The men you just saw are like you. They're in their twenties, thirties, forties, and beyond, and they come from all walks of life and levels of fitness. Some of them were once fit, while others were always heavy; many had tried various diet and exercise routines before and failed, while others were brand new to it all; and some had plenty of time and energy for working out, while others had very little.

What they all have in common, though, is that they used my diet and exercise principles and programs to get into the best shape of their lives. They've replaced unwanted fat with lean muscle and dramatically cut their risk of disease and dysfunction by eating foods they love, doing workouts they enjoy, and taking few, if any, supplements.

I want to introduce you to a few of them whose stories have inspired and touched me, and who are definitive proof that with the right know-how and guidance, anyone can build a body they can be proud of. If they can do it, why not you?

KENDRICK'S STORY



I'm 26 years old and dropped 47 pounds in four months following Mike's *Bigger Leaner Stronger* program, going from 210-to-163 pounds and around 8-to-9 percent body fat.

I've tried a LOT of other programs that didn't work the way I expected. I got bigger and a little bit stronger but never got the definition I desired. For the first time in my life, I got significantly stronger while losing fat. Learning to lose fat the correct way has been such a blessing.

I'll admit I was a bit suspicious of the advice to lift heavy weights while you lose fat, because it goes against all "conventional" wisdom. I'd done several programs which involved doing as many reps as you could in a minute, drop sets, even one in which you did 10 sets of 10 reps of each exercise. I always got burnt out, plateaued, and took forever to recover, and I gained too much body fat along

the way.

With *Bigger Leaner Stronger*, my bench press has increased from 240-to-265 pounds, my squat has gone from 250-to-320 pounds, my deadlift has gone from 295-to-365 pounds, and my military press has gone from 145-to-175 pounds. Those results speak for themselves.

I think anyone who's willing to put forth the work and wants to maximize their potential will benefit from this program. I always had the work ethic to achieve it; I just needed the knowledge. This is just the beginning, and I'm far from done.

I'm much more confident, and my friends are constantly asking me to go to lift with them and set them up on a program and diet. That's been awesome to me! Push yourself, be patient, trust the process, and find out what you are capable of accomplishing.

BRANDON'S STORY



I used Mike's *Bigger Leaner Stronger* program to lose 60 pounds in twelve weeks to prepare for a men's physique competition for men over 50. I took fifth place in the men's open novice division, and my strength increases were massive too. My bench increased from 185-to-285 pounds, and I increased my squat, deadlift, and overhead press too!

Before discovering *Bigger Leaner Stronger*, my excuses and untrained mind (conditioned to give up easily) prevented me from attaining my goals. I used my shoulder surgery, my drinking problem, and my lack of wanting to change my lifestyle as excuses.

Then, I stumbled upon Mike and *Bigger Leaner Stronger*, and my life changed. I used the knowledge Mike shares to overcome anything that was in my way—cravings, urges, and thoughts that had held me back in the past. I haven't

looked back since September of 2016, and now I'm unstoppable.

THIS PROGRAM WORKS! Just apply the principles that Mike has laid out for you, and trust me, the sky's the limit. Don't change, add, subtract, or manipulate anything—just do it like he's laid it out and record your workouts daily.

Remember, this is NOT a quick fix, Hollywood diet, or fad weight loss program. This is the beginning of a lifestyle change you need to maintain for the rest of your life to feel and be happier and healthier. I guarantee you that *Bigger Leaner Stronger* is better than anything you've tried before.

I promote *Bigger Leaner Stronger* and Muscle for Life on a daily basis—so much that it now interferes with my own personal workouts, but that's fine with me. This program saved my life! A BIG and SINCERE thank you to Mike—I owe you big time!

JENKIN'S STORY



In one year of following the *Bigger Leaner Stronger* program, I went from 207-to-142 pounds for a total of 65 pounds of weight loss! For the first six months my strength improved, especially my deadlift. It went from 275-to-325 pounds while I lost weight!

Working out wasn't an issue, but being disciplined and not eating whatever I wanted every day was tough. It took me a solid three months to really make the permanent lifestyle changes I needed to.

I came across Mike's book and saw that it had all the basics of eating for building a great physique. Originally I was only going to follow Mike's nutrition plan, but after reading about the training program, I saw it shared a lot of the same principles I had come across in other programs I liked. I couldn't help but completely agree with what Mike was saying, so I gave that a shot too.

The whole program really spoke to me, and I appreciate how Mike backs everything he says with facts. Mike really connected with me as a reader, and he gained my trust. I felt he had integrity; thus, I was willing to follow his program. And I have only gotten results since.

The program is not a magic pill or quick fix; it does not give you a secret or a shortcut. It merely teaches principles. By abiding by these principles, you'll see results, but it will take time, patience, discipline, and effort—DON'T GIVE UP!

I'm so moved by these outstanding guys, and I hope you are too. What's even more amazing, though, is I hear from people with similar success stories every few days, and I've shared hundreds of them on my website at www.biggerleanerstronger.com/success (worth a gander if you're in need of some get-up-and-go). Transformations like these aren't reserved only for those with elite genetics or superhuman discipline. Anybody can join the club, including you. Who knows, maybe people will be reading about your results one day?

Anything is possible if you want it enough. And in this book, I'll show you the way.

1

The Promise

No matter how old you are, no matter how bad your genetics, and no matter how lost you now feel after trying and abandoning past diets and workout programs … You absolutely, positively can have the lean, muscular, and powerful body that you dream about, and you're about to learn how.

What if you could follow a science-based, time-proven formula for eating and exercising that makes adding 15-to-25 pounds of lean muscle while also shedding handfuls of body fat a breeze?

What if you could build your best body ever without starving or depriving yourself? What if you could still indulge in the "bad" foods that you love, like pasta, pizza, and ice cream? And what if you didn't have to live in the gym or grind through grueling workouts that leave you feeling beaten up and burned out?

What if you could see dramatic results in the mirror in just the first thirty days and know that you've finally found the master key to getting (and staying) big, lean, and strong, now and forever?

Imagine waking up every morning, looking and feeling fit, strong, and powerful. Imagine, in just a couple of months from now, taking your shirt off at the pool or beach with confidence and having dawn-to-dusk energy to tackle your

day. Imagine how relieved you'll be to no longer fret about your waist size or how you look naked. Imagine how satisfied you'll feel, knowing that you're getting a little fitter, stronger, and healthier every day.

All that and more is what awaits you on the *Bigger Leaner Stronger* program, and it won't be as difficult nor as complicated as you may think. Whether you're twenty-one or forty-one, fit or fat, strict or slack, you have the power to transform your body. Ask the many people whose lives have been changed by my books, articles, podcasts, and other work. They accepted my help, and now they're in better shape than ever before. They're the proof that together, we can do the same for you, too.

These may sound like outrageous claims, but this book is a realistic blueprint anyone can follow to realize them. It's a blueprint quite unlike what you normally find in a fitness book, too, because it doesn't involve "hacks," quick fixes, or other unsustainable methods that produce fast but fleeting results. Instead, I'll provide you with nutritional principles and meal plans with enough structure to get results and enough freedom to embrace your eating preferences and lifestyle. This way, you can look forward to every meal that you eat and never feel like you're "on a diet" ever again. I also won't try to force you into a cookie-cutter training program that may or may not meet your needs or liking. Instead, I'll give you three workout routines to choose from—a three-day-per-week routine, a four-day, and a five-day—and teach you how to modify them according to your goals and circumstances. This way, you can enjoy your training and never feel like you're working too hard or not hard enough.

I'll also be your guide the whole way, encouraging you to discover what you're truly capable of; helping you overcome obstacles and setbacks like lack of motivation, diet and exercise plateaus, and nagging aches and pains. I'll show you how to avoid pitfalls like self-doubt, perfectionism, and unrealistic expectations,

and how to achieve your biggest fitness goals in a skillful and sustainable way.

In *Bigger Leaner Stronger*, I'll also prove to you that getting ripped isn't as complicated as you've been led to believe. It doesn't require following a "special" diet like intermittent fasting or keto, using "sophisticated" exercise techniques like muscle confusion or functional training, or filling your cabinets with dubious pills and powders like collagen protein or branched-chain amino acids.

Instead, the real "secret sauce" of the fitness elite can be summed up like this:

1. They control their calorie and protein intake.
2. They mostly eat nutritious foods.
3. They do at least a few hours of strength training per week.

In other words, the gateway to the body you've always wanted is in the fundamentals, not the fringes. The devil's in the details, however, because as you'll learn in this book, there are many more incorrect ways of executing those strategies than correct ones. It's like making music—the process amounts to combining notes into pleasing harmonies, melodies, and rhythms, but simply knowing that isn't enough to create a hit. You need to understand how to work with those elements in very specific ways. Unfortunately, however, there's a lot more misinformation about how to get fit than how to make music. Why is that? Why are long-debunked lies still trafficked by celebrities, influencers, authors, and "experts"?

First, there's a lot of money in it. When millions of people are strongly motivated to solve a problem, brilliant marketers will always find an abundance of things for them to buy. Thus, in the case of the fitness advice industry—magazines, websites, books, podcasts, and so on—much of the information is, shall we say, *inspired* by the need to keep you buying. And one of the best marketing buttons for getting people to buy is "new." By steadily sharing new

suggestions, then—diet, training, and supplementation "tricks," "breakthroughs" and "shortcuts," for example—information sellers can keep you hooked.

That isn't to say that new information is bad, though. Fitness is a sweeping subject with many trails, tunnels, and caverns to explore. Most of them won't get subscribers, however. Your average guy just wants to gain 20-to-30 pounds of muscle in the right places and see his abs, not learn about the nuances of nutrient metabolism or training periodization, and it doesn't take more than a single book (like this one) to teach him how to do that. If the fitness advice industry told it straight, they'd have maybe twenty-five essays, total, that'd teach you basically what you'll learn here.

This book, however, is different, because I have other incentives and payoffs. I've had success as a self-published author and entrepreneur, so my livelihood isn't beholden to publishers, advertisers, or fads. Instead, I get my lifeblood directly from you, and how well I do depends entirely on how well I serve your interests. Thus, in this book, I can go against the grain and recommend science-based diet and exercise strategies that differ from the kind of miracle-solution promises that make editors and marketers salivate but ultimately fail to deliver.

First and foremost, I'm going to teach you the power of effective strength training for gaining and maintaining muscle, and getting and staying lean. I'll also introduce you to eating and training systems that actually work as a long-term lifestyle. For example, I'll show you how to lose fat and gain lean muscle while eating plenty of carbs; how to transform your body with just a few hours of strength training per week and a lot less cardio; and why supplements are entirely optional and often a waste of money because only a small number are scientifically proven to enhance your physique, health, and performance.

That said, *Bigger Leaner Stronger* isn't for everyone. It's not for people who prefer comfortable lies over cold truths. It's not for people who are hunting for

the One True Plan that doesn't require skill, struggle, or sacrifice. It's not for people who always demand a lot for a little. It's for people who know that the only "secrets" that work are those you're willing to work for. It's for people who are ready to screw up their courage and leave their comfort zone. It's for people who want to earn a hearty tomorrow by investing in it today.

Skeptical? You should be. I was too when I first encountered the information and strategies I'll share with you in this book. Take heart, though, because *Bigger Leaner Stronger* doesn't require a leap of faith. Most of what you'll learn has been around for decades, already known to high-level strength and bodybuilding coaches and athletes, but nobody has connected the dots for you like I will here. Moreover, *Bigger Leaner Stronger* works, and fast, so you'll see real, visible improvements in your body within the first thirty days of starting the program, and within a couple of months, people will start asking what the deuce you're doing. You'll see your weight moving in the right direction, your clothes fitting better, and your muscles looking more defined. I promise.

And if, for whatever reason, you're not seeing these types of results, I still have good news for you: It's not because this is just another overhyped hoax that can't or won't work for you. It only means that you need some help with the execution, and I'm always just an email away and happy to assist, so feel free to write to me anytime at mikem@legionsupplements.com.

Remember, too, that tens of thousands of men (that I know of) have used the principles and program in this book to build a fit and strong body they're proud of. So, you're in good hands. And soon you'll be on your way.

If we're to succeed, however, I'll need a few things from you. First, I need you to approach this book with an open mind. Many "gurus" promise similar payoffs as I have and then leave you in the lurch, creating an intoxicating surge of hope that becomes a crushing wave of despair. This time will be different. By

dealing with diet, exercise, and supplementation in entirely new ways, you can finally stop stumbling off and scrambling back onto the wagon, a little worse for wear each time. *Bigger Leaner Stronger* will show you how to escape that terrible tailspin. It's a realistic and sustainable plan that men of all ages and circumstances can win with, but you must temporarily suspend your disbelief and trust in the process.

Second, I need you to show up. Some people buy a book like this with an eye to execution, but for whatever reason, their intentions don't translate into actions. That's the only guaranteed way to fail with the program. So in this book, I'll periodically ask you to stop reading and start doing—to take concrete steps toward your new body and life. If you can take those steps, and then keep taking them, you can reach the promised land.

And third, I'd love to hear from you once you've achieved your first major milestone, like filling out your favorite t-shirt and jeans, waking up full of energy, or thrilling at your growing strength. Admittedly, this is a selfish request, but hearing how people are using my work to get fitter, healthier, and happier is the real pay for my efforts, so I'd love to listen to your story when you're ready to share it. And since And since I always try to lead by example, I'll quickly tell you mine.

2

Who Is Mike Matthews and Why Should I Care?

Opportunity is missed by most people, because it is dressed in overalls and looks like work.

—UNKNOWN

I've made some strong claims about what this book will do, but I'm confident that the *Bigger Leaner Stronger* program will work for you. Not only is the program evidence-based, but it's also the product of my own journey in fitness. After examining much of the important scientific literature, trying scores of workout programs, diets, and supplements, and working with thousands of people, I don't know everything, but I do know what works and what doesn't.

Like most people, I was clueless when I started lifting weights nearly twenty years ago. I picked up some bodybuilding magazines and started working out a couple of hours per day and spending hundreds of dollars on supplements per month. I carried on that way for over seven years, jumping from diet to diet, workout program to workout program, and supplement to supplement. Despite my considerable efforts, I had only gained about twenty-five pounds of muscle and novice-level strength, and I still didn't know how to get and stay lean or break through my training plateaus and build the body I really wanted.

I then started working with personal trainers, but it was just more of the same: Training programs that revolved around "sophisticated" (and mostly useless) training techniques like supersets, functional training, and "muscle confusion"; various brands of "clean eating" that didn't make a difference; and exotic-sounding pills, powders, and potions that merely lightened my wallet and brightened my pee. After giving many months and paychecks to trainers, not much had changed. I still looked the same and still had no idea what to do to get bigger, leaner, and stronger. I enjoyed working out too much to quit, but I wasn't happy with my physique and didn't know what was holding me back.

I finally decided to get "serious" about my fitness and learn the true physiology of muscle growth and fat loss. I tossed the magazines out, ditched the Internet forums, and stopped paying attention to locker room chatter, and started studying the work of leading strength and bodybuilding coaches, consulting veteran natural bodybuilders, and reviewing scientific research.

Before long, I began to realize that getting into outstanding shape is much simpler than many fitness experts would have you believe. It has little to do with a lot of the stuff we see on TV and social media, and read in websites, books, and magazines. This data smog of myths and misconceptions not only makes losing fat and gaining muscle much harder than it should be, it also undermines our self-confidence, strains our relationships, and blunts our desire for self-improvement.

When I finally escaped all of the misinformation and changed what I was doing in the kitchen and gym according to what I was learning, the results were almost too good to believe. I started gaining muscle again. I started gaining strength again. I got a six-pack for the first time. I was getting jacked by spending less time in the gym, doing less cardio, and eating more freely. It felt like I had found the cheat codes to the fitness game.

It wasn't long before friends and family noticed how my body was changing

and asked for advice. I shared what I was doing and helped them shed fat they thought they'd never lose, gain strength and muscle definition they never thought they'd have, and experience levels of self-confidence they'd only dreamed about.

Some of these people started urging me to write a book about my methods. I liked the idea. "What if I'd had such a book early on?" I thought. It would've saved me untold time, money, and frustration, and I would've reached my fitness goals much faster. So, I wrote that book—this book—and self-published it in January 2012.

It sold maybe twenty copies in the first month, but within a couple of months, it was selling twenty copies per day and generating just as many positive emails and reviews from readers. I made notes on how to improve the book based on suggestions and feedback, and I saw the opportunity to write more books on the science of getting healthy and fit.

Fast forward to today, and I've written and published multiple fitness books that have sold over 1.5 million copies, including one for women called *Thinner Leaner Stronger*, a cookbook called *The Shredded Chef*, and a pick-me-up called *The Little Black Book of Workout Motivation*. What's more, dozens of men and women write to me every day and share how my work is helping them build their ideal body faster than they ever thought possible. And that never gets old.

So, are you ready to become one of them? Turn the page and we'll be on our way.

3

How to Use This Book

Nothing in the world is worth having or worth doing unless it means effort, pain, difficulty.

—THEODORE ROOSEVELT

There's an old Chinese proverb that encapsulates the essence of this book. It goes like this: "tell me, I'll forget; show me, I'll remember; involve me, I'll understand."

This isn't a book that you just read and reflect on—it's a book you *act* on. The information I've included and the way I've organized it is intended to help you lift the ink off these pages and use the principles in your life. Thus, I want to share a roadmap that'll help you get the most from *Bigger Leaner Stronger*.

This book contains six parts:

1. In this first part, I've given you a foretaste by introducing you to the *Bigger Leaner Stronger* philosophy, and hopefully, whetted your appetite and inspired you to make a simple decision: "I'll give this an honest look."
2. In the second part, we'll address the "inner game" of fitness and help you cultivate a winning mindset by clarifying goals and purposes,

boosting your confidence, and strengthening your willpower.

3. In the third part, you'll learn how to get and stay fit and healthy without following a restrictive "diet" ever again. Instead, I'll give you a simple but powerful method of meal planning that provides maximum flexibility and sustainability.
4. In the fourth part, we'll get into how to train the *Bigger Leaner Stronger* way, and I'll give you a science-based system for transforming your body as well as ready-made workout routines for you to start doing.
5. In the fifth part, I'll touch on the least important aspect of the program —supplementation— because while most supplements don't do much, some can enhance muscle and strength gain, physical performance, and overall health and wellbeing.
6. In the sixth part, we'll wrap up with additional insights and tactics that'll boost your results as well as pre-designed meal plans and workout templates.

There are also two ways you can approach this book:

1. Read it and then apply what you've learned.
2. Read and apply as you go.

I recommend number two because, after working with tens of thousands of people, I've found that the sooner someone can get going, the more likely they are to *keep* going. So long as we're in motion, the thorny vines of doubt and despair can't seize us. So long as we're *doing*, *thinking* can't stop us.

Therefore, I've inserted implementation steps throughout this book that will ask you to act. I didn't choose these spots arbitrarily—I picked the points where

immediate execution will greatly increase your chances of long-term success. If I were coaching you, this is exactly how I'd start you on the program.

Some of these steps will require writing, and I recommend you use a notebook. Many people like to use a journal for creating their meal plans, tracking their workouts, and reviewing their progress. By keeping everything in one place, you can easily assess and revise your plans as needed, and you can memorialize your transformation. In fact, if you're like many of us meatheads, you'll come to treasure your dog-eared stack of fitness diaries as a testament to your dedication to self-improvement.

If you want to get started as soon as possible, you can simply learn the "hows" of the program, start right away, and then learn the "whys" after. To do this, begin with chapter 10, which will teach you how to create meal plans for losing fat and gaining muscle. Make your first meal plan, start using it, and then go to chapter 14 to find the workout routines. Get going on them (and if you're not sure how, check out the quickstart guide, read the following chapter on how to track your progress, and finally, read chapter 17 to learn about supplementation. Then, come back to this book and read it from the beginning to learn how to get the most from the *Bigger Leaner Stronger* program.

Finally, this book contains a lot of information, including many technical details, but don't let that intimidate you. I've tried hard to ensure this is an accessible and practical handbook for everyone, not an academic textbook for fitness geeks. No matter how green you may be, soon, you'll know more than most people ever will about how to build muscle, lose fat, and get strong.

Whether you take your time to read and absorb everything in this book before starting or skip to the actionable items and learn the rest after you're off the scratch line, if you commit to the program, you'll win. Doubts will dissolve, and you will realize that you're more than capable of upgrading your physique.

And remember, you can email me anytime at mikem@legionsupplements.com if you have any questions or need help.

PART 2

Winning at the “Inner Game” of Getting Fit

The Big Ideas

1. Know your true fitness whys.
2. Envision two dimensions of your ideal body.
3. Create powerful affirmations to stay on track.

4

Have You Found Your Real Fitness Whys?

The discipline of desire is the background of character.

—JOHN LOCKE

In his timeless bestseller *The Inner Game of Tennis*, Tim Gallwey explained that while many people view games purely in terms of their "outer" components—external opponents, challenges, and goals—there's another element at least as important—the "inner game" that takes place inside our minds, involving obstacles like lack of focus and motivation, anxiety and insecurity, and other feelings and attitudes that inhibit performance.

That astute observation is just as fitting to fitness as to any sport or pastime. While most authors, coaches, and experts focus on aspects of the outer game of fitness like diet, exercise, and supplementation protocols, they rarely give much attention to the inner game. This sets people up to fail because simply knowing *what* to do to lose fat and gain muscle isn't enough. You need to then consistently *do* those things for days, weeks, months—ultimately years—if you're going to master your fitness.

For most people, motivation and discipline are the nastiest inner-game barriers—the ones that most undermine their enthusiasm and resolve. I've seen

this unfold many times. People struggle with restrictive diet plans, difficult exercise routines, and unexpected physical and mental challenges and setbacks. Then, as time passes, they see little change in their bodies, so the process feels like all pain and no gain. Given those circumstances, it's no surprise that so many people give up on their fitness goals within the first few months. Sometimes sickness or injury strikes, or they take a week off and then never return. Sometimes, they just "stop caring" about getting fit.

Maybe you've experienced this yourself. I certainly have. Fitness is hard, and no matter your determination, if you can't make it a lifestyle that delivers clear and consistent results, your drive will dry up as sure as water is wet. I don't want this to happen to you. I want to see you solve the puzzle and win the prize. In fact, I want *Bigger Leaner Stronger* to be the fitness program that finally makes all the difference and surpasses your expectations.

To do that, however, you must learn to appreciate the process rather than obsess over the outcome. We live in the Age of Impatience—too many people want four-hour workweeks, six-minute abs, and thirty-second meals. They don't want principles and protocols; they want shortcuts and "secrets." They don't want to plant in the spring, tend in the summer, and harvest in the fall; they want to shirk and slack and reap a bounty that was never sowed.

I wish it were otherwise, but you can't gain ten pounds of muscle in thirty days or add a couple of inches to your biceps or get rid of your gut in a couple of weeks. You have to give a lot to get a lot. That's okay, though, because this teaches you an invaluable life lesson. Fitness is one of those rare things you can't buy, steal, or fake. Lying, cheating, or kvetching cuts no ice in this arena, and there are no privileges or points awarded for status or style. "Working out" is an apt description, because you either put in the work or you get put in your place.

That doesn't mean you must be a fitness fetishist, though. Everywhere you

look on social media, there's another hard-faced hardbody growling about embracing the suck and doing it anyway so you can earn the aftermath. The ball-ache is that approach ultimately holds you back because you can only burn the candle so much before burning out.

On the other hand, there are those who coo in ASMR tones about the need for kindness and care, for embracing your wholeness and radiance, for dropping into your heart space and feeling into your sensations and don't forget to exfoliate and it's okay if you didn't work out today … here's a baby Yoda. Such syrup goes down easily, but mollycoddle yourself too much and your spirit cools and congeals and you slide into stupor.

Too many people lean too heavily toward one of these extremes and neglect the other. They've weaned themselves on either a diet of nothing but oppressively tough love or cloying self-care. The key is balance—in knowing when to flow and when to float, when to brace up and bear down, and when to back off and breathe. Go, go, go versus slow, slow, slow.

How do you find that balance? That's one of life's mysteries, and I wish I had the answer. My guess is there isn't one. It's a matter of learning about ourselves and our capabilities and limits, and the only way to do that is to spend a lot of time at the coalface and reflect on our experiences. How hard can we push before the wheels start to fall off? How much rest do we need to recover?

Regardless of where your Goldilocks zone lies, know this: Self-care is meant to be a sojourn, not hearth and home. Linger there for too long and it's procrastination; leave it out for too long and it's punishment. The trick is knowing how to thread this needle, how to get the work done without ravaging the worker.

Fortunately, you can make as many mistakes as it takes to get into your stride without screwing this fitness thing up. You can garbage down enough food to

tranq a girthy buffalo without blowing it. You can miss a workout or two or five because of injury, travel, or reasons best known to the Dread Lord Cthulhu without wrecking it. In short, you can fail over and over in fitness and still succeed.

How? The only way to lose at this game is to give up. That's the only mortal sin. No matter how many times you go off the rails, so long as you get back on course, you can't mess this up. Fitness isn't a contest with winners and losers. There are no deadlines or duels, distinctions or demerits. Fitness is like love, laughter, and learning—a wellspring of health and happiness to pursue purely for its own sake. Fitness isn't fleeting. It's forever. Delight in the realization that you have all the time you need to find your feet (and abs).

There are many obstacles and pitfalls that can prevent you from achieving your full fitness potential, however, and in this section of this book, I'll help you develop the inner game that's required to overcome them.

To start, I have a question for you: Why are you reading this book? If you don't have a prompt, passionate answer, we have some work to do, because people without clear, inspiring, and attainable fitness goals are always quick to quit. They work out sporadically and barely even break a sweat. They falter when faced with temptations and troubles (pesky neighborhood potlucks!). They want to believe in magic bullets and miracle methods. If you're going to succeed where most of these people fail, we need to first engage in a little soul-searching.

There are many reasons why people follow diet and training regimens. Some enjoy pushing their body's limits. Others are in it to impress peers and potential partners. Many want to boost their self-confidence and wellbeing. Most want to be healthy. These are all terrific reasons to get fit—looking and feeling great, avoiding disease and dysfunction, and so on—but it's important to discover *your* reasons.

What Are Your Fitness Whys?

One of the best parts about being fit is the occasional moment when you do something that makes you stop for a second and think, "Wow, it's awesome I was able to do that." These are experiences that make you beam, put a spring in your step, and sometimes even make your day. And I'm not referring to shallow wins like "turning heads in the gym," but "enjoying my vacation guilt-free," "playing with my kids without getting winded," and "being a good role model for my family."

I've worked with thousands of guys, and here are some of the fitness wins they've shared with me:

- Getting asked for advice in the gym
- Feeling strong and confident
- Looking great naked
- Being more productive
- Feeling more energetic
- Relishing "cheat meals"
- Surprising their doctor
- Loving how their clothes fit
- Setting a good example for their family
- Enjoying sports and outdoor activities
- Beating a disease or dysfunction
- Tackling a new physical hobby

I love these because they're simple, specific, and sincere slices of the good life—the real kind of reasons to get into outstanding shape. What would your list look like? Why do you want what you've just affirmed? Why are you committed

to your fitness? Brainstorm your reasons and write them down until you feel keyed up and ready to get into action, because soon, you'll be working to make them all a reality.

Now that we've contemplated your most personal reasons for getting into great shape, let's talk about the most visceral dimension of fitness: the visual.

What Does Your Ideal Body Look Like?

If you're like most people reading this book, you want to look a certain way. And there's nothing wrong with that. Just because you're motivated by the mirror doesn't mean you're a narcissist. To be sure, there are plenty of puffed-up fitness wackadoos out there, but if we can trade on our vanity a bit to help us feel fantastic, what's wrong with that? Especially when that energy flows to every aspect of our lives, enhancing our ability to love, work, and play. As "Prime Time" Deion Sanders said, "If you look good, you feel good. If you feel good, you play good. If you play good, they pay good."

If you could have it your way, what would your body look like? Words and daydreams are a good place to start, but let's go further and find a couple of pictures of the type of body you want. Maybe it's a movie star or athlete in their prime, like Gerard Butler in *300,* an influencer you follow online, or someone else. Once you've found your pictures, save them on your computer or phone or in the cloud for whenever you need a jolt of inspiration. Some people also like to print and stick them in their fitness journals.

Not sure what to choose because you're not sure what's possible? Start here: How would you have to look to hit the beach shirtless without a hint of insecurity? Find a couple of pictures of that look because that's something everyone can accomplish.

Why bother with pictures of your goal? Because it helps to know that every completed *Bigger Leaner Stronger* workout is a stepping stone to a physique that's as real as this book, not merely a figment of your imagination. What's more, when you take regular progress pictures of yourself (something we'll talk about later in this book), your excitement will mount as you witness your body approaching its prime.

What Does Your Ideal Body Feel Like?

One of the biggest benefits of this journey is the sublime pleasure of inhabiting a strong, fit body rather than a weak, disabled one. Getting fit produces an uprush of energy, mood, alertness, mental clarity, and physical resilience, to name just a few. And then there are the more profound experiences that await, like more intimacy, pride, purpose, satisfaction, and self-actualization.

Let's explore what this might be like for you in the form of affirmations—positive statements that describe desired states of being like "I'm always full of energy" and "My thinking is always quick, clear, and focused." I know it sounds a bit woo-woo, but research shows that creating and reviewing affirmations can benefit you in various ways, including enhancing exercise compliance, working memory, and cognitive performance.

There are four broad categories of health and fitness affirmations that I'd like you to address:

1. Physical. Physical affirmations concern physical energy, beauty, and function. They include statements like "I sleep well every night," "I'm rarely sick," and "I have no significant aches or pains."
2. Mental. Mental affirmations concern cognition—your ability to acquire knowledge and understanding through thinking and experiencing. This can include proclamations like "I focus deeply," "I learn quickly," and "I remember easily."
3. Emotional. Emotional affirmations relate to positive or negative sensations. They can include declarations like "My feelings don't control me," "I can always find inspiration," and "I bounce back easily from setbacks."
4. Spiritual. Spiritual affirmations involve your sense of purpose and

motivation. This can include pronouncements like "I'm always my most authentic self," "I'm always growing," "I make my destiny," and "I know I'll succeed."

And here are several tips for getting the most out of your affirmations:

- Keep them short. This makes them easier to process and remember. A sentence of four or five right words is more effective than a windy passage.
- Start with "I" or "My." Affirmations center on you, so that's where you should start. "I have no shoulder pain" is much better than "The pain in my shoulders has disappeared."
- Write in the present tense, not in the future. For example, "I enjoy my meals and never struggle with hunger or cravings" is superior to "I *will* enjoy my meals and never struggle with hunger or cravings" or "Within three months, I'm enjoying my meals and never struggling with hunger or cravings."
- Don't affirm "wanting" or "needing." You want to focus on being, doing, or having, not desiring.
- Make positive statements. This may require rejecting negative thoughts and attitudes, but your words shouldn't represent this. Think, "I'm disciplined and determined," not "I'm no longer undisciplined and unmotivated," or "I enjoy my training," instead of "I don't dread working out."
- Give your affirmations some emotion by including, "I'm [emotion] about ..." or "I feel [emotion]." For example, you could say "I'm excited to get a little fitter every day." This addition can make your ideas more exciting (thus impactful) and influence how you feel during whatever

you're affirming (for example, actually feeling excited when you go to bed because you know you got a little fitter that day).

- Make your claims believable. Unrealistic affirmations won't do much for you, so make sure you're formulating concepts that you can have faith in. If you're struggling, start with a qualifier like "I'm open to ..." or "I'm willing to believe I can ..."

Are you ready to start scribbling? Great! Begin with one affirmation per category (physical, mental, emotional, and spiritual), and revise them as much as you need to find concepts and phrasings that resonate deeply with you, sparking joy and positivity.

All right. You've formulated your first affirmations. What should you do with them? I like to read them first thing every morning and then revisit them whenever I feel down. The daily repetition reinforces my intentions, and the as-needed reviews help breathe wind back into my sails when I'm losing momentum. Feel free to add more affirmations as you go, too!

Review the work you've done in this chapter whenever you're dragging anchor or need a pick-me-up. Remember it when you're scanning the menu of your favorite restaurant, rubbernecking at the ice cream aisle, and rolling out of bed in the morning like a log off a truck. Return regularly to your personal "whys," physique goals, and affirmations, so when you encounter difficulties or setbacks, you may stagger but you won't stall. You'll always find a way to steady yourself and persevere toward your goals.

The process won't be a cakewalk, though. It'll take time. It'll take patience. It'll take resilience. You can't go from cauldron-bellied to washboard-abbed in

twenty days or bulge your arms or swell your chest in a couple of weeks. Upgrading your physique is a rewarding endeavor, but as I said earlier, you have to give a lot to get a lot.

That's why, in the next chapter, we'll explore the sticky subject of self-discipline. We'll learn how to avoid common willpower mistakes and pitfalls, and how to cultivate the character required to achieve outstanding fitness.

KEY TAKEAWAYS

- There are many reasons why people follow diet and training regimens—but it's important to discover *your* reasons.
- Find pictures of the type of body you want and save them on your computer or phone or in the cloud for whenever you need a jolt of inspiration.
- Imagine what your ideal body feels like for you, and then jot it down as affirmations, which are positive statements that describe a desired state.
- You can organize your health and fitness affirmations into four broad categories: physical, mental, emotional, and spiritual.
    - Physical affirmations concern physical energy, beauty, and function.
    - Mental affirmations concern cognition—your ability to acquire knowledge and understanding through thinking and experiencing.
    - Emotional affirmations relate to positive or negative sensations.
    - Spiritual affirmations involve your sense of purpose and motivation.
- One of the best ways to use the statements you've formulated is to read them every morning before you start your day or whenever you feel down.

The Big Ideas

1. Tackle temptations that sabotage your fitness.

2. Supercharge your ability to stick to your plan.

3. Manage stress properly to make it productive

5

The Art and Science of Upgrading Your Willpower

"Hard choices, easy life. Easy choices, hard life."

—JERZY GREGOREK

According to a 2010 survey conducted by the American Psychological Association, lack of willpower is the number one obstacle people face in achieving their goals. This makes many of them feel guilty about letting themselves and others down and that their lives are, in large part, not under their control. Rather, they feel that many of their actions are dictated by emotions, impulses, and cravings.

On the other hand, studies show that those with higher levels of willpower enjoy many advantages in life—they do better in school, earn more money, have better social and romantic relationships, and are generally happier, healthier, and less stressed.

It's clear, then, that more willpower always trumps less, just as strength of any kind—physical, emotional, spiritual—is always superior to weakness. Regardless of where we generally fall on the willpower spectrum, however, we all have challenges to face, especially those of us who want to get and stay fit. Some of these difficulties are biological—for example, the desire to eat greasy, sugary foods

that our brains recognize as vital to our survival—and others are more uniquely ours. What we find tempting, someone else might find repulsive. And their vices might be as appealing to us as airline food.

Whatever the details, the machinations are the same. Your excuse for skipping the gym—*again*—is remarkably similar to the foodie's justification for bingeing again. How you justify walking through a workout is how someone else eases the guilt of giving in to his cravings for nicotine. It would appear that the struggles of self-discipline are just part and parcel of being human.

Why is self-discipline such a heavy burden for some people though? Why do they give up so easily on goals? Why do they blissfully indulge in so many self-sabotaging behaviors? And what can be done about it? To answer these questions, let's endeavor to better understand the nature of this willpower-devouring beast and how to successfully tame it.

When we say someone has or lacks willpower, we're usually referring to their ability to say "no." They're supposed to study for the exam, but instead accept the invitation to the movies. They're trying to follow their meal plan, but just couldn't resist the apple pie. In short, they have trouble saying "I won't."

There are two other aspects to willpower, though: "I will" and "I want." "I will" power is the other side of the "I won't" coin—it's the ability to do something even when you don't want to, like grinding through a workout, paying the overdue bill, or burning the midnight oil. And "I want" power is the ability to remember the *why* when impulse strikes—the long-term goal or thing you really want—more than the fast food or night on the couch.

If you can win more battles than you lose with your won'ts, wills, and wants, then you can become the master of your destiny. Procrastination licked. Worst habits replaced. Temptations banished.

This ability takes work, though. A real willpower challenge isn't a fleeting,

"wouldn't that be nice" fancy that disappears as quickly as it came. It's more like an all-consuming battle between good and evil raging inside your skull. You feel it physically because when your brain fixates on a promise of reward, it gets bathed in a chemical called *dopamine*, and suddenly, all you can hear is the sweet song of immediate gratification.

Dopamine is a devious little devil too, because it's not designed to make you feel happy and content. Its role is to stimulate you to action, which it achieves by arousing you, sharpening your focus, and revving your drive toward the prize. Furthermore, when dopamine is released, it also triggers the release of anxiety-inducing stress hormones. So the more you think about the thing you want, the more important it becomes, and the more you want it *right now*.

What we don't realize, however, is that this stress we feel isn't caused by not having the dessert, pair of shoes, or Candy Crush trophy. The desire itself causes the stress, like an emotional stick compelling us to obey. Our brain doesn't give a damn about the bigger picture, either. It cares nothing about how we'll feel when we're twenty pounds heavier or a few hundred dollars poorer—it simply identifies sources of pleasure and then raises hell until we give in, even if pursuing those promises will entail risky, chaotic behavior likely to cause more problems than they're worth. Ironically, the rewards that we so ardently desire can escape us time and again, but if there's even an outside chance to finally secure them, we can remain hooked, sometimes to the point of obsession.

Another problem we have to contend with is the *halo effect*. Research shows that the dopamine release triggered by the promise of one type of reward to make us more likely to pursue others. For example, when you look at erotic images, you may be more likely to make risky financial decisions, and if you dream about striking it rich, food can suddenly become very appetizing.

This reward-seeking behavior is especially touchy in today's modern world—

one with many honeypots explicitly engineered to keep us wanting more. Retailers design our in-store experiences to entice us to buy—from what we see when we enter to the arrangement of the aisles, the stocking of shelves, the scents pumped in the air, the free samples offered, the tempo of the background music, and more. Food scientists test hundreds of variations of products to find their "bliss points"—the precise amounts of key ingredients like sugar, salt, and fat needed to produce a blaze of delight in every bite. Social media makers carefully craft algorithms to drown your neurons in dopamine, which explains a lot of the addictive behavior seen among social media users.

When we're constantly awash in dopamine, it's all too easy to feel like one big itch that always needs scratching. And when we consider our neurons, overtargeted and overstimulated by the incessant and inescapable calls for consumption, it's no surprise that the average person is an overweight procrastinator hooked on junk food, entertainment, and social media. Therefore, it takes a rather dramatic shift away from "normal" behavior to avoid these traps and succeed in this brave new world (sorry, had to).

First, we must learn to distinguish between the many toxic rewards dangled in front of us and the genuine ones that produce true fulfillment and satisfaction. Imagine you're in line at your favorite burger joint, craving a thousand-calorie heap of greasy meat, cheese, and bun. The fog clears for a moment, however, and you remember that you're on a diet. Losing the weight matters too. You want to be fit, healthy, and happy. You swore on everything sacred (with a nod to the profane) that you'd see it through this time.

When viewed in that context, the cheeseburger you're about to eat poses a threat to you. It can't force itself down your throat with a few handfuls of french fries in tow, though. It needs your cooperation. So, in this sense, you're the threat. Self-sabotage is the enemy, not ground beef and potato sticks.

How can we tame this wily foe lurking in the murky recesses of our psyche, then? Let's find out.

Don't Fight the Wave—Ride It

You've just hit the couch after a long, tiring day, and your mind begins to wander. Suddenly, a glowing pint of ice cream materializes, and your salivary glands snap to attention. *No*, you say to yourself, *anything but ice cream! Don't think about ice cream!*

The commands don't work, though. The harder you try to banish the vision of the cold, creamy dessert, the more it dominates your consciousness and grips your nervous system. Finally, the only way to extinguish the alarm bells appears to be spooning the stuff down your gullet.

The problem in this scenario isn't the fall into the quicksand, but rather the flailing attempt to escape. Fortunately, there's another way to deal with such an emergency. Research shows that a willingness to think thoughts and feel feelings without acting on them is an effective method of dealing with a wide variety of challenges, such as mood disorders, food cravings, and addiction. On the other hand, trying to suppress negative thoughts and feelings, like self-criticism, worries, sadness, or cravings, can lead to greater feelings of inadequacy, anxiety, and depression, and even overeating.

In other words, instead of "fighting the urge" to do something we know we shouldn't, we should try to"ride the wave" until it crests and peters out, when cooler heads can prevail. This doesn't have to be as difficult as it sounds, either. When a disturbing thought or longing appears, just face it calmly instead of trying to eject it from your mind. You don't have to contemplate its meaning or read into it. You just have to accept that it's there and play it down as something unimportant that'll fade.

"Riding the wave" is particularly relevant to dieting, as research shows that the more you try to suppress thoughts about food, the more likely you are to

struggle with cravings and binge eating. A much better strategy for dealing with hunger and cravings can be found in a study published in the journal *Psychology of Addictive Behaviors* to help people smoke fewer cigarettes. The technique goes like this: first, notice and accept the undesirable feelings; then, remind yourself that while you may not always be able to control where your mind and emotions wander, you can always control how you respond; and finally, before indulging in whatever has your eye, remember the goal at stake and why you're committed to it.

A simple way to implement this tactic is to introduce a mandatory ten-minute wait time before you allow yourself to act on a craving or other impulsive urge. This may not seem like much time, but research conducted by scientists at Princeton University shows that it can make a big difference in how you perceive the situation. The wait not only gives you time to pause and reflect on the matter, but it also tarnishes the allure of immediate gratification. By pushing the reward just ten minutes into the future, you can disarm its most effective weapon for undermining your intentions—urgency.

This approach can help you overcome "I will" challenges as well. If you're dreading something you know you should do, commit to doing it for ten minutes, and then decide whether to continue. Chances are, you'll find that once you're in motion, you'll want to keep going.

Do It for (Future) You

"I want the works
I want the whole works
Presents and prizes and sweets and surprises
Of all shapes and sizes
And now
Don't care how, I want it now
Don't care how, I want it now"

Those words were sung by Veruca in *Willy Wonka and the Chocolate Factory*, and if you listen closely, you can hear millions of people softly singing along every day. They want it all, and they want it now. Food, entertainment, money, love, abs—you name it.

Here's the rub: The longer we have to wait for rewards, the less desirable they become. Economists call this our *time preference*—how much we "discount" a reward's value based on how long it will take to receive it. People with a high time preference discount heavily, caring far more about immediate satisfaction than long-term payoffs or consequences. Those with a low time preference, however, exhibit the opposite behaviors, caring more about future pleasure and choosing to forgo immediate benefits to maximize it.

Some people discount the future more steeply than others, and studies show that the higher someone's time preference is, the worse their self-control will be. In many ways, time preference helps explain why consumer credit card debt is soaring, why fast-food joints are far more profitable than gyms, and why millions of people still light up cigarettes every day. When it comes to the crunch, one bird—or Big Mac—in the hand is often worth many more in the bush.

Like most positive and meaningful things in life, fitness is very much an exercise (heh heh) in time preference. How much discomfort are you willing to endure to enjoy the fruits of a bigger, leaner, and stronger body? How many occasions to eat more and move less are you willing to forfeit in pursuit of larger, delayed opportunities? How well can you keep your eyes on the horizon?

One powerful way to lower time preference and increase the chances for long-term success is to change how we view the nature of today's and tomorrow's rewards. For example, we often don't want to lose something we have, even if we stand to gain something of greater value later. Psychologists refer to this as *loss aversion*, a trap many of us have fallen into when dieting. When we're hovering over a doughnut, it sure seems a lot more desirable in the moment than reaching our weight loss goal a little sooner at some indistinct point in the future. We can turn this psychological quirk to our advantage, however, by reframing willpower challenges. Research shows that before surrendering to desire, if we think about the future outcome at stake and how giving in now will sacrifice progress toward it, we can lower the chances of acting against our best long-term interests.

So when you face the prospect of putting down a pile of piping hot pizza with all your favorite toppings—for the third time that week—take a moment to vividly imagine having achieved your ideal body composition. Really *feel* your rippling abs, bulging biceps, and powerful chest. Envision how proud you are of your new physique when you step out of the shower and look in the mirror. See how well your clothes now fit you. Then think about how digging into the pizza will mean trading all of that for a bloated and bulging body. Chances are the big glutenous grease wheel will suddenly look a lot less appetizing.

You can also institute the mandatory ten-minute wait mentioned earlier. If ten minutes of waiting (and visualizing the long-term desire) isn't enough to squash the desire to indulge, then give in, but not before the ten-minute mark.

"But wait," you might be wondering, "I thought you said I could get into great shape eating pizza?" You can, but as you'll soon learn, it'll need to be the exception, not the rule. You won't *always* have to say no to pizza, but there will likely be times when you do if you want to stay on track.

Another highly effective strategy to change your time preference is called *precommitment*, which entails taking action now to strengthen your dedication to a behavior and ward off any seductions to stray. Much like how Odysseus instructed his men to lash him to the mast to ensure he didn't fall prey to the songs of the Sirens, you too can put safeguards in place, making it much harder to fail. These defenses consist of anything you can do today that makes it difficult and uncomfortable to change your mind tomorrow.

For example, if you have trouble with procrastinating on the Internet instead of working, you can download a program called Cold Turkey (www.getcoldturkey.com) that allows you to block specific websites and applications, or turn your Internet off altogether, for as long as you'd like. If sticking to a meal plan is your struggle, you can precommit by throwing out all junk food in the house and not rebuying it, preparing healthy lunches to bring to work every day, or putting money on the line at a website like www.dietbet.com. If you want to precommit to exercising regularly, you can pay for an annual gym membership instead of going month to month, sign up for an online coaching service, or recruit a friend to work out with you.

Another tool that has helped many thousands of people precommit successfully to all kinds of goals is the website www.stickk.com. Stickk allows you to set a goal and time frame, wager money, and decide what happens with it if you fail. (It could go to a charity, for example, or even an organization you don't like, which can be a stronger incentive.)

Let's All Get Fat and Jump Off Bridges

According to several studies, just 10 percent of people eat enough fruits and vegetables to meet their body's most basic nutritional needs, and just 20 percent exercise enough to preserve their general health and wellbeing. What do they do instead? Thirty-two percent eat fast food at least one-to-three times per week, 70 percent eat more sugar than they should, and 30 percent would rather do housework, file their tax returns, or clean out their garages than give time over to working out.

Statistics like these are supposed to "scare us straight," but they can actually have the exact opposite effect because they remind us that "everyone else is doing it too." And when that's the case, how wrong can it really be?

Even if you're not inclined to think this way, don't assume you're immune to that turn of mind. It's comforting to believe that *we're* not like that, that *we* singularly chart our own course in a vacuum, but this simply isn't true. Extensive research shows that what others do—and even what we *think* they do—has a marked effect on our thoughts and behaviors whether we realize it or not, especially when the people we're observing are close to us.

When we're not sure how to think or act, we tend to look at how other people think and act and follow along, even if subconsciously. We're all instinctively drawn to the allure of safety in numbers. In the marketing world, this effect is known as "social proof," and it's used in myriad ways to sway us. This is why customer reviews and testimonials are vital to every business ("if all these people say it's good, it must be good!"), why companies pay exorbitant sums to secure celebrity endorsements ("if David Beckham likes it, it must be good!"), and why media mentions are so powerful ("if *Forbes* has featured it, it must be good!").

This effect also occurs naturally in our everyday lives. Whenever we tell ourselves that actions are acceptable because other people are doing them too or because of how "normal" they are, we're appealing to social proof. Research conducted by Dartmouth College scientists suggests that we can pick up anything from temporary solutions to long-term habits this way, and they can come from people we do and don't know, including even characters we see in movies.

The reality? Mindsets and modi operandi are far more contagious than most of us realize. Studies show that ...

- Those with obese friends and family members are far more likely to become obese themselves.
- The more a student believes that their peers cheat on tests, the more likely the student is to cheat (even if they're wrong).
- The more someone believes that others underreport income on their taxes, the more likely they are to cheat the IRS themselves.
- The more teetotalers hang out with people who drink alcohol, the more likely they are to start drinking too.
- The more nonsmokers hang out with smokers, the more likely nonsmokers are to start puffing away.
- The more time people spend with those who feel lonely, the more likely they are to feel lonely themselves.

It gets worse, too, because even if overeating, drinking, or smoking isn't your thing, seeing others indulge in these vices can encourage you to give in to other impulses as well. Seeing someone overspend might subconsciously trigger you to overeat. Hearing about someone skipping work might help you justify skipping your workout. Reading about someone cheating on their partner might make it

more acceptable for you to cheat on your diet.

You may have heard that we're the average of the five people we spend the most time with, and an abundance of scientific evidence confirms this to one degree or another. Even if we don't directly adopt the destructive outlooks and activities of those we're around most, the mere presence of these people can influence us in insidious ways, making it harder to do the things we want to do and become the people we want to be.

Remember this if you're just beginning your fitness journey because such "toxic" people are always the first to criticize and question it (and anything positive you might want to do, for that matter). I have a hardline policy for these types of folk—an abrupt and decisive parting of ways—but whatever you do, just don't let them run you into the sand. If you can't avoid contact with such people (coworkers, bosses, relatives, etc.), you can at least defang them by minimizing your interactions and offering them pleasantries over personal matters.

Fortunately, positive attitudes and behaviors are contagious as well, so if we surround ourselves with people who are generally upbeat, uplifting, and possessed of higher than average willpower and self-control, we too can "catch" these traits. Moreover, research shows that simply *thinking* about people with high levels of discipline can temporarily increase our willpower.

Whether you tend to struggle with diet and exercise compliance or not, you can make your life easier by doing three things:

1. Limit your exposure to instances of people failing willpower challenges. That said, while this is generally good advice, watching others lose control can *increase* resolve in some people by prompting them to view such lapses as threats to long-term goals, not as tempting invitations to follow suit. I happen to be one of these types of people. Other people's

flops make me want to work harder to succeed, not falter or fold up. In fact, I view much of what most people consider "normal" as highly negative and undesirable—so much so that a reliable compass for my decision-making is to first ask what "most" people would do in a situation and then consider the merits of doing the exact opposite.

2. Connect with at least one person who's on the same path as you and making progress. You don't even have to physically take the trip together. Regular phone, email, or text check-ins can be enough to encourage each other's success.
3. Read or listen to stories about how others have gotten fit. I interview men and women on my podcast (www.muscleforlife.show) who have used my programs, products, and services to transform their bodies and lives, and every week, I hear from listeners who tell me how motivating they find these interviews because they reinforce my promise that everyone listening can do it too.

The Trap of Being "Good" and "Bad"

Have you ever praised yourself as "good" when you did what you needed to do or stayed strong? Or chided yourself as "bad" when you procrastinated or acted impulsively? Have you ever used virtuous behavior as permission to be sinful? Of course you have. Of course I have. Of course every human who has ever lived has.

Scientists call this phenomenon *moral licensing*, and it can powerfully undermine our self-control. When we assign moral values to our actions, they can feed our desire to simply feel good (enough) about ourselves, even while we compromise our long-term aims or harm others. By being good, we reckon, we "earn the right" to be a little (or a lot) bad. For example, after basking in the glow of a successful training and eating day, we may find ourselves oversleeping and overeating the next, still feeling righteous and in control.

Interestingly, "right" behaviors that can justify the "wrong" don't even have to be related. Studies show that shoppers who pass up a purchase are more likely to feel justified in splurging on indulgent foods; when reminded of their virtue, people tend to donate less to charity; and merely thinking about doing something good can increase the likelihood of immoral or excessive behavior.

In an even stranger feat of mental contortion, when some people imagine what they could've done but didn't, they feel righteous. For instance, after adding healthier items to its menus, McDonald's began selling more Big Macs than ever because, according to research conducted by scientists at Baruch College, the mere opportunity to eat healthily gave people some of the satisfaction of actually doing it, which in turn permitted them to choose the cheeseburger.

The moment we feel an ache for moral permission to stray from our goals or standards, it's all too easy to find that emotional green light. The irony, however, is that all these "licensed" harmful behaviors keep us from achieving what really

matters—a fit body, a robust mind, a long life, a balanced budget, a completed project. We're tricking ourselves into believing that squandering our health, money, time, effort, and opportunities is a "treat" to be "cherished," that self-subversion is okay so long as it *feels* okay.

To escape from this trap, we must first stop moralizing our behaviors. Instead of using fuzzy feelings of "right" and "wrong" and "good" and "bad" to guide our actions, we need to remember *why* we've committed to doing the hard things—like exercising, eating well, and getting enough sleep. We need to view these actions as independent steps necessary for achieving the results we desire, not as noble conduct to counterbalance our sins.

For our purposes here, remember that our goal isn't just effective training or on-target eating. It's looking and feeling fit, strong, and dynamic. It's the pride and satisfaction we feel after every workout. It's the surprise on people's faces when they haven't seen us in a while. It's the newfound intimacy in our love life. So bingeing on ice cream and skipping workouts aren't little "oopsies" that we can erase with the right thoughts. They're threats to our overarching objectives, and while they'll inevitably win some of the battles, we must ensure they don't win the war.

The Crystal Ball of Delusion

Another common psychological trick for abandoning self-control is to justify indulging now with plans for future prudence. For example, research shows that planning to exercise can increase the likelihood of cheating on a diet. In another study, making a future commitment to volunteer three hours per week for a charity doubled people's likelihood of choosing to buy an extravagance like designer jeans over a practical item like an identically priced vacuum cleaner. In another experiment, agreeing to tutor a fellow student decreased the amount of money that people donated to charity.

This type of thinking smacks of moral licensing, but it also introduces another critical problem: The assumption that we'll somehow make different decisions in the future. "Today, I'll eat more dessert," we may say, "but tomorrow, I'll stick to my diet." Or "Today, I'll skip my workout, but tomorrow, I'll train extra hard." Or "Today, I'll binge on my favorite TV shows, but I won't watch any for the rest of the week."

Such optimism would be reasonable if we knew we could implicitly count on ourselves to follow through. But we both know that's not how it goes. We simply give our future selves too much credit, counting on them to be able to do whatever we can't bring ourselves to do now. We're too quick to assume that we'll be more enthusiastic, energetic, willful, diligent, motivated, brave, upstanding—whatever—in a couple of days, weeks, or months, and often for no good reason in particular. When the future finally arrives, that principled, idealized version of ourselves is nowhere to be found, and the new burdens are more demanding than we previously imagined them to be. And so we put them off again, hoping that they'll rescue us next time. In this way, we freight Future Us with an impossible burden of tasks and responsibilities.

Remember—Future You isn't some abstraction whose emotions and desires will be radically different from Present You. When tomorrow comes, you'll likely find yourself in a similar state of mind, and thus, you'll likely respond in a similar way. This is why we can all benefit from improving our ability to connect our present actions with their future consequences. Scientists call this *future self-continuity*, and research shows the better we are at this, the easier it is to get and stay in shape and do the many other creative, constructive things we want to accomplish. Studies show that there are also several simple ways to improve our future self-continuity:

1. You can think about how you'll behave in the future. Just thinking about the future—not even the rewards per se—can strengthen your willpower. Specifically, by imagining yourself in the future, doing what you should be doing or refraining from what you shouldn't be doing, you can increase your chances of success. For example, if you're struggling to stick to a meal plan, you can imagine yourself grocery shopping and eating differently; or if you're dreading the next day's workout, you can envision yourself getting it done despite how you feel. If you're anxious about an upcoming holiday party where you're going to be surrounded by your favorite foods, you can picture yourself having a good time without being excessive.
2. You can write a letter to Future You. In this letter, you should write about what you think Future You will be like, what your hopes for them are, what you're doing for them now that'll pay off later, what they might say about Present You, and even what the consequences of failing now will mean for them down the line. If you want to make this more fun, use the website www.futureme.org, which allows you to

write an email to yourself and choose a future date on which it'll be delivered.

3. You can imagine Future You in vivid detail. To do this, explore the future consequences of your current lifestyle, both good and bad. What will your Future Self look like if you don't commit to stopping the things you know you shouldn't be doing and starting the ones you should? What are the likely physical, mental, and emotional consequences? Disease, regret, shame, ugliness, depression, and loneliness? Don't hold back. Then, consider what'll happen if you succeed in your endeavors. How will you look and feel? Will you be lean, strong, and healthy? Will you feel fulfilled, excited, and thankful? How might these developments affect your life? Will you get more work done and advance your career faster? Will you enjoy more activities with your kids? Will you enjoy your romantic relationships more? Again, explore the possibilities. If you want a comprehensive, evidence-based, and guided method for doing this, check out the Self Authoring Program (www.selfauthoring.com) from Dr. Jordan Peterson.

Like with physical exercise, the more you do these three drills, the stronger your future self-continuity will become, and the better you'll be able to deal with willpower challenges of all kinds, fitness-related and otherwise. Let's start now, shall we?

First, let's write a letter to Future You and explore the angles listed above. Feel free to expand on them, too, following whichever seems most interesting or fruitful.

Next, let's envision Future You as described.

"Oh, Why Not? I'm a Lazy Idiot Anyway!"

What do people tend to do after a relatively minor misstep, like missing a workout or eating too many sweets? Do they shrug it off and move on, or do they berate themselves, catastrophize the affair, and lose the plot?

Unfortunately, the latter is far more common. For many, the vicious cycle of "stumble, sulk, and splurge"—called the *what-the-hell effect* by psychologists—seems inevitable and inescapable. This is how the extra handful of chips can become the whole bag, the nibbles of chocolate can continue until the entire bar is gone, and the glass of wine can be a prelude to the bottle ... or two. Whenever people confronted with a setback say to themselves, "I've already messed up, so what the hell," they're embracing this mentality. They give in, and then, to feel better, they indulge in earnest, which often triggers even worse feelings of shame and regret, which can lead to even greater comedowns.

Well, I have good news for you: You're going to make mistakes on the *Bigger Leaner Stronger* program. You're going to eat too much at parties, stay in bed instead of going to the gym, and give less than 100 percent sometimes. Why is that good news? Because like most everything in life, you don't need to be anywhere near perfect to win in the fitness game. You just have to be good enough most of the time. Perfectionism isn't required, nor is it even desirable, because it often makes the whole process more stressful than it needs to be.

So don't dump on yourself when you mess up. The "damage" is never as bad as you think, and an abusive tirade of self-criticism only makes things worse. For example, many people worry that they've "blown" their diet after a single evening or day of overeating, not realizing that the absolute amount of fat that they can gain from a single meal or day—no matter how much they've eaten—ranges from negligible (a few ounces) to mildly irritating (0.5-to-1 pound).

Therefore, when you goof (and you will), show yourself the same compassion and forgiveness that you'd show a friend. Research shows that this type of response in times of frustration and failure is associated with better willpower and self-control because it helps us accept responsibility for our actions and steam ahead, unfazed.

Nothing Fails Like Success

Once we've set our sights on a goal, what do we crave most? Progress, of course. We want to see positive change and forward movement, which, we hope, will inspire us to keep going.

But that's not how it always goes. Progress can cut both ways. The satisfaction it produces can become complacency, a powerful catalyst for weakening willpower. Instead of reinvigorating us for another charge into the breach, progress can lull us into following one step forward with two back.

This paradox has been demonstrated in a number of studies. For example, scientists at the University of Chicago found that when people were led to believe they were closing in on their weight loss goals, they were 32 percent more likely to choose a chocolate bar for a snack over an apple. I've seen this many times. All too often, people use weight loss progress as an excuse to loosen the dietary reins and hinder further progress.

How can we guard against the slackening effects of success? According to another study conducted by the same University of Chicago scientists, we should avoid getting into the habit of flattering ourselves for all the work we've done. Instead, we should view our wins as evidence of how important our goals are to us and how committed we are to seeing them through. That is, we should discipline ourselves to keep our eyes on the road, not slow down and take in the scenery.

This has been one of my personal "secrets" to success inside and outside the gym. I've always remained more focused on what I still have to do to realize the future I want for myself and my family than on how far we've already come. I've strived to embrace doing and becoming (present and future) over arriving and having (past) and to maintain respect for processes and fundamentals. This philosophy has certainly increased the stress quotient in my life, but the payoff

has been well worth it, and I don't just mean that in a financial sense. In reality, the nonfinancial rewards, which can be summed up in one word—self-actualization—mean a lot more to me than the money.

Moreover, while you may think that this lesson has merely to do with work ethic or persistence, it's deeper than that. Many people think success is like a utopia—an endless beach full of beautiful people frolicking over the golden sands and foamy shores. This is a fantasy. The beach is private and pretty, but it's also a minefield. Navigate it skillfully, and you get to stay and enjoy its luxuries. Make too many false moves, however, and it'll blow you to pieces. The corpses are buried, but they're there, just below the surface—the bodies of the people who committed the cardinal sin: forgetting their broom.

The broom symbolizes the willingness to do any and all work with dignity and pride. Andrew Carnegie, one of the greatest "rags to riches" stories of all time, revered the broom:

> "But if by chance the professional sweeper is absent any morning, the boy who has the genius of the future partner in him will not hesitate to try his hand at the broom.
>
> ...
>
> "I was one of those sweepers myself, and who do you suppose were my fellow sweepers? David McCargo, now superintendent of the Allegheny Valley Railroad; Robert Pitcairn, Superintendent of the Pennsylvania Railroad, and Mr. Moreland, City Attorney."

Whether we're pursuing fitness, finances, or some other fire in our belly, the broom is the key to it all. Lose our broom, or our respect for it, and we're done for.

I know people who learned this in the hardest ways. One guy lost $70 million (cash) in five years flat, one bad investment at a time. All the while, his broom was collecting dust in the closet. Another guy burned through $13 million in legal fees in a misguided and unsuccessful corporate revenge mission. He traded his broom for a club. Yet another went from a humble business owner and millionaire to a megalomaniac with nothing but burned bridges and millions in debt. He now despises the broom and everything it stands for.

The broom also represents the humble appreciation of the challenges of the work at hand. Frank Gehry, one of the most celebrated architects of all time, still values this second lesson of the broom, even after decades of acclaimed work:

> "For me, every day is a new thing. I approach each project with a new insecurity, almost like the first project I ever did. And I get the sweats. I go in and start working, I'm not sure where I'm going. If I knew where I was going I wouldn't do it."

Consider the fact that the man who said that has received over a dozen honorary degrees from various universities, including Harvard, Yale, and Princeton, and has designed some of the most famous buildings in the world. Vanity Fair called him the "most important architect of our age." And every day, what does such an accomplished person do? He picks up his broom and starts sweeping as best he can. Shouldn't we do the same?

The Stress Paradox

Most people think of stress as purely negative, assuming that it should be avoided at all costs. This is mistaken. Our body was designed to handle stress, which is one of the reasons exercise is so beneficial to our health. For example, research shows that acute stress enhances immunity, which in turn accelerates recovery processes and increases resistance to infection.

However, the problems begin when we overdose on stress. Our bodies haven't learned how to effectively cope with *chronic* stress, which keeps us on high alert, accelerates aging, increases susceptibility to disease, and raises levels of systemic inflammation.

What's the solution? It isn't to shy away from or numb ourselves to all forms of stress, but to learn how to manage stress effectively. Two of the most powerful tools for this are rest and relaxation. In fact, it's not a stretch to say that the overall quality and longevity of our lives will depend heavily on how well we can relax. Think of this form of stress management as the "nutritional" aspect of your willpower and self-control. Minimizing the negative side effects of stress in your life not only helps you maintain a healthy level of self-possession, but also opens the door to improvement through "exercising" your discipline.

How do most people try to relax and cope with stress, though? What do they routinely turn to for consolation? Food, alcohol, video games, television, shopping, and surfing the Internet are the usual suspects, and ironically, studies show that the same people using these strategies often rate them as ineffective for reducing stress levels. In some cases, they actually increase stress levels because indulging unhealthy and unproductive impulses leads to guilt, followed by more indulging, followed by more guilt, and so on.

Comfort food is particularly problematic because it spikes blood sugar levels

to provide emotional reprieve, and then crashes them, which, like stress, is a precursor to willpower failures. Research shows that when blood sugar levels are low, we're more likely to give up on difficult tasks, vent our anger, stereotype others, and even refuse to donate to charity.

If we shouldn't turn to feel-good vices to feel better, what should we do instead? An effective way to recover from the stresses of the "daily grind" is to deliberately relax. That's not surprising, of course, but it's not necessarily simple either. For most people (myself included), it's a skill they have to learn.

A band-aid you can quickly apply when stressed is deliberately slowing your breathing to about 10-to-15 seconds per breath, or 4-to-6 breaths per minute. An easy way to do this is to exhale through your mouth slowly and fully with your lips pursed as if you were blowing lightly through a straw. As simple as this technique is, research shows that it can instantaneously boost your willpower and ability to deal with stress.

Relaxation shouldn't be something you "turn on" only when you feel frazzled, however. Studies show that taking time to properly relax every day not only reduces stress hormones and increases willpower but helps preserve health as well. For many people, that means an evening of wine and Netflix, but a much better choice is an activity that elicits a specific type of physiological response. When you're truly relaxed, your heart rate slows, your blood pressure drops, your muscles loosen, and your mind stops analyzing and planning. Everything just slows down.

Let's review thirteen simple and effective ways to enter this state.

1. You can change your perception of stress.

We know that high amounts of stress are associated with impaired mental and physical health and wellbeing, but there's a twist. Research shows that our

perception of stress as harmful is what really gives it teeth. Stated differently, getting overly stressed about stress is what really pushes us into the mud.

Studies show that if we can learn to consciously reappraise stressful situations (choose to look at them differently), however, we can drain them of much of their destructive power. For instance, a frustrating situation doesn't have to be an excuse to rip your hair out. Instead, you can view it as an opportunity to exercise a virtue like patience, tolerance, or resilience. Similarly, a setback is also an opportunity for you to learn what doesn't work, and a painful situation can teach you that you're tougher than you thought.

This reappraisal strategy is far from new. Marcus Aurelius, one of Rome's most famous emperors, had it right a couple of thousand years ago with his *Meditations* when he said the following:

By contrast, things outside our control have no ability to harm us. Acts of wrongdoing by a human agent (torture, theft, or other crimes) harm the agent, not the victim. Acts of nature such as fire, illness, or death can harm us only if we choose to see them as harmful.

. . .

Choose not to be harmed—and you won't feel harmed. Don't feel harmed—and you haven't been.

2. You can get more sleep.

If you sleep too little too regularly, you'll find yourself more susceptible to stress and temptation, lacking the mental energy needed to keep your good habits in play and your bad ones at bay. Indeed, research shows that sleep deprivation causes symptoms similar to ADHD: Distractibility, forgetfulness, impulsivity, poor planning, and hyperactivity. By sleeping enough, however, you can improve fat loss, muscle growth, immunity, longevity, cognition, and even physical

attractiveness.

Sleeping enough really is the ultimate "health hack," so, if you're sleeping less than 7 hours per night, you likely aren't sleeping enough, and if you get at least seven-to-eight hours of sleep per night, you'll immediately enhance just about every aspect of your wellbeing.

3. You can exercise regularly.

If you want a willpower quick fix, this is it. Nothing seems to improve self-control in all aspects of our lives like exercise. Studies show that regular exercise reduces cravings for both food and drugs, increases heart rate variability (a sign of reduced stress), makes us more resistant to stress and depression, and optimizes brain function.

Exercise's effects are immediate, too, and it doesn't even take a lot to boost willpower. For example, research shows that just five minutes of low-intensity exercise outdoors is enough to improve your mental state.

So, the next time you're feeling too tired or short on time to work out, remember the bigger picture. Every workout you do replenishes your willpower and energy, even lighter ones like walks and bodyweight training.

4. You can consume less media.

This shouldn't come as a surprise to anyone these days, but research shows that exposing yourself to a constant barrage of bad news, fearmongering, and morbid reminders of your mortality increases the likelihood of overeating, overspending, and other breakdowns of willpower. By reducing our exposure to the daily drip of pessimism and alarmism, we can also reduce our stress levels.

The maximalist would say to simply swear off media altogether and "go dark," and although an argument can be made for this, I think it's important to at least try to follow what's going on in the world, so I simply limit the time I give to

current affairs (15-to-20 minutes per day during work breaks, usually).

5. You can avoid screens before bedtime.

This is a follow-up tip to the last one because even if you get to bed on time, if you're spellbound by screens all evening, you can sleep poorly. Nighttime light exposure suppresses your body's production of *melatonin*, a hormone that induces sleepiness. This not only makes it harder to fall asleep but also impairs the quality of the sleep you do get.

Although ideal, completely eliminating all exposure to light once the sun goes down isn't exactly feasible. I doubt you're willing to "go dark" come 7 p.m. Fortunately, you don't have to because light's melatonin-suppressing effects depend on intensity. The more intense the light, the more it suppresses melatonin levels.

Thus, a good rule of thumb is to keep all sources of light after dark as dim as possible, to get away from screens altogether at least an hour before bed, and to sleep in complete darkness (and to wear a sleep mask if this isn't possible). If you want to go a step further, invest in a pair of blue-light blocking glasses. They look silly, but research shows wearing them in the hours before bed can improve sleep quality.

6. You can spend less time with tech.

While staring too much at screens at night messes up your melatonin production, it appears that staring at them too much in general can mess up your mind. A number of studies have shown that the more people use and feel tied to their computers and cell phones, the more stressed they generally feel.

In fact, a study published in the journal *BMC Public Health* found that technology overuse is associated with a number of symptoms of poor mental health. Here's a quick summary of their conclusions:

- People who used their cell phones heavily were more likely to complain of sleep disorders and depression.
- People constantly available on their cell phones were the most likely to experience mental health issues.
- People who regularly use the computer late at night were more likely to experience sleep disorders, stress, and depression.
- Frequent computer use without breaks increases the likelihood of stress, sleep problems, and depression.

Researchers aren't clear as to the causes just yet, but the relationship is unmistakable. The more time we spend with our devices, the worse our mental state will likely be.

If you suspect that you spend too much time on social media, email, games, and the rest, and want to see how disengaging from them can benefit your mind, mood, and manner of living, I have a "digital declutter" challenge for you that Cal Newport shares in his book *Digital Minimalism*:

1. First, take a thirty-day break from all non-essential technology, meaning anything you don't absolutely need to keep working and living on a day-to-day basis.
2. During this period, consider what's truly important to you. What are you interested in and what do you value and enjoy outside of the Internet? What should you spend your time doing instead of fiddling with electronics and why? For example, would exercising more make sense? Reading more books? Volunteering more? Learning a new language or instrument? Working more on your side hustle?
3. After thirty days, start reintroducing technologies, but only those that support things you deeply value and are the best ways to affirm those

ideas, experiences, or activities. For instance, you may deeply value your relationships with close friends and family, and while Instagram is one way of staying in touch, a video call once per week may be far more meaningful.

7. You can have more sex.

If you're not having much sex these days, science says you should get busy because it reduces anxiety, stress, and depression and improves mood, happiness, and resilience.

Interestingly, other research indicates that the stress-reducing effects of sex are largely due to improving the quality of people's relationships rather than just the physical release. This helps explain why sex's many benefits seem to be greatest for couples who have been together the longest or are married, and smallest for people in unhappy or uncommited relationships.

8. You can enjoy nice smells.

Aromatherapy is a couple-thousand-year-old method of reducing stress and promoting relaxation. It also has some modern scientific evidence on its side. For instance, scientists at Asia University found that the scent of certain essential oils, like lavender, bergamot, chamomile, and geranium, can lower blood pressure, reduce anxiety, and improve sleep quality.

There are several ways to incorporate this into a daily relaxation routine: a diffuser, candles or fragrance stones impregnated with essential oils, or a small ceramic disc sometimes called a stone diffuser.

9. You can give and get a massage.

I probably don't need to cite research to convince you that receiving a massage is a great way to relieve stress, but one rather interesting study published in the

Australian and New Zealand Journal of Psychiatry found that *giving* a massage has similar effects. That's right: make a deal with your significant other to trade massages before bed, and a whole lot of relaxation is probably going to occur.

The benefits of massage don't stop there, either. Studies show it also reduces pain, anxiety, and depression, and increases immunity as well. It also doesn't take a lot to get the job done—just 10-to-15 minutes of massage is enough to make a difference.

10. You can listen to classical music.

Next time you're stressed, put on some slow, quiet classical music, and before long, you'll be nestled in its soothing embrace. (Some of my personal favorites are Max Richter, Ludovico Einaudi, Beethoven, and Bach.)

Mozart can do more than just chill you out, too. Several studies show that classical music can sharpen your mind, engage your emotions, lower blood pressure, lessen physical pain and depression, and help you sleep better.

11. You can drink green tea.

I'm a big fan of tea and its many health benefits, and here's another reason to drink it regularly: It's a powerful stress buster. Researchers believe the primary way green tea accomplishes this is through its high doses of the amino acid L-theanine and ascorbic acid, which have known anti-stress properties.

As for which types of tea are best for de-stressing, green tea is the most well-researched, but studies show that chamomile, peppermint, lavender, and fennel tea can also help you unwind as well.

12. You can go for a walk (and ideally in nature).

When you review the daily routines of many of history's greatest thinkers and innovators, you'll quickly notice how many of them valued long walks in nature.

Beethoven spent his afternoons walking in the Vienna Woods and found his best inspiration always came while walking. Tchaikovsky was adamant about his twice-a-day walks, which he felt were essential for his health and creativity. Thomas Jefferson advised his nephew that "there is no habit you will value so much as that of walking far without fatigue," and took regular walks around his Monticello estate well into old age.

Science shows they were onto something. In a study conducted by Heriot-Watt University scientists, just 25 minutes of walking in an urban park was enough to noticeably reduce frustration and improve mood. In another study, published in the journal *Scientific Reports* with 20,000 participants, researchers found that those who spent at least two hours in nature every week were significantly happier and healthier than those who spent less time in nature. Finally, other research shows that immersion in and even viewing pictures of nature improves health outcomes for sick people, increases healing after surgery, and reduces blood pressure, resting heart rate, and all-cause mortality.

13. You can take a hot bath.

For thousands of years, hot baths have been used to ease pain, aid in relaxation, and ward off and treat disease. In fact, the word *spa* comes from the Latin *sanus per aquam*—"health through water"—which was an ancient Roman remedy for battle-weary soldiers.

Modern medical research has confirmed that regular dips in hot water are indeed healthful and restorative. Scientists at Loughborough University found that an hour-long soak in a hot tub causes improvements in blood sugar control and other markers of metabolic health similar to an hour-long bike ride.

Human nature is full of paradoxes, and willpower and self-control are no exceptions. We're drawn to both delayed and immediate gratification in the forms of long-term satisfaction and short-lived delight. We're inherently susceptible to urges but also have the power to resist them. We're often oscillating between contradictory emotions like frustration, anxiety, and doubt and happiness, calm, and certainty.

While we may or may not be able to fundamentally change our personality through strengthening our willpower, we certainly can improve our ability to conquer our day-to-day challenges and enjoy more mindfulness, effectiveness, and confidence.

What's more, fitness is a fantastic training ground for willpower and self-control because if you can summon the spirit to push through a punishing workout when every ounce of you wants to quit, to squash cravings when goodies are within arm's reach, and to stay committed to a fitness routine despite the vicissitudes of life, then chances are you'll also have what it takes to meet important work deadlines, live within your means, and realize your other ambitions for self-development and growth.

If you're to do all of that, however, you'll need more than a strong will. You'll also need an efficient system for channeling and amplifying that vital and limited energy so it can sustain a lifestyle that, as Earl Nightingale put it, allows you to progressively realize worthy goals or ideals. The fulcrum of such a system consists of habits, which is the topic of the next chapter.

KEY TAKEAWAYS

- "Riding the wave"—the willingness to think thoughts and feel feelings without acting on them—is an effective method of dealing with a wide variety of challenges, such as mood disorders, food cravings, and addiction.
- A successful strategy that's helped people deal with hunger and cravings involves three steps: notice and accept the undesirable feelings, remind yourself that while you may not always be able to control where your mind and emotions wander, you can always control how you respond, and remember the goal that's at stake and why you committed to it.
- A simple way to implement this tactic is to introduce a mandatory ten-minute wait time before you allow yourself to act on a craving or other impulsive urge.
- If you're dreading something you know you need to do, commit to doing it for 10 minutes and then deciding whether to continue.
- *Precommitment* entails taking action now to strengthen your dedication to a behavior and ward off any seductions to stray.
- You can make your life easier by limiting your exposure to instances of people failing willpower challenges, by joining forces with at least one person who's on the same path as you and making progress, and by reading or listening to stories about how others have gotten fit.
- You can improve your future self-continuity by thinking about how you'll behave in the future, writing a letter to Future You, and imagining Future You in vivid detail.
- Like most everything in life, you don't need to be anywhere near perfect to win in the fitness game—you just have to be good enough most of the

time.

- We should view our wins as evidence of how important our goals are to us and how committed we are to seeing them through, not as excuses to relax our efforts.
- Some simple and effective ways to deliberately relax are changing your perception of stress, getting more sleep, exercising regularly, consuming less media, avoiding screens before bed, spending less time with tech, having more sex, enjoying aromatherapy, giving and getting a massage, listening to classical music, drinking green tea, going for a walk (and ideally in nature), or taking a hot bath.

The Big Ideas

1. Understand the anatomy of habits.
2. Hijack the habit loop to beat bad habits.
3. Build better habits in three simple steps.

6

The Easy Way to Build Good Habits and Break Bad Ones

The chains of habit are generally too small to be felt until they are too strong to be broken.

—SAMUEL JOHNSON

When you woke up today, what did you do? Did you go straight to the shower? Did you check your email first, or Facebook and then email? When did you brush your teeth? Which shoe did you tie first—the right or left? What route did you drive to work? What did you do once you got to your desk?

If you take a moment to reflect on these questions, you'll quickly understand why the cliché that we humans are creatures of habit exists: We all have ingrained patterns of behavior. In fact, according to a 2006 study conducted by Duke University, over 40 percent of people's daily actions don't stem from conscious decisions, but unconscious conventions.

In many cases, this mechanism is useful because it saves us mental energy. We don't need to decide newly each day which shampoo to use or time to go to bed. However, other habits that develop without our "permission" can be quite troublesome. For example, research shows that many families that eat fast food

regularly didn't originally intend to eat as much as they do. The monthly habit eventually became a weekly one, which eventually became a biweekly one, until finally they were eating junk food every day. Similarly, thirty minutes of TV per day can become sixty, and then one hundred. Skipping one workout per week can lead to skipping twice per week, and ultimately culminate in quitting. The occasional drink or two can turn into the regular nightcap.

Thankfully, the power of habits can be applied to positive outcomes, too. As Dan John shares in his book *Attempts*, a little bit of the right things, like exercise, often enough, over the long haul, can produce greatness. Forty-five minutes of exercise several days per week, if done long enough, can transform your body and health. Thirty minutes of reading per day, over time, can make you into an expert in just about anything. A few hours of focused and uninterrupted "deep work" per day, over a career, can produce a legacy.

Harnessing our habits is difficult, though. Our biological hardware isn't oriented toward mechanically repeating the same actions countless times until we've achieved something extraordinary. It's drawn to cheap thrills and primrose paths. Fortunately, however, with the right practices and plenty of patience and perseverance, we can gain control of our habits and ensure they're nudging us toward, and not away from, our long-term fitness and life goals.

You can read entire books on this topic (*The Power of Habit* is a good primer), but in this chapter, I'll give you a crash course on building good habits and breaking bad ones, based on the most important and effective information and evidence currently available.

The Anatomy of Habits

A habit is an automatic behavior or response that's triggered by a situation, person, or environment. You probably know habits are formed through repetition, but there's also a specific pattern in play that goes like this:

1. Cue—We're cued (prompted) by a time, location, event, situation, emotion, person, smell, etc.
2. Response-—We perform an action or routine in response.
3. Reward—We feel or receive some sort of reward.

We experience this "habit loop" in many ways, and often, we move through it automatically, regardless of motivation, stress levels, attention, or other factors. A simple example of it is walking into a dark room, feeling the desire to turn on the light to feel safe or simply be able to see better (cue), flicking the light switch (response), and feeling relieved and comforted by the ability to see our surroundings (reward).

The more we go through this process in a particular way, the more automatic it becomes, until it's a default path of almost no resistance that requires very little cognitive effort to complete. This mechanism is useful when driving, let's say, because our reflexive ability to operate our car allows us to also safely have a conversation on the phone or listen to a song, but it becomes detrimental when passing a fast food joint on the way home stirs a craving for greasy grub that sends us veering off the road (and our meal plan) to dig in.

Simply understanding this three-part model of habit formation and continuation is the first step to mastering it. Next is learning how to commandeer this system and use it to consciously break bad habits, and create and reinforce good ones.

Beating Bad Habits

As Charles Duhigg explains in *The Power of Habit*, with time and effort, almost any habit can be reshaped by doing the following:

1. Identify the routine
2. Experiment with rewards
3. Isolate the cue
4. Have a plan

Let's review each of these actions in more detail.

1. Identify the Routine

This is usually simple enough because the routine is the behavior you want to change. It could be skipping breakfast and buying a donut at work, staying up too late at night on your phone, or heading to the couch after dinner instead of going for a walk.

2. Experiment with Rewards

Rewards are powerful drivers of behavior because they satisfy desires and cravings, but they aren't always obvious. Do you want the morning donut solely because it's delicious? Or are you just as driven by the social interaction or the temporary boost in energy you feel afterward?

Such subtleties can be difficult to tease out through reflection alone, but with simple experimentation, we can discover and exploit them for more constructive ends. This process may take some time—several days or longer—so have patience and don't pressure yourself to make any meaningful changes just yet. Instead, think of yourself as a scientist observing your behavior and collecting data.

To begin, when you feel the urge to do a routine action, try changing the action, the reward, or both. Do you want the donut, or do you just want something sweet? Maybe you're just hungry and want some food? Or is it more about mingling before heading to your desk to start working?

To find out, you could test a few different routines when you feel the urge to get a pastry: One day, you could buy a different sweet food like a cookie or candy bar and eat that instead. The next day, you could choose something nutritious, like an apple. The following morning, you could try a cup of coffee instead. Finally, you could see how skipping food and beverages altogether and just stopping by your friend's office impacts your habit loop.

Exactly what you do instead of eating the donut isn't important, so long as you're testing a plausible theory about what's causing the donut dilemma. If it's the desire for something sweet, the cookie should satisfy. If it's just hunger, the apple should take the edge off. If it's a spark of energy, coffee will work nicely. And if hobnobbing is the culprit, the office drop-in should suffice.

As you test different rewards, Duhigg recommends that following each variation of routine or reward, you should write down the first three things that come to mind. They can be feelings, random thoughts, thoughts about how you're feeling, or even just words that spring to mind. Then, set an alarm for fifteen minutes, and when it's up, see whether you still feel an urge for the original reward.

It's important to write down three things after the routine because it forces you to focus on what you're thinking and feeling at that crucial moment of consummation, and it helps you to remember exactly what happened when you're done with a round of experimentation and seeing what you can glean from the experience. As for the 15-minute wait, this helps you determine whether you've pinpointed the proper reward. If, as in the curious case of the morning pastry,

fifteen minutes have passed after eating an apple and you still want the donut, you can assume the craving isn't motivated by the need for calories. If chinwagging with a colleague doesn't hush the hankering, then the reward isn't the need for interaction either. On the other hand, if after a cup of coffee, you find it easy to get your day started, then you've identified the reward underlying the loop—a boost of energy—as well as a frictionless way to eat fewer calories.

By testing different rewards, you can isolate your true cravings, which is critical in reconstructing a habit.

3. Isolate the Cue

Many habits can be more easily adjusted when you understand the cues that trigger them, but such prompts are often difficult to identify because there's just too much going on. Are we skipping our weekend workouts because we're tired? Or because we're at home instead of at the office? Or because we feel lazy and just want to relax? Or because we're with our family and don't want to leave? Or all of them together?

To help find the signal among the noise, we can use a simple system of organizing cues into five categories and looking for patterns in our experiences:

1. Location. Have you ever walked into your kitchen, seen a bag of chips on the counter, and eaten a few handfuls just because they were there? Then you know the sway location can have over our behavior. In fact, many habits derive much of their persistence almost solely from the environments they're performed in. Often, changing the environment (keeping the chips in the pantry, or better, not buying them in the first place) is the easiest way to discourage or encourage specific behaviors.
2. Time. Many of our daily actions are dictated by the clock. We wake up, go to the bathroom, take a shower, brush our teeth, etc. We stop

working midmorning to mosey on over to the kitchen or cafeteria. We hit the couch after dinner and plug into one of a dozen streaming services vying for our attention. Sometimes it's not the time that matters per se, but what happens or how we feel at that time—two other cues we'll discuss below.

3. Emotional state. Feelings are powerful drivers of behavior, and for most of us, this cue appears more commonly in bad than good habits. Many people say they eat, shop, or scroll on social media when they feel bored or depressed, but not many report showing up for morning workouts or abstaining from late-night snacking because of their feelings. Instead, we often perform good habits *despite* how we feel.
4. Other people. As we discussed in the previous chapter, the people we surround ourselves with can have a marked effect on how we think, feel, and behave. Put simply: If you spend too much time with people who motivate you to do things you know you shouldn't, like eating or drinking too much, sleeping and moving too little, and so forth, making substantial changes in your life will be extremely difficult.
5. Immediately preceding action or event. Many habitual actions are responses to other things that happen, like a phone, social media, or email notification, a discussion or argument, or a person arriving or leaving. Understanding this can help you untangle unwanted habits. For instance, you might turn off all app notifications to undercut the habit of checking your phone too often, or avoid drinking alcohol while eating to work on the habit of eating and drinking too much.

There are cues that don't fit neatly into these buckets (thoughts, smells, and hunger), but research shows that most habits are prompted by the five

phenomena listed above.

So, if you're trying to figure out the cue for late-night snacking, you'd write down answers to the following five questions when the urge strikes:

- Where are you? (on the couch)
- What time is it? (10:45 p.m.)
- What's your emotional state? (annoyed)
- Who's with you? (nobody)
- What action or event preceded the desire? (scrolling through Instagram)

The next time:

- Where are you? (in bed)
- What time is it? (10:35 p.m.)
- What's your emotional state? (bored)
- Who's with you? (girlfriend)
- What action or event preceded the desire? (watching TV)

The third instance:

- Where are you? (kitchen)
- What time is it? (10:22 p.m.)
- What's your emotional state? (fatigued)
- Who else is around? (nobody)
- What action or event preceded the urge? (texting with a friend)

This reconnaissance could uncover many potential patterns, but in the case of night eating, a common cue is simply staying up too late. The reason for this is

threefold: it gives you more time to get hungry again after dinner; it makes you more likely to undersleep, which generally increases appetite; and it usually involves spending more time doing things associated with eating (like screentime).

4. Have a Plan

Once you've identified the reward and cue for a routine (habit) you want to change, you can rewire it to serve your interests by following the right plan.

It starts with the cue. If you can't or don't want to eliminate it altogether, you must find a new way to respond to it that leads to the reward you crave. If lethargy is the cue driving the habit of eating a morning donut and energy is the reward of this loop, then sticking with the cup of coffee may be all the plan you need. "When I get to work and feel sluggish, I'll drink a cup of coffee to get a boost of energy." Or, in the case of the nocturnal noshing, removing the cue by going to bed early may be the easiest way to prevent excessive food consumption. "When I've finished getting ready for bed, I'll go right to sleep instead of staying up later."

In any case, reprogramming habit loops will require making deliberate decisions to behave differently. An easy way to do this is with what psychologists refer to as *implementation intentions*, which are statements about how, when, and where you'll act. Implementation intentions can be expressed in different ways, but a common formula is "When situation X occurs, I'll respond by doing Y." For example ...

- "Every Tuesday, Thursday, and Saturday, I'll wake up at 7 a.m., drink a cup of coffee, and do a workout" will be far more effective than "I'll work out a few times per week."

- "Every evening after dinner, I'll sit on the couch without my phone and read twenty pages before turning on the TV" in place of "I'll read every night."
- "Every weekday, I'll eat a portion of lean protein and three servings of colorful vegetables for dinner" is preferable to "I'll eat more healthy food."
- "Every Sunday morning, after I eat breakfast, I'll prepare food for the next three days" will help you follow your meal plan better than "I'll do more meal prep."
- "Every day, I'll keep a bottle of water with me to drink from and refill it whenever it's empty" will outdo "I'll drink more water."

Don't underestimate the effectiveness of this practice. In a study published by scientists at the *British Journal of Health Psychology*, 91 percent of participants who created implementation intentions for exercising ("During the next week, I will partake in at least twenty minutes of vigorous exercise on [DAY] at [TIME OF DAY] at/in [PLACE]") worked out at least once per week. This is compared to just 38 percent of participants who read a few paragraphs from a random novel and 35 percent of participants who read a pamphlet on and were told how exercise benefits your heart.

Over one hundred published studies prove the power of implementation intentions to influence many positive behaviors, ranging from breast self-examination to dietary adherence and more. The conclusion is crystal clear: by explicitly stating what, when, and where you'll do something, you're much more likely to actually do it. Such statements are far more reliable for regulating behavior than depending on inspiration or willpower, because they function as a trigger-and-response mechanism that doesn't require deliberate supervision.

You can take this technique even further and put your important commitments on your calendar so they stay top of mind. At the beginning of every day, when you check your calendar, you can review your pledges and improve your chances of following through.

You can also use your phone's assistant to receive daily reminders of your obligations so you don't accidentally forget. If your task is eating a nutritious homemade lunch every day, you can set recurring reminders in your phone to prepare the food the night before and grab it in the morning on your way out. Such spadework may sound too simple to be effective, but don't discount its importance. The timely reminder or mundane preliminary, like getting into your workout clothes first thing on Saturday morning so you're dressed for your lunchtime workout, can often be the factor that ensures follow-through.

Such preparatory measures enhance our stick-to-itiveness by reducing the amount of *activation energy* required to act. This chemistry term refers to the minimum amount of energy needed to trigger a chemical reaction, or more broadly, as the minimum amount of effort necessary to start or change something.

Generally, the more mental or physical energy it takes to get in motion or shift our course, the more susceptible we'll be to the peaks and valleys of our willpower and motivation levels. When determination is high, follow-through is about as hard as a haircut, but when it's low, implementation feels like trying to play the piano with boxing gloves. By taking simple steps to protect our habits from the wax and wane of these fickle feelings, however—like those I've just shared with you—we can greatly boost our consistency and long-term results.

Another type of scientifically proven tool you can use to further reduce the activation energy of changing habits is the *if–then statement*, which looks like this: "If X happens, then I'll do Y." The psychology behind this phrase echoes the

what–where–when statements (a specific stimulus and response), allowing you to accommodate life's inevitable contingencies and curveballs so you can make the right decisions when things go sideways.

In a scenario given earlier, you've decided to wake up at 7 a.m., drink coffee, and work out every Monday, Wednesday, and Friday. To complement this implementation intention with if–then statements, consider what may frustrate your objective and what you can do about it. For example ...

- If I go to bed too late, then I'll still get up at 7 a.m. and train.
- If I can't train at 7 a.m. for whatever reason (early meetings or calls, errands, etc.), then I'll go to the gym after work.
- If I can't train after work for whatever reason (personal appointments, kid stuff, etc.), then I'll train on Saturday or Sunday at 9 a.m.

Or what about if-then statements for the what–when–where statement, "Every weekday, I'll eat a portion of lean protein and three servings of colorful vegetables for dinner."

- If I don't have time to cook dinner, then I'll get lean protein and a couple of servings of vegetables from the hot bar at the grocery store.
- If I have to eat out for dinner, then I'll eat just one piece of bread, order lean protein and vegetables, and skip dessert.
- If I have a social event to attend in the afternoon, then I'll have a protein shake beforehand so I'm not tempted to overeat.

If I eat a different dinner or overeat, then I'll forgive the deviation and follow my meal plan the next day.

Every what–when–where statement can be fortified with if-then

companions, and probably should be, because there'll always be unforeseen difficulties and complications. This process of using thought exercises to stress test your plans is akin to running mental simulations, and research shows that it can increase your ability to overcome obstacles and achieve your goals.

Amending habits takes practice and patience. It's easy to forget to do new routines or backslide into our old ways, but by remaining focused on the process of making steady improvements, we can produce remarkable change.

Creating Good Habits

Often, establishing a new desirable habit can be just as thorny as breaking an undesirable one. In fact, if New Year's resolutions are any indicator, people struggle at least as much with doing more of the right things as doing fewer of the wrong ones.

According to a YouGov survey conducted in December 2020, 50 percent of people who intended to make New Year's resolutions wanted to do more exercise or improve their fitness, 46 percent wanted to lose weight, 44 percent wanted to save more money, 39 percent wanted to improve their diet, 21 percent wanted to pursue a career ambition, 18 percent wanted to spend more time with their family, 14 percent wanted to take up a new hobby, 13 percent wanted to spend less time on social media, 10 percent wanted to give up smoking, and 4 percent wanted to drink less alcohol.

Many of these intentions are difficult to realize because they entail passing up immediate and certain rewards (languor, food, shopping, etc.) for the hope of future payoffs (fitness, financial security, etc.). And habits can take a lot longer than 22 days to reach automaticity. Research conducted by scientists at University College London shows that it can range from 18 to 254 days, and in this study, the median was 66 days. Clearly, pep alone often isn't enough. You also need a plan.

Here are three simple ways to greatly increase the stickiness of new habits:

1. Start small
2. Stack
3. Celebrate

Let's go over each.

1. Start Small

Most people think they bungle their attempts to build new habits because they lack the motivation or willpower required to make lasting change. This is rarely true. They only lack the right approach.

As you know, willpower seems to function much like a muscle. It starts the day fresh, and with exertion, becomes fatigued, but then adapts to the stimulus to become a little stronger. That also means that our resolve has its limits—that we can demand more of it than it's capable of (or, perhaps more accurately, that we can access under normal circumstances). Thus, when someone is unable to successfully go from zero exercise to an hour per day and then chastises themself as a failure, this is about as sensible as trying to squat 315 pounds on their first day in the gym and feeling stung when they get stuck.

What should they do instead? Reduce the difficulty, of course, and the same goes for habit building. By starting small—very small, if necessary—you can more or less guarantee a smooth progression toward bigger and more significant transformations.

B.J. Fogg, Stanford University researcher and author of *Tiny Habits*, calls these tiny actions "smallest behaviors that matter," or *SBTMs* for short, and they have four features:

1. They can be done at least once a day.
2. They take less than 30 seconds.
3. They require little effort.
4. They're relevant to the full desirable behavior.

The idea is to start with something so simple that you can't say no to it. Think ten push-ups per day, not a full workout, and one serving of vegetables per day, not a whole new meal plan. Just how small *you* should start with a new habit,

including those that form the foundation of the *Bigger Leaner Stronger* program, will depend on your circumstances. Ten push-ups or one serving of vegetables per day may be so small and easy that you get no sense of satisfaction from it, in which case you'd want to increase the difficulty to something that involves a little (but not *no*) effort.

Many people I've spoken with have shared with me that they started the program by implementing parts of the diet or training component first (eating less "junk" before following a set meal plan, doing three workouts per week, not five, etc.), then once that was in hand, gradually adding the rest.

2. Stack

A simple way to get a running start with new habits is to place them into the slipstream of existing ones already ingrained. This technique is known as "habit stacking," and it uses a special type of implementation intention that works like this: " After [CURRENT ACTION], I will [NEW ACTION]." For instance, "After I drink my cup of coffee in the morning, I'll go for a 15-minute walk." Or "After I get out of my work clothes, I'll change right into my workout clothes." Or "Before I watch sports on Sunday afternoon, I'll prepare my lunches for the week."

Habit stacking works so well because it allows you to piggyback on behavior patterns that are already automatic. We can make momentum our ally. Once we're in motion, it's easier to stay in motion—even if we're going in a new direction.

You can grow your habit stacks over time, too, adding as many actions to the chain as you need. For example, I have a well-established habit stack for my morning routine: I wake up at 6 a.m., go to the bathroom, drink some water, make myself a cappuccino (my favorite part), read for 45-to-60 minutes, do 10

minutes of stretching, get dressed, make a protein shake, eat a banana, brush my teeth, and go to the gym. I have another stack for when I get home: Take my post-workout and greens supplements, check my calendar and create my to-do list for the day, and start answering work messages. I have a stack for driving (call a friend, family member, or coworker), taking scheduled phone calls (if it'll be at least twenty minutes, take it on my upright bike and get my thirty minutes of daily cardio done), and 9 p.m. pre-bed routine (floss, brush my teeth, and spend some time with my wife). In fact, so many of my daily activities are pre-programmed according to a game plan that most of the time, it feels like I wake up and the rest just takes care of itself.

Regardless of how you plan on using habit stacking, the key is selecting the right cue—the right place in your daily routine to insert the new habit you want to adopt. If you'd like to read in the morning like me but can't escape the household chaos for more than a few minutes at a time, consider another time when it'll be easier to execute, like after you take a shower.

According to Fogg's research, when selecting cues for habit stacking, you should strive to match the physical location, frequency, and theme or purpose. To pair the location, ensure that the new habit can be performed in the same place as the existing one. As for frequency, if you want to do something new every day, but try to couple it with something you do every other day, on average, that'll cause hiccups. And as to theme or purpose, the more compatible the components of a habit stack are, the better it'll function. Thus, don't try to combine flossing your teeth with sweeping the garage or eating vegetables with watching TV.

To find the right cues for new habits you want to adopt, start with writing out a list of your current habits (the things you do every day). For example:

- Get out of bed.

- Go to the bathroom.
- Drink some water.
- Brush your teeth.
- Get dressed.
- Make coffee.
- Eat breakfast.
- Take your kids to school.
- Start working.
- Etc.

You can also make another list of things that happen to you each day without fail, like …

- The sun comes up.
- You get a phone call.
- You get hungry.
- The sun sets.

With these two lists, you should have no problem finding places to introduce new habits into your lifestyle.

Remember too that implementation intentions work best when they're highly specific and immediately doable. "When I take a break in the afternoon, I'll go for a 15-minute walk" may not be clear enough to succeed. Do you go for a walk before you eat a snack or after? And where are you going to walk? Better would be, "When I get off the computer around 3 p.m., I'll go for a 15-minute walk around the block."

You can create if–then statements as well, if needed. For instance, as you're roadmapping your new system of habits, you may already foresee eventualities

that should be addressed up front. Or maybe you don't realize until after you get underway that there are more wrinkles than you had anticipated.

3. Celebrate

Unsurprisingly, research shows that a habit can form quickly when we immediately feel good after we perform an action. This reward is the glue that makes habits stick. Therefore, whenever we're trying to change or adopt a new habit, we need to ensure it results in a feeling of success and satisfaction, even if it's slight.

This step doesn't have to be elaborate, either. Briefly doing, saying, or even thinking about something that brings joy is often enough—a fist pump, or "awesome" said aloud or subvocalized. What you do to rejoice doesn't matter nearly as much as the immediacy and emotional intensity of the celebration.

You now know how to win at the inner game of fitness:

1. Understand your whys.
2. Cultivate your willpower.
3. Control your habits.

If you can implement and ultimately internalize that action plan, you can develop the mindset and lifestyle that'll allow you to discover and express your full fitness potential. It won't be easy, though. It'll take discipline and endurance, which are hard and always will be. We're not wired for restraint and regularity, but for stimulation and novelty. Self-command and stamina require conscious strengthening, just like a muscle.

So don't mistake any difficulty as a sign that the undertaking isn't worth it.

The difficulty is the point. The difficulty signals its worth. Tribulation isn't a sign that you don't belong in the arena; the struggle to overcome it is how you prove you're worthy.

Fortunately, once you get your arms around the outer game of getting fit—like you will by the end of this book—the process is a lot more gain than pain and fun than frustration. And that's exactly what we'll tear into next, starting with sifting the art and science of everyone's least-favorite four-letter word: diet.

KEY TAKEAWAYS

- Habits are formed through repetition in a pattern known as a "habit loop" that goes like this: cue, response, reward.
- You can reshape any habit by identifying the behavior you want to change, experimenting with rewards, isolating the cue, and having a plan.
- If you can't or don't want to eliminate the cue altogether, you must find a new way to respond to it that leads to the reward you crave.
- Implementation intentions are statements about how, when, and where you'll act that help you make deliberate decisions to behave differently, and a common formula is "When situation X occurs, I'll respond by doing Y."
- To further reduce the activation energy of changing habits, use if–then statements such as "If I go to bed too late, then I'll still get up at 7 a.m. and train".
- When creating good habits, start by committing to something so simple that you can't say no to it, then once that's in hand, gradually increase the difficulty.
- To get a running start with new habits, place them into the slipstream of existing ones already ingrained using a technique known as "habit stacking." For example, "After I drink my cup of coffee in the morning, I'll go for a 15-minute walk."
- Whenever you're trying to change or adopt a new habit, ensure it results in a feeling of success and satisfaction, even if it's slight.

PART 3

The Last Diet Advice You'll Ever Need

The Big Ideas

1. Understand the importance of energy balance.

2. Diet properly and your metabolism will be fine.

3. Ignore rigid dietary dogmas and fads.

7

The 10 Absolute Worst Diet Myths and Mistakes

The road to nowhere is paved with excuses.

—MARK BELL

Since time immemorial, the highest hallmark of physical status and attractiveness has been a lean, toned, athletic body—a stamp of the ancient heroes, gods, and goddesses. A fit body is still idolized today, but with obesity rates over 42 percent here in America (and still rising), one could assume that looking like a paragon of fitness requires youth, top-shelf genetics, or superordinary effort, discipline, and sacrifice. This isn't true. The know-how is easy enough to grasp—you'll learn everything you need in this book—but it doesn't require as much grit as you may think.

To help you better understand the real science of getting and staying fit, lean, and strong, I want to first disabuse you of some of the biggest diet myths and lies that tie people up in knots:

1. "Calories in versus calories out is bad science."
2. "Calories in versus calories out are all that matters."
3. "Dieting can 'damage' your metabolism."

4. "Carbs and sugars make you fat and unhealthy."
5. "You have to 'eat clean' to be fit and healthy."
6. "Vegan and vegetarian diets are healthiest."
7. "You have to eat huge to get big."
8. "You should or shouldn't skip breakfast."
9. "Frequent eating speeds up your metabolism."
10. "Eating at night can cause weight gain."

Let's hit these fictions head-on. If you've bought into any of them yourself (like I once did), you may be skeptical, even scornful, of what else I have in store for you and thus less likely to give the program a fair shake. Think of this chapter like a construction project—clearing away the dietary debris before we pour the foundation and build the superstructure.

Myth #1

"Calories In Versus Calories Out Is Bad Science"

"Calorie counting doesn't work," the overweight MD says in his latest bestselling book. "It's a relic of our ignorant dietary past," the pretty woman who has been skinny her entire life tells the morning show hosts. "It's time we realized food quality is what matters, not calories," the former triathlete turned guru says on his popular podcast.

The sales pitch sounds sexy. Eat the right foods and you can "unclog your hormones" and "supercharge your metabolism," and your body will take care of the rest. This is music to many people's ears who want to believe they can get lean and fit without ever having to restrict or even pay attention to *how much* they eat, only *what*.

This is malarkey. In fact, it's worse than that. It's a blatant lie because as far as your body weight is concerned, how much you eat is *far* more important than what you eat. Don't believe me? Just ask Kansas State University Professor Mark Haub who lost 27 pounds in 10 weeks eating Hostess cupcakes, Doritos, Oreos, and whey protein shakes. Or a science teacher, John Cisna, who lost 56 pounds in six months eating nothing but McDonald's. Or Kai Sedgwick, a fitness enthusiast who got into the best shape of his life following a rigorous workout routine while eating McDonald's every day for a month.

I don't recommend you follow in their footsteps (the nutritional value of your diet does matter, particularly for health), but their stories speak to an indisputable point: You can improve your *body composition*, or how much muscle and fat you have, while eating copious amounts of junk food. The key to understanding how this works—and to understanding what really drives weight loss and gain—is *energy balance*, which is the relationship between energy intake (calories eaten) and output (calories burned).

Technically, a *calorie* is the energy needed to raise the temperature of one kilogram of water by one degree Celsius. Various foods contain varying numbers of calories. For example, nuts are very energy dense, containing about 6.5 calories per gram, on average. Celery, on the other hand, contains very little stored energy, with just 0.15 calories per gram. If you added up the calories of all the food you ate in a day and then compared that number to how many calories you burned, you'd notice one of three things:

1. You ate more calories than you burned. (You'll gain weight.)
2. You ate fewer calories than you burned. (You'll lose weight.)
3. You ate about the same number of calories as you burned. (You'll maintain your weight.)

This is the first law of thermodynamics at work, which states that energy in a system can't be created or destroyed but can only change form. This applies to all physical energy systems, including the human metabolism, which is the series of physical and chemical processes that keep us alive—namely, the production of energy as well as the creation, maintenance, and destruction of cells and tissues. When we eat food, its stored energy is transformed by our muscles into mechanical energy (movement), by our digestive systems into chemical energy (body fat), and by our organs into thermal energy (heat).

These scientific axioms explain why every controlled weight-loss study conducted in the last century has found that meaningful weight loss requires energy expenditure to exceed energy intake; why decades of obesity research has concluded that energy intake exceeding expenditure is the main driver of weight gain; and why bodybuilders dating back just as far—from the "father of modern bodybuilding" Eugen Sandow, to the sword-and-sandal superstar Steve Reeves, to the iconic wunderkind Arnold Schwarzenegger—have been using this knowledge to control their body composition as desired.

Your checking account is a trite but apt metaphor for how this process works. If you "deposit" (eat) more calories into the account than you "spend" (burn), you create a *positive energy balance*, and your body will "save" (store) a portion of the surplus energy as body fat. If you put fewer calories into the account than you spend, however, you create a *negative energy balance* or *energy deficit*, and your body will turn to its "energy savings" (body fat, mostly) to make up for the shortfall and get the energy it needs to keep functioning. And what happens if you *consistently* make small withdrawals from your savings account? That's right—the total gets whittled down to lower and lower levels, and in the case of fitness, this is what makes you leaner and leaner. We're lucky our body works this way, too, because if it didn't and we missed a meal, our energy supply

would run out and we'd die. Luckily, when food energy isn't available, our body can break down body fat (and other tissues when necessary) and use the constituents to produce cellular energy.

Now, none of this means you have to meticulously count calories to lose weight, but you do have to understand how the calories that you eat influence your body, and manage your eating according to your goals. What you don't have to do, however, is follow a fad diet that eliminates foods or even entire food groups from your diet. This can work for some people because forbidding the delicious, high-calorie foods that people love—the ones that are easiest to overeat—and prescribing less palatable, lower-calorie fare, can lower calorie intake enough to produce weight loss. But it can fail just as easily if it doesn't result in a large enough calorie reduction to produce the energy deficit to achieve weight loss.

Put differently, the people who have lost weight with restrictive diets didn't succeed because they stopped eating carbs, sugar, gluten, or any other bogeyman per se—they lost weight because doing away with certain foods or substances helped them keep "calories out" over "calories in" long enough to reduce their weight. This is why studies show that people who eat the least lose the most weight and those who eat the most lose the least (or even gain weight), regardless of the dietary protocol used, which has included Mediterranean, vegan, vegetarian, paleo, Weight Watchers, Slimming World, South Beach, Best Life, Atkins, DASH, and others.

So, know this: No matter what type of diet you follow, you'll lose weight only when you consistently eat fewer calories than you burn, and the only way to fail to lose weight is eating too many calories too consistently, not flouting or following the wrong arbitrary eating rules.

Still skeptical? I understand. Maybe you've heard that recent scientific

research has contested or even confounded the energy balance model, or maybe you know of individual stories that seem to refute it, like people who apparently can't lose weight with a very low-calorie diet. "Bill ate 1,000 calories per day and worked out like a fiend for a month and *gained* weight. How can energy balance explain that? Hormones?" Puzzling indeed, until you realize the culprit is almost always human error, not haywire hormones or metabolic hoodoo. Virtually every time, such people don't understand what hormones are—chemicals transported by the blood or other bodily fluids to cells and organs, where they cause some action or have some specific effect—and are making one or more of the three most common weight loss mistakes:

1. Underestimating calories in. Most people are bad at estimating and even tracking how many calories they're eating. For instance, studies show that people can think they're eating 800 calories per day (starving themselves), when it's actually 1,200, 1,500, or more. This mistake is all too easy to make when you review the delicious foods many people like to eat. Take a single slice of pizza or cheeseburger, for instance, which contains 300-to-400 calories or about as many as you burn in a 30-minute jog or 60-minute weightlifting workout.
2. Overestimating calories out. Physical activity, including exercise, doesn't burn as many calories as many people think. In a York University study, people overestimated how many calories they burned during vigorous exercise by 72 percent on average. This error alone could prevent weight loss even when energy balance is understood and eating is well regimented.
3. Overeating in the extreme. Too many "cheat meals," or worse, "cheat days," can make meaningful weight loss impossible. Let's say that you

ate about 300 fewer calories than you burned Monday through Friday, creating a total calorie deficit of 1,500 calories. Huzzah! That's enough to lose a half a pound of fat! Saturday, however, is your cheat day, and so you eat about 1,000 more calories than you burn. Sunday is a cooldown, but you still eat a few hundred more calories than you burn. What just happened here? Run the numbers. You've just wiped out the cumulative calorie deficit (and thus the fat loss) of five days of dieting in two days, and now you're back to square one.

Along with complete ignorance of energy balance, these three missteps are responsible for most weight loss failures and explain the unreliability of diets that deal in rules and restrictions instead of hard science and numbers. Again, you can lose weight without counting calories, but it can be difficult to maintain a consistent calorie deficit if you have no idea how many calories you're eating and burning day-to-day.

Another popular argument against the primacy of energy balance is the claim that the human body is far more complex than the simple heat engine that powers our lawn mower or car, so it's silly to assume they operate according to the same principles. This is pseudoscientific sophistry. The human body is certainly more intricate than the combustion engine, but that doesn't confound the weight of nearly 100 years of evidence in the scientific literature. Energy balance works just as I've described in the lean and obese, the healthy and diseased, the old and the young.

In the body composition show, energy balance is the hero. It isn't the whole cast, though, which brings me to the next myth ...

Myth #2

"Calories In Versus Out Are All That Matters"

For all its majesty, the "CICO" (calories-in-calories-out) model has feet of clay: it accounts for how much energy you're eating and burning, but not where the energy is coming from.

As critics of energy balance will often point out, this detail is vital because our body doesn't process all calories the same way. Protein, for example, has very different effects in the body than carbohydrate or fat, which explains why some foods are more conducive to fat loss and muscle building than others (but don't have special properties that make them "fattening" or "slimming").

CICO also neglects the nutritional dimension of eating, which must be considered for long-term health, fitness, and vitality. While it's possible to get and stay lean eating like a 12-year-old boy on vacation, body composition isn't a perfect barometer for health. Just as you can be slightly overweight and healthy if you eat well and exercise regularly, you can also be fit but unwell if your body is starved of key nutrients because you've forsaken fruits, vegetables, whole grains, and high-quality protein for cookies, chips, pizza, and ice cream.

Therefore, if you want to optimize your body composition and your health, a carefully controlled diet of junk food won't cut it. Instead, you'll want to follow a system that scientists refer to as "flexible dieting," which you'll learn about in the next chapter.

Myth #3

"Dieting Can 'Damage' Your Metabolism"

If you poke around online, you can find many stories about people, typically

women, who say they aren't losing weight with extremely low-calorie dieting and hours of exercise every week because of metabolic abnormalities caused by calorie restriction.

Essentially, the story goes like this: Dieting dramatically decreases your metabolism, eventually halting fat loss, and if you go too far, you'll need to follow a lengthy "recovery" protocol to fix the "damage" to have a healthy metabolism again. Hence, the term "metabolic damage." Also, when your body is experiencing the physiological adaptations that apparently cause metabolic damage, it's said to be in "starvation mode." This apparently kicks in the first day of your diet and gets progressively worse as time goes on.

How true are these claims, though? While your metabolism does decrease slightly while dieting, the drop isn't nearly as dramatic, significant, or prolonged as many people make it out to be. "Metabolic damage" isn't real, has never stopped someone from losing weight, and doesn't need to be "fixed" with special diet techniques. "Starvation mode" is sort of real, but it isn't nearly as dramatic as it's often portrayed, and naturally resolves itself after you stop dieting.

A salient example of this comes from a scientific study published in the *American Journal of Clinical Nutrition*. In this experiment, 171 overweight women lost an average of 26 pounds in five months, and the researchers measured the participants' body composition and *resting metabolic rate* (the number of calories your body burns at rest in 24 hours) four times over a two-year period. The scientists found that after five months of dieting, the women's RMRs had only dropped about 50 calories per day, and a year later, when measurements were taken again, their metabolisms had recovered to where they were before.

Other studies have shown that the metabolic decline associated with dieting, including long periods of very low-calorie dieting, ranges from less than 5 to about 15 percent. Furthermore, it took about a 10 percent reduction in body

weight to produce the larger, double-digit drops, and most of the research on the matter was conducted with people who made every mistake in the book—they ate too few calories and too little protein and did no resistance training.

So, while scientists have known for decades that dieting checks metabolic rate, which can make it harder (but not impossible) to continue losing weight, they don't refer to this effect as metabolic damage. Instead, it's known as *metabolic adaptation*.

As you know, energy balance alone explains why meaningful weight loss requires that you eat fewer calories (less energy) than you burn for an extended period of time, and why the essence of effective dieting is "eat less and move more." What many people don't realize, however, is their daily energy expenditure can change significantly once they start dieting. By reducing "calories in," "calories out" also naturally drops because restricting your calories doesn't just cause weight loss—it also leads to a cascade of unfavorable changes in hormones like leptin, ghrelin, thyroid hormones, and testosterone, which all work to reduce energy expenditure and increase energy intake.

This counterpunch can be frustrating when you want to lose fat, but it's also been vital to our survival as a species. For most of human history, calories were scarce—we'd regularly go for days without eating—and so our body adapted to favor the storage of excess calories as fat. In other words, humans physically and psychologically evolved to endure famine by storing and conserving calories, and when you decide to diet to get rid of your belly fat, your body responds the same way as if you were truly starving—it employs "defense mechanisms" to keep you alive. Thus, calorie restriction acts as a physiological tripwire that increases hunger, decreases motivation to move and exercise, and reduces metabolic rate, and these phenomena make getting leaner increasingly difficult as time goes on.

The important thing to remember, though, is that while metabolic

adaptation can slow weight loss, it can never stop it entirely. Additionally, these adjustments quickly reverse themselves once you're no longer in a calorie deficit, and there are simple steps you can take to mitigate metabolic downsides of dieting like doing strength training, using mild-to-moderate calorie restriction, and eating enough protein (all of which are part and parcel of the *Bigger Leaner Stronger* program).

Myth #4

"Carbs and Sugars Make You Fat and Unhealthy"

People love simple explanations and compelling conspiracies, and these two quirks explain the popularity of most mainstream fad diets. The three-step formula is simple:

1. "It's not your fault you're overweight and unhealthy. Jerks keep saying it's because you eat too much and move too little, but they're wrong. You're a victim of bad science and worse food."
2. "New research shows you what to blame. Behold the heinous methods or molecules responsible for all your woes. Take your weapon! Strike them down with all your hatred and your journey toward the dark side will be complete!"
3. "Avoid this at all costs and you'll live happily ever after. Renounce this hobgoblin, escape its wrath, dream your quest and bring your visions into fantasy ... blah blah blah green tea infinity ... will that be cash or credit?"

These emotion-based tactics are how marketers sold us on low-fat dieting a

decade ago and how they sell us on low-carb and low-sugar dieting today. Cut the vile carbohydrates and sugars out of your life, they say, and the pounds will just melt away.

It all sounds so neat until someone like me comes along and points out examples of the glitches in the matrix, like randomized controlled trials (the gold standard of scientific evidence) that have found no difference in weight loss whatsoever between low- and high-carb and low- and high-sugar diets. For instance, scientists at the following institutions found similar results:

- An Arizona State University study found no difference in weight or fat loss between people consuming 5 and 40 percent of their calories from carbohydrate for ten weeks.
- A study published in the journal *Hypertension* found no difference in weight or fat loss between people consuming 4 and 30 percent of their calories from carbohydrate for six weeks.
- A Harvard School of Public Health study found no difference in weight loss between people consuming 65, 45, and 35 percent of their calories from carbohydrate for two years.
- A study published in the *Journal of the American Medical Association* found no difference in weight or fat loss between people who consumed 50 and 25 percent of their calories from carbohydrate for one year.
- A Duke University study found no difference in weight or fat loss between people consuming 4 and 43 percent of their calories from sugar for six weeks.
- A study published in the *International Journal of Obesity* found no difference in weight loss between people consuming 5 and 10 percent

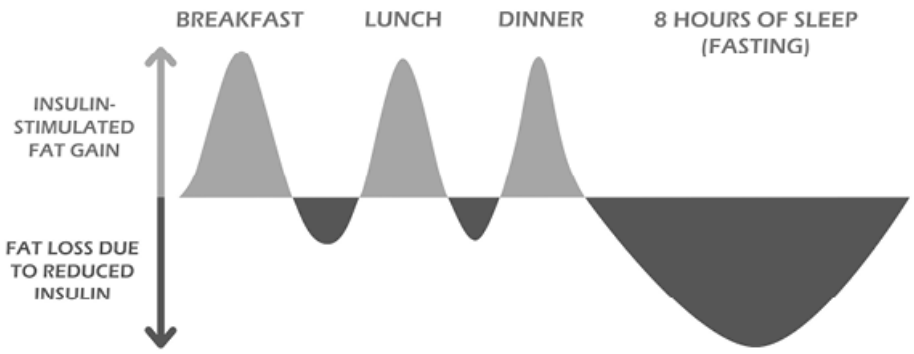
of their calories from sugar for eight weeks.

Although the anti-carb and anti-sugar hysteria has reached a frenzied pitch in the last several years, as the studies above show, it's completely unwarranted. Carbs and sugars aren't nearly as dangerous or fattening as you've been told. They don't make you fat or unhealthy—only overeating can do that—and ironically, most people achieve their fitness goals faster with *more* carbohydrate in their diet, not less.

"But what about insulin?" people often ask me. "Don't carbs spike insulin levels, and doesn't that make you fatter?" Not quite. Eating carbs (among other foods) triggers insulin production, and insulin triggers fat storage (among other physiological reactions), but fortunately, none of that can make you fat—only overeating can.

Insulin is a hormone that causes muscles, organs, and fat tissue to take in and use or store nutrients like *glucose* (a sugar in many carbs and an important source of energy in organisms) and *amino acids* (the molecules that make up proteins). When you eat many types of food (including protein), insulin levels rise, and this allows your body to use some of the energy provided by the meal to power the many physiological processes that keep you alive. Depending on the circumstances, your body can also use some of the energy to increase its fat stores, which is why some people call this state the body's "fat-storing mode." Then, after you've finished digesting, absorbing, burning, and storing food you ate, insulin levels fall to a low, baseline level, and your body must now rely primarily on fat stores for energy. This state is often called the body's "fat-burning mode."

You flip between these states every day, storing small amounts of fat after most meals and then burning small amounts in between them. Here's a simple way to visualize this mechanism:



The lighter portions represent the periods when you've eaten food, giving your body energy to use and store, and the darker parts show when food energy is absent and body fat must be burned instead. Insulin's role in this system makes it an easy scapegoat for unwanted fat gain, but this makes about as much sense as blaming the mailman for receiving too much junk mail. Without the excess calories needed to physically grow fat cells, no amount of insulin or insulin-producing foods can significantly increase body fat levels.

In the final analysis, you have nothing to fear from insulin, so you can achieve peak health and fitness while eating many of your favorite foods. If you're still skeptical, maybe because some experts claim otherwise, that's okay, because before long you'll have firsthand proof in the way of rapid muscle gain and fat loss as well as improvements in just about every health biomarker you care to monitor.

Myth #5

"You Have to 'Eat Clean' to Be Fit and Healthy"

The cult of "clean eating" is more popular than ever these days, and while its heart is in the right place, its brain is a muddle.

For starters, many "gurus" disagree on which foods are "clean" and which aren't—one's trash is another's treasure—so there are many sects to choose from

that prescribe wildly different protocols. Will you go vegan or carnivore, for instance? Keto or fruitarian? Gluten-free, non-GMO, raw, or organic, or all of the above? Choose carefully, because no matter which philosophy you pick, someone will claim it'll eventually ruin your body, mind, and soul.

Second, binary labeling of food as "clean" or "dirty," "healthy" or "unhealthy," or "good" or "bad" is simplistic and inaccurate. Just as it's incorrect to stamp high-calorie foods as "fattening" and low-calorie ones as "slimming," it's incorrect to designate more nutritious foods as "good" and less nutritious ones as "bad." Instead, it's far more useful to think of foods in terms of their key characteristics —calories, *macronutrients* (nutrients required in relatively large amounts—specifically, protein, carbohydrate, and fat), and *micronutrients* (nutrients required in trace amounts). Also, instead of judging whether an individual meal is healthy, it's more constructive to think about whether your diet on the whole is healthy, which depends on energy balance (is your calorie intake calibrated properly?), macronutrient balance (are you eating enough protein, carbohydrate, and fat?), and micronutrient balance (are you getting enough vitamins, minerals, and other nutrients to preserve health?).

For example, if you get most of your daily calories from nutritious foods like high-quality protein, fruits, vegetables, legumes, whole grains, nuts, seeds, etc., your diet is healthy regardless of which foods the remaining calories come from. Ironically, this style of eating produces a diet that looks fairly clean because it adopts what's useful in the clean eating paradigm. The more accurate rendition of this myth, then, is, "You have to eat clean-ish to be fit and healthy."

Myth #6

"Vegan and Vegetarian Diets Are Healthiest"

There's no denying that plant foods are vital for maintaining optimal health and performance. A large body of evidence shows that people who eat higher amounts of fruits, vegetables, and whole grains are generally healthier and more likely to live longer, disease-free lives than those who don't eat enough of them.

Fruits and vegetables provide essential vitamins and minerals, but they also contain other types of phytonutrients that benefit health in various ways. Two good examples of this are *sulforaphane* and *anthocyanins*, phytonutrients found mainly in broccoli and blueberries and known to reduce cancer risk, cellular wear-and-tear, and cholesterol levels, but are not found on food labels, which list only vital-to-life nutrients. Many of these additional nutrients aren't in multivitamins or other supplements, either, so pills and powders can't replace the real McCoy.

Fiber is another (often overlooked) component of plant foods that deserves attention. Fiber is an indigestible type of carbohydrate found in many types of foods, including fruits, vegetables, legumes, nuts, and grains. Its importance has been known for a long time—the ancient Greek physician Hippocrates, who famously said "let food be thy medicine, and medicine be thy food," recommended whole-grain breads to improve bowel movements—and modern scientific research has confirmed that eating enough fiber increases your chances of living a long and healthy life.

While healthy eating necessarily includes a wide variety of plants and vegetables, ranging from dark, leafy greens to garlic to cruciferous vegetables to cereal grains and more, that doesn't mean eating *only* those foods is ideal, and especially if you're trying to improve your body composition.

For instance, gaining muscle is trickier as a vegan or vegetarian because it's

difficult to eat enough high-quality, well-absorbed protein. This partly explains why studies have shown omnivores tend to have more muscle than vegetarians and vegans. You can work around this issue by carefully choosing certain plant foods that are rich in suitable protein, but even then, you'll likely have to eat quite a lot of these foods or include supplements in your regimen to meet your daily protein needs.

Studies show that certain micronutrient deficiencies are more common among vegans than omnivores, too, including vitamin B12, omega-3 fatty acids, zinc, calcium, and others. You can mitigate this problem by consuming a variety of different micronutrient-dense foods, but you'll also need to supplement strategically.

There's another (and arguably better) way to get enough premium protein and key nutrients, though: Simply include animal foods in your diet, and specifically red meat, poultry, seafood, dairy, and eggs. Some call this a "flexitarian diet."

That's controversial advice nowadays, though. Take red meat, for instance, which many plant-based apostles liken to cigarettes. Just a serving or two of red meat per day, they say, is enough to markedly impair your health and wellness. A cursory review of the scientific evidence doesn't offer much insight. Some studies seem to show that red meat is indeed bad for you, others suggest it's benign, and others indicate it's actually good for you.

Unriddling the matter would require a book unto itself, but the long and short is twofold:

1. The argument against eating red meat relies on evidence that shows an association between red meat (particularly processed red meat) and cancer and heart disease. It's worth noting, though, that much of this

research is based on relatively low-quality studies involving sedentary, often overweight people following a standard Western diet awash with highly processed foods. Thus, it's hard to say how eating red meat or even processed red meats would affect active, lean, healthy people who are also eating lots of fresh fruits, vegetables, and whole grains.

2. The highest quality research available on red meat eating shows that it isn't unhealthy and that reducing red meat consumption has no significant benefit. Moreover, meat also provides a good source of protein and nutrients difficult to get with vegetarian and vegan diets, like iron, zinc, and creatine.

Dairy and eggs are two more animal foods often wrongfully attacked as unhealthy. Despite claims to the contrary, studies show that milk doesn't cause weight gain or cancer, doesn't degenerate your bones, and doesn't contain unhealthy amounts of hormones, pus, or blood. Research also shows that eggs don't raise cholesterol levels or increase the risk of heart disease.

While you can get and stay fit, lean, strong, and healthy on a vegan or vegetarian diet, it's easier to mess up with this kind of diet rather than with a well-designed omnivorous one.

Myth #7

"You Have to Eat Huge to Get Big"

You've probably heard that big muscles require a big appetite. You've probably also heard that "bulking" is unnecessary and even counterproductive because of excessive fat gain. There's a measure of truth in both of these claims. If you want to maximize muscle building, you need to eat quite a bit of food, but if you

follow a cookie-cutter bodybuilding bulking program, you'll probably gain a lot more fat than muscle.

Why? Because how much food you eat (energy balance) strongly affects muscle growth. Put simply: your body's "muscle building machinery" runs best when energy is abundant, so if you don't eat enough calories every day, you'll struggle to get bigger and stronger.

When you restrict your calories for fat loss, you won't gain much muscle or strength because *muscle protein synthesis* rates drop, hormones sour, and workout performance suffers. (That is, unless you're new to resistance training, but more on that later.) Thus, if you want to build muscle effectively, you need to make sure you're not consistently in a calorie deficit, even if it's slight. Instead, you want to do the opposite: regularly feed your body slightly more energy than it burns (provide it with a "calorie surplus").

How big of a calorie surplus is required to grease the skids, though? This is where people often go astray. They mistakenly assume that more food means more muscle and so try to eat like a Clydesdale—20, 30, or even 40 percent more energy than they burn every day. Unfortunately, research shows that the "sweet spot" for goosing muscle gain is eating between 5–10 percent more calories than you burn, and eating more than that will only cause you to gain more fat, not muscle.

"But won't even a small calorie surplus like 5 or 10 percent cause fat gain, too?" you may be wondering. Yes, it will, but the fat will accumulate slowly, at about the same rate as muscle (approximately 1:1 ratio). "But why not instead strive to eat the exact number of calories you burn every day to avoid any fat gain whatsoever?" First, because a calorie surplus is inherently anabolic—to a degree, overeating alone stimulates muscle growth. Second, because your total daily calorie expenditure is a small moving target that's basically impossible to find, let

alone hit. The best we can do is accurately estimate calories out using methods like those you'll learn in this book and then overshoot it by a bit (5–10 percent) to ensure we're not habitually in a calorie deficit.

Fortunately, however, any fat added while "lean gaining" is easily stripped with a proper "cutting" phase (and you'll learn all about those techniques soon), and this process of alternating between phases of lean gaining to get bigger and stronger and cutting to get rid of excess body fat (while preserving lean mass) is how you transform your physique. With every round, you get a little fitter, until finally, you have the look (size, symmetry, and definition) that you want.

Myth #8

"Frequent Eating Speeds Up Your Metabolism."

You've probably heard that you should eat many small meals when trying to lose weight to "stoke the metabolic fire," accelerate fat loss, and better control your appetite. The theory is simple: When you eat, your metabolism speeds up as your body processes the food. Thus, if you eat every few hours, your metabolism will remain in a constantly elevated state, right? And nibbling on food throughout the day should help with appetite control, right?

While this may seem plausible, it didn't pan out in scientific research. In an extensive review of diet literature, scientists at the French National Institute of Health and Medical Research looked at scores of studies comparing the metabolic effects of a wide variety of eating patterns, ranging from one to seventeen meals per day. They found no meaningful difference between nibbling and gorging, because small meals caused small, short metabolic increases, while large meals caused larger, longer increases. Therefore, when viewed in terms of 24-hour energy expenditure, eating pattern had no significant effect. Another study

published in the *British Journal of Nutrition* echoed this finding. In this case, there was no significant difference in weight, fat, or muscle loss between groups of people eating three meals and three meals plus three snacks per day.

And what about meal frequency and appetite? This can go both ways. Scientists at the University of Missouri found that after twelve weeks of dieting to lose weight, increasing protein intake improved appetite control, but meal frequency (three versus six meals per day) had no effect. A similar experiment was published in the journal *Obesity*, investigating the effects of meal frequency and protein intake on perceived appetite, satiety (fullness), and hormones. They also found that higher protein intake led to greater feelings of fullness, but surprisingly, six meals resulted in generally lower levels of satiety than three. On the other hand, research shows that some people are less satiated on three meals per day versus more and that increasing meal frequency can also increase feelings of fullness and make it easier for people to stick to their diets.

As you've likely gathered by now, in many ways, the best dietary protocol is the one you can stick to, and that's very true in the case of meal frequency. Most people I work with enjoy eating four-to-six meals per day (I'm the same way), but some enjoy eating just two or three meals per day.

Myth #9

"You Should or Shouldn't Skip Breakfast."

Breakfast is a controversial meal these days. Some "experts" say it's vital for preserving health and preventing weight gain while others claim skipping it entirely is the trick to staying lean and healthy.

Scientific research cuts both ways as well. For example, one study conducted by the Harvard School of Public Health found that men who regularly skipped

breakfast had a 27 percent higher risk of heart attack or death from heart disease. Another study found that skipping breakfast was associated with a higher risk of weight gain. On the other hand, an extensive review of the literature published in the *American Journal of Clinical Nutrition* concluded that missing breakfast has either little or no effect on weight gain. In fact, the data showed that breakfast eaters tend to consume more calories than those who skip it.

What's the deal, then? Does skipping breakfast make it easier or harder to lose weight or does it have no effect whatsoever? And what about muscle building? Does it help or hinder those of us looking to get fitter and stronger? Let's take a closer look at the studies I just referenced and see what we can learn.

The first thing that jumps out in the Harvard research is the people who didn't eat breakfast were generally hungrier later in the day and ate more at night. Eating food at night isn't a problem per se (see the next myth), but research shows that meal skipping can lead to overeating and an increase in total overall energy intake. Overeating leads to weight gain, of course, and as overweight people are at an increased risk of heart disease, skipping breakfast can be *associated* with an increased incidence of cardiac events and heart disease. But that doesn't mean that skipping breakfast *causes* heart attacks.

How about the review study? The researchers found only a handful of rigorous, well-executed studies on the effects of eating and not eating breakfast, and you have to go all the way back to 1992 to find the only long-term, carefully controlled trial that randomly assigned people to routinely eat or skip breakfast and then measured the effect on their body weight. In the 1992 study, conducted by scientists at Vanderbilt University, eating or skipping breakfast had no significant effect on weight loss. What mattered weren't breakfast habits but overall eating habits and dietary compliance, which merely confirms what metabolic researchers have been saying for decades—*when* you eat is far less

important than *what* and *how much*.

We can find more support for these findings in the research available on the *intermittent fasting* style of dieting. In case you're not familiar with it, intermittent fasting revolves around eating (feeding) and not eating (fasting) on a regular schedule, with particular emphasis on fasting.

With a normal type of diet, you eat food every few hours, from, let's say, 8 a.m. until 9 p.m. That is, every day you eat food intermittently for about 13 hours and eat nothing for about 11 hours. With intermittent fasting, however, you flip this around by eating food intermittently for, with some methods, about 8 hours and eating nothing for about 16 hours. For example, with intermittent fasting, you might start eating every day at 1 p.m. and stop at 9 p.m. In fact, that exact protocol ("16/8") is particularly popular among bodybuilders who enjoy intermittent fasting, and it's basically just skipping breakfast.

And what about scientific evidence? Several studies show that intermittent fasting is just as effective for improving health and body composition as eating more frequently. That said, contrary to the claims of many intermittent fasting fanatics, it also doesn't appear to offer any inherent advantages.

The bottom line is this: If you enjoy breakfast, eat it, and if you enjoy skipping it, skip it. Many people like eating breakfast because they just like breakfast food. Others find that a hearty breakfast helps perk them up or reduce hunger levels throughout the day. On the other hand, just as many people prefer to skip breakfast and "break the fast" closer to lunch because they aren't hungry or don't like standard morning fare. The key is knowing what works best for you.

Myth #10

"Eating at Night Can Cause Weight Gain."

For years, personal trainers, weight loss gurus, and fake doctors on the Internet have claimed that your metabolism is fastest early in the morning and gradually slows throughout the day before bottoming out in the evening. Thus, they claim, you should eat more calories in the morning and fewer calories in the evening to avoid fat gain. "Eat breakfast like a king, lunch like a prince, and dinner like a pauper," as the old saw goes.

Well, research published in the *European Journal of Clinical Nutrition* shows that metabolic rate doesn't change significantly while you sleep, much less in the evening hours beforehand, and studies show that in people who are a healthy weight, metabolic rate can actually increase slightly while you sleep. What's more, even if metabolic rate did decline as the day wore on, this wouldn't make eating at night more fattening than eating earlier in the day because of ... you got it ... energy balance. The only way to gain weight (fat) is to eat more calories than you burn, regardless of when or what you eat.

Now, while eating at night doesn't cause weight gain, that doesn't mean there aren't any benefits to eating less at night. Specifically, research shows that eating more calories earlier in the day can lead to less hunger and cravings. In this way, eating a large breakfast and successively smaller meals can help with weight loss or maintenance for some people (see the previous myth). Conversely, however, other studies show that in some people the opposite is true—they're generally less hungry when they eat smaller meals during the day and a large dinner.

As with eating breakfast, eating larger meals at night is purely a matter of personal preference. Do what works best for you.

In reading this chapter, you've taken your fitness knowledge to a whole new level—a level very few people, including many doctors, athletes, and even scientists, rarely achieve. And we're just getting warmed up (pun intended)!

In the next chapter, I'll build on everything you've learned here and introduce you to the easiest and most effective approach to dieting in the world: *Flexible dieting*. In fact, as you'll soon see, it barely even qualifies as a "diet" in the way that most people understand the term because it enables you to transform your body eating foods you want to eat, seven days per week. Even better, once you experience the power of flexible dieting firsthand, you'll thrill at the realization that you're now immune to the diet frustrations and anxieties plaguing most people, and that you're finally free to develop a positive, healthy relationship with food.

Big promises I know, but flexible dieting will deliver on them in spades. Keep reading to find out how.

KEY TAKEAWAYS

- Energy balance is the relationship between energy intake (calories eaten) and output (calories burned), and it's the basic mechanism that regulates weight gain and loss.
- If you consistently consume more calories than you burn you'll gain weight. If you consistently consume fewer calories than you burn you'll lose weight.
- No individual food can make you fatter. Only overeating can.
- The three most common weight loss mistakes are underestimating calories in, overestimating calories out, and overeating in the extreme.
- Some foods are more conducive to fat loss and muscle building than others, but none have special properties that make them "fattening" or "slimming."
- "Metabolic damage" isn't real, has never stopped someone from losing weight, and doesn't need to be "fixed" with special diet techniques.
- "Starvation mode" is sort of real, but it isn't nearly as dramatic as it's often portrayed, and naturally resolves itself after you stop dieting.
- Carbs and sugars don't make you fat or unhealthy—only overeating can do that—and ironically, most people achieve their fitness goals faster with *more* carbohydrate in their diet, not less.
- It's more useful to think of foods in terms of their key characteristics—calories, macronutrients, and micronutrients—than to label them as "clean" or "dirty," "healthy" or "unhealthy," or "good" or "bad."
- Your body's "muscle building machinery" runs best when energy is abundant, so if you don't eat enough calories every day, you'll struggle to get bigger and stronger.

- The “sweet spot” for goosing muscle gain is eating between 5–10 percent more calories than you burn, and eating more than that will only cause you to gain more fat, not muscle.
- Meal frequency has no significant effects on metabolic rate or weight loss.
- Skipping breakfast doesn’t appear to offer any inherent advantages, so if you enjoy breakfast, eat it, and if you enjoy skipping it, skip it.
- Eating at night doesn’t cause any more weight gain than eating during the day.

The Big Ideas

1. Control your calories to control your weight.

2. Eat a high-protein diet to improve your body composition.

3. Eat a balance of carbs and fat that works for you.

8

Welcome to the Wonderful World of Flexible Dieting

There's more to life than training, but training is what puts more in your life.

—BROOKS KUBIK

How important is your diet to achieving your health and fitness goals? Some people say it's everything. Others say it's second-fiddle to exercise, genetics, or some other factor. Still others say it's 70, 80, or even 90 percent of the game.

I say it's 100 percent. And training correctly? That's another 100 percent. Having the right attitude is 100 percent too. And let's not forget getting enough sleep and rest, which is also 100 percent. I know, we're at 400 percent so far, but that's my point—we need a new paradigm because the foundation of fitness consists of pillars rather than puzzle pieces. If you undermine just one of them enough, the whole structure collapses. You won't lose fat or gain muscle effectively if you don't control your calories and macros. You won't recover from your training if you don't give your body the necessary nutrition. You won't get stronger if your workouts don't follow certain principles and patterns. You won't enjoy the process if you have the wrong expectations. You won't be able to stick to the plan if you don't maintain good sleep habits. The list yammers on.

That's why I want to help you go all out in your pursuit of your best body. That's why the *Bigger Leaner Stronger* program will likely be different from any you've tried before. And that's why I want you to give 100 percent to each part of it to achieve 100 percent of the gains. Let other people train at 60 percent, diet at 30 percent, and focus at 20 percent. Soon, they'll be in your rear-view mirror, and who knows, maybe you'll inspire them to pick up the pace.

Metaphors aside, the straight answer to how important diet is in your fitness journey is this: It either works for you or against you, enhancing or degrading your bottom-line results. No matter what you do in the gym, you'll never get the full payoff unless you also know what to do in the kitchen. This explains why people can sweat blood in their training, yet look like they've never even seen a barbell or bicycle, let alone spent countless hours with them.

The first step to successful eating is understanding its proper relationship to training and your body composition. You use diet to control your body fat level and boost muscle growth, and you use training to gain and maintain strength and muscle. Many people get this mixed up. They think that they work out mostly to burn calories and fat, which often leads them into the hamster wheel of exercising like mad to cancel out all of their eating—an unrewarding, unsustainable, and often unhealthy lifestyle that often ends in a total wipeout.

That won't happen to you on this program. With *Bigger Leaner Stronger*, you'll learn how to use food and training to satisfy both your wants and your body's needs. And that means you'll no longer dread "dieting," which often feels more like self-sacrifice than self-improvement. My program educates you on how your metabolism works and gives you the tools to manage it effectively, while most diets resort to browbeating, fearmongering, and food restriction instead. If you want to get fit, they proclaim, kiss just about everything you like to eat goodbye. Grains, gluten, sugar, refined carbs, red meat, processed foods, dairy,

caloric beverages—it's all gotta go. All your toys. Throw them all into the fire.

If you've contemplated such programs and thought *maybe I'm not up to this*, *maybe I'm not tough or dedicated enough*, or *maybe that beach body isn't really worth it* … . I have good news for you. The *Bigger Leaner Stronger* diet isn't one of those ordeals, and you absolutely are up to it, you absolutely have what it takes, and looking and feeling better than you ever have before is absolutely worth it.

In fact, this diet defies how most people understand the word because what kind of "diet" has you be *less* strict about the foods you eat? How can you eat bucketfuls of delicious carbs every day while "dieting"? Which self-respecting "dieter" could eat candy or fast food with a clean conscience?

The answer is an evidence-based system of eating known as "flexible dieting," and it has four steps:

1. Control your calories.
2. Eat plenty of protein.
3. Eat plenty of nutritious, relatively unprocessed foods.
4. Eat a balance of carbohydrate and fat that works for you.

If you can follow that simple formula (which, you may have noticed, contains nothing about *only* or *never* eating certain foods), you can transform your body and develop a healthy, enjoyable, and sustainable lifestyle and relationship with food while eating meals you enjoy every day.

Let's take a closer look at each of these points.

1. Control your calories.

This is the first step because it's the most important part of the equation, especially for improving body composition. Trying to lose fat, maximize muscle growth, or maintain your physique without an accurate and reliable way to control calorie intake is about as straightforward as shaving with an axe. Possible, but not prudent.

Furthermore, each of those three goals requires that you calibrate your calories differently.

- To lose fat, you must consistently eat fewer calories than you burn. This creates the calorie shortfall required for shrinking your body's fat stores, and without such an energy deficit, nothing much will change—no matter what else you get up to in the kitchen or gym.
- To maximize muscle growth, you must consistently eat more calories than you burn. When you restrict your calories to lose fat, you get leaner, but you also impede muscle building. Thus, if you want to gain muscle and strength as quickly as possible, you need to ensure you're not eating too few calories, and the easiest way to do this is to intentionally eat slightly more calories than you burn. If you're now wondering about fat gain ("Won't that make me fatter?"), you're on the ball, and you're correct—a calorie surplus does indeed cause a fat gain. Fortunately, however, when you know how to do it correctly (as you will on the *Bigger Leaner Stronger* program), you can minimize the unwanted fat storage.
- If you want to maintain your body fat level and slowly add muscle and strength, you want your calorie intake to match your calorie expenditure. Practically speaking, exactness isn't possible because your

calories in and out are moving targets that you can't pinpoint perfectly. You don't need to hit either of those bullseyes, though—you only need to be accurate *enough* and adjust your meal plan based on how your body responds.

2. Eat plenty of protein.

A protein is a naturally occurring compound that's composed of one or more long chains of amino acids. Proteins are an essential part of all organisms and are used to create body tissues such as muscle, hair, and skin, as well as various chemicals vital to life.

While the scientific search for the "One True Diet" continues, there's one thing we know for certain: it's going to be high in protein because study after study has confirmed that high-protein dieting is superior to low-protein dieting in multiple ways, including ...

- Faster fat loss
- More muscle gain
- Faster metabolism
- Less hunger
- Stronger bones
- Better moods

Protein is even more important for us gym-goers because working out increases our need for amino acids, and high-protein eating is also essential for preserving lean mass while dieting to lose fat.

3. Eat plenty of nutritious, relatively unprocessed foods.

You can stay fit, healthy, and vital while hitting the drive-thru or greasy spoon now and then or having some sugar or "empty calories" regularly, but not if too much of your diet consists of nutritionally bankrupt fodder. Over time, this can lead to nutritional deficiencies that cause health problems, impair mental and physical performance, and even blunt fat loss and muscle gain.

An illustrative example of the importance of nutrition is the mineral zinc, which is found in foods like beef, seeds, and legumes and is required for proper thyroid function. As the thyroid produces hormones that influence metabolic rate, if your body doesn't get enough zinc, your metabolism can slow down.

A University of Massachusetts case study shows how significant this can be. In this experiment, two zinc-deficient college women received 26 milligrams of zinc per day for four months, and just four months later, one woman was burning 194 more calories and the other was burning 527 more calories per day than before. Those numbers represent the amount of energy burned in about thirty and sixty minutes of moderately intense cardio and, all other things being equal, would speed up fat loss by about ½ and 1 pound per week—from just correcting a zinc deficiency.

Now granted, such a case study can't support sweeping claims about the relationship between nutrition and metabolism, but it still demonstrates the importance of providing our bodies with a spectrum of vitamins, minerals, and other crucial molecules. Baking is a good analogy for this dimension of eating. What happens if you underdose the leavener? Or overdose the flour? Or forget the fat? You can do everything else right and still cook up a clunker because the gods of baking are merciless tyrants. Without all of the right components in the right quantities, instead of cookies, you get wretched lumps of charred tree bark.

Likewise, the recipe for robust health and fitness calls for a lot more than mere calories. Take fiber, for example, which is often overlooked, even among "healthy" eaters. Fiber is an indigestible carbohydrate found in many types of plant foods, and studies show that eating enough of it reduces the risk of many types of disease and is linked to living a longer, healthier life. Accordingly, the USDA recommends 25 grams of fiber per day for women up to age 50 and 38 grams for men (unfortunately, however, American adults eat slightly less than 20 grams per day, on average).

Another benefit of eating mostly whole, minimally processed food is that it requires more energy to process. About 20 percent of the calories in whole-grain bread with lightly processed cheddar cheese are burned during digestion, for example, compared to only 11 percent of the calories in white bread with highly processed American cheese.

This phenomenon is referred to as the *thermic effect of food* (TEF). While trivial in a single meal, eating mostly highly processed, low-TEF foods can become a significant metabolic disadvantage. You may burn several hundred more calories per day just by choosing less processed fare—instead of a bagel for breakfast, some homemade muesli; instead of chips for a snack, some crispy potato wedges; and instead of white pasta at dinner, whole-wheat instead. Bit by bit, day by day, these differences add up.

This third element of flexible dieting reflects the importance of nutrition and understanding what you *should* be eating regularly as opposed to what you *shouldn't*. It feels fuzzy, though—what do I mean by "nutritious" and "relatively unprocessed" foods? And how much is "plenty"?

Nutritious or relatively unprocessed foods haven't been extensively pre-prepared or pre-packaged. They're in, or close to, their natural states, and you prepare and cook them yourself. Think lean protein, fruits, vegetables, whole

grains, legumes, nuts, seeds, and oils. Such foods are rich in vitamins, minerals, fiber, and other nutrients, and while they may have undergone mechanical processing (cleaning, cutting, portioning, heating, or freezing), there's minimal (or no) chemical or compositional treatment (adding sugar, salt, preservatives, hydrogenated oils, or flavor enhancers).

As for "plenty," the rule of thumb is to eat at least 80 percent of your calories from the "good stuff," and so long as you do that, you can use your remaining calories (20 percent or less) for your favorite treats, if you desire. Hence the "flexible" in "flexible dieting."

4. Eat a balance of carbohydrate and fat that works for you.

If you're like most people, this final principle of flexible dieting offers a welcome counterpoint to many current dietary dogmas. High-carb, low-carb, high-fat, low-fat—these choices don't much matter for improving your body composition, but they do for enjoying your diet. And so, with *Bigger Leaner Stronger,* you can eat carbs—even lots of carbs, if you prefer—including the ones you like to eat. You'll eat dietary fat, too, to support health, feel fuller, and enhance flavor, but you won't have to eat nearly as much as people following, let's say, a ketogenic or paleo diet (unless you want to eat that way).

To better understand why restricting carbs is unnecessary (and is actually counterproductive when you're physically active), let's discuss what a carbohydrate is and what happens in your body when you eat it.

Whether we're talking about the natural carbs (sugars) found in fruit or vegetables or the processed ones found in candy or soda, carbs of all types are composed of carbon, oxygen, and hydrogen and are turned into a simple molecule in the body called *glucose*, which the blood transports to cells for use. The key difference between these forms of carbohydrate is the *conversion rate*—the soda contains *sucrose* (table sugar), which turns into glucose quickly, whereas the fruit and vegetables contain *fructose* and *starch*, slower-burning sugars.

Some people say the slower a carbohydrate is digested, the "healthier" it is and vice versa—hence the dietary protocols that revolve around eating only "slow" carbs. This is mostly wrong. To be sure, vegetables are slow carbs that we should eat regularly, but a baked potato is a "fast" carb that's rich in vital nutrients; white rice—a nutritive "fast" grain—isn't associated with negative health effects; and a Snickers Bar is a slower carb than oatmeal, one of the greatest grains for human health. So, a slow carb isn't always a high-quality one and a fast

one isn't always junk.

The primary reason we can't replace carbs like potato, rice, and oatmeal with sugary delights isn't because of how quickly or slowly we assimilate them, but because the body needs a lot more than just glucose to survive, let alone thrive. It also needs vitamins, minerals, fiber, and other substances not found in junk food and beverages. Hence the association between eating large amounts of *added sugar*, like sucrose and fructose, and obesity, metabolic and health conditions, and nutritional deficiencies. People who eat too much added sugar tend to eat too little nutritious food, which harms their health. That doesn't mean that we need to completely avoid nutrient-poor carbs, though, or restrict our consumption of all forms of carbohydrate, especially those with health benefits like whole grains, fruit, vegetables, and legumes.

What's more, if you're healthy and physically active, you'll likely do better with a moderate- or even high-carb diet, not a low-carb one (especially if you're doing strength training). Your body creates a substance from carbohydrate known as *glycogen*, stores it in the muscles and liver, and uses it for fuel during intense exercise. When you cut your carbs, your body's glycogen reserves fall, and studies show that this hurts workout performance and processes related to muscle repair and growth. The situation is exacerbated by the fact that, when you're exercising regularly, restricting carbs also raises your cortisol and lowers your testosterone levels, which further hinders performance and recovery. It's no surprise, then, that research shows athletes following a low-carb diet recover slower and gain less muscle and strength than those eating more carbs.

Now you know why, in the next chapter, I'll invite you to eat a considerable amount of carbs regardless of whether you're looking to maximize fat loss or muscle gain. And if you're tempted (but timid) to accept my offer because of the specter of weight gain (and worse) that you've been warned about elsewhere, just

follow the plan exactly as it's laid out, and within mere weeks, the results will be undeniable, and your fears will be routed.

Let's now talk about dietary fat, the darling du jour of the diet industry, the star of runaway hits like paleo and ketogenic diets. According to some nutrition experts, a high-fat diet is a near cure-all. No matter what may ail you—obesity, lethargy, brain fog, acne, you name it—they claim the solution is simple: just eat more dietary fat (and, incidentally, less carbohydrate).

As you've likely already guessed, I don't share their enthusiasm for this slippery little molecule. Although you need to eat a fair amount of dietary fat to stay healthy, you have no reason to follow a high-fat diet unless you enjoy it, and even then, you may want to consider dialing it back. Let's talk about why, starting with a discussion of the two different fats found in food: triglycerides and cholesterol.

Triglycerides

Most of the dietary fat that we eat is in the *triglyceride* form, which is composed of three molecules of one or more *fatty acids* and one molecule of *glycerol* (a colorless, odorless, syrupy, sweet liquid).

Triglycerides are found in many foods, including dairy, nuts, seeds, meat, and more, and come in two types: Those that are liquid at room temperature (unsaturated) and those that are solid (saturated). Both are necessary, bolstering health and fitness in many ways—absorbing nutrients from food, creating hormones, supporting skin and hair health, etc.—but there are important differences between them that you should know about.

Saturated Fat

From the 1950s until just recently, officialdom has claimed that eating saturated fat increases the risk of heart disease—full stop. We now know this isn't true.

While prominent nutrition and cardiology researchers still maintain that eating *too much* saturated fat can impair heart health, low-to-moderate intake appears to pose no risk to our tickers.

The diet and supplement industries have made mountains of hay with this reversal, spawning all manner of high-fat diets and supplements that allow or even encourage eating nearly unlimited amounts of saturated fat. The problem, however, is that much of the scientific literature used to promote these regimens has been criticized by respected researchers as deeply flawed, and some studies still show a weak but consistent relationship between high saturated fat intake and heart disease. Thus, many scientists still stand by the generally accepted dietary guidelines for saturated fat intake, which is less than 10 percent of daily calories, and I agree—nobody can credibly claim that eating large amounts of saturated fat can't endanger your health.

Unsaturated Fat

Unsaturated fat comes in two forms: *monounsaturated* and *polyunsaturated*. Monounsaturated fat is liquid at room temperature and solidifies when cooled, and some of the better sources include nuts and their oils, olive oil, seeds and their oils, avocado and its oil, and many types of meat, poultry, and seafood. This is one of the best forms of fat you can eat—studies show that monounsaturated fat can reduce the risk of heart disease, and it also appears to be responsible for some of the health benefits associated with the Mediterranean diet, which involves eating a lot of olive oil.

Polyunsaturated fat is liquid at room temperature and remains so when cooled, and common sources include safflower, sesame, and sunflower seeds and oils, corn, many nuts and their oils, and fish such as salmon, sardines, tuna, trout, mackerel, and herring. There are two types of polyunsaturated fats known as

omega-3 and *omega-6* fatty acids—designations that refer to their chemical structures. We must get these fatty acids from our diet (our body can't produce them from other substances), which is why they're referred to as *essential fatty acids*.

These molecules produce many effects in the body, and while the chemistry is complex, omega-6 fatty acids generally cause "bad" (but sometimes necessary) things to happen, including higher levels of inflammation and insulin resistance, and omega-3 fatty acids generally cause "good" (but sometimes inappropriate) things to happen, including less inflammation and more insulin sensitivity.

Some people think that means they should avoid omega-6 fatty acids and load up on omega-3s, but the key is striking the right balance. While scientists once believed that it was important to keep omega-6 fatty acid intake relatively low, the latest research suggests that eating enough omega-3 fatty acids is more important than restricting omega-6 intake. Specifically, a ratio of omega-6-to-omega-3 intake between 2:1 and 1:1 appears to be optimal, which for most people would require at least 3-to-6 grams of omega-3 fatty acids per day.

Unfortunately, though, the average American doesn't approach this target. What with the large amounts of vegetable oil, butter, meat, and eggs low in omega-3 fatty acids and the paltry amount of fatty fish (the best dietary source of omega-3 fatty acids) in his diet, Joe Six Pack eats less than 2 grams of omega-3 fatty acids per day on average and maintains a ratio of omega-6-to-omega-3 in the range of 15-to-17:1. Eventually, this can play havoc with his health, increasing the risk of heart disease, cancer, osteoporosis, and inflammatory and autoimmune diseases.

Thankfully, the solution is straightforward: swap several weekly servings of poultry, pork, or red meat for fatty fish (salmon is my personal favorite) or supplement with fish oil (which we'll talk more about later in this book).

Cholesterol

Cholesterol is the other form of fat found in food, and it's a waxy substance present in all cells that's used to make hormones, vitamin D, and chemicals that aid digestion.

Like saturated fat, cholesterol has long been maligned as harmful to heart health, but recent research has shown that this isn't accurate. A crucial detail is how cholesterol is delivered to cells. It's carried by molecules known as *lipoproteins*, which are made of fat and proteins. There are two kinds of lipoproteins:

1. Low-density lipoprotein (LDL)
2. High-density lipoprotein (HDL)

Although similar in structure, these molecules have different effects in the body. Studies show that high levels of LDL in your blood can lead to an accumulation of fat in your arteries and increase the risk of heart disease, which is why people generally refer to it as the "bad" type of cholesterol. HDL, on the other hand, carries cholesterol to your liver and other organs, which flushes it from your body, earning it the reputation as a "good" molecule.

Scientists are still studying how lipoproteins work, but the current weight of the evidence shows that you want to keep your LDL levels relatively low and your HDL levels higher. Despite what you may have heard, the best ways to do this aren't eating a low-cholesterol diet and avoiding saturated fat. The recipe is fourfold:

1. Regular exercise (especially intense exercise, including strength training)
2. Healthy (relatively low) body fat levels

3. Good sleep hygiene
4. Not smoking

Studies also suggest that out of the four, regular intense exercise most benefits cholesterol levels, which points to a theme in the fitness literature—in some ways, exercise alone can neutralize at least some of the downsides of other unhealthy habits, including a poor diet. It would be a stretch to call exercise a natural panacea, but it may be as close to one as we'll ever have.

By now you likely see why flexible dieting allows you to find a balance of carbohydrate and dietary fat that works for you. Once you screen the evidence and separate the facts from the falsehoods, you realize that this approach not only reflects the science of how these macronutrients affect your health, performance, and body composition, but also allows you to enjoy your diet—the critical component of long-term success. In other words, flexible dieting is one of the only styles of eating that works as well in reality as it looks on paper.

Time and again, headlines announce that "diets don't work." According to one study or expert or another, if an eating practice qualifies as "dieting," it won't help most people lose significant amounts of weight and keep it off. You've probably at least entertained the idea yourself based on your own experiences with fad diets that have come and gone over the years.

This is like saying that economics or politics "don't work"—reductive and flat-out ridiculous when you understand that the problems aren't with the framework but the forms. Most diets don't work simply because they suck—they heavily restrict calories, which makes you miserable; they neglect protein, which means more hunger and muscle loss; they forbid foods, which frets your nerves;

and they lack an exit ramp for returning to normal eating, which is why so many people regain some, much, or even all the fat they lose.

None of that means that all forms of dieting *must* suck and fail, though. As you'll soon see for yourself, your diet can be effective, enjoyable, and sustainable, no matter your goals and tastes. With flexible dieting, you can secure a 360-degree win. And in the next chapter, you'll learn how, including figuring out your calories, macros, and food choices, culminating in your first *Bigger Leaner Stronger* meal plan that you can implement right away, while you read the rest of this book.

KEY TAKEAWAYS

- You use diet to control your body fat level and boost muscle growth, and you use training to gain and maintain strength and muscle.
- "Flexible dieting" has four steps: control your calories, eat plenty of protein, eat plenty of nutritious, relatively unprocessed foods, and eat a balance of carbohydrate and fat that works for you.
- Controlling your calories is the first and most important part of controlling your weight and improving your body composition.
- To lose fat, you must consistently eat fewer calories than you burn.
- To maximize muscle growth, you must consistently eat more calories than you burn.
- If you want to maintain your body fat level and slowly add muscle and strength, you want your calorie intake to roughly match your calorie expenditure.
- A high-protein diet is superior to a low-protein diet in multiple ways, especially for people looking to improve their body composition.
- Eating plenty of nutritious, relatively unprocessed foods prevents nutritional deficiencies that cause health problems, impair mental and physical performance, and even blunt fat loss and muscle gain.
- A good rule of thumb is to eat at least 80 percent of your calories from nutritious, unprocessed foods, and use your remaining calories (20 percent or less) for your favorite treats, if you desire.
- If you're healthy and physically active, you'll likely do better with a moderate- or even high-carb diet, not a low-carb one (especially if you're doing strength training).

- While eating too much saturated fat can impair heart health, a low-to-moderate intake of less than 10 percent of daily calories appears to pose no risk to our tickers.
- Eating enough omega-3 fatty acids is more important than restricting omega-6 intake. Specifically, a ratio of omega-6-to-omega-3 intake between 2:1 and 1:1 appears to be optimal, which for most people would require at least 3-to-6 grams of omega-3 fatty acids per day.
- The current weight of the evidence shows that you want to keep your LDL levels relatively low and your HDL levels higher, which is best done by exercising regularly, maintaining relatively low body fat levels, getting enough sleep, and not smoking.

The Big Ideas

1. Calculate how many calories to eat every day.

2. Calculate how much protein, carbohydrate, and fat to eat every day.

3. Eat plenty of nutritious foods you like and allow for treats.

9

The Easiest Way to Calculate Your Calories and Macros

You are right to be wary. There is much bullshit. Be wary of me too, because I may be wrong. Make up your own mind after you evaluate all the evidence and the logic.

—MARK RIPPETOE

With the flexible dieting formula you learned in the last chapter, you can transform not only your physique but also your relationship with food. You can lose fat and gain muscle while giving yourself permission to eat a wide variety of foods you enjoy without guilt or shame. You can steadily follow a meal plan without being consumed by it. You can nourish your body without neglecting your needs.

There's a catch, however: It involves some math. It's simple stuff—basic arithmetic—and I'll carefully explain and illustrate each step, but if figures make your eyes glaze, take your time with this material. Not only do I know with certainty that you can master the numbers (I've yet to have anyone drop out of this class), I also have a contingency plan: Ready-made meal plans for losing fat and building muscle. You won't want to follow these templates for long (you'll want to customize them), but they can help you get into your stride.

To give you a foretaste of the *Bigger Leaner Stronger* method of flexible dieting, let's walk through what a typical day on this program might look like. As you'll see, I'll ask you to use measuring utensils to ensure you don't accidentally over- or undereat, but in time, as you become attuned to proper portion sizes, you'll use the spoons, cups, and scale less and less until you don't need them at all (except in special circumstances—like dramatic changes in your meal plan).

So, here we go: You wake up and mix a scoop of protein powder with some water, milk, or a milk substitute, and you eat a banana. That holds you over until the middle of the morning, when you enjoy a cup of low-fat Greek yogurt (or, if you're like me, Icelandic skyr) and 10 halves of walnuts. Next is lunch, where you have 5 ounces of chicken or fish on a homemade salad topped with tomato, carrot, and three tablespoons of your favorite dressing. A couple of hours later, it's midafternoon snack time, and you eat a cup of low-fat cottage cheese topped with ½ a cup of blueberries on a buttered English muffin. Then, at dinner, you cook up a tasty meal of another 5 ounces of chicken or fish, a cup of coconut rice, and a vegetable medley. Finally, to end on a (sugar) high note, you savor 3 ounces of your favorite dark chocolate. Your actual mileage will vary, of course, but this example gives you a flavor of the straightforward nature of the program.

In the previous chapter, you learned the framework of flexible dieting, and now, you'll learn the fieldwork—the practical formula for fitting it to your goals and lifestyle. This process has four steps:

1. Calculate your calorie target.
2. Calculate your macro targets.
3. Choose your foods.
4. Create your meal plan.

Let's review each, starting with the first element: calories.

How Many Calories Should You Eat Every Day?

Suppose you're planning to drive across the country with a friend, and they say they want to do it without looking at the gas tank. They insist you should only stop to refuel when you feel like stopping, and only pump as much (or as little) gas as you feel like pumping. What would you think?

What if they noticed your side-eye and snapped back with something like, "I refuse to be a slave to the fuel meter! I'll drive however I want!" Or "Someone on Instagram said you don't have to worry about fuel levels if you use organic, gluten-free, low-carb, non-GMO gasoline." Or "Who wants to constantly keep an eye on fuel levels? There has to be a better way." What would you do next? Gather up your toys and go play with someone else, right?

The point is, when someone says they want to improve their body composition without even considering their calorie intake, they're acting just as cracked. You don't have to count calories per se (although there are good reasons to do so, at least initially), but if you want to greatly increase your odds of success, you need to ensure that your energy balance serves your goals.

Fortunately, you don't need a degree in Excel or even a special app to figure out how many calories to eat—just the calculator in your phone. First, you need to decide what you want to do with your physique. Here are your basic options:

- Cutting—If you want to get leaner, you need to consistently eat fewer calories than you burn. Fitness folk call this a "cutting phase."
- Lean gaining—If you want to maximize muscle and strength gain, you need to consistently eat slightly more calories than you're burning. This is a "lean gaining" phase.
- Maintaining—If you're happy with your body composition, and you want to slowly (and for more experienced weightlifters, *very* slowly)

add muscle and strength without gaining fat, you need to consistently eat about the same number of calories you're burning. This is a "maintenance phase."

To get the body you want, all you have to do is use lean gaining phases to focus on adding muscle mass to your frame (along with some fat) and then cutting phases to strip away that fat (while retaining the newly gained muscle). Think of it like sculpting, where you add clay (lean gaining) and then mold it (cutting) until you're ready to present your creation's final form (maintenance). This is how you transform your physique.

Maintenance can feel like quite a luxury, too, because the fitter you get, the more lenient you can be with your diet and training without losing ground. It takes a lot more time, effort, and precision to build a great physique than to maintain one, so once you're ready to flip on the cruise control, you'll be able to eat more loosely, allowing your calories and macronutrients to fluctuate more than usual. If you desire, you can also scale back your strength training to try other types of exercise that may interest you, like calisthenics, circuit training, Olympic weightlifting, martial arts, cross-training, or something else.

This phased approach to bodybuilding is vitally important, but understood and followed by few. Instead, most people try to gain muscle and lose fat at the same time and wind up failing on both fronts. Why? "Body recomposition," as it's known, is only possible to any significant degree when you're new to strength training and your body is hyper-responsive to it. Once your "newbie gains" have expired, however—six to eight months for most people—the only way to continue gaining muscle and strength is lean gaining, which always entails at least some fat gain as well, and the only way to lose fat is still cutting.

Whenever someone comes to me stuck in a body composition rut, this

phenomenon is almost always one of the reasons. They haven't faced up to the fact that the rules have changed and they no longer can have their cake and eat it too. They're usually eating maintenance calories (and often think they're eating far more than they really are), and as they err on the side of undereating to prevent fat gain, they're also frequently in a slight calorie deficit. When this is the case, they'll remain stalled no matter what they do in the gym.

Remember this advice as you watch your abs fade while you're in a lean gaining phase. You'll have them back soon enough (along with a slightly fitter physique). Furthermore, realize that body fat going up when you're lean gaining is a good sign because most people gain about as much muscle as fat. Thus, you know you're succeeding when your body weight, body fat, and strength are all rising.

How Many Calories You Should Eat When Cutting

We've established that you must consistently eat fewer calories than you burn to lose significant amounts of fat, but how many fewer? Ten percent? Twenty percent? More?

Some experts counsel mild calorie restriction and modest workouts to lean out over many months. The advantages of this "slow cutting" approach supposedly include less muscle loss, better workouts, as well as less hunger and fewer cravings. This isn't entirely wrong, only wrongheaded. Slow cutting can feel easier than a more aggressive approach, but the upsides are minor in most people and offset by a major drawback: duration.

Slow cutting is, well, *slow*, and this can cause more trouble than eating less food every day. For instance, by eating 10 percent fewer calories than you burn every day instead of 20 percent fewer, you're halving your rate of fat loss and doubling the duration of your cutting phase. And if you're like most people, the longer you're restricting your calories, even slightly, the more likely you are to stumble over or even succumb to life disruptions, dietary slip-ups, scheduling snafus, etc. So when cutting, we don't want to take the scenic route and risk losing our way. Instead, we want to zip directly toward our destination, with as few stops as we can comfortably manage. This way, we get faster results while spending less time restricting our calories and more time enjoying more food.

Fortunately, when you know what you're doing, you can achieve rapid fat loss without sacrificing muscle, suffering in the gym, or sinking your metabolism. This mostly depends on how many calories and how much protein you eat. As for the calories, my recommendation for cutting phases is an aggressive-but-not-reckless calorie deficit of 20-to-25 percent.

This isn't a range I picked from thin air. Research shows that, when

combined with resistance training and a high-protein diet, eating 75-to-80 percent of the calories you're burning every day strikes an ideal balance between fat loss and muscle preservation—even in lean athletes who are more susceptible to losing muscle while cutting.

How many calories is that for you, though? This is where many evidence-based fitness guys like me would begin sharing formulas for calculating *basal metabolic rate* (how much energy you burn at rest) and then *total daily energy expenditure* (your basal metabolic rate plus additional energy burned through physical activity and digesting food).

This approach has advantages, especially with people who are unusually active, but chances are you're not one of those people, so we can take a shortcut. When cutting, eating between 8 to 12 calories per pound of body weight per day creates a 20-to-25 percent calorie deficit in most people. If a guy weighs 210 pounds and wants to lose fat rapidly, he should eat between 1,700 (210 x 8, rounded up by 20 calories) and 2,500 (210 x 12, rounded down by 20 calories) calories per day.

You should choose the low or high end of the range based on how physically active you are.

- If you're sedentary (little to no exercise or vigorous physical activity), you'd have to choose the lowest number (8) to lose a significant amount of fat. This can work (a calorie deficit is a calorie deficit, after all), but it makes for a meager meal plan ("poverty macros" as bodybuilders like to say) and all but guarantees muscle loss. Furthermore, if you simply eat more without also exercising, you're now "slow cutting," which is also suboptimal. This is one of the many reasons I don't recommend solely

restricting calories for weight loss—including exercise, particularly strength training, makes the process more productive and enjoyable.

- If you're lightly active (one-to-three hours of exercise or vigorous physical activity per week), you should choose the middle of the range (9-to-10 calories per pound of body weight per day). This, for instance, would be appropriate if you're following the 3-day *Bigger Leaner Stronger* program and don't do any cardio to speed up your fat loss.
- If you're moderately active (four-to-seven hours of exercise or vigorous physical activity per week), you should choose the highest number (12). This would be a good place to start if you're following the 3-day program and doing an hour or two of cardio per week or if you're following the 5-day program with no cardio.
- If you're very active (eight or more hours of exercise or vigorous physical activity per week), you'll likely want to eat more than 12 calories per pound of body weight per day when cutting—14 or even 16 depending on how much energy you're generally burning every day. This is a case where more detailed metabolic calculations are recommended, and to make it easy for you, I've created a total daily energy expenditure calculator that you can find at www.biggerleanerstronger.com/TDEE.

And just like that, we've demystified an element of dieting—calorie intake—that stymies so many, ranging from the halfhearted weekend warrior to the hard-charging athlete. But how can the answer be so artless? How can such a simple method work?

It starts with a little-known fact: Your basic metabolic needs for survival drive most of the body's energy expenditure, not physical activity. For instance,

the brain alone accounts for about 20 percent of the basal metabolic rate, and excluding muscle, the other major organs are responsible for another 60 percent. To give that context, my basal metabolic rate is about 2,100 calories per day, so nearly 1,700 of those calories are just the table stakes of staying alive.

Soon after scientists discovered how energy hungry our organs are, they realized that with the right data, they could produce a mathematical model for estimating energy expenditure in people who didn't move around much. Next, they set out to expand their computations to include an estimation of the additional calories burned from moving around. To do this, researchers exhaustively explored the energy costs of various types of activities and used their findings to slowly increase the accuracy and utility of their predictive formulas. Eventually, all they needed to approximate someone's all-in energy expenditure was their gender, weight, and amount of light, moderate, and vigorous physical activity per week.

This line of work was (and still is) unknown to most, but it gained currency among the more scientifically minded bodybuilders, who started to use it to refine their nutrition plans for more fat loss and muscle growth. Then, astute practitioners developed guidelines based on patterns they observed, like 8-to-12 calories per pound of body weight per day for cutting and others, which I'll also share with you in this chapter.

So, while these dietary signposts are simple, don't mistake them as simplistic. Additionally, remember that no matter how involved calculating calorie intake is, it only gives you a reasonable starting place, to be adjusted based on how your body responds (something we'll discuss later in this book).

How Many Calories You Should Eat When Lean Gaining

As you learned in chapter 7, the easiest way to maximize muscle building is to intentionally eat more calories than you burn. You don't want to overdo this, though, because after a point, eating more only makes you fatter, not fitter. What is this threshold? Research suggests that it's somewhere around 110 percent of your total daily energy expenditure. This means that you can gain almost as much muscle (but a lot less fat) eating 10 percent more calories than you burn every day as you can at 20 or 30 percent more.

So that's my recommendation for calibrating calories when lean gaining: Maintain a calorie surplus of about 10 percent. For most people, this comes out to 16-to-18 calories per pound of body weight per day.

As with cutting, you should choose your number based on how physically active you are.

- If you're sedentary (no exercise or vigorous activity), you shouldn't be lean gaining because extra calories without resistance training just makes you fatter.
- If you're lightly active (one-to-three hours of exercise or vigorous activity per week), start with 16 (and try to do at least three strength training workouts per week).
- If you're moderately active (five or more hours of exercise or vigorous activity per week), go with 17, and then if you aren't steadily gaining weight and strength, move up to 18.
- If you're very active (eight or more hours of exercise or vigorous physical activity per week), use my total daily energy expenditure calculator (www.biggerleanerstronger.com/TDEE) to determine how many calories you should eat when lean gaining.

How Many Calories You Should Eat When Maintaining

Twelve-to-sixteen calories per pound of body weight per day is the sweet spot for most people when maintaining.

- If you're sedentary, choose the lowest number (and start training!).
- If you're lightly active, 13-to-14 should work.
- If you're moderately active, 15-to-16.
- If you're very active, use my total daily energy expenditure calculator (www.biggerleanerstronger.com/TDEE) to determine how many calories you should eat when maintaining.

And that's it for calories. Before you can use what you've just learned to create effective meal plans, though, we need to address your protein, carbohydrate, and fat intake (macros).

How Much Protein You Should Eat

There's a lot of conflicting advice on protein intake. Zealous meat eaters and weightlifters often suggest sky-high amounts, up to 2 grams per pound of body weight per day. On the other hand, many fitness experts, especially those following plant-based diets, advocate a much lower amount, ranging between 0.5 and 0.8 grams per pound of body weight per day.

Fortunately, a large amount of research has been done on the protein needs of physically active people. For example, the authors of a review published in the Journal of the International Society of Sports Nutrition explain that 0.55-to-1 gram of protein per pound of body weight per day—or 25-to-40 percent of daily calories—is adequate for supporting muscle gain so long as calories are at or above expenditure (maintenance or lean gaining). And when calories are restricted for fat loss, the authors suggest a higher protein intake: 1-to-1.5 grams per pound of body weight per day. That said, the higher end of that range isn't necessary unless you're lean and muscular and working to get very lean (a bodybuilder preparing for a show, for instance).

For the rest of us, 1-to-1.2 grams of protein per pound of body weight per day when cutting is adequate, and if someone's very overweight, 0.6-to-0.8 grams per pound of body weight per day is sufficient. And when lean gaining and maintaining, 0.8-to-1 gram of protein per pound of body weight per day is fitting. In all cases, protein intake usually comes out to around 30-to-40 percent of daily calories.

To put ideal protein intake into perspective, here's the protein content of popular high-protein foods:

- A full chicken breast: ~55 grams of protein
- A 3-ounce steak: ~24 grams of protein

- A 3-ounce salmon filet: ~19 grams of protein
- A cup of low-fat Greek or Icelandic (my favorite) yogurt: ~15 grams of protein
- An ounce of Parmesan cheese: 11 grams of protein
- An egg: 6 grams of protein
- A cup of pinto, mung, or fava beans: ~14 grams of protein
- A cup of green peas: 8 grams of protein
- A cup of cooked rice or quinoa: ~7 grams of protein
- A scoop of whey protein powder: ~20 grams of protein
- A protein bar: 15-to-20 grams of protein, depending on the brand

As you can see, high-protein eating is fairly straightforward—a couple of servings of meat or fish per day along with some dairy, legumes, or whole grains, and maybe a scoop or two of protein powder if needed. This may be more protein than you're used to eating, but in my experience, most people settle into the routine within a month or two. If you're concerned because you're not big on meat, seafood, and dairy (the best whole-food sources of protein), you can lean more on sources of high-quality plant protein (beans, spelt, chickpeas, lentils, and seitan are my top picks) or protein powders.

That's it for figuring out how much protein to eat.

How Much Carbohydrate You Should Eat

Unlike protein, you have a lot of flexibility with your carbohydrate intake because it has a smaller impact on your body composition. If you prefer restricting your carbs or even following a low-carb diet, we can make that work.

However, as eating more carbs, not less, works best for most physically active people, especially those doing regular strength training, I recommend you start with 30-to-50 percent of your daily calories from carbohydrate when cutting, lean gaining, or maintaining. This comes out to about 0.75-to-2 grams of carbs per pound of body weight per day for most people.

And how do you turn "percent of daily calories" into "grams of carbs per day"? Easy. Let's say you're a 200-pound man starting a cutting phase. You've just determined your daily calorie target is 2,100 calories, and you want to get 40 percent of those calories from carbs. First, you'd multiply 2,100 by 0.4 to get 840. Then, since each gram of carbohydrate contains about 4 calories, you divide the calories of carbs by 4 to determine how many grams to eat every day. So, in our example, you'd divide 840 by 4, which comes to 210 grams of carbs per day.

Now, if you like lower-carb living, you can adjust that number downward—15-to-20 percent of daily calories from carbs works well for many people because it gives enough room for a healthy amount of fruits and vegetables—or upward to 50 or even 60 percent if you particularly enjoy carbs, so long as you ensure that the majority of those calories are nutritious and that your protein and fat intake doesn't drop too low.

Also, if you drastically cut your carbs, you'll need to increase either your protein or your fat intake to ensure you don't eat too few calories. I could make an argument for increasing just your protein intake (more satiety and possibly more muscle growth and less fat gain), but it won't likely work because most low-

carbers want to "trade" carbs for more fat, not protein, and that's fine so long as it doesn't send saturated fat soaring. Anyway, to successfully adjust your carbohydrate intake downward, you only need to know how to turn additional calories allotted to protein and fat into grams. Like carbohydrate, protein also contains about 4 calories per gram, and fat has about 9 per gram, so here's a simple way to work the numbers:

1. Calculate your daily calories from carbs.
2. Calculate your daily calories from protein.
3. Add them together and subtract the sum from your total daily calories.
4. Assign the remaining calories to protein and fat as you see fit.

To illustrate this, say you're 200 pounds again and plan on eating 2,100 calories and 200 grams of protein per day with 20 percent of calories from carbs. To determine grams of carbs per day, you do the following: 2,000 x 0.2 = 420 and 420 / 4 = 105. To turn 200 grams of protein into calories: 200 x 4 = 800. Then, to add them together and subtract the sum from total daily calories: 420 + 800 = 1,220, which is deducted from 2,100 for 880 calories remaining to give over to protein or fat, and if you wanted to allocate them all to fat: 880 / 9 = 98 grams (which you could round up to 100). Thus, your macros would be 200 grams of protein, 105 grams of carbs, and 95 to 100 grams of fat, for a total of approximately 2,100 calories.

How Much Fat You Should Eat

You don't need nearly as much dietary fat for optimal health as the fat freaks claim. Studies show that 20-to-30 percent of daily calories from fat—0.2-to-0.4 grams of fat per pound of body weight per day for most people—works well. If you're following a low-carb diet, however, you can eat a lot more fat than that—up to 50-to-60 percent of daily calories depending on your preferences.

There are two ways to tally your daily fat target:

1. Figure out your protein and carb targets first, and see how many calories that leaves for fat.
2. Figure out your protein and fat targets first, and see how many calories that leaves for carbs.

If you care more about eating enough carbs than splurging on fat, go with number one, and if eating oodles of fat is more important to you than carbs, number two. Either way, the procedure is the same: turn the two macros you've worked out into calories, subtract the sum from your total daily calories to learn how many you have left for the final macro, and change those remaining calories into grams.

For instance, if you weigh 170 pounds and are starting a lean gaining phase, you may work out your calories and macros like this:

- 170 x 17 = 2,890 calories per day, rounded up to 2,900
- 170 grams of protein per day, which is 170 x 4 = 680 calories
- 2,900 x 0.5 = 1,450 calories from carbs per day, which is 1,450 / 4 = 362 grams, rounded down to 360

- 680 + 1,450 = 2,130 subtracted from 2,900 = 770 calories per day remaining for fat, which is 770 / 9 = 85 grams

Once you've finished calculating your macros, you can quickly confirm you've done it correctly by multiplying your protein and carbs by 4 and fat by 9 and checking the result against your daily calorie target. They won't match exactly, but they should be within 5 percent or so of each other. If the result is off by a significant margin, however, review your math because there's a mistake.

To help you better understand all of this in practice, let's review some more examples of calorie and macro targets for different dietary preferences (primarily adjustments to carbohydrate and fat), weights, goals, and activity levels. Also, I'll continue rounding up and down for the sake of simplicity—62 grams of fat becomes 60, for instance, 278 grams of carbs becomes 280, and so forth.

Weight	Goal	Activity Level	Calories	Protein	Carbs	Fat
170 lbs	Cutting	3 hours/week	1,700	170	130	55
190 lbs	Cutting	6 hours/week	2,300	190	230	70
210 lbs	Cutting	2 hours/week	2,100	210	170	65
230 lbs	Cutting	5 hours/week	2,800	230	290	80
250 lbs	Cutting	8 hours/week	3,000	200	370	80
150 lbs	Lean gaining	5 hours/week	2,700	150	320	90
170 lbs	Lean gaining	3 hours/week	2,700	170	340	75
190 lbs	Lean gaining	8 hours/week	3,400	190	435	100

So much for the arithmetic of dieting. Now let's thrash out food choices.

What Kind of Foods You Should Eat

I wish that we could supercharge our body and mind by eating just a few portions of special foods every day, but it just ain't so. No individual food transforms our health or fitness. Only a lifestyle can do that—one that prioritizes proper nutrition, training, sleep hygiene, and rest and recovery.

Unfortunately, grifters never let pesky facts foil a ripe opportunity for fraud, and so we have the "superfood" bonanza, starring greens, grains, seeds, fruits, and other predominantly plant-derived foods. This craze has motivated many people to eat better, but it has also sowed confusion about how the body works and how to boost health and wellness. Here's the score: you need to eat several portions of fruit and vegetables every day to get adequate nutrition, including vitamins, minerals, and fiber. And if your macros allow it, a daily portion of whole grains and legumes is also advisable.

This element of eating is nonnegotiable. Just as you can't reliably manage your weight without an understanding of energy balance or reliably gain muscle without a high-protein diet, you can't maintain optimal long-term health without eating an abundance of fruits and vegetables. No exceptions.

Eating a variety of fruits and veggies—especially colorful ones—is also helpful, because some contain more of certain nutrients than others. Most of your calories should come from a combination of these types of foods and whole grains (brown rice, corn, oats, quinoa, barley, etc.), healthy fat (oils—especially olive oil—nuts, seeds, and avocado are excellent), legumes (beans and peas), and tubers (potatoes and other root vegetables).

As for protein sources, the worthiest are animal foods like meat, fish, eggs, dairy, whey, and casein. While many plant foods contain protein, and sometimes a fair amount per serving, it's often less conducive to muscle gain than animal

protein because of deficiencies related to their amino acid profiles, bioavailability, or both. That isn't to say you can't get fit on a plant-based diet—you absolutely can—only that it's trickier and requires studious micromanagement of your diet.

For all the diversity of food choices I've just given you, you may have noticed that I didn't include any caloric beverages. This is intentional because they often have little nutritional value and don't satiate like food, making them a generally poor source of calories. Drink 500 calories of fruit juice or soda, which contains little or no nutritional value, and you can be hungry an hour later, whereas 500 calories of nutrient-dense protein and veggies will keep you full for hours. Little wonder, then, that studies show that sugary drink consumption is associated with weight gain and an increased risk of several medical problems, including diabetes and an increased risk of obesity.

That said, you don't have to completely abstain from all caloric beverages. Except in the case of whole milk, a source of protein and healthy fat, you just have to regard caloric drinks as treats—the next and final element of *Bigger Leaner Stronger* meal planning we'll discuss.

What should you drink, then? Water, of course. In fact, drinking enough water is a simple shortcut to better health and performance. Research shows that dehydration hurts cognition, mood, and endurance, causes constipation, and may even increase the risk of heart disease.

As for how much water you should drink, if you simply drink whenever you're thirsty, you'll stay well hydrated. While you can follow more rigid formulas (eight, 8-ounce glasses of water per day, 0.5 to 1 ounce of water per pound of body weight per day, one liter per hour of exercise, etc.), research shows it's unnecessary even when you're exercising in hot, humid conditions. If you're thirsty, drink water until you aren't thirsty, and if you aren't thirsty, don't drink any. This holds true even if you're a heavy sweater or exercising in hot, humid

conditions.

While we're talking about hydration, let's address the common claim that caffeinated beverages like pre-workout supplements, coffee, and tea are dehydrating. Caffeine does have a slight diuretic effect, but studies show that it's negligible even at high doses (up to 500 milligrams per day) and doesn't significantly impact hydration status. So, good news: your daily dose of jitter juice counts toward your water intake.

What About Treats?

As no individual food can harm your health—only your diet on the whole can—no foods are off limits with flexible dieting, no matter how "unhealthy" they supposedly are. For instance, my favorite dainties are dark chocolate, ice cream, pancakes, pastries, and pasta.

To make room for such sweetmeats, you can use up to 20 percent of your daily calories for them when cutting, lean gaining, or maintaining, and these calories can come from whatever mixture of protein, carbs, and fat you'd like.

Let's see how this might work with our 2,100-calorie cutting plan. In this case, we'd have 420 calories to give over to goodies, which provides plenty of options—a pint of low-calorie ice cream, a bar of chocolate, five Reese's cups, nearly a bag of lower-calorie chips, or a fistful of Oreos. And yes, I'm saying you can eat any of those things every single day when cutting if you'd like with absolutely no downsides.

Should You Cut, Lean Gain, or Maintain?

If you feel your *body fat percentage* (the portion of your body weight that's fat) is too high and above all you want to get leaner, you want to cut first. There's no reason to lean gain—which will require gaining some fat—when your primary concern right now isn't gaining muscle as quickly as possible. The same goes if you're very overweight—you also want to cut first even if your long-range goal involves gaining a fair amount of muscle mass. This is the healthiest and smartest way to begin your fitness journey.

If you're thin or lean and want to focus on getting bigger and stronger, you want to lean gain. What's more, if you're new to strength training, research shows that for the first couple of months of lean gaining, you'll gain very little fat.

Lastly, if your body fat is in a normal range, I recommend you decide to cut or lean gain based on your body fat percentage. If you're 15 percent body fat or higher, start *Bigger Leaner Stronger* with a cutting phase to get down to 10-to-12 percent. There are two reasons for this:

1. You'll like how you look more. At least half of the reason you're reading this book is to look good, and there's nothing wrong with that. When you get above 15 percent body fat, abs and vascularity disappear and you start feeling overweight, which can be demotivating. You may start to question why you're working so hard to look like *that*. By setting a ceiling for your body fat percentage, however, you'll find it easier to stay the course.
2. Your cutting phases will be easier. The higher your body fat percentage is at the beginning of a cut, the longer and more difficult it is to get lean because even when you do everything right, more time restricting calories generally means more hunger, cravings, and other pesky side

effects of dieting. Thus, by never letting your body fat level get too high, you'll always have shorter and more manageable cuts.

Finally, if your body fat percentage is between 10-to-15 percent, you can cut or lean gain based on what's most exciting to you. If you've never felt big and strong, and you most want to move more weight and take up more space, start with lean gaining. And if you've never seen your abs and want to check that box before committing to a calorie surplus, start with cutting.

You're probably not sure how to determine your body fat percentage, so let's review that, starting with some visuals.



As you can see, you look "athletic" around 15 percent body fat, "ripped" around 10 percent, and "shredded" around 8 percent. To figure out your approximate body fat percentage, simply compare your physique to those in the chart and find the one that's the closest fit.

Many people want to measure their body fat even more accurately, and while this isn't necessary to get into great shape (or even useful, really, because ultimately all that matters is what we see in the mirror), my preferred method for this is called *skinfold testing*. You should know, however, that skinfold testing tends to work best with people who have average or above-average amounts of body fat (15 percent and higher). In leaner people, it tends to underestimate body fat percentage. For example, according to the method I'll share, I'm close to 5 percent body fat, but in reality, I'm around 9 percent.

There are different protocols for skinfold testing, but all involve using a device called a caliper to pinch and measure the thickness of the skin and fat underneath in several places. The procedure that's most popular among bodybuilders is known as the Jackson/Pollock 3-Site method, and as the name implies, it calls for measuring in three locations—the right pec (chest muscle), abdomen, and right thigh. Then, you use mathematical formulas for translating those measurements into an estimated body fat percentage.

Although such a technique sounds far more precise than comparing yourself to pictures, if the skinfold measurements aren't taken properly, it can produce wildly incorrect results. And remember that even when it's done perfectly, skinfold testing only allows you to more precisely pinpoint your body fat percentage from, say, "13-to-16 percent" (looking at pictures) to "14-to-15 percent" (using skinfold testing).

Fortunately, however, skinfold testing is an easy skill to grasp. Let's discuss the key points, starting with the fundamentals.

- Take all measurements on the right side of your body, standing, and with your muscles relaxed (don't tense or flex your muscles because this skews measurements).
- Pinch your skin by placing the thumb and forefinger about two inches apart, firmly pushing them into your skin and then together, and then gently pulling the fold of tissue from your body.
- Measure a pinched fold by placing the tips of the caliper in the middle of it between your fingertips (perpendicular to it), close the caliper until it clicks (and no further), and note the measurement. (Some calipers don't click when optimal pressure is applied, but the company Accu-Measure makes a "click" type that I like.)
- Take each round of measurements three times and calculate the average for each location. This means you'd take the first measurements of your right pec, abdomen, and thigh, then a second set of measurements followed by a third, and then you'd figure out the average number for each spot. So, if your first pec measurement is 15 mm, your second is 12 mm, and your third is 13 mm, you'd use the average of ~13 mm to figure out your body fat percentage. As you become more skilled at skinfold measurements (and aware of their limitations), you can take single measurements and dispense with the averaging.

As for how to take each of those measurements:

- For the pec measurement, pinch a diagonal skinfold halfway between your nipple and the nearest edge of your armpit.
- For the abdomen measurement, pinch a vertical fold of skin one inch to the right of your belly button.

- To take a thigh measurement, pinch a vertical fold of skin in the center of the front of your thigh halfway between your kneecap and the point where the top of your thigh meets your hip.

Once you have your pec, abdomen, and thigh averages, plug them into the following formula to calculate your body fat percentage:

Body density = 1.10938 – (0.0008267 x sum of skinfolds) + (0.0000016 x square of the sum of skinfolds) – (0.0002574 x age)

Body fat percentage (%) = (495 / body density) – 450

If you'd rather skip the math, you can find a simple calculator I created at www.biggerleanerstronger.com/bodyfat.

With respect to maintenance phases, most people use them primarily to maintain ideal body fat levels for extended periods of time (summer, for example). As you neither lose fat nor enhance muscle building when maintaining, it's usually not a suitable way to start the *Bigger Leaner Stronger* program.

How Long Should You Cut, Lean Gain, and Maintain?

The duration of a cut depends on how much fat you want to lose and how quickly you lose it, and when lean gaining, on how lean you are when you start and how quickly you gain fat. Here are general directions to follow:

- Unless you have a good reason to stop sooner (personal preference, for instance), your cutting phases should end when you're around 10-to-12 percent body fat. Don't bother getting leaner unless you have a special reason because it's a brute to sustain (for most people, it requires a high level of activity and dietary discipline, including meticulous calorie tracking). Thus, when cutting, you can go for as long as it takes to reach your body fat target—eight-to-twelve weeks in most cases, but your first cut can take significantly longer if you have a lot of fat to lose.

Also, some people claim you shouldn't restrict calories for more than a few weeks or months at a time before taking a break because it can harm your metabolism, but such claims aren't supported by scientific research. That said, if you need to cut for longer than eight weeks to reach your goal, you don't need to restrict your calories every day from the beginning until the end. In fact, you probably shouldn't. If you have a lot of fat to lose, it may take several months, and while you could just gut it out, it's better to split up the cut into alternating periods of restricting and raising your calories.

An effective and evidence-based method of doing this is to restrict calories for six-to-eight weeks and then increase food to a maintenance level for five-to-seven days. These calorie increases give you a break from the stresses caused by dieting, which is why they're referred to as diet breaks. Here's how to do a diet break correctly:

1. Calculate your daily maintenance calories by multiplying your body weight by the appropriate number (as discussed earlier in this chapter). Fourteen calories per pound of body weight works well for most people who are working out at least a few times per week.
2. Determine how much you should increase your calories by subtracting your current daily calorie intake from your maintenance calorie target.
3. Add food to your meal plan until you reach your new calorie target, and to amplify the benefits of the break, get at least half of these extra calories from carbs.
4. Follow your new meal plan for five-to-seven days.
5. Return to your cutting plan for the next six-to-eight weeks.

Let's see how this would work for a 200-pound man cutting on 2,200 calories per day. After seven weeks of dieting, he notices his energy levels have fallen dramatically and he's wrangling with high levels of hunger and cravings. Time for a diet break. To do this, he first figures out his maintenance calories (200 x 14 = 2,800 calories), and then how much more he needs to eat every day (2,800 – 2,200 = 600 calories). He loves carbs, so he decides to use all of the calories toward carbohydrate, which allows for 150 grams to be added to his meal plan (600 / 4 calories per gram of carbohydrate). He likes oatmeal and fruit at breakfast, so he plugs that in along with a couple of slices of bread with his normal lunch and a cup of rice with dinner. After seven days of enjoying the extra food, he cuts it out of his plan to resume his cut at 2,200 calories per day.

- Unless you have a good reason to stop sooner, your lean gain phases should end when you're around 15-to-17 percent body fat. This gives

you plenty of time to gain muscle and strength—at least twelve-to-sixteen weeks, usually—and ensures you don't get too fat, which you'll likely regret when you have to cut.

With each cycle of lean gaining and cutting, you'll move a little closer to your ideal physique. When you've finished a cut, you can follow it with maintenance if you're happy with your physique or want to enjoy your muscle definition before lean gaining; or you can move directly into a lean gaining phase if you want to be bigger and stronger and are ready to get after it.

This is also why maintenance is best suited to people who have gained a considerable amount of muscle and now want to stay lean for a while (or even indefinitely). Remember when you're in a maintenance phase, not much changes with your body composition, which is great if you have the body you want, but not if you still have work to do.

It may not seem like it yet, but you've just taken a major stride toward a bigger, leaner, and stronger you and cleared one of the largest hurdles—the science of using food you like to lose fat, gain muscle, and optimize health and wellbeing.

You now understand how to figure out how many calories to eat every day and to translate them into protein, carbs, and fat. You also know which types of food to indulge in and which to limit; when to cut, lean gain, and maintain; and how to execute each of these dietary modes skillfully. We still have a gap to bridge, though, and that's how to use everything you've learned to create a concrete meal plan that provides the right amount of calories, macros, and nutrition for your goals while also accommodating your palate, schedule, and eating patterns.

All that and more in the next chapter!

KEY TAKEAWAYS

- If you want to get leaner, you need to consistently eat fewer calories than you burn. This is a "cutting phase."
- If you want to maximize muscle and strength gain, you need to consistently eat slightly more calories than you're burning. This is a "lean gaining" phase.
- If you're happy with your body composition, and you want to slowly (and for more experienced weightlifters, very slowly) add muscle and strength without gaining fat, you need to consistently eat about the same number of calories you're burning. This is a "maintenance phase."
- For cutting phases, use an aggressive-but-not-reckless calorie deficit of 20-to-25 percent. For most people, this means eating 8-to-12 calories per pound of body weight per day.
- For lean gaining phases, maintain a calorie surplus of about 10 percent. For most people, this comes out to 16-to-18 calories per pound of body weight per day.
- Twelve-to-sixteen calories per pound of body weight per day is the sweet spot for most people when maintaining.
- 1-to-1.2 grams of protein per pound of body weight per day when cutting is adequate, and if someone's very overweight, 0.6-to-0.8 grams per pound of body weight per day is sufficient.
- When lean gaining and maintaining, 0.8-to-1 gram of protein per pound of body weight per day is fitting.
- Start with 30-to-50 percent of your daily calories from carbohydrate when cutting, lean gaining, or maintaining. This comes out to about

0.75-to-2 grams of carbs per pound of body weight per day for most people.

- Get 20-to-30 percent of daily calories from fat, which is 0.2-to-0.4 grams of fat per pound of body weight per day for most people.
- You need to eat several portions of fruit and vegetables every day to get adequate nutrition, including vitamins, minerals, and fiber. And if your macros allow it, a daily portion of whole grains and legumes is also advisable.
- The best protein sources are animal foods like meat, fish, eggs, dairy, whey, and casein.
- Use up to 20 percent of your daily calories for treats when cutting, lean gaining, or maintaining, and these calories can come from whatever mixture of protein, carbs, and fat you'd like.
- If you feel your body fat percentage (the portion of your body weight that's fat) is too high and above all you want to get leaner, you want to cut first.
- If you're thin or lean and want to focus on getting bigger and stronger, you want to lean gain.
- If your body fat is in a normal range, I recommend you decide to cut or lean gain based on your body fat percentage. If you're 15 percent body fat or higher, start with a cutting phase to get down to 10-to-12 percent, and if your body fat percentage is between 10-to-15 percent, you can cut or lean gain based on what's most exciting to you.
- Unless you have a good reason to stop sooner (personal preference, for instance), your cutting phases should end when you're around 10-to-12 percent body fat.

- Unless you have a good reason to stop sooner, your lean gain phases should end when you're around 15-to-17 percent body fat.

The Big Ideas

1. Calculate your calories and macros.

2. Create your meal plan in layers.

3. Enjoy “cheat meals” the right way.

10

The Bigger Leaner Stronger Meal Plan

Don't measure yourself by what you have accomplished, but by what you should have accomplished with your ability.

—JOHN WOODEN

How did you learn to ride a bike? You probably started with training wheels, right? Then, once you felt sturdy, you ditched the stabilizers and rode freely. This gradual approach works well with diet and exercise as well. You start with a set of simple and clear instructions, and after logging enough meals and workouts, you'll be ready to add more moving parts.

This chapter gives you one of those training wheels in the form of an effective method of meal planning as well as done-for-you examples that you can use to guide your efforts and get you started right away. Then, in the next section of this book, we'll install the second wheel by learning the elements of effective strength training and reviewing precise training plans that'll show you how to get to making gains.

Now, there are many ways to create a meal plan, but the best fulfill five criteria. The ideal meal plan:

1. Controls your calories.
2. Controls your macros.
3. Includes foods you like.
4. Provides adequate nutrition.
5. Follows a schedule you like.

The more a meal plan meets these standards, the more likely it is to work, and when it accomplishes them all, results are virtually guaranteed. The *Bigger Leaner Stronger* method does just that.

With my system, you'll create a single-day eating plan that hits your calorie and macro targets with mostly nutritional foods and then you'll follow the plan every day. That doesn't mean you'll have to eat the same foods every day, though —you can include as much or as little variety in your plan as you'd like. For instance, you may want to create several options for one or more meals (breakfast or dinner A, B, or C) and choose whichever is most appetizing when you wake up or get home from work. If you're like most people I've worked with over the years, though, you'll discover you don't need much or any diversity in your daily routine when you get to eat stuff you like. What's more, you'll also probably come to appreciate the ease and convenience of a "samey" style of eating. In any event, you'll get to choose the approach that works best for you.

You can also create two (or more) meal plans if that better meets your needs. For instance, some people like to eat fewer and different meals on the weekends (skipping breakfast, usually), and this can easily be addressed with weekday and weekend plans.

The *Bigger Leaner Stronger* Method of Meal Planning

The *Bigger Leaner Stronger* method of meal planning is simple. There are just four steps:

1. Calculate your calories and macros.
2. Choose how many meals to eat.
3. Choose what to eat and drink.
4. Choose when and how much to eat and drink.

And voilà, you have a first-class meal plan to get your teeth into. Let's unravel each of these points.

1. Calculate your calories and macros.

As you learned in the previous chapter, this only requires two steps: First figuring out your daily calorie target based on your body composition goal and activity level and then turning that target into the proper proportions of macros. The following charts sum up both calculations.

	Cutting	Lean Gaining	Maintaining
Sedentary (little or no exercise or vigorous physical activity)	8 calories per pound of body weight per day	You shouldn't be lean gaining.	12 calories per pound of body weight per day
Lightly active (one-to-four hours of exercise or vigorous physical activity per week)	10 calories per pound of body weight per day	16 calories per pound of body weight per day	14 calories per pound of body weight per day
Moderately active (five-to-eight hours of exercise or vigorous physical activity per week)	12 calories per pound of body weight per day	18 calories per pound of body weight per day	16 calories per pound of body weight per day
Highly active (Nine or more hours of exercise or vigorous physical activity per week)	14+ calories per pound of body weight per day	20+ calories per pound of body weight per day	18+ calories per pound of body weight per day

	Protein	Carbohydrate	Fat
Cutting	1-to-1.2 gram per pound of body weight per day. When very overweight, 0.6-to-0.8 grams per pound of body weight per day. In both cases, it typically comes out to 30-to-40 percent of daily calories.	15-to-60 percent of daily calories. The sweet spot for most people is 30-to-40 percent, which typically comes out to 0.75-to-2 grams per pound of body weight per day.	20-to-55 percent of daily calories. 20-to-30 percent works well for most people, which is typically 0.2-to-0.4 grams per pound of body weight per day.
Lean gaining	0.8-to-1 gram per pound of body weight per day	15-to-60 percent of daily calories. The sweet spot for most people is 30-to-40 percent, which typically comes out to 0.75-to-2 grams per pound of body weight per day.	20-to-55 percent of daily calories. 20-to-30 percent works well for most people, which is typically 0.2-to-0.4 grams per pound of body weight per day.

Maintaining	0.8-to-1 gram per pound of body weight per day	15-to-60 percent of daily calories. The sweet spot for most people is 30-to-40 percent, which typically comes out to 0.75-to-2 grams per pound of body weight per day.	20-to-55 percent of daily calories. 20-to-30 percent works well for most people, which is typically 0.2-to-0.4 grams per pound of body weight per day.

Remember these guidelines are meant to be useful jumping-off points, not dietary diktats. You may need to modify my directions based on your actual circumstances and how your body responds (which we'll talk about later).

Before you keep reading, let's take a moment to work out your calories and macros now based on how much time you plan to give to your *Bigger Leaner Stronger* workouts every week (three- to-six hours is what most people commit to), and write them down. Refer back to the previous chapter as needed to refresh your memory on the procedure. Also, if you're not sure what ratio of carbs and fat will work best for you, start with 30-to-40 percent of daily calories from each.

2. Choose how many meals to eat.

You learned in chapter 7 that eating more or fewer meals has no appreciable effect on metabolism or fat loss. You also learned that effects on appetite and satisfaction vary. That means you should set up your meal schedule according to your inclinations and lifestyle. What matters most is eating the right amount of calories and macros every day, not eating per a specific pattern.

Most people settle on four-to-six meals per day (breakfast, lunch, and dinner, with a snack before and after lunch), but some like eating just two (lunch and dinner) or three (breakfast, lunch, and dinner). That said, try to eat at least two meals per day (and three or more is probably better) because research shows that eating just one meal per day can make it harder to gain muscle.

To help you choose the right meal frequency for you, consider the following points:

- **Your appetite.** When scheduling your meals, account for when you're hungriest, and you'll find it much easier to stick to your meal plan and get results. For instance, if you wake up famished, don't force yourself to eat a light breakfast or skip it. By extension, if you're like me and your appetite doesn't rev up until lunchtime and you prefer large dinners, feel free to plan your meals in line with this. (I train in the morning, and I have a cappuccino and banana thirty minutes before and a scoop of protein after, and I don't eat again until around noon). One meal I don't recommend skipping, however, is dinner because it can result in nighttime hunger that disrupts sleep. But it can be sparse if that works best for you—a serving of protein and vegetables is fine.
- **Your timing preferences.** Appetite aside, what are your favorite meals of the day? Do you love breakfast food the most? Or is a delicious

dinner more your style? Bear this in mind when creating your meal plan. For instance, my favorite meal of the day is dinner because I like to eat light throughout the day and then enjoy a large portion of my calories at the end of the day, when I'm done working.

- **Your schedule.** How many sit-down meals do you have time for? Is it easy for you to get in your few squares or are you too busy once your day begins? Consider the meal structure that'll best fit your routine, and don't be afraid to eat irregularly, go for several hours without eating, or supplement one or two proper meals with several smaller snack-sized ones throughout the day.

Next, come up with a meal schedule that makes sense to you, and write it down.

3. Choose what to eat and drink.

Once you know how many calories and macros to eat and when you'd like to eat every day, you're ready to figure out what foods to eat. The first step is listing your favorite types of …

- **Protein:** Foods that are mostly protein, like red meat, poultry and seafood; high-protein dairy like Greek yogurt, skyr, cottage cheese, etc.; egg white; high-protein plant foods like seitan, tofu, tempeh, etc.; and protein powder and bars.
- **Nutritious carbs:** Relatively unprocessed foods that are mostly carbohydrate, like fruits, veggies, legumes, tubers, and whole grains.
- **Healthy fat:** Relatively unprocessed foods that are mostly fat, like olive oil, avocado, nuts and nut butters; seeds; whole egg; and full-fat dairy, like regular yogurt, cheese, butter, and milk.
- **Treats:** More highly processed and less nutritious foods and caloric beverages, often with added sugar, like white bread and pasta, soda, energy drink, fruit juice, candy, breakfast cereal and bars, and pastry.

For example, my lists would look like this:

- **Protein:** pork chops, chicken breasts, New York strip steaks, 80/20 and 90/10 ground beef, skyr, low-fat cottage cheese, egg whites, and whey and vegan protein powder
- **Nutritious carbs:** onions, garlic, broccoli, mushrooms, peppers, carrots, cauliflower, green beans, peas, Brussels sprouts, strawberries, bananas, blueberries, potatoes, sweet potatoes, rice, quinoa, oatmeal, and black beans

- **Healthy fat:** avocados, olive oil, sesame oil, walnut oil, walnuts, whole milk, whole egg, and fish oil (supplement)
- **Treats:** ice cream, white pasta, dark chocolate, pastry, and pancakes

If a food you want to include in your meal plan isn't listed above, you can still eat it—just classify it correctly. And what about recipes, restaurants, alcohol, and other drinks (coffee, tea, juices, etc.)? They too can be incorporated in your meal plan. Let's talk about how.

Recipes

You can include recipes in your meal plans, but you need to know the calories and macros in each serving ("one serving of Beef Stroganoff for dinner with 350 calories, 20 grams of protein, 45 grams of carbs, and 10 grams of fat"). Therefore, it's best to choose recipes that provide macros or that are simple enough for you to work out the macros yourself (using an online database like www.cronometer.com).

"Fitness-friendly" recipes also contain common ingredients that are easy to measure and enhance with low- or zero-calorie spices and additions, and they produce food that's easy to portion out and that keeps well if leftovers are in the plan.

Less suitable recipes don't list macros and include ingredients or steps that make the macros difficult to grasp or separate into distinct servings, like gourmet dishes with sauces or loaded nachos, for example. You can still make those types of meals, but they're not worth the hassle right now if you're new to all of this. Save them for later, when you're a more experienced meal planner.

Fortunately, there's essentially an endless supply of recipes online and in cookbooks that fit the bill for every meal and macro. The free bonus material that comes with this book (www.biggerleanerstronger.com/bonus) includes twenty of

my favorite recipes from my flexible dieting cookbook *The Shredded Chef* (which contains 125 in all) and there are hundreds more at my blog (www.biggerleanerstronger.com/recipes). Gina Homolka's website www.skinnytaste.com as well as her *Skinnytaste* cookbooks are also terrific resources.

Here are examples of recipes that work well with the *Bigger Leaner Stronger* style of meal planning:

Cheesy Baked Egg Toast

Serves 1

Per serving:

- 330 calories
- 27 grams of protein
- 24 grams of carbs
- 14 grams of fat

Ingredients:

- 2 slices whole-wheat bread
- 2 large eggs
- 1 tablespoon butter
- Salt and freshly ground black pepper, to taste
- 1/4 cup shredded cheddar

Instructions:

1. Place a baking sheet in the middle of the oven and preheat the oven to 250°F (175°C).
2. Place the bread on a cutting board. Press the back of a teaspoon into the bread to form a rectangle-shaped depression in each slice. Take care not to rip the bread.
3. Season the bread with salt and pepper.
4. Crack an egg into the depressed rectangle shape in each slice of bread

and season the top of each egg.

5. Spread butter around the edges of the bread.
6. Sprinkle the cheese around the buttered edges of the bread.
7. Bake the bread for 10 minutes or until the cheese is beginning to brown and the egg is cooked.
8. Remove from the oven and serve.

Spiced Caribbean Oatmeal

Serves 4

Per serving:

- 292 calories
- 17 grams of protein
- 31 grams of carbs
- 12 grams of fat

Ingredients:

- 3 cups 2% milk
- 1 cup quick-cooking oats
- 1 tablespoon unpacked brown sugar
- 1/2 teaspoon ground cinnamon
- Pinch of ground nutmeg
- 1/4 cup ground pistachios
- Pinch of salt
- 1 (6-ounce) container 2% Greek yogurt

Instructions:

1. Combine the milk, oats, sugar, cinnamon, nutmeg, pistachios, and salt in a medium pot and warm over medium-high heat.
2. Bring the mixture to a boil, stirring constantly until the oatmeal is cooked, about 2 minutes.
3. Remove from the heat, swirl in the yogurt, and serve.

High-Protein Salad

Serves 4

Per serving:

- 228 calories
- 14 grams of protein
- 30 grams of carbs
- 7 grams of fat

Ingredients:

Salad:

- 1 can (15-ounce) kidney beans
- 1 can (15-ounce) lentils
- 3 cups arugula
- 4 tablespoons capers

Dressing:

- 1 tablespoon caper brine
- 2 tablespoons peanut butter
- 1 tablespoon tahini
- 1 tablespoon tamari (or soy sauce)
- 1 tablespoon balsamic vinegar
- 2 tablespoons hot sauce

Instructions:

1. In a bowl, whisk together all of the dressing ingredients until they're combined.
2. In a separate bowl, mix together the arugula, beans, lentils, and capers.
3. Top the salad mix with the dressing and serve.

Vanilla Protein Crepes

Serves 4

Per serving:

- 214 calories
- 17 grams of protein
- 5 grams of carbs
- 4 grams of fat

Ingredients:

- 1 cup old-fashioned oats
- 2/3 cup vanilla protein powder
- 2 teaspoons baking powder
- 1 teaspoon ground cinnamon
- 3 egg whites
- 1 cup milk
- 1/2 cup applesauce
- 1 teaspoon vanilla extract

Instructions:

1. In a bowl, mix together the old-fashioned oats, protein powder, baking powder, and cinnamon.
2. Add the remaining ingredients and blend until the mixture is smooth.
3. Coat a nonstick skillet with cooking spray and warm over medium-low heat.
4. Pour the batter into the skillet a 1/2 cup at a time until the crepe

reaches desired size and thickness.

5. When bubbles begin to form on top, flip the crepe.
6. Once fully cooked, remove the crepe and repeat the process until all the batter is used.
7. Serve the crepes rolled up with fresh fruit, yogurt, or plain.

Chunky Chicken Quesadilla

Serves 2

Per serving:

- 315 calories
- 30 grams of protein
- 28 grams of carbs
- 9 grams of fat

Ingredients:

- 1 (6-ounce) boneless, skinless chicken breast, trimmed of fat
- 1 tablespoon low fat sour cream
- 2 (8-inch) whole-wheat tortillas
- 1/3 cup salsa
- 1 cup shredded lettuce
- 1/3 cup shredded low-fat cheddar cheese

Instructions:

1. Coat a medium nonstick skillet with cooking spray and warm over medium heat. Add the chicken and cook for 3-to-5 minutes per side. Once fully cooked, transfer the chicken to a cutting board.
2. Spread the sour cream onto 1 tortilla. Slice the chicken breast and layer it over the sour cream, topping with salsa and lettuce. Sprinkle with cheese and top with the other tortilla.
3. Recoat the skillet with cooking spray and warm over low heat. Cook the quesadilla until golden, about 3 minutes per side, using a large

spatula to carefully flip it.

4. Remove from the skillet, slice, and serve.

Zesty Italian Meatball Wrap

Serves 6

Per serving:

- 329 calories
- 37 grams of protein
- 31 grams of carbs
- 7 grams of fat

Ingredients:

- 1 pound lean ground chicken
- 1/2 cup whole-wheat bread crumbs
- 1/2 onion, finely diced
- 1/2 cup shredded parmesan cheese
- 1 large egg, whisked
- 1 tablespoon garlic, minced
- 1/2 tablespoon fresh parsley, chopped
- 1 teaspoon Italian seasoning
- 1 teaspoon lemon zest
- 1/2 teaspoon sea salt
- 1/2 teaspoon freshly ground black pepper
- 1 1/2 cups marinara sauce
- 6 Italian herb flatbreads (or whole-wheat tortillas)
- 1 large green bell pepper, diced
- 2 small tomatoes, diced
- 1/2 cup banana peppers, sliced

Instructions:

1. Preheat the oven to 400°F (200°C). Coat a baking sheet with nonstick cooking spray.
2. Place the ground chicken, breadcrumbs, onion, parmesan cheese, egg, garlic, parsley, Italian seasoning, lemon zest, salt, and pepper in a bowl; mix with your hands to combine.
3. Using a spoon, form the mixture into 24 equally-sized meatballs
4. Place the meatballs evenly apart on the baking sheet and top each with 1 teaspoon of marinara sauce.
5. Bake the meatballs in the oven for 15-to-18 minutes or until fully cooked.
6. Spread 2 tablespoons of marinara sauce on each flatbread.
7. Place 4 meatballs in each wrap and top with bell pepper, tomatoes, and banana pepper, roll up and serve.

Beef Stroganoff

Serves 4

Per serving:

- 301 calories
- 29 grams of protein
- 6 grams of carbs
- 16 grams of fat

Ingredients:

- 8 ounces egg noodles
- 1 pound beef tenderloin, trimmed of fat and sliced
- Salt and ground pepper, to taste
- 2 tablespoons extra-virgin olive oil
- 1/2 medium onion, sliced
- 4 ounces white mushrooms, sliced
- 1 tablespoon cornstarch
- 1 (10.5-ounce) can condensed beef broth, divided
- 1 teaspoon Dijon mustard
- 1 clove garlic, peeled and minced
- 3 tablespoons white wine
- 1/2 tablespoon Worcestershire sauce
- 2 tablespoons low-fat sour cream
- 2 tablespoons reduced-fat cream cheese

Instructions:

1. Bring a large pot of lightly salted water to a boil over high heat. Cook the noodles to package instructions. Drain and reserve.
2. Meanwhile, season the meat with salt and pepper. Warm the oil in a large skillet over medium heat. Add the beef; brown on all sides and then push to one side of the pan.
3. Add the onion and mushrooms, cooking until tender, about 3-to-5 minutes. Push to the side with the beef.
4. In a small bowl, combine the cornstarch with 2 tablespoons of the cold beef broth. Add to the skillet and mix with the juices in the pan to deglaze.
5. Pour in the remaining beef broth. Bring to a boil, stirring frequently. Reduce the heat to low and stir in the mustard, garlic, wine, and Worcestershire sauce. Cover with a tight-fitting lid and simmer for 10 minutes.
6. Two minutes before the beef is done, stir in the sour cream and cream cheese. Stir well and allow the beef to finish cooking in the sauce.
7. Let the meat rest for 5 minutes, then serve.

Parmesan-Crusted Pork Chops

Serves 4

Per serving:

- 256 calories
- 42 grams of protein
- 6 grams of carbs
- 6 grams of fat

Ingredients:

- 1/4 cup 2% milk
- 1/4 cup grated parmesan cheese
- 1/4 cup seasoned breadcrumbs
- 1/4 teaspoon salt
- 1/4 teaspoon pepper
- 1/4 teaspoon garlic powder
- 4 (6-ounce) boneless pork chops, 1/2-inch thick and trimmed of fat

Instructions:

1. Preheat the oven to 375°F (190°C). Coat a baking sheet with nonstick cooking spray.
2. Place the milk into a shallow dish. Into another shallow dish, add the cheese, breadcrumbs, salt, pepper, and garlic powder, and mix to combine.
3. Dip each pork chop into the milk and then coat it in the breadcrumb mixture.

4. Place the breaded chops onto the prepared baking sheet.
5. Bake the chops in the oven until fully cooked, about 9-to-11 minutes on each side, then serve.

Classic Tiramisu

Serves 8

Per serving:

- 227 calories
- 5 grams of protein
- 21 grams of carbs
- 13 grams of fat

Ingredients:

- 1 (8-ounce) container reduced-fat cream cheese, softened
- 1/2 (8-ounce) container mascarpone cheese
- 1/2 cup Truvia
- 1 tablespoon brewed espresso or Kahlùa (optional)
- 24 ladyfingers
- 1 cup espresso, brewed and cooled
- 1 tablespoon Kahlùa
- 1 tablespoon packed brown sugar
- 1 1/2 teaspoons unsweetened cocoa powder
- 1/2 ounce bittersweet chocolate, grated

Instructions:

1. To make the filling, add the cream cheese and mascarpone in a large bowl, using a handheld mixer to beat at medium speed until smooth. Add the Truvia and 1 tablespoon of brewed espresso or Kahlùa, if desired, beating at medium speed until blended.

2. Cut the ladyfingers in half lengthways. Arrange 24 ladyfinger halves, cut-side up, in the bottom of an 8-inch square baking dish.
3. In a small bowl, mix the cup of espresso, tablespoon of Kahlùa, and brown sugar. Pour half the mixture over the ladyfingers to soak them; then spread a layer of half the cream cheese filling on top.
4. Repeat the procedure with the remaining 24 ladyfinger halves, soaking with the remaining espresso mixture and then topping with the remaining cream cheese filling.
5. In a small bowl, combine the cocoa and grated chocolate; sprinkle evenly over the top of the filling.
6. Cover with plastic wrap and chill for 2 hours or overnight. Cut and serve.

Maple-Raisin Bread Pudding

Serves 4

Per serving:

- 277 calories
- 10 grams of protein
- 50 grams of carbs
- 3 grams of fat

Ingredients:

- 2 cups 1/2-inch cubes French bread
- 1 cup skim milk
- 2 large eggs
- 2 1/2 teaspoons vanilla extract
- 4 tablespoons pure maple syrup, divided
- 1/3 cup raisins

Instructions:

1. Preheat the oven to 350°F (180°C). Spread the bread pieces onto a baking sheet, ensuring they don't touch. Toast in the oven, stirring after a couple of minutes. Bake until golden, about 5 minutes, and then let cool.
2. In a large bowl, whisk together the milk, eggs, vanilla, and 3 tablespoons of maple syrup. Use a spatula to stir in the raisins and gently fold in the toasted bread. Cover and refrigerate for at least 30 minutes or up to 4 hours.

3. Preheat the oven to 325°F (160°C). Coat 4 small ramekins with cooking spray and divide the mixture between them. Evenly space the ramekins in an 8” x 8” baking pan filled with 1 inch of hot water.
4. Bake the bread pudding until set, about 45-to-50 minutes. Drizzle the remaining 1 tablespoon of maple syrup over the top and serve.

Restaurants

The byword of sustainable dieting is flexibility, so we shouldn't have to omit eating out from a meal plan, even when cutting. We should acknowledge the drawback, though—eating restaurant meals makes it harder to control your calories and macros.

For example, a steak often has at least an additional 120-to-150 more calories from oil and butter that's sopped up during cooking. The calories in plain pasta and potato can easily be equaled or surpassed by a hearty sauce. Many desserts have 25-to-50 calories per tablespoon, and even vegetable dishes can pack a punch because of high-fat extras like butter, oil, and cheese. None of that means you should quit going to your favorite restaurants, though. You just need to be smart about it. Let's learn how to do that.

1. Try not to eat out more than once per week when cutting.

Even when you make prudent choices, restaurant meals are a bit of a black box when it comes to calories and macros, making it easy to overeat. You can minimize the chances of this happening by confining yourself to one restaurant meal per week.

An exception to this rule, however, is in the case of a restaurant where you can accurately estimate the calories and macros of what you're eating (usually because the meals are standardized with numbers provided or are very simple). In this case, you can eat out whenever you'd like without issue.

For example, I had no problems eating a salad with chicken and a side of rice prepared by a local restaurant (the ingredients were exactly what I'd use if I were making the meal myself). So long as you can find the calories and macros of each component online and the portions are fairly regular, healthier fast-food chains that allow you to layer your meals (first rice, then beans, then protein, then

veggies, etc.) can also work well.

However, be careful with traditional restaurants that give calories or macros on their menus or websites (as well as online food databases that include regular restaurant food). Unfortunately, many establishments underreport these numbers, so you should add 20 percent to the purported total calories to be "safe."

2. Try not to eat out more than twice per week when lean gaining or maintaining.

Many people pay close attention to their calories and macros when cutting and then loosen the reins when lean gaining or maintaining, because losing fat feels more urgent than slowing fat gain or steadying body composition.

However, this hurts your long-term progress. Overeating too often when lean gaining causes you to gain fat too quickly, forcing you to start a cutting phase earlier than you'd like, before you were able to gain much muscle or strength. This mistake particularly impedes more experienced weightlifters who can't gain size or strength when cutting and need to spend at least six-to-nine months of the year lean gaining to make significant improvements in their body—an impossible task if they don't control their eating.

Overeating during a maintenance phase is less of a dilemma. You're usually not concerned with further developing your physique because you understand that progress is mostly paused. The point of maintenance, however, is to stabilize your body composition, so if you eat too much too often, a yo-yo pattern will likely result—body fat levels visibly rise, followed by calorie restriction to return to the desired look, followed by a new period of fat gain, etc. This isn't harmful per se, but it can be irksome and defeats the purpose of maintaining.

You need to eat prudently when lean gaining and maintaining, and limiting

yourself to a couple of restaurant meals per week can help with this. That said, the exception noted earlier applies here as well: if you can correctly account for the calories and macros in the meals, you can eat out as much as you'd like. And how do you know if you're correctly adding the calories and macros up in these meals? Assuming you're accurate in your estimations of other (non-restaurant) meals, if your weight isn't moving in the right direction, those restaurant meals are the prime suspect.

3. Try to stick to meals that can be measured.

The most meal-plan-friendly meals are simple and easily quantified, like grilled pork chop with Brussels sprouts and sweet potato wedges, chicken kale salad with croutons and dressing on the side, and grilled salmon fillet with baked potato and grilled asparagus.

If you have a restaurant or three that you like to frequent, you can rotate between a few of your favorite meals that you've appraised. This way, you don't have to bother with trying to figure out calories on the fly, or worse, refusing to eat out because you have to "stick to your meal plan."

By the same token, try to avoid fancier dishes that are difficult to estimate calorically, like stuffed meats and pastas, casseroles, and sauce-heavy eats like goulash, panang chicken curry, salads drenched in dressing, and so on. This means that some restaurants should be avoided if possible. Take Olive Garden, for instance, where just about everything on the menu is some kind of heavy casserole or pasta or meat swimming in cheese, butter, and sauce. If you must eat in such a restaurant, don't fret (it's just one meal after all), but do try to keep your calories within reason (don't eat a pile of bread, a mountain of creamy pasta, and a platter of desserts).

Alcohol

According to many fitness folk, drinking even lightly or sporadically inexorably leads to weight gain. Thus, if you want to be in shape, they say, you have to give up booze. Happily, research shows otherwise. For starters, studies show that moderate drinking is associated with *lower* body weights, not higher, and can result in *more* weight loss when dieting, not less. What's more, research shows that calories from alcohol itself don't affect body fat levels in the same way other calories do. After analyzing the diets of almost 2,000 adults, scientists at the University of São Paulo found that an increase in calories from alcohol alone didn't result in the weight gain that would've occured if those calories had been from food. In fact, when compared to nondrinkers, drinkers took in an average of 50-to-130 more calories per day (from alcohol) and yet weren't any fatter than their alcohol-free counterparts. It's almost like the extra calories simply "didn't count."

There are two likely reasons for these observations. First, alcohol can decrease appetite, which helps with weight loss, and improves insulin sensitivity, which can help with fat burning. Second, the body can't directly convert alcohol into body fat like it can with food because it's processed differently. The chemistry is complex, but you can think of it like this: Wood contains about 4 calories per gram, but no amount of wood shavings on our salads or sawdust in our shakes can make us fatter because our bodies don't produce chemicals for digesting and absorbing them. My point? The calories that matter are those that our bodies can process and use, and calories in alcohol don't matter in the same way as those in food.

That said, alcohol can contribute to fat gain in a unique way: By blunting fat burning and enhancing the conversion of carbs and fat into body fat. In other

words, while alcohol doesn't provide usable calories for the creation of body fat, it constipates our body's fat-burning mechanisms and augments its fat-making ones. This can't significantly impact your body fat levels without a regular calorie surplus (lean gaining, for instance), but it should be noted. Therefore, you can drink alcohol without scuttling your fitness goals, but you should follow three tips:

1. Limit yourself to no more than two servings per day while cutting and maintaining (because of satiety and health) and one serving of alcohol per day while lean gaining (because of the calorie surplus).
2. Choose lower-calorie wines, beers, and spirits over higher-calorie drinks like heavy beer, cider, and fruity cocktails.
3. Consider alcohol a treat (part of the 20 percent of daily calories you can use indiscriminately) and include the calories in your meal plan.

Other Beverages

You've probably guessed this by now, but if a beverage contains calories (like a soda, juice, latte, or sweetened tea), it must be included in your meal plan like a food. The same goes for stuff we like to splash into our drinks like creamer, milk, sugar, syrup, and so on—a jot here and tittle there may seem negligible, but together, they can add up.

This fact highlights a problem with many beverages: Despite containing calories, they're not nearly as satisfying or nutritious as many foods. Thus, generally speaking, the fewer calories you drink every day, the better you'll do cutting, lean gaining, and maintaining. I don't want to leave you high and dry, though, so how about this: Try not to drink more than 10 percent of your daily calories. That way, you'll have calories for your morning caffeine ritual (the best way to start the day) with some leftover liquid calories to spare.

All right, you now have all of the information you need to complete this third step of the *Bigger Leaner Stronger* method of meal planning—choosing what to eat and drink. Let's do that now! Write a few of your top choices for each food group.

Proteins

Relatively unprocessed foods (except in a few cases) that are mostly protein, like red meat, pork, poultry and seafood; high-protein dairy like Greek yogurt, skyr, cottage cheese, etc.; egg white; high-protein plant foods like seitan, tofu, tempeh, etc.; and protein powder and bars.

Nutritious Carbs

Relatively unprocessed foods that are mostly carbohydrate, like fruits, veggies, legumes, tubers, and whole grains.

Healthy Fat

Relatively unprocessed foods that are mostly fat, like olive oil, avocado, nuts and nut butters; seeds; whole egg; and full-fat dairy like regular yogurt, cheese, butter, and milk.

Treats

More highly processed and less nutritious foods and caloric beverages, often with added sugar, like white bread and pasta, soda, energy drink, fruit juice, candy, breakfast cereal and bars, and pastry.

Recipes

It's best to choose recipes that provide macros or that are simple enough for you to work out the macros yourself

Restaurant Meals

The most meal-plan-friendly meals are simple and easily quantified, like grilled pork chop with Brussels sprouts and sweet potato wedges, chicken kale salad with croutons and dressing on the side, and grilled salmon fillet with baked potato and grilled asparagus.

4. Decide how much to eat.

Now that we've identified your favorite foods, it's time to turn them into delicious meals that meet all of your nutritional needs.

Before we get started, however, take a moment to download the free bonus material that comes with this book (www.biggerleanerstronger.com/bonus) because it contains meal plan templates as well as ready-made meal plans that'll help you follow along and apply what you learn here.

Overview

The most efficient way to build a meal plan is in layers. First, you add your primary sources of protein; then, your nutritious carbs, starting with three-to-five portions of vegetables; then, your primary sources of healthy fat; and finally, your treats. To keep the math manageable, you subtract the calories and macros added from your daily targets as you move through each step in the sequence (to see how much more food you can eat), and then, if needed, you massage portions and possibly foods to hone your numbers.

The calories and macros in your meal plan don't have to match your targets perfectly, either. They just have to be close enough, and your calories and protein matter far more than your carbs and fat. I recommend tweaking your plan until you're within 5 percent or so of your calorie target and 10 percent of your protein target, allowing your carbs and fat to shift up or down to make meals that you want to eat.

There are two ways to achieve this level of precision in meal planning:

1. You can look up the foods you want to eat in an online database like www.cronometer.com (my favorite) and then use a food scale and utensils to learn the proper portions of non-prepackaged foods and

beverages. Once learned, you can put the tools away if you trust your eye enough or continue using them for assurance until you do. For example, say your meal plan calls for 100 grams of dry oatmeal (that you cook) with 50 grams of blueberries and 4 ounces of whole milk. Odds are you don't instinctively know how much food that is, so in the beginning, you'd use your food scale and measuring cup to learn, and then go by sight once you've got it.

2. You can use an informal method of eyeballing foods that doesn't require any software or hardware, like comparing portions to your palm, fist, and thumb. For instance, the meal above may be expressed as a "fist" of dry oatmeal, a "half-fist" of blueberries, and a half-cup (which you can easily visualize) of whole milk.

Both approaches can work, but as the first style is the most accurate, that's what I'll teach you in this book. It's also more involved, of course, which may make it seem more daunting, but as you'll see, it's more like learning to paint by numbers.

Also, to help you fully understand each step of creating a *Bigger Leaner Stronger* meal plan, we'll build one as we go for a fictitious meathead named Joe Bro, who weighs 220 pounds and wants to lose fat, and who plans on eating the following number of calories and macros every day:

- 2,200 calories
- 200 grams of protein
- 200 grams of carbs
- 65 grams of fat

Let's start with step one—working out your (and Joe's) sources of protein.

1. Add your protein.

In this step, you want to add primary sources of protein until you meet most (80 percent or so) of your daily protein needs. Why not 100 percent? Because your carbs and fats will add some protein as well, and this should bring you close to 100 percent (and if it doesn't, you can easily adjust meals as necessary).

As for how to distribute your protein throughout your meal plan, I have a couple of recommendations:

1. Try to eat 30-to-40 grams of protein at a time. While it's untrue that our bodies can only absorb small amounts of protein (many people claim 20-to-30 grams), there is an upper limit to how much muscle growth can occur from a single serving of protein. Similarly, there's also a minimum amount of protein required to reach that ceiling. The goal, then, is to eat enough protein to maximize muscle growth but not so much that some of it is essentially "unusable" and thus eliminated. In most people, this serving-size sweet spot is around 30-to-40 grams of protein. Don't feel beholden to this range, though. If, for example, a meal you want to eat contains 60 grams of protein or even more, help yourself, but try not to make this a habit.
2. Eat at least three servings of protein every day. Research shows that this is better for controlling appetite and likely better for muscle building.

Once you've added the primary sources of protein to your meal plan, it's time for a bit of calorie and macro housekeeping. Specifically, subtract the total calories and grams of protein, fat, and carbs in the foods that you added from your daily targets to see how much of each you have left. You've done this step correctly when you've reached about 80 percent of your protein needs and left yourself with plenty of carbs and fat to keep working with (if you add too much

high-fat protein, for instance, you may not have enough fat left for the other foods you want to eat).

Let's see how all this would look for Joe, whom we met earlier. Recall that he weighs 220 pounds and wants to lose fat, and his daily calorie and macro targets are:

- 2,200 calories
- 200 grams of protein
- 200 grams of carbs
- 65 grams of fat

Here's how his meal plan could begin:

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr, 2% fat, plain	300 g	223	34 g	10 g	5 g
Lunch (12 p.m.)	Chicken, breast, skinless, boneless, grilled	150 g	248	47 g	0 g	5 g
Snack (3 p.m.)	Cottage cheese, 2% fat, plain	220 g	178	24 g	10 g	5 g
	Whey protein	30 g	105	23 g	2 g	0 g
Dinner (6 p.m.)	Beef, sirloin, trimmed of fat	150 g	275	45 g	0 g	9 g
Total			1,029	173 g	22 g	24 g
Remaining			1,171	27 g	178 g	41 g

As you can see, Joe has added four servings of protein, totaling 1,029 calories, 173 grams of protein, 22 grams of carbs, and 24 grams of fat.

Let's continue to the next meal planning step and start adding carbs.

2. Add your nutritious carbs.

A healthy diet requires eating several servings of nutritious carbs (vegetables in particular) every day, so this step has two parts:

1. Adding vegetables to your meal plan.
2. Adding other nutritious carbs, like fruits, whole grains, legumes, and tubers.

You have many popular foods to choose from too:

Vegetables	Fruits	Whole Grains, Legumes, and Tubers
Artichokes	Apples	Amaranth
Arugula	Apricots	Barley
Asparagus	Bananas	Beets
Beet greens	Blackberries	Black beans
Bell peppers	Blueberries	Black-eyed peas (cowpeas)
Bok choy	Cantaloupe	Black rice
Broccoli	Cherries	Buckwheat
Brussels sprouts	Cranberries	Brown rice
Cabbage	Dates	Bulgur
Carrots	Figs	Cannellini beans
Cauliflower	Grapes	Cassava (aka yuca, arrowroot)
Celery	Grapefruit	Chickpeas (garbanzo beans)

Chives	Honeydew	Corn
Collard greens	Kiwifruit	Cranberry beans
Cucumber	Mandarin oranges (clementines, satsumas, tangerines)	Einkorn
Eggplant	Mangoes	Farro
Endive	Nectarines	Fava beans
Fennel	Oranges	Great northern beans
Green beans	Papayas	Japanese sweet potatoes
Garlic	Peaches	Kidney beans
Jicamas	Pears	Lentils
Kale	Pineapple	Lima beans
Kimchi	Plums	Millet
Kohlrabi	Raspberries	Mung beans
Leeks	Strawberries	Navy beans
Lettuce	Watermelon	Oatmeal
Mushrooms		Parsnips
Mustard greens		Peas
Okra		Pinto beans
Onions		Popcorn
Pickles		Quinoa
Pumpkin		Red beans
Radishes		Red potatoes

Rhubarb		Spelt
Sauerkraut		Sweet potatoes
Shallots		Taro
Seaweed		Teff
Spinach		Vitelotte potatoes
Squash, butternut		White potatoes
Squash, spaghetti		Wild rice
Squash, yellow		
Swiss chard		
Tomatillos		
Tomatoes		
Water chestnut		
Watercress		
Zucchini		

As you did with protein, in this step, add carbs until you've reached about 80 percent of your daily target. This way, you should have enough room when it comes time to add treats.

When adding vegetables, aim for at least three fist-sized portions per day and try to get to six. Also, for bonus points, include at least one serving of dark leafy greens like spinach, lettuce, kale, or collard greens because of their abundance of nitrates as well as other beneficial compounds, including isothiocyanates and potassium. Most people like to work greens into lunches and dinners (salads and side dishes are popular), but do it however you prefer. A veggie omelet can be a

great breakfast, for instance, and a fruit-and-spinach smoothie can make for a perfect snack. Additionally, you can eat more than one serving of vegetables in a single meal, up to all of them, if you'd like. Splitting up your vegetables across different meals has no advantages or disadvantages.

Once you've put your vegetables on paper, other nutritious carbs are next, with another round of tallying the calories and macros added and subtracting them from what you have left for the rest of your meal plan. At this point, you should have little (if any) of your protein unused, a bit of your carbs, and much of your fat.

Let's see how this step could work for Joe (new additions are italicized).

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr, 2% fat, plain	300 g	223	34 g	10 g	5 g
	Banana	152 g	135	1 g	35 g	0 g
Lunch (12 p.m.)	Chicken, breast, skinless, boneless, grilled	150 g	248	47 g	0 g	5 g
	Bok choy	150 g	20	3 g	3 g	0 g
	Bell pepper	100 g	18	1 g	5 g	0 g
	Onion	70 g	28	1 g	6 g	0 g
Snack (3 p.m.)	Cottage cheese, 2% fat, plain	220 g	178	24 g	10 g	5 g
	Whey protein	30 g	105	23 g	2 g	0 g
	Whole wheat bread (2 slices)	86 g	218	10 g	37 g	3 g
Dinner (6 p.m.)	Beef, sirloin, trimmed of fat	150 g	275	45 g	0 g	9 g
	Sweet potato	300 g	228	4 g	53 g	1 g
	Tomato	100 g	18	1 g	4 g	0 g
	Asparagus	100 g	20	2 g	4 g	0 g

Total	1,714	196 g	169 g	28 g
Remaining	486	4 g	31 g	37 g

Now we're ready to work healthy fats into the meal plan.

3. Add your healthy fat.

Now it's time to allot most (about 80 percent) of the fat you have remaining to nutritious fatty foods that you like, with an emphasis on unsaturated fat. The most popular choices are:

- Oils (and olive oil in particular) and butter
- Nut butters
- Nuts and seeds
- Cheese
- Milk
- Egg
- Avocado

When you're done adding healthy fat to your meal plan, subtract their calories and macros from your remaining totals.

Let's do this now for Joe (new additions are italicized).

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr, 2% fat, plain	300 g	223	34 g	10 g	5 g
	Banana	152 g	135	1 g	35 g	0 g
	Almonds	20 g	121	4 g	4 g	11 g
Lunch (12 p.m.)	Chicken, breast, skinless, boneless, grilled	150 g	248	47 g	0 g	5 g
	Bok choy	150 g	20	3 g	3 g	0 g

	Bell pepper	100 g	18	1 g	5 g	0 g
	Onion	70 g	28	1 g	6 g	0 g
	Sesame oil dressing	14 g	124	0 g	0 g	14 g
Snack (3 p.m.)	Cottage cheese, 2% fat, plain	220 g	178	24 g	10 g	5 g
	Whey protein	30 g	105	23 g	2 g	0 g
	Whole wheat bread (2 slices)	86 g	218	10 g	37 g	3 g
Dinner (6 p.m.)	Beef, sirloin, trimmed of fat	150 g	275	45 g	0 g	9 g
	Sweet potato	300 g	228	4 g	53 g	1 g
	Tomato	100 g	18	1 g	4 g	0 g
	Asparagus	100 g	20	2 g	4 g	0 g
Total			1,959	200 g	173 g	53 g
Remaining			241	0 g	27 g	12 g

4. Add your treats.

As you learned in chapter 9, up to 20 percent of your daily calories can come from less-than-nutritious foods and beverages, including alcohol, desserts, and highly processed carbohydrates such as white bread and pasta, bagels, pastries, etc. Here's how to include these foods in your meal plan:

1. Establish how many "junk calories" you can add to your meal plan.
2. Decide what you'd like to eat and drink.
3. Add treats to your meal plan.
4. Adjust other meals as needed to round out your numbers.

Let's go over each step.

1. Establish how many "junk calories" you can add to your meal plan.

First, calculate 20 percent of your daily calories by multiplying your daily calorie target by 0.2. This is the maximum amount of lower-quality calories that you can eat every day. (You don't *have* to eat this many if you don't want to, but you shouldn't be afraid to either.)

Next, compare this to how many calories you have remaining in your meal plan. If you have between 10 and 20 percent of your calories left, you can move ahead with adding treats. If you have less than 10 percent left, though, you'll probably want to make some more room for treats before continuing (a square or two of chocolate or tablespoon or two of ice cream isn't much of a dessert). To free up calories for indulgences, simply shave your carbs or fats (but not protein, unless you're over your target). Because each gram of carbohydrate has 4 calories and each gram of fat has 9 calories, it won't take major changes. Small adjustments add up quickly.

For instance, if Joe only had 100 calories left for treats (about 5 percent of his total calories), he could eat one fewer slice of bread to gain 109 calories and ditch the banana at breakfast to get back another 135 calories, giving him 344 calories to play with in this step.

2. Decide what you'd like to eat and drink.

Once you have enough calories to add treats, choose a few options from your list that you'd most enjoy, and research online how many calories they contain per serving. Decide how to proceed based on what you find. Would you like to commit all of your calories in this step to just one? Or maybe split them between two or all three?

If you find that your choices contain too many calories, however, you can make further adjustments to your other meals to reclaim more calories for this

step (so long as you don't go above 20 percent of your daily total), or you can go back to your list of favorite foods to look for something else that'll work.

For example, suppose that Joe has his full 20 percent allowance remaining for this step (440 calories). He loves apple fritters, but learns that each one contains about 500 calories. So rather than eating ¾ of a fritter for dessert, he considers alternatives and settles on an ice cream sundae instead because he'll find it more satisfying.

3. **Add treats to your meal plan.**

Work this over until you're happy with all of your meals, and if, after adding your treats, you still have calories left over, fit them into other meals as desired. For instance, if Joe were left with 200 calories, he could add a few squares of dark chocolate, two-to-three cups of low-calorie ice cream (such as Halo Top), or one slice of peach cobbler (about 100 grams).

Speaking of Joe, let's finish his meal plan (final additions are italicized).

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr, 2% fat, plain	300 g	223	34 g	10 g	5 g
	Banana	152 g	135	1 g	35 g	0 g
	Almonds	20 g	121	4 g	4 g	11 g
Lunch (12 p.m.)	Chicken, breast, skinless, boneless, grilled	150 g	248	47 g	0 g	5 g
	Bok choy	150 g	20	3 g	3 g	0 g
	Bell pepper	100 g	18	1 g	5 g	0 g
	Onion	70 g	28	1 g	6 g	0 g
	Sesame oil dressing	14 g	124	0 g	0 g	14 g
Snack (3 p.m.)	Cottage cheese, 2% fat, plain	220 g	178	24 g	10 g	5 g
	Whey protein	30 g	105	23 g	2 g	0 g

	Whole wheat bread (2 slices)	86 g	218	10 g	37 g	3 g
Dinner (6 p.m.)	Beef, sirloin, trimmed of fat	150 g	275	45 g	0 g	9 g
	Sweet potato	300 g	228	4 g	53 g	1 g
	Tomato	100 g	18	1 g	4 g	0 g
	Asparagus	100 g	20	2 g	4 g	0 g
	Dark chocolate	30 g	175	2 g	15 g	11 g
Total			2,134	202 g	188 g	64 g
Remaining			66	-2 g	12 g	1 g

Let's see how he did. He ended with 2,134 calories—3 percent from his target (great)—hit just about 100 percent of his protein needs (perfect), and wound up with slightly fewer carbs and fat (no problem). In other words, by following the system you just learned, Joe made a meal plan that'll work like gangbusters, and you can too!

Also, just to ensure that you understand what effective and well-designed meal plans look like, I've included in the bonus material that came with this book several examples of various sizes who want to cut and lean gain. You can refer to these meal plans as models when you're building your own, or if you see one that you like, you can simply follow it—an easy workaround if you're not inclined to take the time to create a custom meal plan yet.

If you want to follow one of the premade plans, there are a few things you need to know, though:

1. When cutting and lean gaining, start with the plan closest to your current body weight, rounding down. If you're 241 pounds and want to cut, for instance, go with the 230-pound plan and not the 250-pound one; and if you're 178 pounds and want to lean gain, use the 170-

pound plan.

2. When cutting, once you're within 5 pounds of the next-lowest plan, switch to it if you want to keep cutting. If you've used the 230-pound plan to cut down to 215 pounds, switch to the 210-pound plan to keep losing fat.
3. When lean gaining, once you're within 5 pounds of the next-highest plan, switch to it if you want to keep lean gaining. If you've lean gained up to 185 pounds, switch to the 190-pound plan.

Making Your Meal Plans More Appetizing

If the prospect of eating several servings of meat and seafood every day is already getting you down, don't worry—with some simple seasoning tips, you can take your chow from "weak" to "wow." The three best tools are:

1. Salt
2. Acids
3. Spices and seasonings

Let's review each.

Salt

For decades, we've been told to heavily restrict salt and sodium (a mineral vital to our health) to reduce the risk of high blood pressure. Recent research indicates this may not be necessary, however. An extensive review published in the *Cochrane Database of Systematic Reviews* concluded that "there is little evidence for long-term benefit from reducing salt intake," and a subsequent analysis by scientists at the University of Exeter involving the review of more than 6,000 subjects found that in people with normal or high blood pressure, there's no evidence that cutting salt intake is an effective means of reducing the risk of heart attack, stroke, or death. Furthermore, research published in the journal *JAMA Internal Medicine* found that a lower consumption of sodium may actually be associated with a greater risk of dying from heart disease.

One reason for these findings is that scientists have learned that you don't get the whole story by looking only at sodium intake. You must also take potassium intake into account, as well as the relationship between sodium and potassium consumption. A study conducted by researchers at the Centers for

Disease Control and Harvard University found that people with the highest ratio of sodium to potassium were twice as likely to die of a heart attack and had a 50 percent higher risk of death from any cause than people with the lowest ratio.

This is a major public health problem because while the National Academy of Medicine (formerly the Institute of Medicine) recommends 4.7 grams of potassium per day, a study published by the Centers for Disease Control found that the average potassium intake among American adults was between 2.3 and 3 grams per day. That makes insufficient potassium intake one of the most common nutrient deficiencies in Western diets, which is why the 2015–2020 Dietary Guidelines for Americans labels potassium as a "nutrient of public health concern." (And unsurprisingly, many of the best sources of potassium are the types of foods most people avoid—fruits and vegetables like bananas, lentils, avocados, spinach, beans, squash, potatoes and sweet potatoes, mushrooms, and peas.)

While some people are more sensitive to sodium than others because of genetic factors and can benefit from sodium restriction, nobody should nearly or wholly eliminate this elemental compound from their diet. As long as you're not eating a large amount of prepackaged or highly processed foods or egregiously oversalting your meals, and so long as there's enough potassium in your diet (and there will be if you follow my plan), you're unlikely to run into any sodium-related problems.

With that understanding, let's sing some of salt's culinary praises because it's the superstar of zero-calorie food enhancers—just a dash enhances the flavor of meat and fish and elevates any other seasonings it's paired with. To salt effectively, however, think "small and slow." When you're salting finished food, sprinkle a little evenly throughout and taste before adding more; and when you're cooking, keep a salt shaker on the counter and add some at every stage of the process,

tasting regularly. If you're making a soup, for example, add a little salt every time you add something new to the pot that isn't already salty, and sample the broth often. You should also experiment with different types of salts. Or, simply add salt toward the end of the process, when all of the other ingredients are added.

Kosher salt, sea salt, and table salt all have slightly different flavor profiles and can be applied in a number of different ways, including brining, rubbing, and seasoning.

Acids

If you remember some of your high-school chemistry, you know that acids and bases sit on opposite ends of the pH scale. In cooking, acidic foods have a sour, tangy, bright, or sharp taste, and they can be used to enhance the flavor of savory or sweet foods. A beef stew, for instance, may benefit from a splash of red wine vinegar, while a bit of lemon juice added to a fruit salad will make its sweetness pop.

Marinades also benefit greatly from acids. Instead of slapping some seasonings onto a tough cut of meat, turn them into a marinade by incorporating an acid like vinegar, tomato juice, or citrus juice, which will help the meat absorb the flavoring more deeply.

Spices and Seasonings

There are as many spice and seasoning options available to you as there are cuisines in the world, and different pairings take dishes in different directions. For example, with various spice combinations, you can turn a stir-fry into an Asian, jerk, or even Italian variation. There are many flavor profiles to experiment with, and my favorite categories include:

- "Warm" flavors: cardamom, cinnamon, nutmeg, curry, and turmeric

- “Hot” flavors: black pepper, red pepper flakes, cayenne pepper, sriracha, and Tabasco
- “Green” flavors: basil, parsley, sage, rosemary, tarragon, mint, and thyme
- “Pungent” flavors: garlic, garlic powder, onion powder, scallion, mustard, and horseradish
- “Zesty” flavors: ginger, orange zest, and lemon zest
- “Smoky” flavors: chipotle, chili powder, smoked salt, and diced bacon
- “Nutty” flavors: walnuts, sesame, soy, pecans, pine nuts, and pistachios
- “Sweet” flavors: sugar, molasses, maple syrup, and honey

Watch out for “dead” herbs and spices, too, by checking your use-by dates regularly and replacing those that have expired. You can also know if a spice or herb is still good by giving it a quick smell. No scent equals no flavor.

What About "Cheat Meals"?

It feels great to just let go from time to time. To stop stressing and striving and just live for the moment. To just "be human." With meal planning, that means "cheating" (or "free" or "normal" eating as some call it)—no calorie or macro counting and no caring about nutrition. There are many opinions on this. Some people say that even slight departures from the plan can be problematic. Others are of a mind that straying is okay so long as you stay away from certain foods. Still others say it's fine—even valuable—to routinely splurge like a competitive eater.

My take is different. You can enjoy "off-plan" eating without harm, with meals including just about anything you want to eat, but you can't indulge in regular binges without paying the piper. Here's how many fit people approach the matter: They follow their meal plans "perfectly" (good enough) for a week or so, then have something special—usually, but not always, in a "cheat meal" instead of in multiple meals throughout a day—and then return to their meal plan. They also usually stick to this practice for the same reason that foresters burn away fallen branches and undergrowth to prevent a wildfire. By regularly immolating your dietary demons when they're just kindling, you can prevent them from becoming fuel for a raging inferno. There's scientific support for this habit too. Studies have shown that people who take these regular "diet breaks" tend to lose more fat over time, because by and large, this leads to better compliance and fewer blunders.

So, to those who forbid cheat meals, I say *pshaw*.

The proverbial devil is in the details, though, so here's how to "cheat" without ruining your diet:

1. Do it just once or twice per week.

2. Try not to surpass your maintenance calories.
3. Drink alcohol sensibly.

We'll review each of these points, but first, let's dump the term "cheating" because it's too explicitly and implicitly negative for the benign, even beneficial, activity of untypical eating. Let's talk about *treating* instead, which better captures the essence of the deed.

What constitutes "treating"? There are two varieties: Eating more calories (of any kind, nutritious or not) in a day than planned, and replacing most of your nutritious calories with nonnutritious ones. The drawbacks of over-treating are obvious. Overeat too often when cutting, and you'll stop losing fat, and when lean gaining, you'll start gaining fat too quickly. Eat too much junk food, and you'll develop nutritional insufficiencies, and possibly deficiencies, that'll erode your health. There's a balance you must find, then, that allows you to productively wander from your meal plan. Let's learn how to do that.

1. Do it just once or twice per week.

Treating doesn't have to spike your calorie intake for the day, but it often will. If you do that several times per week, you can easily erase your calorie deficit when cutting and halt fat loss, or swell your calorie surplus when lean gaining and gain too much fat.

Limit yourself to just one or two "treatings" per week, however, whether as single larger meals or multiple smaller ones throughout the day, and you'll find the right balance.

2. Try not to surpass your maintenance calories.

Many people don't know just how calorie-laden their favorite treat meals are, especially restaurant meals soused in butter, oil, and sugar to maximize

satisfaction. For example, between 2011 and 2014, scientists at Tufts University analyzed 360 dinner entrees at 123 restaurants in San Francisco, Boston, and Little Rock. They found that the dishes had, on average, about 1,200- to-1,500 calories depending on the cuisine (American, Italian, and Chinese food were the highest).Think that's bad? In another study, scientists at the Center for Science in the Public Interest found that the Cheesecake Factory offered a 2,780-calorie bruléed French toast with a side of bacon that packed 93 grams of saturated fat (yipes!) and 24 teaspoons of sugar, as well as a creamy farfalle pasta with chicken and roasted garlic that weighed in at 2,410 calories and 63 grams of saturated fat. Remember too that those numbers are just for entrees, and when you add in some bread and an appetizer and dessert (as many people do), the calories shoot into the stratosphere. It should surprise no one, then, that a study published in the *European Journal of Clinical Nutrition* concluded that there's little caloric difference between full-service and fast-food dining.

The moral here is simply to exercise good judgment when treating because if you overdo it, you can set yourself back further than you realize. For example, here are approximate calorie counts for the most popular treat foods:

- Bacon cheeseburger: 595 calories per burger
- Creamy pasta: 593 calories per cup
- French fries: 498 calories per large serving
- Deep-dish pizza: 480 calories per slice
- Loaded nachos: 1,590 calories per plate
- Traditional cheesecake: 400 calories per slice
- Ice cream: 270 calories per half cup
- Pecan pie: 541 calories per slice
- Chocolate chip cookies: 220 calories per large cookie

As you can see, when you start combining items like these, the calories add up remarkably fast. A bacon cheeseburger with a side of fries followed by a cup or two of ice cream, for instance, stretches toward 2,000 calories or more, depending on circumstances (serving sizes, home-cooked or fast food, etc.).

For all these reasons, we should set reasonable boundaries for treating, and I recommend trying to end treat days around maintenance calories (your total daily energy expenditure), which is generally 14-to-15 calories per pound of body weight. You can surpass this ceiling by a small margin—5-to-10 percent—without issue, but if you eat much more than that, you may not like the result (especially if you do it repeatedly).

Another strategy to keep your calories reined in when treating is to eat mostly protein leading up to (and after) your treat meal—a tactic referred to as "calorie borrowing." Suppose you have a treat meal planned at your favorite restaurant and you'd like to eat about 1,500 calories, which is about triple the calories in the dinner on your meal plan. You can create the room needed for this meal by stripping carbs and fat from other ones (leaving just the protein), thereby freeing up hundreds of calories to spend in your treat meal. For example, if you normally eat oatmeal with protein powder and walnuts for breakfast (around 500 calories), you could do a double-sized protein shake instead (around 200 calories). Then at lunch you could turn your homemade cheeseburger into a hamburger lettuce wrap, leave the butter off your broccoli, and skip the potato wedges, saving a couple hundred more calories. Finally, instead of high-protein yogurt with muesli and fruit in the afternoon, you could eat just the yogurt (maybe with some protein powder mixed in, if desired). With these simple changes, you've created a large "calorie buffer" that you can enjoy in your treat meal without blowing by your maintenance calories.

Calorie borrowing can also help you enjoy your vacations and holidays more,

when you want (or are expected) to eat a lot of calorific food. By prioritizing high-protein snacks and reserving most of your daily calories for your larger, sit-down meals (ideally just one or two), you can skillfully navigate these periods. Make this dietary strategy the exception, not the rule, however, because if you overuse it, you can develop a perverse preoccupation with meal manipulation, not a healthy and sustainable relationship with food. So vacations and holidays apart, only borrow calories when you treat yourself.

3. Drink alcohol sensibly.

As you learned earlier, the "diet-friendly" way to drink alcohol is to:

1. Limit yourself to no more than two servings per day while cutting and maintaining, and one serving of alcohol per day while lean gaining.
2. Choose lower-calorie wines, beers, and spirits over higher-calorie drinks like heavy beer, cider, and fruity cocktails.
3. Consider alcohol a treat (part of the 20 percent of daily calories you can use indiscriminately) and include the calories in your meal plan.
4. Remain mindful of the calories in the alcoholic beverages, especially when drinking more than usual as part of a treat meal.

One final note about treat meals: They aren't necessary as a matter of course. If you assign up to 20 percent of your daily calories to indulgences, you may not feel the need for regular treat meals. Instead, you may want to reserve your off-plan meals for traveling, special social occasions, etc.

What About Pre- and Post-Workout Nutrition?

Years ago, my pre- and post-workout meals were sacred, inviolable rituals. A protein shake before and after every workout was crucial, I believed, and especially after training, when your body's "anabolic window" was rapidly closing and with it your opportunity for muscle and strength gain.

Chances are you've heard something similar. Bodybuilders and gym bros alike have been saying as much for decades. How important are these meals, though? Does eating before or after workouts actually matter? The long story short is this: eating before and after workouts isn't vital, but it's not entirely without merit, either.

Let's start with pre-workout nutrition.

Should You Eat Protein Before You Work Out?

If you haven't eaten protein in the three-or-four hours preceding a strength training workout, then it's a good idea to eat 30-to-40 grams before you train. If you *have* eaten protein in the last few hours, though, then you don't need to eat more. You can just eat after your workout.

As far as muscle building goes, eating protein does two vital things:

1. It stimulates the creation of muscle proteins (muscle protein synthesis) and suppresses the breakdown of muscle tissue (muscle protein breakdown).
2. It provides your body with the raw materials (amino acids) needed to build muscle tissue.

That's why you need to make sure that you eat enough protein every day if you want to maximize muscle growth.

While there's evidence that eating protein before a resistance training workout can magnify its effects on muscle protein synthesis rates, the effects don't appear to be strong enough to establish that eating protein before a workout is clearly superior to not doing so. Instead, pre-workout protein should be viewed in the context of your entire diet. If you haven't eaten protein three-to-four hours before your workout, your body's muscle protein synthesis rates are going to be low. This means that your body's muscle-building machinery will be idle, waiting for the next feeding of protein to kickstart it into action.

Think of any time where this apparatus lies dormant as lost production time. Your body could have been building muscle if only it were given the right stimulus and supplies. Ideally, then, you'd eat another serving of protein more or less immediately after muscle protein synthesis rates bottom out (every few hours,

generally). By doing this, you'd effectively keep muscle protein synthesis humming all day (and if you eat some protein before bed, you can keep the process going while you sleep.)

If you go into a workout several hours after eating, you're letting muscle building remain paused that much longer, and you're possibly experiencing a weaker anabolic response to the training. Moreover, if you then wait too long to eat after the workout, muscle protein breakdown rates will shoot up and exceed synthesis rates, which can result in muscle loss.

Should You Eat Carbs Before You Work Out?

Your body is highly efficient at tapping into glycogen stores during exercise to meet the heightened demands for glucose, so if you're eating at least a moderate-carb diet (20-to-30 percent or more of your calories from carbs), eating carbs before you train won't make much of a difference in your performance. An exception, however, is when you're going to work out at the end of an extended (8+ hour) fast. In this case, you'll likely have more energy and strength if you eat 30-to-40 grams of carbs 30-to-60 minutes before training.

As for what type of carbohydrate you should eat before training, anything will do: Fruit, starch, simple sugars, etc. Choose whatever you enjoy most and is easiest on your stomach. My favorite pre-workout carbs are nutritious whole foods like oatmeal, bananas, blueberries, strawberries, white potatoes, sweet potatoes, and rice.

Should You Eat Fat Before You Work Out?

You can, but you don't need to. There are several theories about how eating fat before a workout can improve performance, but the scientific literature disagrees. A good summary of the research on the matter can be found in a study conducted by scientists at Deakin University. Here's their conclusion:

> Thus, it would appear that while such a strategy can have a marked effect on exercise metabolism (i.e., reduced carbohydrate utilization), there is no beneficial effect on exercise performance.
>
> Chalk up yet another strike against high-fat, low-carb dieting.

Should You Eat Protein after You Work Out?

It's a good idea to eat 30-to-40 grams of protein within an hour or so of finishing a workout. Remember that after we finish training, muscle protein breakdown rates rise, quickly surpassing synthesis rates. Muscle gain can't occur until this reverses (synthesis rates outstrip breakdown rates), and eating protein causes exactly that by:

1. Providing the amino acid *leucine*, which directly stimulates muscle protein synthesis.
2. Stimulating the production of insulin, which suppresses muscle protein breakdown rates.

These observations explain why many people talk about the post-workout "anabolic window." The idea here is that once you've finished a workout, you need to eat within a certain amount of time (30-to-60 minutes, generally) to maximize muscle gain. If you don't, the story goes, you'll gain less (or no) muscle from the workout.

This theory is inaccurate because the size of this window depends on when you last ate protein. If you haven't eaten protein in three or more hours before you work out, then it's more important to eat protein soon after you finish (your body can't start building muscle until you eat). If you have eaten protein within a couple of hours of starting your workout, however, then the timing of your post-workout meal is less important. Your body will still be processing the food you ate earlier, so in this case, you could eat immediately after your workout, but better would be waiting a couple of hours.

Should You Eat Carbs After You Work Out?

Maybe, but not for the reason we're often told, which is to spike insulin levels and thus enhance muscle growth in various ways. Unfortunately, studies suggest that this doesn't work. Adding carbs to post-workout meals doesn't appear to accelerate muscle gain because protein alone provides enough insulin to curb muscle breakdown rates, which is the primary way insulin can directly affect muscle growth.

That said, adding carbs to your post-workout meal will keep insulin levels elevated for longer, which is desirable from a muscle-building standpoint because of insulin's effects on processes related to muscle loss. This is one of the reasons why high-carb diets are better for gaining muscle than low-carb ones—research shows that high-carb diets result in generally higher insulin levels, leading to lower muscle protein breakdown rates and therefore more muscle growth.

It's also worth noting that the body won't store carbs as fat until glycogen stores have been refilled, which is why people often recommend eating your most carb-rich meals immediately after you work out. How much this can actually benefit your body composition over time is debatable, but it certainly won't hurt.

Should You Eat Fat After You Work Out?

Sure, if you want to. Some people claim that you shouldn't, though, because it slows down the digestion and absorption of the protein and carbs that your body so desperately needs. While it's true that adding fat to a protein- or carb-rich meal slows down the rate at which food is cleared from the stomach, it's not true that this makes for less effective post-workout nutrition. For example, several studies have shown that the fat content of a meal has no effect on glycogen replenishment rates, and that whole milk may be more anabolic than nonfat milk.

Holy fishsticks—that was a bonafide bison bludgeoner of a chapter, so here's a hat tip for hanging in there. I hope I've done my job well too, because if I have, you now know exactly how to use all kinds of food you like to build muscle, lose fat, and optimize your health and well-being. You should also be ready to put all your newly acquired knowledge into action by creating your first *Bigger Leaner Stronger* meal plan (and remember that you can find a digital template for this in the bonus material that comes with this book).

Why not do that right now? And then, once you're happy with your meal plan, why not start using it right away? That's right—it's time to officially start the program and set your course for a bigger, leaner, and stronger you!

Then we'll carry on and learn how to use the elements of effective strength and cardiovascular training to design relatively short but highly effective workouts that will transform your body faster than you thought possible.

KEY TAKEAWAYS

- Try not to eat out more than once per week when cutting, or more than twice per week when lean gaining or maintaining, and when ordering in a restaurant, stick to meals that can be measured.
- Limit yourself to no more than two servings of alcohol per day while cutting and maintaining (because of satiety and health) and one serving per day while lean gaining (because of the calorie surplus).
- Consider alcohol a treat (part of the 20 percent of daily calories you can use indiscriminately) and include the calories in your meal plan.
- If a beverage contains calories (like a soda, juice, latte, or sweetened tea), it must be included in your meal plan like a food.
- The most efficient way to build a meal plan is in layers. First, you add your primary sources of protein; then, your nutritious carbs, starting with three-to-five portions of vegetables; then, your primary sources of healthy fat; and finally, your treats.
- If you want to treat yourself without ruining your diet, limit yourself to one or two treat meals per week, try not to surpass your maintenance calories on treat days, and drink alcohol sensibly.
- If you haven't eaten protein in the three-or-four hours preceding a strength training workout, then it's a good idea to eat 30-to-40 grams before you train. If you *have* eaten protein in the last few hours, then you don't need to eat more.
- If you're eating at least a moderate-carb diet (20-to-30 percent or more of your calories from carbs), eating carbs before you train won't make much of a difference in your performance. An exception, however, is when you're going to work out at the end of an extended (8+ hour) fast. In this

case, you'll likely have more energy and strength if you eat 30-to-40 grams of carbs 30-to-60 minutes before training.

- Eating fat before you work out has no beneficial effect on exercise performance, but you can if you want to.
- If you haven't eaten protein in three or more hours before you work out, then it's more important to eat protein soon after you finish (30-to-40 grams within an hour of finishing is a good target), but if you have eaten protein within a couple of hours of starting your workout, then the timing of your post-workout meal is less important.
- Adding carbs to your post-workout meal will keep insulin levels elevated for longer, which is desirable from a muscle-building standpoint because of insulin's effects on processes related to muscle loss.

PART 4

The Last Exercise Advice You'll Ever Need

The Big Ideas

1. Your genetics can’t stop you from getting fit.
2. Many people can build muscle and lose fat at the same time.
3. Cardio is optional when dieting.

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The 10 Absolute Worst Exercise Myths and Mistakes

I am building a fire, and every day I train, I add more fuel. At just the right moment, I light the match.

—MIA HAMM

Nine of ten people you see in the gym don't train correctly. I could write an entire book cataloguing the most common mistakes, but here's a small sampling:

- They spend too much time on the wrong exercises.
- They undertrain and overtrain muscle groups.
- They use poor form, especially on the more technical exercises.
- They use too little or too much weight.
- They rest too little or too much in between sets.

In fact, what most people do in the gym doesn't even qualify as training but merely exercise. What's the difference? Exercise is physical activity done for its own sake—to burn calories or improve energy levels or mood—whereas training is a systematic method of exercising done to achieve a specific, longer-term goal, like increased strength, muscle definition, or athleticism.

There's nothing inherently wrong with exercise (it beats sitting on your keister), but only training can give you the body you really want. While exercise can make you healthier, it guarantees nothing in the way of fat loss or muscle gain, the two biggest physiological levers you need to know how to work to build your best body. Unfortunately, most gym-goers don't understand this, and that's why weeks, months, and years can go by with them doing the same old exercises, lifting the same old weights, and sporting the same old bodies.

There are many training myths and mistakes that contribute to this plight, but in this chapter, we'll confront the ten that make building muscle far more difficult than it should be.

1. "Some guys just don't have the genetics to get big and strong."
2. "Strength training makes you stronger but not bigger."
3. "You can't build muscle and lose fat at the same time."
4. "You should change exercises frequently."
5. "Strength training is dangerous."
6. "You must use bands, machines, and other contraptions."
7. "Exercise doesn't help you lose fat."
8. "Strength training isn't effective for losing weight."
9. "You have to do a lot of cardio to get and stay lean."
10. "You don't need to do isolation exercises."

Let's knock the starch out of these fallacies, starting with number one.

Myth #1:

"Some Guys Just Don't Have the Genetics to Get Big and Strong"

For many, "genetics" is an unpalatable word, associated with things we want to change but can't, like height, beauty, and athleticism. I won't blow smoke—muscle building is heavily influenced by genetics, and there are hard limits to how much muscle we can gain.

There are many physiological reasons for this, but you can get a fairly accurate estimate of your muscle-building potential by analyzing your bone structure. Research shows that people with larger bones tend to be more muscular than people with smaller frames. Bigger-boned people also tend to have higher testosterone levels and gain muscle faster when they start lifting weights. The point: burly people have more genetic potential for size and strength than bony ones.

Now you're probably wondering how you measure up. Two good indicators of your overall bone structure are the circumferences of your wrists and ankles. Height being equal, people who have wider wrists and ankles tend to be naturally more muscular and have a higher potential for muscle growth than those with narrower ones. Now, if you're like me and don't need to pull out the measuring tape to know that you have slender bones, I have good news: even people with shoddy bodybuilding genetics can gain more than enough muscle and strength to look and feel fantastic.

Most guys only need to gain 20-to-25 pounds of muscle and reach intermediate-level strength to look and perform like scorchers—something literally any guy can do with the right plan, no matter how skinny and weak he is when he first touches a barbell. It doesn't have to take a lifetime, either. It'll happen faster in some people than others, but for most guys, it requires no more than three years of consistent training.

Myth #2:

"Strength Training Makes You Stronger but Not Bigger"

If there's one mainstream misconception that causes more harm to men's physiques than any other, it's this one—the idea that heavy weightlifting is mostly for gaining strength and not muscle. If you primarily want to get bigger, we're told, you should always use lighter weights and do more reps (8-to-12 reps per set is often promoted as the promised land).

This is incorrect. Backward, actually, because the most reliable way to get big is to get strong, and the best way to do that is to lift heavy weights. There are several reasons for this that we'll discuss in more detail in the next chapter, but they can be summarized like this: Heavy weightlifting produces large amounts of tension in your muscle, causing a great activation of *muscle fibers* (collections of long, threadlike strands called *myofibrils* that contract). And as you'll soon learn, generating higher levels of tension in your muscles over time is the single most effective way to stimulate muscle growth. This explains why your number one goal as a natural weightlifter should be to increase your whole-body strength. And the most effective way to do that is to lift heavy weights with exercises that involve multiple joints and muscle groups (*compound exercises*).

That doesn't mean that you should *never* lift lighter weights or that you can't gain muscle with them, though—only that your bias should be toward higher- and not lower-intensity training. For this reason, in the *Bigger Leaner Stronger* program, you'll train in just two rep ranges: 4-to-6 reps for compound exercises, and 6-to-8 reps for *isolation exercises* (exercises that involve a single joint and focus on one muscle group).

"Wait a minute," you may be thinking. "If that's true, then how can some people be way stronger than they look?" You may answer steroids, superior

genetics, or flawless technique, and while these things can be factors (especially with strength athletes), the most important one is something less understood: anatomy.

While we all have the same muscles, all located in the same general regions, they're attached to our skeletons in different ways. These discrepancies are usually small—only a centimeter or two—but they can translate into huge differences in natural strength. The mechanisms are highly technical, but they add up to greater mechanical advantage. Because muscles function as levers, where they attach to your bones greatly impacts how much force they can produce and thus how much weight they can move. Studies show that, thanks to anatomical differences, strength can vary by as much as 25 percent among people with identical amounts of lean mass.

Some people's muscles and bones are also arranged in a way that allows them to lift far more than you'd expect based on their size. Short upper arms give an advantage on the bench press (the bar doesn't have to move as far), long arms and short legs are ideal for the deadlift (for the same reason), and short femurs improve your squat strength.

Fortunately, as in the case of genetics and muscle building, whether we were born to move mountains of weight bears little on our ability to get into great shape (only on our prospects as a strength athlete). If you're reading this book to build a strong, muscular, lean, and healthy body, take comfort because none of that requires an anatomical leg up.

Myth #3:

"You Can't Build Muscle and Lose Fat at the Same Time"

Yes, you absolutely can. Well, *most* people can at least, and you're probably one of

them. Here's why:

1. If you're new to weightlifting (less than one year of proper training or 15 pounds of muscle gain) or are getting started again, you shouldn't have any trouble building muscle and losing fat at the same time.
2. If you have at least six-to-eight months of effective training under your belt and have gained at least 10 pounds of muscle and aren't coming off a long break, you probably can't do both and will have to optimize your regimen for one or the other (muscle gain or fat loss).

Why are those the rules? Why can't everyone gain muscle and lose fat at the same time, regardless of their circumstances? Physiologically speaking, fat loss and muscle growth have "irreconcilable differences" that stem from their relationship to the body's energy balance. When you maintain a calorie deficit, your body fat levels drop, but so does your body's ability to create muscle proteins. Testosterone levels also decline and cortisol levels rise when calories are restricted for extended periods of time. This consequence makes it easier to lose muscle while dieting and partly unravels why most people can't gain muscle and lose fat at the same time—by restricting our calories to lose fat, we also restrict muscle growth.

This isn't so with people new to resistance training, though. They can get bigger and leaner at the same time. When you first start weightlifting, you can gain muscle at a very fast rate because your body is hyperresponsive to it, so many guys can gain up to 20 pounds of muscle in their first year of strength training, whereas the best someone like me could hope for is a pound or so of muscle gain over the next year.

This newbie gains phase generally lasts six-to-eight months for most people, and it can easily overpower the muscle-related disadvantages of a calorie deficit.

In other words, cutting can still slow down muscle growth when you're new, but it can't halt it altogether. Eventually this blessing fades, however, and with it goes your ability to "recomp" (*recomposition*) your body. From that point on, you'll need to cut when you want to lose fat and preserve muscle, and lean gain when you want to gain a substantial amount of muscle. (Maintenance is the medium state—no fat loss or gain and minimal or no muscle growth.)

Myth #4:

"You Should Change Exercises Frequently"

How many times have you heard that you need to constantly change your workout routine to continue making progress? That you have to "confuse" and "shock" your muscles into growth by regularly subjecting them to new exercises and workouts?

This sounds sensible. If we want to improve something, whether it be a skill or some aspect of our fitness, we have to continually push boundaries and tackle new challenges. Wouldn't that imply, then, that we'd have to regularly subject our muscles to new types of physical demands? That doing the same workouts every week would result in stagnation? While it's true that doing the *exact same* workouts again and again will lead to a slump, the "muscle confusion" theory misses the forest for the trees. Your muscles have no cognitive abilities. They're not trying to guess what workout you'll do today and can't be "confused" by fancy workout programming. Muscle tissue is purely mechanical. It can contract and relax. Nothing more.

That said, there's validity to the basic premise that muscles won't keep getting bigger and stronger unless they're *forced* to. Where muscle confusion goes astray, however, is with the *type* of stimulus it emphasizes. You can change up your

workout routine every week—heck, every day—and still hit a plateau because "change" doesn't cause muscle growth. *Progressive tension overload* does, and more so than any other single training factor.

This term refers to increasing the amount of tension your muscles produce over time, and while there are several ways to accomplish this, the most effective one (and the one that forms the nucleus of *Bigger Leaner Stronger* training) is to progressively increase the amount of resistance your muscles have to work against. In other words, the key to gaining muscle and strength isn't merely changing movement patterns, rep ranges, or rest intervals—it's *making your muscles work harder*. And that's exactly what you're doing by gradually increasing resistance levels (loads) in your training.

Bigger Leaner Stronger will also involve less exercise variety than many mainstream bodybuilding programs. Unless you have to change exercises sooner because of injury, equipment availability (hotel gym, for instance), or other obstacles, you'll do the same exercises every week for eight weeks at a time. And some exercises—the most important ones—will never get benched.

Some people think that a rigorous procedure like this sounds less enjoyable than a more diverse one, but that's only until they learn how effective it is. By not making frequent changes to exercises, you have enough time to get attuned to your routine + hone your exercise technique + accurately track your progress = an equation for remarkable results (fun!).

Myth #5:

"Strength Training Is Dangerous"

Many people think strength training (heavy weightlifting) is dangerous, and I understand why. When you compare deadlifting, squatting, and bench pressing

large amounts of weight to other forms of exercise, like jogging, cycling, or calisthenics, strength training looks daunting.

Poke around on the Internet, and you'll find plenty of material to feed this perception. Personal stories range from the tame—mild joint and muscle aches and the like—to the downright horrific, with some long-time powerlifters and bodybuilders so incapacitated that they can't even tie their shoes until the ibuprofen kicks in.

And so strength training has been saddled with a bum rap for decades now. Thankfully, the tide is turning and strength training is gaining currency, but many people still think that its risks far exceed its rewards. Ironically, however, research shows that when done properly, strength training is remarkably safe. In one review of twenty studies, Bond University scientists found that bodybuilding produced an average of just one injury for every 1,000 hours of training—nearly four years of training five days per week. Researchers also noted that most of the injuries tended to be minor aches and pains that didn't require any type of special treatment or recovery protocols. In most cases, a bit of extra R & R won the day.

As you'd expect, more intense and technical types of weightlifting, like CrossFit, Olympic weightlifting, and powerlifting, result in more injuries, but fewer than you might think. These activities produced just 2-to-4 injuries per 1,000 hours of training, whereas studies show that sports like ice hockey, football, soccer, and rugby have injury rates ranging from 6-to-260 per 1,000 hours, and even long-distance runners can expect about 10 injuries per 1,000 hours of pavement pounding. Therefore, you're about 6-to-10 times more likely to get hurt playing beer league sports than by following a strength training program like *Bigger Leaner Stronger*.

You'll get a bigger payoff with strength training, too, because it delivers a number of health and fitness benefits that you can't get from other types of

exercise. Here's a short list of what a well-designed strength training routine can do for you:

- Stronger and healthier joints
- More muscle mass
- Better heart health
- Improved brain health
- Greater longevity and quality of life
- More bone density
- Faster metabolism
- Improved flexibility

And those perks are just the highlight reel. When you compare the upside of strength training to the long odds of getting hurt and the mildness of most of the injuries that do occur, the choice is clear: to strength train is far smarter than not to.

Whether we realize it, we make these types of judgment calls every day. Every time we step into a car, take the stairs instead of the elevator, or play with a pet, we're accepting a certain amount of risk. The only surefire way to avoid this element of living would be to never leave our beds—but even then, we'd have to contend with fallout associated with lack of physical activity like muscle loss, impaired sleep, and an increased risk of heart disease, type 2 diabetes, and cancer. All we can do, then, is assess outcomes and probabilities of situations we face and try to tilt the scales in our favor as much as we can. And as to strength training, it's easily disarmed with proper programming, technique, and recovery, which you'll learn about in this book.

Myth #6:

"You Should Always/Never Do This Type of Workout Split"

A *workout split* refers to how your workouts are organized in terms of which exercises you do and which muscle groups you train in each session, and everyone seems to have a different opinion on what works best. Some people claim that the traditional bodybuilding method of training one major muscle group in each workout is optimal. Others decry the *body-part split*, however, and beat the drum for something else, like the *full-body split* or *upper/lower split*. Still others disagree with all of that and are convinced that you should organize your training around "movement patterns" or some other feature or factor. Separating the sheep from the goats can be difficult, too, because you can find "sciencey" explanations for many of these assertions. The main problem, however, is that all of these opinions focus on brushstrokes instead of the big picture.

Ultimately, it isn't a workout split that drives muscle growth—your biceps don't care if they get trained in an "arms," "pull," or "upper-body" workout. Your muscles will grow when you do the right amount of the right exercises with the right amount of weight and post-workout rest and recovery. Your workout split is just a tool that helps you accomplish those ends, not a target unto itself, and therefore, no single workout split is "best" for everyone, under all circumstances, at all points in time. For example, if your goal is to maximize the development of your upper body muscles while still growing your lower body, the best workout split for you will look very different than if your goal is to compete in a powerlifting competition (which requires tremendous lower-body strength). There are other factors to consider when choosing a workout split, too, such as other demands and obligations in your life, training experience, and personal preferences, etc.

Now, if you're like most guys reading this book, your goal is probably similar to the one I just outlined—a lot more upper-body muscle and strength, with enough lower-body development to maintain good proportions—and you probably also care about staying injury free and highly engaged in your home and work life. And to do all of that, you don't have to follow one particular workout split (many can do the trick), but you must follow a few non-negotiable training tenets that I'll share in the next chapter.

And as for which type of workout split is used in *Bigger Leaner Stronger*, it's a mash-up of the *push/pull/legs* and *upper/lower* splits—essentially a push/pull/legs routine with extra chest, arms, and shoulders work to grow those muscle groups as fast as we can. This scheme works extremely well for people new to proper strength training, but you should know that it may not always be the best way for you to train, especially if you want to get as big and strong as your genetics will allow. Eventually, you'll likely need to modify this program or use another workout split altogether to reach your ultimate goals, (which is why I wrote a follow-up book to this one called *Beyond Bigger Leaner Stronger*).

Myth #7:

"Exercise Doesn't Help You Lose Fat"

When most people want to start losing weight, they start doing cardiovascular exercise, and although they may wind up wearing out their running shoes and giving their spin bikes countless sweat showers, they rarely see meaningful changes in the mirror or on the scale. Researchers have long noted the same thing in studies. When people exercise for a few weeks or months—even vigorous exercise like high-intensity interval training—they lose less weight than you'd expect, and sometimes none.

A study published in the journal *Obesity Reviews* found that people who followed a cardio program (either moderate- or high-intensity) only lost about 2 pounds of fat after 12 weeks on average. That's 0.2 pounds per week, an amount so small it's hard to even measure. Even more eye-opening are the results of a review study published by Queen's University. After analyzing thirty-one studies, they found that weight loss outcomes got *worse* as the duration of exercise increased—people who followed an exercise plan for several months lost just 7 pounds on average, and people who followed an exercise plan for six months or more lost almost nothing.

As a result, many journalists, doctors, and fitness authorities have declared that exercising for weight loss is like trying to bail out a boat with a sieve—a fruitless endeavor. Or at least the *ordinary* way of exercising is futile, some say, but if you give them $49.99 in 78 easy hourly installments, they'll give you access to their proprietary patent perpetually pending breakthrough training techniques guaranteed to melt belly fat faster than a roided hornet—*ahem* ... I digress.

All of this anti-exercise rhetoric has a soft underbelly, however, because none of the research used to support it controlled people's calorie intake properly. And what do you think happens when you put a bunch of overweight people on an exercise program without addressing anything related to nutrition and lifestyle? They burn more calories—whoop whoop—but then get hungrier and eat more—womp womp.

This "compensatory eating response" associated with exercise is cause for constant hand-wringing ("You just eat more after you work out! Oh the humanity!"), but it's merely a natural, healthy, and necessary response to increased energy expenditure. After all, if we didn't get hungrier after strenuous exercise, humans would've starved to death long ago.

Luckily, while this reaction to exercise isn't under our control (appetite *will*

increase), how we act on it is. Moreover, if you follow the nutrition plan I gave you in the last section of this book, you'll significantly dampen the unwanted appetitive effects of frequent exercise. So, while exercise alone doesn't guarantee anything in the way of weight loss, what happens when you do a few hours of the right kind of exercise per week *and* eat properly (as you'll learn how to do in this book)? You lose fat, you lose it quickly, and you enjoy the process.

Myth #8:

"Strength Training Isn't Effective for Losing Weight"

This misbelief has a kernel of truth in it because while literally correct, it's also misleading. Strength training is indeed a bad way to lose *weight*, but when combined with proper dieting, research shows that it's a fantastic way to lose *fat* faster while preserving (or gaining!) muscle.

A Duke University study illustrates this point perfectly. Researchers recruited 196 obese or overweight men and women ranging from 18-to-70 years old and split them into three groups:

1. Group one did three one-hour resistance training workouts per week.
2. Group two jogged three days per week at a moderate intensity for about 45 minutes per session.
3. Group three did both the resistance training and cardio workouts.

After eight months, guess which group lost the most weight? No, it wasn't group one or three. It was number two, the cardio-only group. The kicker? That was also the only group that lost muscle as well. And guess who lost the most fat while also gaining muscle? That's right, group number three—the resistance

training and cardio group. In other words, adding resistance training to the cardio workouts resulted in less *weight* loss due to muscle gain but more *fat* loss.

There's another myth related to this one that's worth addressing here: Higher-rep (and lower-weight) training is better for fat loss than lower-rep (and higher-weight) training, mostly because doing more reps burns more calories than doing fewer reps. This can seem intuitively true because higher-rep training typically feels harder than lower-rep work, but research shows otherwise—the differences in energy expenditure between, say, doing 20-rep and 10-rep sets are negligible. What's more, although strength training doesn't burn that many calories (300 to 400 calories per hour, usually), it can boost the number of calories you burn after your workouts and raise your basal metabolic rate over time. Studies show that heavier weights and fewer reps (7 reps or fewer more per set) produces better metabolic effects than lighter weights and more reps, too.

We also have to remember that the primary reason to include strength training in a fat loss regimen isn't calorie or fat burning but preserving (or gaining) muscle mass while you lose fat. This improves body composition, of course, but it also helps you lose fat faster because the less muscle your body breaks down for energy, the more body fat it must burn instead. And again, the best way to do this is to lift heavy weights.

Myth #9:

"You Have to Do a Lot of Cardio to Get and Stay Lean"

You've probably heard that you must sacrifice excessive amounts of time to the treadmill or StairMaster to look good. Allow me to disabuse you of such nonsense. When it comes to improving your body composition, cardio is a mixed blessing. It contributes to your fat loss efforts by burning energy, but not as much

as you may think. For instance, guess how much energy 30 minutes of vigorous running burns? About 300-to-500 calories depending on how much you weigh. And guess how easy it is to eat that right back? A handful of nuts, a cup of yogurt, and an apple does the trick. Or if you're the more indulgent type, a modest-sized chocolate chip cookie with a glass of milk.

My point isn't that you shouldn't eat these foods when you want to lose weight or that the energy you burn during cardio doesn't matter—only that cardio just doesn't burn as much energy as we wish it did (just like strength training).

Another reason to limit cardio when you want to build a lean and powerful physique is it can interfere with muscle and strength gain in two ways:

1. In the short term, by making you more generally fatigued, which makes it harder to progress in your training.
2. In the long term, by disrupting cell signaling related to muscle growth.

That doesn't mean you should completely shun cardio, though. It does have health benefits, including some that you don't get from strength training, and it can help you maintain a higher total daily energy expenditure, which means faster fat loss and easier weight maintenance. And once you learn how to do cardio correctly, you can enjoy these benefits without suffering any of the downsides. More on that soon.

Myth #10:

"You Don't Need to Do Isolation Exercises"

As you learned a moment ago, compound exercises are fantastic for gaining

muscle and strength. However, some people put them on a pedestal as *all* you need to fully develop every *major muscle group* (a group of muscles highly involved in pushing, pulling, and squatting) in your body. Isolation exercises, they say, may be fun, but they're superfluous if you do enough squatting, bench pressing, deadlifting, and overhead pressing.

You can find research to support this idea. Studies conducted by scientists at the Federal University of Goiás, the University of the Amazon, Santa Cecília University, and elsewhere have found that adding isolation exercises to compound exercises didn't significantly increase muscle growth or strength in untrained and trained men and women.

As researcher and writer Greg Nuckols has noted in an unpublished meta-analysis, though, most of these studies were conducted in such a way that made it almost impossible for isolation exercises to show benefits. When Nuckols analyzed the results of seven studies on this topic, he found that isolation plus compound exercises increased muscle size by about 3.8 percent versus 3 percent with just compound exercises. That wasn't *statistically significant* (large enough to indicate a cause-effect relationship), but it would be practically significant when considered in the context of months and years of continued training. Think of it this way: If I told you that you could increase muscle growth by 27 percent by spending an extra 20 to 30 minutes in the gym each week doing a few relatively easy exercises, would you do it? Does Dolly Parton sleep on her back?

Another reason to include isolation exercises in a strength training routine is working your muscles in several different ways—in different directions and at different angles—produces better results than just one or two ways. In a study conducted at the University of São Paulo, for instance, researchers found that despite doing the same amount of weekly volume, people who did a combination of lower-body exercises that included the Smith machine squat, deadlift, leg press,

and lunge gained more strength and experienced more balanced and proportionate muscle growth than people who only did the Smith machine squat. The same effect has been noted in several other studies as well:

- Researchers at Londrina State University found that training with three different exercises produced more symmetrical and complete growth of the thighs, biceps, and triceps than training with one exercise.
- Scientists at the Federal Institute of Sudeste of Minas Gerais found that six months of bench pressing produced consistent growth of the chest muscles, but not the triceps, which plateaued after about eight weeks. This suggests that adding triceps exercises would've produced more triceps growth.
- A research team at the University of Tokyo found that squats produced very little growth of the *rectus femoris* (a muscle in the middle of your thigh), which also suggests that including an isolation exercise that targets this muscle, like the leg extension or Bulgarian split squat, would be beneficial.

To summarize my case for doing isolation exercises:

1. Isolation exercises allow you to continue training specific muscle groups when it's no longer practical to do so with a compound exercise. For instance, your chest and shoulders will probably be bushed after several sets of bench and dumbbell pressing, but your triceps may be up to a few sets of an isolation exercise. Or, while your low-back and forearms are typically shagged after just a few sets of deadlifts, your lats and hamstrings aren't.

2. Isolation exercises allow you to train a muscle group in different positions and through different ranges of motions, which likely improves muscle growth. For example, bench pressing and overhead pressing (compound exercises) train your triceps in a very different position than triceps extensions or dumbbell pullovers (isolation exercises).
3. Doing the same three or four exercises every week for months on end gets boring, and boring workouts tend to be less productive than engaging ones.
4. Repeating the same exercises in the same way for long periods of time probably increases the risk of *repetitive stress injuries* (a gradual buildup of damage to tissues from repetitive motions), especially when you start using heavier weights.

While the lion's share of your gains will come from compound exercises, by supplementing them with the right isolation exercises (like you will in *Bigger Leaner Stronger*), you'll get even more muscle and strength out of your training.

You've just learned some of the most fundamental lessons in the muscle-building racket: lift heavy weights, progressively overload your muscles, use free weights, constrain cardio, and do a combination of compound and isolation exercises.

You've also learned the biggest reasons why so many people flounder in the gym: they lift light weights, change exercises too often, prioritize the wrong exercises, and do too much cardio.

We'll carry this discussion further in the next chapter, where you'll get the *Bigger Leaner Stronger* playbook for building workouts that build muscle and

strength.

KEY TAKEAWAYS

- Exercise is physical activity done for its own sake—to burn calories or improve energy levels or mood—whereas training is a systematic method of exercising done to achieve a specific, longer-term goal, like increased strength, muscle definition, or athleticism.
- The most effective way to increase your whole-body strength is to lift heavy weights with exercises that involve multiple joints and muscle groups (*compound exercises*).
- If you're new to weightlifting (less than one year of proper training or 15 pounds of muscle gain) or are getting started again, you shouldn't have any trouble building muscle and losing fat at the same time.
- If you have at least six-to-eight months of effective training under your belt and have gained at least 10 pounds of muscle and aren't coming off a long break, you probably can't do both and will have to optimize your regimen for muscle gain or fat loss.
- The key to gaining muscle and strength is making your muscles work harder by gradually increasing resistance levels (loads) in your training.
- Your workout split is just a tool that helps you do the right amount of the right exercises with the right amount of weight and post-workout recovery, not a target unto itself, and therefore, no single workout split is "best" for everyone, under all circumstances, at all points in time.
- The primary reason to include strength training in a fat loss regimen isn't calorie or fat burning but preserving (or gaining) muscle mass while you lose fat, which improves your body composition and helps you lose fat faster because the less muscle your body breaks down for energy, the more body fat it must burn instead.

- Doing cardio has health benefits, including some that you don't get from strength training, and it can help you maintain a higher total daily energy expenditure, but it doesn't contribute to fat loss as much as you may think.
- It makes sense to include isolation exercises in your strength training routine because they allow you to continue training specific muscle groups when it's no longer practical to do so with a compound exercise, they allow you to train a muscle group in different positions and through different ranges of motion, which likely improves muscle growth, they keep workouts fresh and engaging, and they help you avoid repetitive stress injuries.

The Big Ideas

1. Use an effective mix of intensity, volume, and frequency.
2. Achieve progressive overload.
3. Use proper form.

12

The Ultimate Strength Training Plan for Men

Seest thou a man diligent in his business? He shall stand before kings.

—PROVERBS 22:29

By the end of the Second World War, military hospitals were overrun with soldiers who had sustained serious orthopedic injuries like bone fractures, breaks, and muscle, tendon, and ligament tears. The situation was grim. There simply weren't enough physicians to treat everyone in a timely manner, and their moldy rehabilitation protocol of rest, heat, and gentle exercise put patients on a long road to recovery—up to six-to-nine months was common.

In 1945, however, an Army doctor named Thomas DeLorme began experimenting with a new rehab technique to help servicemen heal faster—one that had transformed his own health. As a child, DeLorme was stricken with rheumatic fever, a serious disease that often killed children and caused permanent heart damage in those who survived. After four months of enforced bed rest, doctors told DeLorme that his heart was now too weak for him to ever do any strenuous activity again. "I was determined to prove the medicos wrong," DeLorme later said, "and immediately upon leaving my sick bed I started a

comeback campaign."

That campaign was strength training, and it not only helped DeLorme turn the corner and conquer the aftermath of his illness, but it also won him modest recognition as a strength athlete. His competition highlights included a 250-pound clean and jerk, 240-pound bent press, 160-pound curl, and 503-pound deadlift.

Although World War Two dashed DeLorme's Olympian ambitions, he continued with his strength training routine while serving his country—unusual in medical circles at the time because most physicians agreed that extreme effort of any kind was unhealthy, particularly for the heart. DeLorme, however, knew otherwise, and when he wanted to figure out how to help injured servicemen heal faster, he hypothesized that strength training could bear fruit.

After an inaugural success with a patient recovering from knee surgery, DeLorme knew he was on the right track. He designed a new rehab regimen that consisted of multiple sets of strength training exercises with weights that the patient could lift for 10 reps. Once they could complete several 10-rep sets (for a time, it was seven) with a given weight, they'd increase the load and repeat the process. DeLorme called his system "Heavy Resistance Exercise," and it worked wonders, consistently restoring strength and function to men who had completed traditional physical therapy and resigned themselves to permanent disability. The good news spread quickly, and DeLorme's breakthrough became the standard of care for both military and civilian physical therapy.

DeLorme continued his research into the curative effects of strength training, refined his training programs, and published multiple academic texts on the science of resistance training, including the seminal book *Progressive Resistance Exercise: Technic and Medical Application*—works widely read by other medical professionals that legitimized strength training as a healthy and effective

form of exercise.

This tale illustrates one of the laws of muscle and strength gain: To get bigger and stronger muscles, you must gradually increase the amount of tension they can produce. Technically, this refers to one of the three primary "triggers" or "pathways" for muscle growth:

1. Mechanical tension—Force produced in muscle fibers by stretching and contracting. There are two types of mechanical tension: "passive" and "active" tension. Passive tension occurs when your muscles are stretching during a resistance exercise, and active tension occurs when they're contracting.
2. Microscopic damage to muscle fibers caused by high levels of tension. Whether muscle damage directly causes muscle growth or is just a byproduct of mechanical tension is unclear, but at this time, it belongs on the list.
3. Cellular fatigue—A number of chemical changes that occur in and around muscle fibers as they contract repeatedly. When you take a set to *muscular failure*—the point where you can no longer move the weight despite giving maximal effort—or close to it, this causes high amounts of cellular fatigue.

Out of these three drivers of muscle growth, studies show that mechanical tension is the most important, meaning that it produces a stronger muscle-building stimulus than muscle damage and cellular fatigue. Thus, for strength training to be maximally productive, it must underline mechanical tension more than muscle damage and cellular fatigue. Consequently, then, what do you think is more effective for gaining muscle and strength: Training with heavier or lighter loads? That's right—heavier loads, which produce more mechanical tension than

lighter ones and thus activate more muscle growth (which leads to more strength).

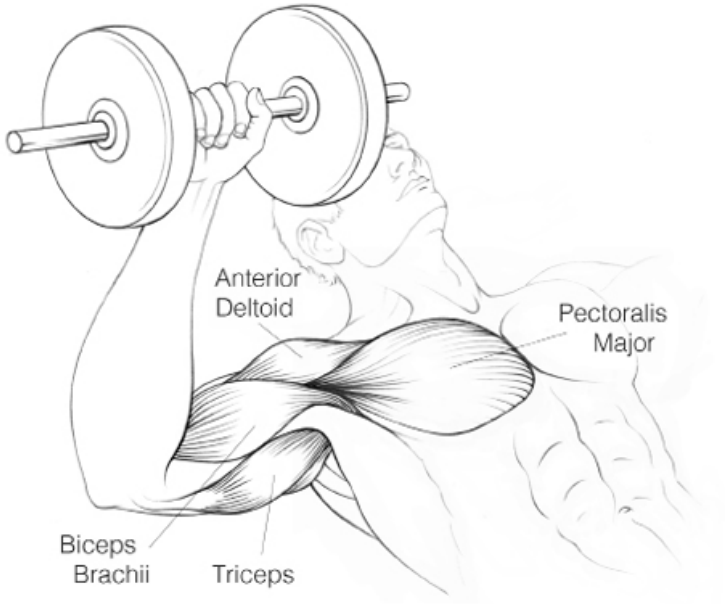
Merely lifting heavy weights isn't enough to keep driving muscle growth, though. To do that, we must steadily increase mechanical tension levels in the muscles over time (progressive overload). This explains why DeLorme's methods were so effective. Week by week, month by month, and year by year, his patients handled heavier and heavier loads that generated more and more mechanical tension in their muscles.

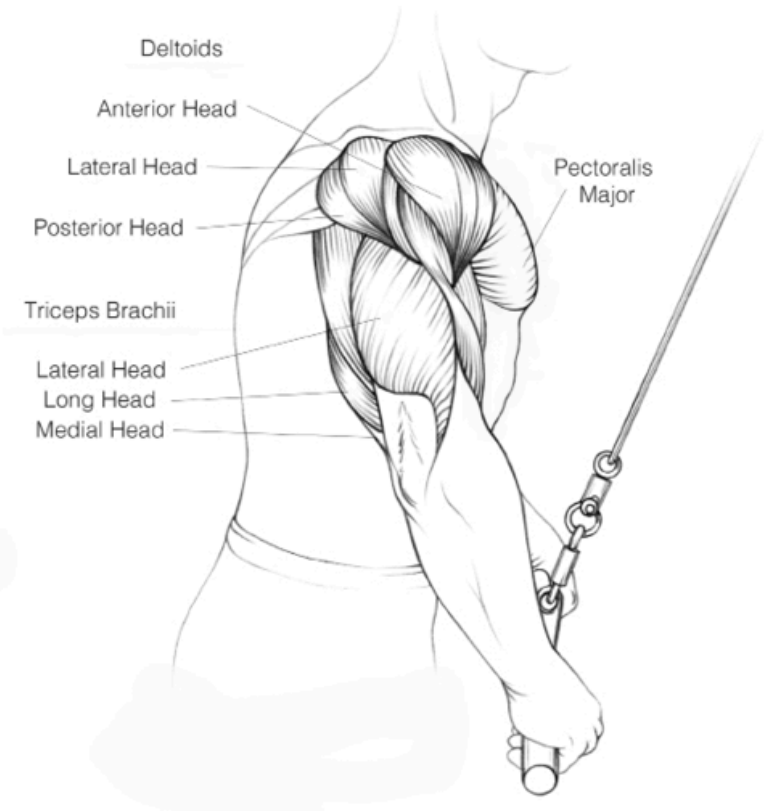
In *Bigger Leaner Stronger*, I'll have you do the same, and as you learned in the last chapter, we'll concentrate on three types of movements that have been strength training staples for over a century now:

1. Pushing
2. Pulling
3. Squatting

Let's learn about each and why they're so vital for building an exceptional body.

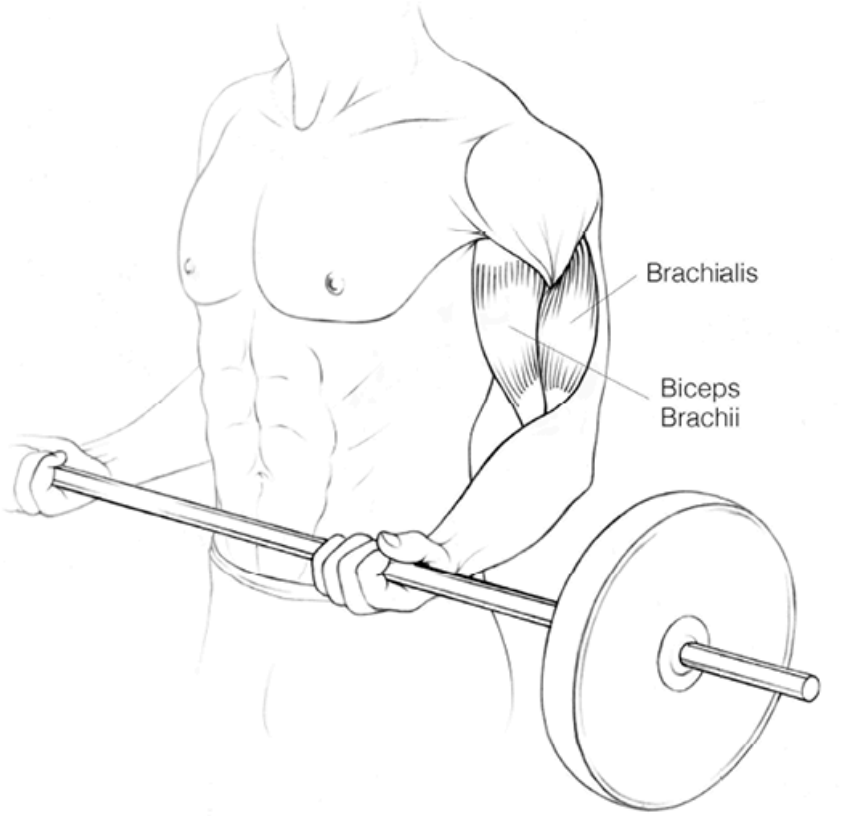
Pushing against resistance—like you do with the push-up, dumbbell bench press, and barbell bench press—is a fantastic way to gain muscle and strength in your upper body, because it engages some of the largest muscle groups above the waist, including your pectorals (chest), deltoids (shoulders), and triceps (arms).

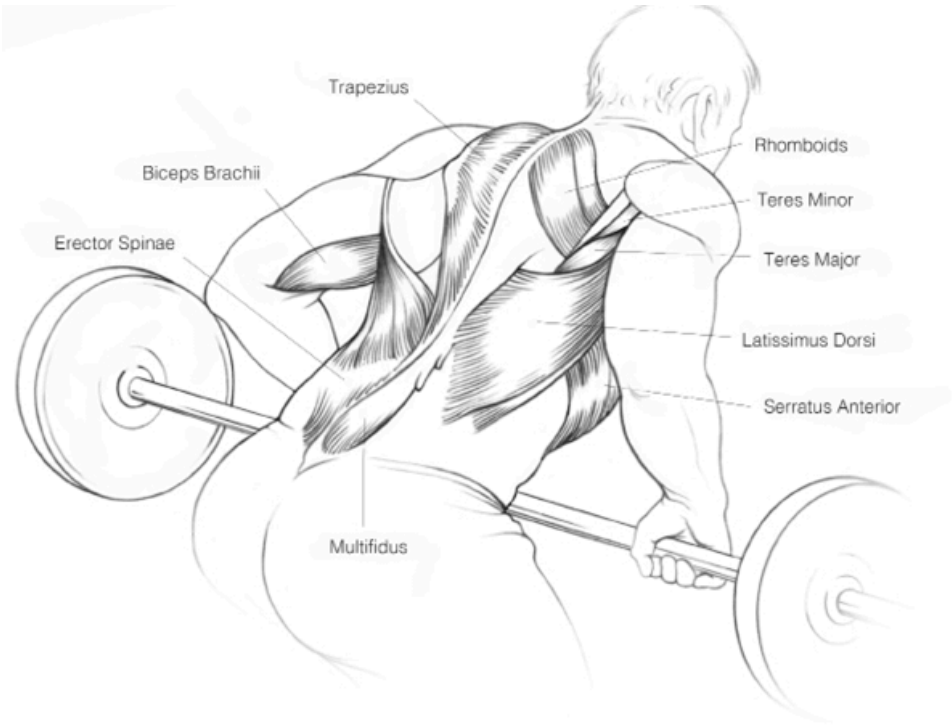




Pulling against resistance is another one of the best ways to develop upper-body strength and muscularity. Pulling something toward your body—like with the

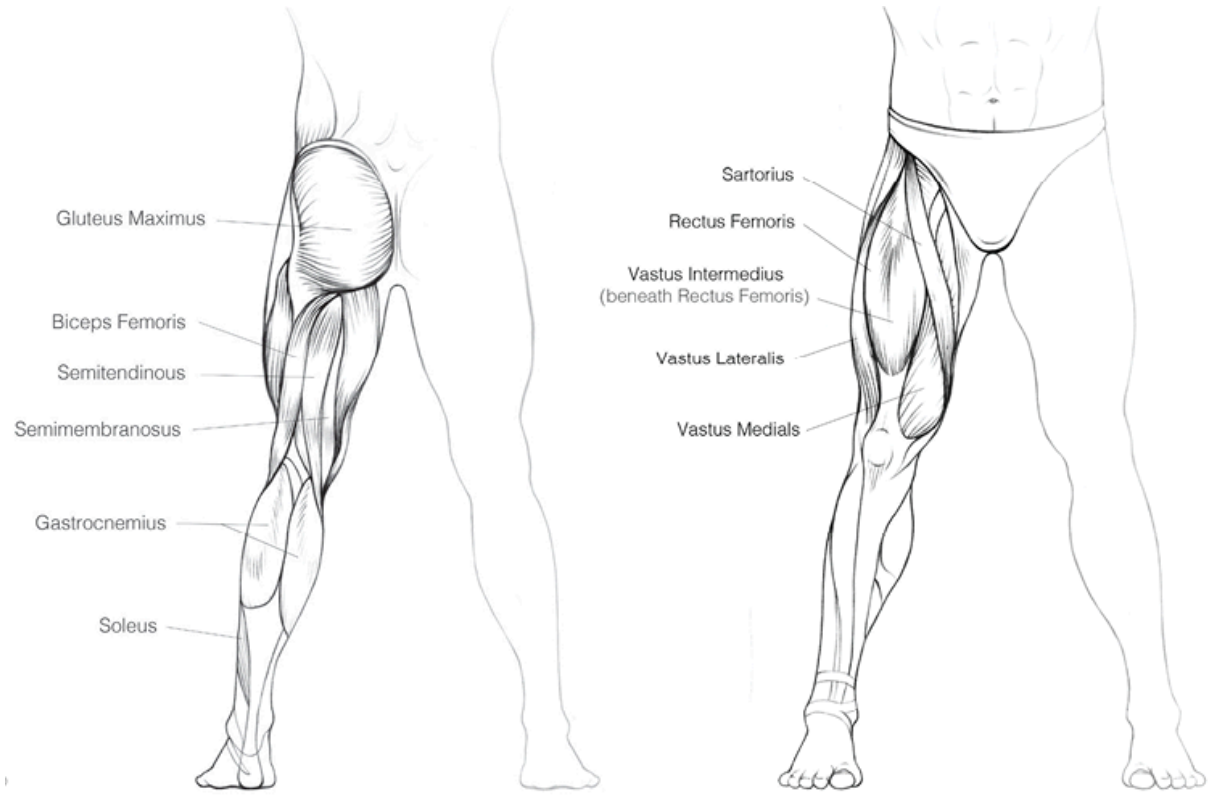
pull-up, dumbbell row, and barbell row—recruits the largest muscle group in your torso, the back muscles, as well as your biceps (arms).





Squatting against resistance is the absolute best way to develop a strong and powerful lower body. When you squat, you activate the largest muscle group in your entire body, the quadriceps, along with every other muscle below the waist, including the hip muscles (hip flexors), butt muscles (glutes), hamstrings, and calves.

In fact, the squat is nearly a whole-body exercise because it also trains your core muscles (abs and obliques) as well as your back muscles (especially your erector spinae)—everything except your chest, arms, and shoulders, basically.



This push-pull-squat paradigm embodies an important part of the *Bigger Leaner Stronger* philosophy: We don't want to just "do things that work" in the gym—we want to do what works best. In other words, we want to spend our time and effort on the exercises and training techniques that produce the most results and avoid those that offer less.

If this strategy sounds familiar, it's probably because you've heard of it before: *The Pareto Principle*. Named after the economist Vilfredo Pareto, this principle states that in many realms, roughly 20 percent of the causes generate roughly 80 percent of the effects. We can observe this axiom in operation all around us—about 20 percent of criminals commit 80 percent of crime, around 20 percent of patients are responsible for 80 percent of healthcare spending, and 15 percent of baseball players deliver 85 percent of the wins.

The Pareto Principle also applies to training. Out of the thousands of fitness

maxims and methods you can follow, just a few of them produce a bucketful of gains, and the rest are either unproductive or unnecessary unless you're nudging your genetic ceiling for muscularity and strength. *Bigger Leaner Stronger* epitomizes these paramount principles, and its system can be expressed as a simple formula:

3–5 | 9–12 | 9–12 | 75–85 | 1–3 | 2–4 | 8

No, that isn't a secret code you have to break. It's the "secret" to building the body you've always wanted. Here's what it means:

- Do **3-to-5** strength training workouts per week.
- Do **9-to-12** *hard sets* per major muscle group per week.
- Do **9-to-12** hard sets per workout.
- Use **75-to-85** percent of one-rep max.
- End most hard sets **1-to-3** reps shy of muscular failure.
- Rest **2-to-4** minutes in between hard sets.
- Take it easy every **8** weeks.

Let's review those directions one at a time and learn how to combine them into a strength training workout routine that really works.

3–5

Do Three-to-Five Strength Training Workouts Per Week

The next time you're chasing the wind on (delete your) Instagram, search the hashtag #nodaysoff, and you'll meet countless fit people preening themselves on their dogged determination. Their effort is often commendable, but not prudent. Intense training six or seven days per week is a high road to injury and burnout

(especially when cutting, when people tend to be most gung-ho).

Strength training is hard. It batters your joints, tendons, and muscles. It redlines your nervous system. It hammers your bones. These effects are healthy and necessary for forcing the body to get fitter, but they also heap up fatigue that reduces strength, power, and focus—a phenomenon known as *overreaching*. Research suggests that this response to training may partly be a mental or emotional state rather than a purely physical one, but it's real, and you need to know how to manage it. If you ignore your body's warnings and press on, overreaching can spiral toward *overtraining*, and you can develop symptoms like …

- Persistent soreness, fatigue, and weakness
- Trouble sleeping
- Loss of appetite
- Unintended weight loss (and not in the good way)
- Irritability, anxiety, and restlessness
- Abnormal heart rate
- Inability to focus
- Depression

To avoid this abyss, I recommend three-to-five strength training workouts per week, which is enough to achieve even the loftiest fitness ambitions without flaming out. This is why I'll give you three *Bigger Leaner Stronger* routines to choose from: a three-day-per-week, four-day, and five-day routine.

A caution: Once you're on the program and seeing results, you'll probably start to feel that your "rest days" are missed opportunities to gain a little more muscle and strength. Remember, however, that rest and relaxation are vital components of the program because muscle isn't built in the gym—that's where

you give it the signal to grow. It's only during downtime that your body can recover from your training and become fitter and stronger than before.

9–12

Do 9-to-12 Hard Sets Per Major Muscle Group Per Week

Many people think that training frequency (how often you train a major muscle group) is a major factor in muscle building—more important than *volume* (the total amount of time your muscles spend contracting, as measured in different ways for different reasons, including total reps, hard sets, and weight x reps) and *intensity* (the degree of tension your muscles produce in each rep). To them, it's black and white: more frequency is always better, and if you're not training each major muscle group at least two-to-three (or more) times per week, you're not going to get very far.

This makes for good social media snippets, but it's too broad of a brush. How frequently you can and should train muscle groups depends on several things:

- Your fitness goals
- Your workout schedule
- Your workout intensity
- Your workout volume

For instance, if you can lift weights three days per week and are more interested in developing your upper body than your lower body, it doesn't make sense to do three whole-body workouts per week. Instead, you'll want to focus on your upper-body muscles.

The relationship between intensity, volume, and frequency is fairly complicated, and there are many viable ways to calibrate these variables. However, there's also a universal rule that governs the results: As volume and intensity go up in individual workouts, the frequency of those workouts must go down. In other words, while you can squat or bench press three times per week, you can't do 10 hard sets (sets taken close to muscular failure) per workout.

What's more, research shows that training frequency isn't nearly as important for gaining muscle and strength as intensity and volume. In fact, frequency can be viewed simply as a tool for providing our muscles with enough weekly volume to stimulate growth—a target that varies widely based on training experience, fitness level, and genetics. According to a recent study published in the *International Journal of Sports Physiology and Performance*, optimal volume for gaining muscle appears to be in the range of 10-to-20 hard sets per major muscle group per week with moderately heavy weights (60 percent of one-rep max or higher) and adequate rest in between sets, with the lower number of sets being suitable for beginners and the higher end for advanced weightlifters.

Practically, that means that someone new to strength training doesn't need to do more than 9-to-12 hard sets per major muscle group per week to make significant gains, whereas intermediate and advanced weightlifters may need to do upward of 15-to-20 hard sets for a major muscle group per week to continue making progress. Curiously, studies also show that when novices increase their volume from the lower end of the 10-to-20-set range to the higher end, they don't gain muscle and strength any faster because their bodies simply can't build muscle any faster. Additional volume, then, results in heavily diminished returns or even backsliding.

Thus, as the *Bigger Leaner Stronger* program is intended for men who have yet to gain their first 20-to-30 pounds of muscle, it entails 9-to-12 hard sets per

major muscle group per week—the appropriate amount of volume for that goal. Eventually, however, that's not enough volume to keep progressing, which is why in the sequel to this book, *Beyond Bigger Leaner Stronger*, intended for intermediate and advanced weightlifters, the volume increases to around 15 hard sets per major muscle group per week.

9–12

Do 9-to-12 Hard Sets Per Workout

Just as there's a point of diminishing returns for weekly volume per muscle group, there's also one for volume in an individual workout. Research shows that this threshold is likely between 8-and-10 hard sets per muscle group, depending on how heavy the resistance is and how fit you are. This limitation helps us understand why frequency becomes more important as you get bigger and stronger. If you have to do, say, 15 hard sets for your chest per week to add muscle and strength without doing more than 10 hard sets in an individual workout, you have to split that volume up into at least two workouts.

In *Bigger Leaner Stronger*, though, you'll warm up and do 9-to-12 hard sets per workout, which may sound suspicious to you. That's all? How can that work when many other bodybuilding programs call for 15-to-20 or more sets per workout? A couple of reasons.

First, many bodybuilding routines use "special" techniques like supersets, drop sets, giant sets, and whatnot that greatly increase volume. While such devices can be effectively incorporated into a training routine, they're not necessary (especially for beginners) nor as generally effective for gaining muscle and strength as *straight sets* (traditional weightlifting sets). As *Bigger Leaner Stronger* workouts only use straight sets, they have less volume than programs

with different types of sets.

Second, unless you're a seasoned weightlifter, you only need to do 9-to-12 hard sets per major muscle group per week to maximize muscle growth, and when programmed correctly, 9-to-12 hard sets per workout is enough to accomplish this.

Third, and most importantly, most bodybuilding programs use lighter weights and higher rep ranges. This approach allows for large amounts of volume, but as you learned in the previous chapter, it's not optimal for *naturally* gaining muscle and strength. With steroids, however, the rules change, and a lot more than the average drug user would care to admit.

75–85

Use 75-to-85 Percent of One-Rep Max

In *Bigger Leaner Stronger*, you'll use two different loads in your workouts: on *primary exercises*, which you'll learn about in the next chapter, around 80-to-85 percent of one-rep max (4-to-6 reps per set), and on *accessory exercises*, around 75-to-80 percent of one-rep max (6-to-8 reps per set).

Why these particular loads and rep ranges? Why not the more common prescription of lighter weights and more reps? Or the coming trend of heavier weights and just a few reps? Contrary to what you might expect, it's not because the *Bigger Leaner Stronger* way is fundamentally superior for gaining muscle than those other styles. In fact, research shows that a breadth of rep ranges can produce significant amounts of muscle growth.

For example, in a study conducted at Lehman College and Victoria University, scientists reviewed 21 studies that compared training with heavier weights (60-plus percent of one-rep max) and lower reps versus lighter weights

(less than 60 percent of one-rep max) and higher reps. They found that men and women gained about the same amount of muscle regardless of what rep range and weights they used, so long as they took each set close to muscular failure (the point at which they could no longer lift the weight despite exerting maximum effort). Other studies have also found no difference in muscle growth when using 10 versus 30 reps per set, 8-to-12 versus 20-to-25 reps, 8-to-12 versus 2-to-6 reps, or 3 versus 7 reps.

And so, while there are a few exceptions to this rule—sets of 1 rep with very heavy weights and sets of 30+ reps with very light weights aren't effective for building muscle—anything between 4-to-20 reps per set can produce about the same amount of muscle growth.

Notice, however, that I've been speaking hypothetically about the utility of different rep ranges—*technically* a variety of rep ranges can work equally well for building muscle so long as they're taken close to muscular failure. But in practice, some rep ranges are more equal than others. For instance, go do a 15-or-20-rep set of barbell squats that ends a rep or two shy of muscular failure. And then imagine having to do a couple more sets like that plus a few 15-to-20-rep sets for a few more exercises, like the leg press and dumbbell lunge. I do not like this workout, Sam-I-Am.

The affliction of higher-rep training isn't just physical, either, because such workouts also require a lot of time. Sets of more reps take longer to complete, of course, but they also cause high levels of fatigue, forcing you to rest longer between sets to catch your breath. This approach to training, then, is like trying to dig a well with a spoon. Why make our job more difficult than it needs to be?

Another reason I want you to do between 4 and 8 reps per set is research shows that using heavier weights for fewer reps is better for gaining strength than using lighter weights for more reps, and, well, this is *Bigger Leaner Stronger*, after

all. Moreover, as you move from a novice to an intermediate trainee and beyond, the most reliable way to get bigger is to get stronger, so attuning yourself (physically and psychologically) to this style of training will pay big dividends now and even more later.

"But wait," you might be thinking, "if strength eventually drives growth, why not train with even heavier weights, like a powerlifter?" A good question with a simple answer: Because that style of training often doesn't provide sufficient volume to maximize muscle gain. A dramatic example of this is a study conducted by scientists at the The University of Mississippi, which found that squatting as much weight as possible for five sets of a single rep twice per week caused *zero* leg muscle growth after 10 weeks. And even when there's enough volume for hypertrophy (growth), training with near-maximal weights batters your joints and amasses fatigue. For instance, a study published in the *Journal of Strength and Conditioning Research* found that 7 sets of 3 reps per workout and 3 sets of 10 reps (with each set taken close to failure) were equally effective for muscle building, but after eight weeks, the participants in the former group were experiencing symptoms of overtraining, joint pain, and fatigue, whereas those in the latter group reported no such problems and were eager to continue training. Thus, when choosing rep ranges, we must consider both the theoretical validity and the practicality and sustainability of different approaches.

Oh and in case you're wondering why I'm calling for 4-to-6 reps with some exercises and 6-to-8 reps with others, it's mostly because many exercises lend themselves better to lower or higher rep ranges. For example, a set of 10 reps of the barbell deadlift that ends close to muscular failure is murderous, but a set of 6 reps isn't. And similarly, a tough set of 6 reps of, say, the dumbbell side raise can be all sorts of awkward, but a set of 10 reps isn't.

Also, if you're finding all this talk about one-rep maxes and rep ranges

daunting, fear not. No math will be required for figuring out your training weights. Instead, I'll show you how to easily and quickly determine your starting weights for the program and then how to increase them as you get stronger.

1–3

End Most Hard Sets 1-to-3 Reps Shy of Muscular Failure

According to many hardbody hardos, if you're not "training insane" by regularly pushing to muscular failure, you'll "remain the same." This is dippy. Taking sets to muscular failure is better for gaining muscle and strength than breezing through your workouts, but it also places a lot of strain on the body (especially with primary exercises) and invites injury.

What about training *close* to muscular failure, though? How does that compare to going all the way? Research shows that it appears to be just as effective. In a study published in the *Journal of Strength and Conditioning Research*, thirty-two untrained men were split into two groups:

1. Group one was to take each set to muscular failure.
2. Group two was to take each set to *volitional fatigue* (the point where a set becomes uncomfortable, usually a few reps short of muscular failure).

Both groups performed three sets of leg extensions at 80 percent of one-rep max twice a week for twelve weeks. By the end of the study, the data showed that while group one experienced higher levels of muscle activation, both groups gained almost exactly the same amount of strength and muscle.

A similar outcome was also seen in another study with untrained women

and biceps curls, but why? Why didn't the additional muscle activation result in more muscle or strength? One plausible explanation is that all of the subjects were new to resistance training. Thanks to the newbie gains phenomenon that you learned about in chapter 11, it's possible that volitional fatigue was enough to whip muscle-building into overdrive and that the extra muscle activation was superfluous, like watering your grass after a rainstorm.

What happens with more experienced lifters, though? Can they benefit from training to muscular failure? Scientists at East Tennessee State University investigated this. When a group of experienced weightlifters did three workouts of 12 sets per week, taking all sets to muscular failure, they performed worse in every way versus a similar group of weightlifters who ended each set a couple of reps short of muscular failure.

There are two reasons for this:

1. Taking a set to muscular failure isn't any more *anabolic* (promoting tissue building) than ending close to muscular failure. When you push a muscle cell close to its limits, it triggers a cascade of signals that lead to more muscle endurance, strength, and (often) size. Thus, the last few reps of a set influence muscle growth more than the first few, when the weight moves quickly, but there's little difference between the stimulus produced by the very last and second- or third-last reps.
2. Training to muscular failure causes disproportionately more fatigue, soreness, and wear and tear than to near-muscular failure. The more often you train to muscular failure, the less likely you are to fully recover from your training, especially with a program like *Bigger Leaner Stronger*, which involves heavy weights and a moderate amount of volume.

So, while it's fine to push to muscular failure periodically if you're feeling lusty, especially on isolation exercises, virtually all of your training should only verge on it. Most sets of primary exercises should end with 1-to-2 good reps still in the tank, and most sets of accessory exercises with 0-to-1 good reps remaining.

Surprisingly, knowing how to manage the difficulty of hard sets is one of the greatest unsung "secrets" of successful strength training. Many people don't even get close to muscular failure in their workouts and wonder why nothing changes, and many others meet failure far too often and wonder why everything always hurts. You now understand how to thread this needle.

2–4

Rest 2-to-4 Minutes in Between Hard Sets

People often keep rest periods in between sets short or even skip them because they're in the gym to move and sweat, not dilly-dally. This works if you're there to burn calories, but if you want to gain muscle and strength, it's a mistake.

In strength training, you push your body to the limit and then back off, and if you don't rest enough in between sets, you won't be able to give maximum effort in your workouts. Several studies make this clear. In a review of 35 strength training studies, scientists at the State University of Rio de Janeiro found that 3-to-5 minutes of rest between sets allowed for more reps over multiple sets and produced more strength gain. In another study published in the *Journal of Strength and Conditioning Research*, the researchers concluded the following:

> The findings of the present study indicate that large squat strength gains can be achieved with a minimum of 2 minutes' rest between sets, and little additional gains are derived from resting 4 minutes between sets.

Here's what I'd like you to do: rest around 2 minutes (slightly more is okay if your heart's still beating quickly or you don't feel ready for your next set yet) in between hard sets for smaller muscle groups like the shoulders, triceps, and biceps (isolation exercises, usually) and around 3 minutes (again, a little more is fine if necessary) between hard sets for larger muscle groups like your back and legs (generally compound exercises).

If you haven't trained like this before, it'll likely feel weird or even wrong at first—like you're sitting around more than you're working out. Remind yourself that the lulls are part of the process, though, and rest easy (literally) knowing that it's helping you get more out of your training, not less.

Also, make sure that you actually *rest* in between sets so your muscles are ready for another go. That means loitering—not doing plyometrics, isometrics, or even light cardio. You should also watch your time (the timer or stopwatch on your phone works well for this) so you don't accidentally rest too little or too much. Otherwise, suit yourself while you rest, but you'll probably enjoy your training more if you stay off the Internet, social media, and e-mail, and instead focus on how your workout is going, how you're feeling, and how you'd like your next set to turn out (studies show that picturing a successful set can enhance performance!).

8

Take It Easy Every 8 Weeks

In many ways, building a strong and fit body is just like building anything else worthwhile—the more you put into it, the more you get out of it ... to a point. You have to train hard to get the body you want, but piss and vinegar alone won't bring it off. Train too intensely too often with too few breaks, and your wages will

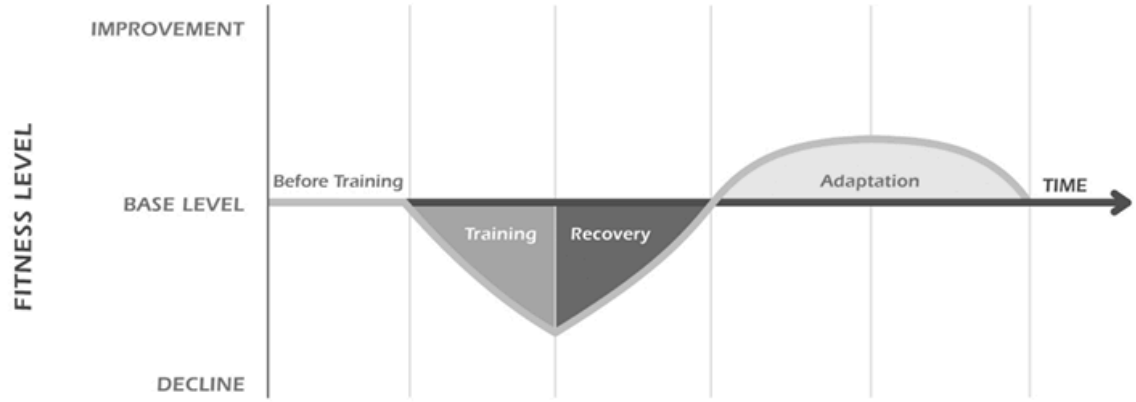
include plateaus, overuse injuries, and soggy workouts.

This makes plain why top level athletes of all stripes—from hockey players to cyclists to powerlifters—include planned periods of extra rest and recovery in their training schedules. In fact, they place great importance on recuperation because the consequences of neglecting it are too severe (chronic underperformance, nagging injuries, career-ending catastrophes, etc.).

The weight given to recovery is based on research on how the body deals with physical stress. Here's the basic outline:

1. You provide a stimulus.
2. You remove the stimulus.
3. Your body adapts.

In strength training, the muscle is in the first step (training), but the magic is in the third step (adaptation, also known as "gains" in gym-goer argot). It's the adaptation response to the training stimulus that makes us bigger and stronger, not the stimulus itself. Hence the bodybuilding adage that muscles don't grow in the gym. They grow *after* the workouts, when your body heals and improves, *if* you give it the chance (with enough of number two—removing the stimulus). Here's how this pattern looks visually:



With strength training, just as it's necessary to only do so much in each workout and each week, it's also smart to ease up every four-to-ten weeks depending on various factors including your age, training experience, training intensity, overall stress levels, sleep hygiene, and more. By doing this, you avoid constipating or even canceling adaptation by demanding more of your body than it can deal with.

In *Bigger Leaner Stronger*, you'll do this by *deloading* every eight weeks—a technical term for a periodic reduction in workout intensity or volume (or both).

Now that we've gone through the *Bigger Leaner Stronger* formula for effective strength training, let's discuss other aspects of the program that are vital to making it work.

How to Achieve Progressive Overload

A couple of chapters ago, you learned that progressive tension overload is the primary mechanical driver of muscle growth. No matter what exercises you do, how often you train, how heavy the weights are, or any other factor related to workout programming, you *must* achieve progressive tension overload to make gains.

There are a few workable ways to implement progressive tension overload in a strength training routine, and one of the best methods is known as *double progression*. With double progression, you work in a certain rep range (4-to-6, for example), and once you hit the top of that rep range for a certain number of sets in the same workout (one, two, or three, usually), you increase the weight (normally by 5 or 10 pounds). Then, if you can at least hit the bottom of your rep range with the new, heavier weight, you work with it until you reach the progression target (number of *top-rep sets*) again, increase the weight again, and so forth. As you may have gathered, it's called double progression because you strive to first increase reps (volume) and then "cash in" that progress to increase weight (intensity).

To see how this works in action, let's say you're doing a *Bigger Leaner Stronger* workout that calls for 3 sets of barbell bench pressing in the 4-to-6-rep range. On your first set, you get 6 reps with 135 pounds, which is the progression target for the program (1 top-rep set), so it's time to increase the weight. You add 10 pounds to the bar (145 pounds), rest a few minutes, and get 4 reps on your next 2 sets of the exercise in that workout. Because you're still within your target rep range (4-to-6 reps) with 145 pounds, you'd keep working with it until you get 6 reps for 1 set, and then you'd increase the weight to 155 pounds.

Here's how this could look over the course of four weeks.

	Set 1	Set 2	Set 3
Week 1	135 x 4	135 x 4	135 x 5
Week 2	135 x 6 (target)	145 x 4 (increase)	145 x 4
Week 3	145 x 4	145 x 5	145 x 5
Week 4	145 x 6 (target)	155 x 4 (increase)	155 x 4

And what if you can't reach the bottom of your rep range with the new, heavier weight? For example, what if you moved up to 145 pounds on the barbell bench press and only got 2 or 3 reps? You'll find out in the next chapter, along with directions for dealing with other eventualities.

How to Use a Proper Range of Motion

Range of motion refers to how much you *flex* (bend) and *extend* (straighten) your joints when you do an exercise. For example, when you curl a dumbbell, *flexion* occurs when you raise the dumbbells (bending your elbows), and then *extension* occurs when you lower them (straightening your elbows). With strength training exercises, there's a limit to how much you can safely and comfortably flex and extend your joints, like your knees and hips in the squat and deadlift, and your elbows in the bench press. A proper range of motion in a strength training exercise, then, is one that moves the affected joints through their full and normal range of flexion and extension (beyond which, you can cause injury).

For example, with the barbell bench press, a full range of motion requires that you lower the bar until it touches your chest (full elbow flexion), and then press it upward until your arms are straight (full elbow extension). And with the chin-up, you should lift yourself until your chin is above the bar (full elbow flexion), and then lower yourself until your arms are straight (full elbow extension).

When strength training, using a full range of motion is important because it's more effective for gaining muscle and strength than a shorter range of motion (because it makes your muscles work harder), and it also can reduce the risk of joint pain and injury by distributing the stress produced by the exercise over the entirety of your joints. For instance, the first several inches of a squat places a large amount of stress on the tendons below your knees, but as you keep moving downward, the load shifts to other tissues.

In the next chapter, I'll introduce you to all of the exercises you'll be doing in your *Bigger Leaner Stronger* workouts and show and explain a full range of motion for each of them.

How to Use Proper Form

Exercise form refers to how well you execute the movement, and while using a full range of motion is an important component of proper form, you also need to do it in a specific and controlled manner. With any type of deadlift and squat, for example, you want the bar (or dumbbell) to move straight up and down (you don't want it to sway toward or away from you), and with a standing biceps curl, it's important that your back remains more or less motionless (as opposed to swinging forward or backward). You should also always feel that your muscles are actively working to produce motion, not just using gravity or momentum. When squatting, instead of relaxing your quadriceps and allowing your torso to drop toward the floor, you want to keep your lower-body muscles tight as you lower yourself. Similarly, with the pull-up, instead of swinging your knees and torso to help on the way up and then free-falling down (as many Crossfitters do), you want to keep your legs still as you smoothly pull yourself up and lower yourself down.

An important prerequisite to using proper form is using the proper amount of weight—enough to effectively challenge your muscles but not so much that it forces you to sacrifice your form. I'll explain how to establish your training weights later in this section of the book.

At bottom, proper form consists of moving the right amount of weight through the right range of motion with the right technique. And again, I'll spell it all out for every exercise in the program in the next chapter.

How Hard Your "Hard Sets" Should Be

You learned earlier in this chapter that you should end most hard sets 1-to-3 reps shy of muscular failure. Here's exactly what to do:

- End all hard sets of bodyweight exercises 1 rep shy of muscular failure (0 good reps left in the tank).
- End all hard sets of *primary exercises* 2-to-3 reps shy of muscle failure (1-to-2 good reps left). In *Bigger Leaner Stronger*, the primary exercises are those that train the most muscle mass and allow you to use the most weight. As you'd expect, they're mostly compound exercises.
- End all hard sets of *accessory exercises* 1-to-2 reps shy of muscular failure (0-to-1 reps left in the tank). Accessory exercises are those used to further train muscle groups, bring up stubborn muscles, and help prevent and correct muscle imbalances or weaknesses that may limit your progress on your primary exercises. They're usually isolation exercises.

With bodyweight and accessory exercises, you can work close to muscular failure because it's less taxing and risky than with primary exercises. If your workout calls for dumbbell side raises, you'd end each hard set at the point where you feel you can't complete another rep without compromising your form (swinging your hips forward, raising the dumbbells only halfway, etc.). And with, let's say, the military press, you'd work in each hard set until you feel you could do 1 or 2 more good reps before your technique skids.

And how do you know how close you are to muscular failure in a set? It's mostly a matter of cut-and-try, but it doesn't take long to learn. As an easy way to get attuned faster, try this: As a set starts to feel hard, ask yourself, "If I absolutely

had to, how many more reps could I get with good form?" Your intuitive answer will usually be accurate, especially as you gain experience.

How to Use a Proper Rep Tempo

There are two schools of thought about how fast you should perform strength training exercises—fairly slowly and fairly quickly. People in the slow-training camp often say their style is superior because it increases the amount of time muscles remain under tension. However, while *time under tension* does contribute to muscle growth, you still need to create high levels of tension by using heavy weights, and this is where slow-tempo training miscarries—to slow down your rep tempo and remain in the right rep range, you have to use less weight. This is why studies consistently show that using a faster rep tempo produces better results than using a slow rep tempo. So that's what I'll recommend.

Specifically, a "1-0-1" rep tempo works well, which has you complete the first part of a rep in about one second ("one, one thousand"), pause momentarily, and then go back to the starting position in about one second. In a biceps curl, for example, you'd curl the weight up in one second, pause briefly, and then lower the weight down in one second.

Don't fret about following this tempo perfectly. So long as you're moving through each rep in a swift but controlled manner, you're doing it right.

How to Avoid Injury

Many people believe that most strength training injuries are caused by overdoing it in a single set or workout, but this isn't true. The most common cause of injury is more insidious: It's failing to fully recover from previous workouts. An everyday scenario might be that your elbow feels stiff the day after an upper-body workout, but you brush it aside and keep training. A few weeks later, your elbow is hurting during and after bench pressing. "No pain, no gain," you think, and keep plugging away. A few more weeks and … well, now your elbow always hurts.

These repetitive stress injuries (RSIs) are the scourge of athletes everywhere because they're not severe enough to sideline you but can significantly sour your performance. Fortunately, resolving RSIs generally requires nothing more than a bit of rest. Indeed, that's the only way to heal an RSI once it has taken root—you must stop doing whatever caused it (and will continue to irritate it) as well as anything else that bothers the affected area. This normally means staying away from an exercise or type of movement pattern for a week or two, but sometimes you have to stop training a muscle group entirely until the pain is gone.

Strength training isn't dangerous, but it's a demanding physical activity, and if you do enough of it, you're bound to develop an RSI here and there (especially as you get stronger and the weights get heavier). You can take simple precautions to prevent them, however, and (touch wood) completely ward off more serious problems.

If It Feels Bad, Don't Do It

If you feel sharp pain when you're doing an exercise, end the set immediately. I'm not referring to residual muscle soreness from your last workout or the burn you feel as you approach failure, but something that makes you wince at least a little. This is a warning that something is wrong. Don't ignore it.

The same goes for "strange" feelings during an exercise. The initial stage of an RSI doesn't always involve pain. Instead, your knee feels "weird" on the last couple of reps of squatting or your elbow feels "funny" or your shoulder "tight" when pressing. Sensations like these aren't always danger signals, but it's better to be careful with them than cavalier. So consider pain or strange feelings as warnings and pay attention.

Sometimes it's hard to tell whether the discomfort you feel is normal or not, though. Asking yourself the following two questions can help.

1. Is the pain or strange sensation on one or both sides of my body? When you're doing a *bilateral exercise* (one that trains both sides of the body simultaneously) properly, one side shouldn't hurt more than the other. If that's the case, it's more likely a sign to stop rather than muscle fatigue.
2. Is the pain in a joint or other specific spot in my body? These are the pain or strange feeling issues you're most likely to experience, not muscle aches or stiffness. Localized pain will suddenly strike in a joint or tendon.

When a feeling of pain or something strange strikes, stop training, rest for a couple of minutes, and try the exercise again. If there's no improvement, do another exercise that feels okay, and come back to the original one in your next

workout. If it's still problematic, stay away from it until you're in the clear.

Progress Gradually

Soldiers have a slogan that goes like this: Slow is smooth, smooth is fast. This mindset is essential for succeeding in strength training as well. One of the easiest ways to get hurt is through overwork. You feel particularly strong one day or want to turn heads in the gym or set a PR, so you ignore the plan and load the bar with more weight than ever before. This is almost always a mistake because it can lead to compromised form, place an inordinate amount of strain on your joints, tendons, and ligaments, and impair recovery.

Slower progress, however, is smooth, and smooth progress is fast. For instance, if you're new to strength training, you'll probably be able to add weight to most exercises every week or two for the first few months. After you've been at it for a year or so, however, adding weight to the most difficult exercises once or twice per month is commendable.

A winning benchmark for strength training is "progress is progress." Sometimes you'll race ahead and sometimes you'll crawl, but if you're moving forward, you're doing well.

Maintain Good Form

"Everyone knows" they should use proper form, but it can be tempting to cut corners when you want to get that extra rep or two. The occasional infraction on an isolation exercise is acceptable (a sloppy final rep on a biceps curl or shoulder raise, for example), but regular violations aren't. The substance of strength training is methodically moving heavy loads through full ranges of motion with proper technique, not haphazardly heaving as much weight as possible.

Don't rob Peter to pay Paul by cheating on your form. Learn proper technique and always strive to maintain it.

You can study strength training and muscle building for hundreds of hours and barely scrape the surface. The mechanics are complex and involve hundreds of factors and functions, but fortunately, you don't need an advanced degree to understand what to do in the gym to get strong and fit.

In fact, you now possess a powerful plan for long-term fitness success: a moderate dose of moderately difficult strength training workouts that are tough enough to produce results but not so rough that you feel agonized, exhausted, or burned out.

The strength training strategy I've just shared with you has enough horsepower to radically transform your body and health, and enough versatility to satisfy just about any circumstances and preferences. So, if you've had a falling-out (or five) with fitness, I hope to help you fall in love with it again. And if this is your first foray, you're going to have a lot of fun.

To get started, however, you'll need to know the grammar of strength training exercise selection contained in the next chapter.

KEY TAKEAWAYS

- The three primary "triggers" or "pathways" for muscle growth are mechanical tension, muscle damage, and cellular fatigue.
- Mechanical tension is the most important driver of muscle growth and thus, for strength training to be maximally productive, it must underline mechanical tension more than muscle damage and cellular fatigue.
- Training with heavier loads produces more mechanical tension than lighter ones and thus activates more muscle growth (which leads to more strength).
- To keep driving muscle growth, we must steadily increase mechanical tension levels in the muscles over time (progressive overload).
- Three-to-five strength training workouts per week is enough to achieve even the loftiest fitness ambitions without flaming out.
- Unless you're a seasoned weightlifter, you only need to do 9-to-12 hard sets per major muscle group per week to maximize muscle growth.
- Rest around 2 minutes in between hard sets for smaller muscle groups like the shoulders, triceps, and biceps and around 3 minutes between hard sets for larger muscle groups like your back and legs.
- Take a deload—a technical term for a periodic reduction in workout intensity or volume (or both)—every eight weeks.
- With double progression, you work in a certain rep range, and once you hit the top of that rep range for a certain number of sets in the same workout, you increase the weight. Then, if you can at least hit the bottom of your rep range with the new, heavier weight, you work with it until you reach the progression target again, increase the weight again, and so forth.

- When strength training, using a full range of motion is important because it's more effective for gaining muscle and strength than a shorter range of motion, and it also can reduce the risk of joint pain and injury by distributing the stress produced by the exercise over the entirety of your joints.
- Proper form consists of moving the right amount of weight through the right range of motion with the right technique.
- End all hard sets of bodyweight exercises 1 rep shy of muscular failure (0 good reps left in the tank), end all hard sets of primary exercises 2-to-3 reps shy of muscle failure (1-to-2 good reps left), and end all hard sets of accessory exercises 1-to-2 reps shy of muscular failure (0-to-1 reps left in the tank).
- A faster rep tempo of "1-0-1" produces better results than using a slow rep tempo.
- To resolve a repetitive stress injury, stop doing whatever caused it (and will continue to irritate it) as well as anything else that bothers the affected area until the pain is gone.
- If you feel sharp pain when you're doing an exercise, end the set immediately. This is a warning that something is wrong. Don't ignore it.
- A winning benchmark for strength training is "progress is progress." Sometimes you'll race ahead and sometimes you'll crawl, but if you're moving forward, you're doing well.

The Big Ideas

1. Concentrate on mastering a relatively short list of effective exercises.

2. Use primary exercises to maximize muscle and strength gain.

3. Use accessory exercises to boost muscle and strength gain and prevent muscle imbalances.

13

The Best Exercises for Building Your Best Body Ever

There is no reason to be alive if you can't do the deadlift!

—JÓN PÀLL SIGMARSSON

Out of the many strength training exercises you can do, a few dozen are the real breadwinners, and out of those, a few stand head and shoulders above the rest (remember the Pareto Principle?). This streamlines our training because it allows us to ignore most of what we see people doing in the gym (and elsewhere) and concentrate on mastering a relatively short list of exercises.

In this chapter, I'll share those blue-ribbon exercises with you, separated into two categories:

1. Primary exercises
2. Accessory exercises

In the last chapter, you learned that primary exercises are mostly compound exercises and will produce the bulk of your results on the program because they train the most muscle mass and allow you to use the most weight.

As great as the primary exercises are, however, they don't adequately train all of the major muscle groups that we want to develop, especially the more stubborn ones like the shoulders, biceps, and lats. Accessory exercises will further stimulate these muscle groups, which will enhance their growth, boost performance on primary exercises, and help prevent muscle imbalances that can slow your progress. We'll also organize the *Bigger Leaner Stronger* exercises according to whether they train push, pull, or squat muscles (and note which ones), which encompasses every major muscle group in our bodies.

So, here's the full list of the *Bigger Leaner Stronger* exercises:

	Pushing Muscles	**Pulling Muscles**	**Squatting Muscles**
Primary Exercises	Barbell Bench Press	Barbell Deadlift	Barbell Squat
	Incline Barbell Bench Press	Barbell Romanian Deadlift	Barbell Front Squat
	Dumbbell Bench Press	Barbell Row	Leg Press
	Incline Dumbbell Bench Press	One-Arm Dumbbell Row	
	Dip	Chin-Up	
	Close-Grip Bench Press	Pull-Up	
	Standing Barbell Overhead Press		
	Seated Barbell Overhead Press		
	Seated Dumbbell Press		

	Arnold Dumbbell Press		
Accessory Exercises	Triceps Pushdown	Lat Pulldown	Dumbbell Lunge (Walking or In-Place)
	Seated Triceps Press	Seated Cable Row (Wide- and Close-Grip)	Dumbbell Bulgarian Split Squat
	Lying Triceps Extension ("Skullcrusher")	Barbell Curl	Leg Curl (Seated or Lying)
	Dumbbell Side Lateral Raise	Alternating Dumbbell Curl	
		Dumbbell Hammer Curl	
		E-Z Bar Curl	
		Barbell Rear Delt Row	
		Dumbbell Rear Lateral Raise (Standing or Seated)	
		Machine Reverse Fly	

The lingo can seem formidable if you're new to all of this, but don't worry—all of these exercises are easy to learn and eventually master, regardless of your athletic talent. Some of the exercises may seem miscategorized, too. Why is the dumbbell rear lateral raise a "pull" exercise when it's not a pulling motion, and why are calf raises "squat" exercises? Because we're bracketing exercises based on the primary muscle groups involved in pushing, pulling, and squatting—not merely the movements themselves—and there are different ways of training these muscles. Although the shoulders are generally considered "push" muscles, the rear

delts are more engaged when pulling than pushing (and the opposite is true of the anterior delts), for instance.

You'll notice that several of the exercises train one side of your body at a time, such as the one-arm dumbbell row, alternating dumbbell curl, and lunges. These are known as *unilateral exercises*, and though the pictures only show me training one side of my body, one *full* rep is one rep for both sides (left and right). For instance, you've completed one set of 10 reps of the one-arm dumbbell row when you've done 10 reps for each arm.

You may also wonder why there aren't any "core" or "ab" exercises in the lineup. There are! They're called the squat, deadlift, bench press, chin-up, pull-up, one-arm dumbbell row, and others. My point is while there are no *direct* ab exercises in the program, like crunches, planks, and situps, many of the compound exercises in it heavily train your abdominals, obliques, and other core muscles.

For instance, a study conducted by scientists at Nord University measured the core muscle activity of twelve powerlifters and national level sprinters during barbell back squats and weighted planks. The researchers found that on average, the barbell back squat activated the rectus abdominis ("ab muscles") and external oblique muscles just as effectively as the weighted plank and the erector spinae muscles significantly more. Rectus abdominis activation also increased with each rep of the squat, but not the weighted plank. Thus, the researchers concluded, " ... we recommend targeting core muscles by integrated high-intensity exercises [such] as the squat instead of an isolated and isometric core exercise, especially for athletes."

That said, if you'd like to include some ab exercises in your workouts to speed up your core development (or if you just like doing them), you can. Simply add 2 or 3 sets of an ab/core exercise of your choice to the end of your legs and pull

days, after you've finished all of your prescribed exercises. Here are my favorite exercises for this:

- Sit-ups (especially weighted)
- Cable Crunch
- Lying Leg Raise or Boat Pose
- Captain's Chair Leg Raise
- Plank
- Hanging Leg Raise
- Abdominal Rollout

The same is true of the calves—while there are no *direct* calf exercises on this program, many lower body exercises also train the calves indirectly. If you want your calves to develop faster, though, add 3 sets of calf exercises to the end of your pull and lower body workouts (6 sets per week). I also recommend splitting the volume between straight-knee (such as the leg press or standing calf raise) and bent-knee calf (such as the seated calf raise) exercises because they train your calves in slightly different ways.

Let's now review how each of the *Bigger Leaner Stronger* exercises look and work. Also, to help you further, I've included links to video demonstrations of all of the exercises you'll learn about in this chapter in the free bonus material that comes with this book (www.biggerleanerstronger.com/bonus).

The *Bigger Leaner Stronger* Primary Pushing Exercises

Barbell Bench Press



Lie on a bench so your eyes are directly under the bar, tuck your shoulder blades back and grab the bar with your hands slightly wider than shoulder-width apart. Wrap your fingers and thumbs around the bar and bend your wrists backward just enough to allow the bar to settle into the base of your palm, but not so much that they're folded back at right angles. Place your feet on the floor about shoulder-width apart and directly under your knees. Gently arch your back by pushing your chest toward the bar (while keeping your shoulders and butt on the bench), pull your shoulder blades together and toward your butt (imagine you're "putting your shoulder blades into your back pockets"), and move the bar from the hooks to directly over your chest.

Squeeze the bar as hard as you can, and lower it until it touches your chest, keeping your elbows about 6-to-10 inches from your ribs. Then, push the bar up and return to the starting position, keeping your shoulder blades and elbows tucked. When you're on the last rep of a set of the barbell bench press, finish it

fully (arms straight), and then slam the bar back into the upright metal bars, and lower it onto the hooks. Don't try to press the bar directly back into the hooks because if you miss, it can fall on your face.

We could leave the bench press instructions at that and you'd be primed for success, but let's go a little further and answer some common questions I get about proper technique.

"How wide should I grip the barbell?"

Use a medium grip that's just a little wider than your shoulders, about 18-to-24 inches, depending on your build. If you go too narrow, you'll shift the emphasis to the triceps instead of the pecs, and if you go too wide, you'll reduce the range of motion and effectiveness of the exercise as well as increase the risk of irritating your shoulders.

A good way to check your grip width is to have a friend look at your forearms at the bottom of a rep. Your grip is correct when your forearms are straight up-and-down (perpendicular with the floor), not angled inward to or outward from your torso.

"How much should I arch my back?"

Enough to fit a hand in the gap between your back and the bench, but not so much that you experience discomfort or can't keep your butt and shoulders on the bench. You'll occasionally see powerlifters using an exaggerated arch to reduce the range of motion of the exercise and lift heavier weights, but this reduces the effectiveness of the exercise for muscle building.

"Is the bench press bad for my shoulders, elbows, and wrists?"

Not if you use proper technique. A study published in the *Strength and Conditioning Journal* supports this. Researchers found that the chances of injury

on the bench press are low so long as you:

1. Use a medium grip that's a little wider than shoulder-width apart.
2. Keep your arms at about a 30-to-60-degree angle relative to your torso.

The main reason the bench press has copped a bad rap is simply due to its popularity, especially among men who don't know what they're doing (typically using too much weight and bad form). These circumstances naturally produce more injuries than those exercises performed less frequently with lighter loads. If you avoid these mistakes, however, and progress prudently, your joints have nothing to fear from the bench press.

"Can I bench press with a shoulder injury?"

Probably, but it depends on the nature and severity of the injury. If you experience shoulder pain (or any pain, really) during the exercise, you should stop bench pressing until you can do it without pain. If a week or two of resting your shoulder doesn't fix the problem, it's a good idea to see a sports medicine doctor so you can better understand what's wrong and what to do about it.

If you had an injury in the past that limits your shoulder, elbow, or wrist mobility or causes discomfort while bench pressing, you may be able to successfully ease into the exercise by starting with lighter-than-usual weights and gently working toward a full range of motion. You'd be surprised how a few weeks of low-intensity strength training can get "stuck" joints moving. That said, if your restriction doesn't allow you to lower the bar all the way to your chest, that's fine. Don't try to force it—lower the bar as far as you comfortably can, and otherwise perform the exercise as described.

"How do I keep my shoulder blades in the proper position?"

If your shoulder blades tend to "untuck" or shift out of place while bench

pressing, here are a few helpful cues for maintaining upper-back tightness:

- Pulling your shoulder blades into your back pockets
- Pushing your back into the bench, rather than pushing the bar toward the ceiling
- Squeezing the bar as hard as possible

"Can I use a thumbless (suicide) grip?"

You can, but I don't recommend it. The "thumbless" or "suicide" grip involves positioning your thumbs next to your index fingers instead of wrapping them around the bar. With this grip, the bar can slip out of your hands and crash down on your chest, or worse, your neck.

"Can I put my feet on the bench instead of the floor?"

You can, but I don't recommend it unless you have good reason to do so (an injury that makes arching your back painful, for instance). Some bodybuilders like to bench press with their feet on the bench because they think it helps isolate their chest muscles, but all it really does is make your upper body less stable, limiting how much weight you can press and making it harder to maintain proper form.

Instead of elevating your feet while you bench press, find a position on the floor that's comfortable for you by moving them closer or farther away from your torso until it feels right.

"How long should I pause when the bar touches my chest?"

About a half second or so—any longer will just reduce your performance on the exercise. Don't make the opposite mistake, either, and let the bar bounce off of your chest. This will allow you to get an extra rep or two, but it'll also make the exercise less effective and increase the risk of injury.

If you tend to lower the bar too quickly, imagine there's an egg resting between your pecs and try to tap it with the bar without breaking it. Don't overcorrect here, however, and start lowering the bar too slowly (two seconds or more to reach your chest) because it'll make the exercise less effective.

"Why do my shoulders hurt when bench pressing?"

This can happen for various reasons, but a common one is flaring the elbows out to the sides of the body, which puts the shoulders in a compromised position. The solution is simple—keep your elbows at a 30-to-60-degree angle relative to your torso throughout the entire movement, or about 6-to-10 inches or so from your sides. Don't bring your elbows in closer than this, however, because it'll slash your stability and strength and possibly irritate your elbows.

"Is it okay that the bar doesn't move straight up-and-down?"

Yes! The bar should start directly above your shoulders, follow a slightly curved path toward your nipples, touch your chest near them, and then move back to the starting position. As you get stronger, this "J-curve" will become more pronounced, and the bar will start to move toward your shoulders immediately after you start pushing it off of your chest.

Incline Barbell Bench Press



Perform the incline barbell bench press the same way you do the barbell bench press, except first adjust the bench to roughly a 45-degree angle.

"Do I need to incline bench press?"

No, but it's a good idea if you want to build powerful and proportionate shoulder and chest muscles. The reason for this has to do with the anatomy of your primary chest muscle—the *pectoralis major*, or pecs. Its main function is to bring the upper arm across the body, and unlike most other muscles, its fibers aren't all aligned in the same direction. There's a *sternocostal head*, which attaches the sternum (breastbone) and ribcage to your upper arm, and a *clavicular head*, which attaches your collarbone to your upper arm.

This subtle anatomical anomaly changes how the two portions of your pecs respond to exercises. For instance, certain exercises, like the flat and decline bench press, emphasize the larger sternocostal head of the pecs, while others, like the incline and close-grip bench press, emphasize the smaller clavicular head. Notice

that I said *emphasize*, not *isolate*, because all movements that involve one head do, to some degree, involve the other as well. That said, you want to dedicate some of your chest training to focusing on the clavicular head for two reasons:

1. It's a small, stubborn part of the pec that takes its sweet time to grow.
2. The exercises that are best for developing it also happen to be great for developing the sternocostal head.

For many years, I did nothing but flat and decline pressing, and this gave my pecs an unbalanced, "bottom-heavy" look. After a year or so of focusing almost exclusively on incline pressing, however, the imbalance was corrected, and my chest had a much more even, aesthetic look. This is why all of the *Bigger Leaner Stronger* workout routines include some incline pressing.

Close-Grip Bench Press



Perform the close-grip bench press the same way as the barbell bench press, except use a slightly narrower grip, which emphasizes the triceps muscles. Specifically, you position your hands directly over your shoulders or just inside

shoulder-width apart. This will put your hands about 10-to-16 inches apart for most men. If your shoulders or wrists feel uncomfortable at the bottom of the movement (when the bar is touching your chest), simply widen your grip by about the width of a finger on each side and try again. Repeat until it's comfortable.

"Will putting my hands even closer together train my triceps even more?"

Probably not. Using a super narrow grip (usually less than 10 inches or so) isn't better than a shoulder-width grip, and it often causes wrist and shoulder pain.

"Why does the bar curve more when I close-grip bench press?"

Bench pressing with your hands closer together changes the biomechanics of the exercise, so the bar naturally wants to touch your chest at or even slightly below your nipples. This is fine.

Dumbbell Bench Press



Sit on the edge of the bench with the dumbbells resting vertically on your thighs.

Then lean back and gently push your thighs (and the dumbbells) toward your chest. Continue to roll back onto the bench until you're lying flat and holding the dumbbells at either side of your chest. Then, keeping your shoulder blades and elbows tucked, push the dumbbells straight up until your arms are straight. Lower the dumbbells by reversing the motion and return to the starting position.

When you're finished with a set of the dumbbell bench press, you can either lower the dumbbells to your chest and then drop them to the floor, or bring your legs up and toward you into a sitting position, push the dumbbells onto your thighs, and then swing your legs and dumbbells downward and your torso upward to rise into a sitting position. I prefer the latter method—it's more difficult but ensures I don't damage the equipment or make a ruckus in the gym.

"Do I need to touch the dumbbells at the top of each rep?"

No, but you can if you want to. Some people like to do this to ensure they use a full range of motion, but you can achieve the same thing with the dumbbells apart by ensuring your elbows are straight at the end of each rep.

Incline Dumbbell Bench Press



Perform the incline dumbbell bench press the same way you do the dumbbell bench press, except first adjust the bench to roughly a 45-degree angle.

Dip



Hold a dip bar with your hands underneath your shoulders and thumbs wrapped around the bars. Flex your glutes and hamstrings to keep your legs from swinging, lower your body until your upper arms form a 90-degree angle, and then push upward and return to the starting position.

Standing Barbell Overhead Press



Adjust the hooks in a power rack or squat rack so the bar is at the height of your midchest. Face the bar and grip it with your palms facing forward and thumbs wrapped around the bar, slightly wider than shoulder-width. Hold the bar low in your hands, closer to your wrists than your fingers. Your wrists should be bent just enough to allow the bar to settle into the base of your palms, but not folded back at a 90-degree angle.

With the bar in your hands, bend your knees slightly and lean forward, positioning the bar on top of your shoulders and upper chest. Then, stand up, lifting the bar off the rack, and take one step back with each foot. Take a deep breath into your stomach, arch your upper back enough so the bar is directly over your midfoot and your head is slightly behind the bar, squeeze your glutes and the bar as hard as you can, and push it straight up.

As soon as the bar passes your eyes, push your head and chest forward and

under the bar. Keep pushing the bar toward the ceiling until your arms are straight, and then lower it and return to the starting position.

Seated Barbell Overhead Press



Perform the seated barbell overhead press the same way as the standing barbell overhead press, except do it seated on a utility or vertical bench (or an adjustable one set to the upright position) in the middle of a squat rack instead of standing. Most people also prefer to spread their feet farther apart and push themselves back into the bench throughout each rep to increase stability.

Many people also wonder which exercise is better—the standing or seated overhead press. While the standing press engages more total muscle mass in your body, the seated press has several advantages over the standing variation:

1. It emphasizes the shoulder muscles more.
2. It allows for slightly heavier weights.
3. It's easier to perform correctly.

Thus, I recommend alternating between standing and seated pressing, which is what you'll find in the year's worth of workouts in the bonus material that comes with this book (www.biggerleanerstronger.com/bonus).

Seated Dumbbell Press



Adjust a bench to an upright position (most people prefer a slight incline of about 75-degrees or so), and push the dumbbells up with your knees, one at a time. Tuck your shoulder blades back and down, push the dumbbells up until your arms are straight (but not locked), and lower them to the starting position.

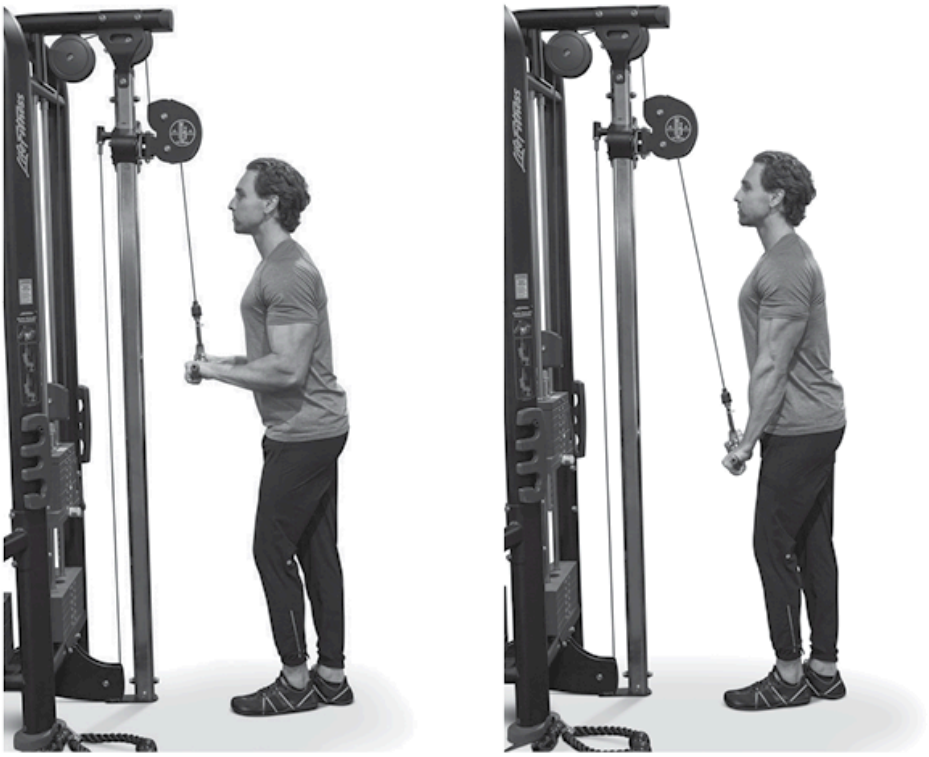
Arnold Dumbbell Press



Perform the Arnold dumbbell press the same way you do the seated dumbbell press, except begin and end each rep with the dumbbells below your chin and in front of your chest with your palms facing you. The handles of the dumbbells should be at about the height of your collarbone. As you push the dumbbells upward, rotate your wrists outward so that you finish the rep in the same position as the dumbbell shoulder press, and then reverse the motion and lower the dumbbells back to the starting position.

The *Bigger Leaner Stronger* Accessory Pushing Exercises

Triceps Pushdown



You can use several different handles for this exercise, but I recommend the rope handle because it tends to be the most comfortable. That said, you can also use a straight metal handle, EZ bar, or V bar.

Move the cable pulley to its highest position, attach the handle, and stand in front of it with your feet about shoulder-width apart. Grab the handle with both hands, take a step back so that your hands are supporting the weight, and lean slightly forward while pulling your elbows back toward your torso, so that your shoulders are over your hands. You can also put one foot behind the other if that's more comfortable for you. Push the handle straight down until your arms are

straight, keeping your elbows at your sides, and then raise your hands and return to the starting position.

Seated Triceps Press



Sit on an upright bench and hold one end of the dumbbell with your palms. Look at a spot on the floor 6-to-10 feet in front of you, press your back into the bench, and lower the dumbbell behind your head as far as your flexibility allows, ideally so your wrists are just below your elbows. Once you've reached the proper depth, push the dumbbell toward the ceiling and return to the starting position. Try to minimize upper-arm movement during the exercise (keep your elbows on either side of your forehead), as shifting your arms increases the chances of knocking your noggin with the dumbbell and transfers the weight from your triceps to your shoulders.

Lying Triceps Extension ("Skullcrusher")



Load an EZ bar with weights and place it in front of a bench. Sit on the edge of the bench, lean forward, and grab the EZ bar with both hands about 6-to-10 inches apart and your palms facing away from you. Hoist the EZ bar onto your thighs and then simultaneously lean backward and use your thighs to push the EZ bar over your chest, just like you would when preparing for the dumbbell bench press.

Plant your feet firmly on the floor about shoulder-width apart, pull your shoulder blades down and back, and arch your back. Then, extend your arms to push the EZ bar straight up in the air over your chest. Take a breath of air, and lower your hands toward your face until the bar is directly over your forehead (hence the name, "skullcrusher"). Then return to the starting position.

If this exercise irritates your elbows, modify it by lowering the bar over the top of your head until it almost touches the bench at the bottom of each rep. This takes some of the stress off of your elbows and increases the range of motion of the exercise (which is generally good for muscle growth).

Dumbbell Side Lateral Raise



Holding a dumbbell in each hand, stand with your feet about shoulder-width apart. Take a breath of air, lift your chest slightly, and then extend your arms out in an arc toward the ceiling with your palms facing the ground. Keep raising the dumbbells until your arms are roughly parallel to the floor, then lower them to return to the starting position.

The *Bigger Leaner Stronger* Primary Pulling Exercises

Barbell Deadlift



Load the bar with plates. If you aren't strong enough (yet!) to use at least one 45-pound plate on either side of the bar, create a platform to ensure that the bar is about 8-to-10 inches off the floor. To do this, create two stacks of plates underneath the loaded ends of the barbell. Or, if your gym has lighter-weight plates that are the same diameter as 45-pound plates, use those. Position your feet so they're slightly narrower than shoulder width apart with your toes pointed slightly out, and move the bar over the middle of your feet so that it's under or slightly behind your shoulders when you bend over and grab it. If you're taller or skinnier, this will probably place the bar against your shins. If you're shorter or thicker, it'll place it somewhere around the middle of your feet.

To get into the starting position, push your hips backward and bend at the waist and knees until you can touch the bar. Grip the bar just outside your shins with both palms facing down and press your upper arms into your sides as if you were trying to crush oranges in your armpits. Your arms should be straight and locked, with enough room on the sides for your thumbs to clear your thighs as you ascend and descend. Make sure your head is in a neutral position. Don't look up at the ceiling or down at the ground. Take a deep breath into your stomach, brace your abs, squeeze the bar as hard as you can, and straighten your back by pushing your hips slightly upward. The bar shouldn't rise off the floor at this point, but you should feel tension in your lower back, hips, and hamstrings. Some weightlifters refer to this as "taking out the slack" because it minimizes unnecessary movement after you begin the pull.

To initiate the pull, squeeze the bar as hard as you can (try to crush it), explode upward by pushing through your heels to "break" the bar off the floor (don't start slowly—"attack" the bar), keeping your spine straight, lower back slightly arched, and core tight. If you set up correctly, your hips and shoulders should rise at the same rate. The bar should move up your shins, and once the bar passes your knees, push your hips into it as you continue to stand up. Try to keep the bar on as vertically straight of a path as possible because any deviations are just going to slow you down and make it harder to maintain good form. The bar shouldn't move noticeably toward or away from you. When you're fully upright, your chest should be out and shoulders down, and you shouldn't lean back, hyperextend your lower back, or shrug the weight up, as shown in the second picture above.

To lower the bar to the floor, push your hips backward and, keeping your lower back slightly arched and your core tight, let the bar slide straight down your thighs until it clears your knees. Then, maintaining your grip on the bar, allow it

to drop to the floor, and adjust your body as needed to get into the proper starting position for the next rep. Don't try to lower the bar slowly or quietly. The entire descent should take one-to-two seconds or less.

"Is the deadlift bad for my back?"

Few people argue that the deadlift isn't an effective strength and muscle builder, but some claim it's bad for your joints, especially your lower back. This seems plausible at first glance because it places a lot of stress on the lower-back, but is that actually bad? Let's see what the scientific literature has to say.

Scientists at the University of Valencia investigated effective ways to train the spinal erector muscles (aka the *erector spinae*), which run down both sides of your spine and play a major role in preventing back injuries. Researchers split 25 people into two exercise groups:

1. Group one performed bodyweight exercises like lumbar extensions, forward flexions, single-leg deadlifts, and bridges.
2. Group two performed two weighted exercises, barbell deadlifts and lunges, using 70 percent of their one-rep max.

Muscle activity was measured using *electromyography*—a technique of measuring and analyzing electrical activity that occurs in the muscles when they contract. The deadlift most activated the spinal erector muscles, winning by a long shot. The researchers concluded, then, that the deadlift is an effective way to strengthen these muscles, which can reduce the risk of lower back injury.

Another insightful study on the matter was conducted by scientists at the University of Waterloo to determine how much strain the deadlift puts on the entire back and how likely it was to produce injury. The researchers recruited four competitive powerlifters and had them do two exercises:

1. Lean as far forward as possible, bending over from the waist and then returning to an upright posture (fully flexing and extending their backs). This allowed the researchers to measure the limits of the powerlifters' natural range of motion.
2. Deadlift a weight close to their one-rep max (400-to-460 pounds).

The researchers used real-time X-ray imaging (called *videofluoroscopy*) to watch the lifters' spines while they completed both tasks. Many deadlifting injuries occur because of too much lower-back bending, which can pull vertebrae out of position, pinch and compress intervertebral cartilage discs, and strain lower-back ligaments. So the scientists measured each of those things—how much the participants' vertebrae moved, how much their spinal discs pinched and compressed, and how much their lower-back ligaments stretched.

The examination found that the deadlift was completed within the weightlifters' natural range of motion, and that there were no signs of excessive shifting or pinching of the intervertebral discs or stretching of the lower-back ligaments. Thus, the researchers concluded that the deadlift is a fantastic exercise for strengthening your entire back, including your lower-back, and doesn't force an unnatural range of motion or put excess strain on your spine or ligaments.

So, why does the deadlift have a reputation as a one-way trip to Snap City? Three reasons:

1. Bad form. For instance, how many people have you seen who pulled with a rounded lower back? This is a major no-no, as it shifts much of the stress away from the powerful spinal erector muscles to the vertebrae, intervertebral discs, and ligaments.
2. Using too much weight. Like the bench press, the deadlift is an "ego exercise" for many people, so instead of gradually advancing on it over

weeks, months, and years, they try to force the issue and add weight too quickly. This makes it harder to maintain proper form and can put too much stress on your tendons, ligaments, and other connective tissues (also true of all exercises).

3. Physical therapists, chiropractors, and sports medicine doctors have conditioned many people to believe that the back is inherently brittle and that popping a spinal disc is as easy as cracking a knuckle. Thus, when people experience even mild discomfort when deadlifting, they assume the worst. While this philosophy has been a boon to back surgeons, research shows that the spine and its supporting structures are remarkably resilient, that debilitating back injuries are rare, and that many cases of back pain may be rooted in a psychological rather than a physical condition. For instance, many people with back pain have no evidence of spinal injury in an MRI scan, and others with detectable damage have no pain. That said, if you experience pain while deadlifting (in your lower back or elsewhere), you should avoid the exercise until you can do it without pain.

"Can I deadlift with a back injury?"

It depends on the nature and severity of your situation. As a general rule, if deadlifting doesn't aggravate the injury, then it's okay (and maybe even beneficial) to do. If it *does*, however, then you should avoid deadlifting until you can pull without pain. That said, if you're recovering from a serious trauma like a back surgery, a torn ligament, or a broken vertebrae, talk to your doctor before deadlifting to know if it's safe.

"Should I deadlift if I've hurt my back doing it in the past?"

If I had a nickel for every time someone has told me they can't deadlift because

they've "tweaked their back" before, I'd be making money in a very odd way. Just about everyone who has taken strength training seriously has offended their lower back deadlifting at least once or twice, but fortunately, this doesn't mean they "aren't made for the exercise" or need to swear it off. While traumatic injuries can preclude deadlifting (at least temporarily), most people haven't suffered such misfortunes. So, as long as the exercise isn't currently causing you pain, past troubles will likely have little impact on future performance.

"Should I breathe in and out throughout each rep?"

Absolutely not. Many personal trainers will tell you to breathe in while descending and out while ascending to ensure you get enough oxygen. This isn't only physiologically wrong, it's basically impossible to do as you get stronger.

Instead, use the *Valsalva maneuver* to control your breathing, which involves forcefully breathing out against a closed windpipe. This traps air in your lungs and creates pressure inside your abdomen, known as *intra-abdominal pressure*, which stabilizes your torso, allowing you to lift more weight and probably reduce the risk of injury as well. Here's how to do it:

1. Before you start your first rep in a set, take a deep breath into your stomach of about 80 percent of your maximum lung capacity. Your belly should feel "full" but not so much that you have trouble keeping your mouth closed when the rep gets hard.
2. Press your tongue against the roof of your mouth, and without letting any air escape, try to breathe out through your mouth. You should feel your abdomen, back, and jaw tighten.
3. Pull to a standing position.
4. Start lowering the bar, and once it passes your knees, breathe out as you let it fall to the floor.

5. Repeat for each rep.

An important caveat: The Valsalva maneuver increases your blood pressure more than continuous breathing. Studies show that this isn't inherently dangerous, and holding your breath is instinctive when lifting heavy weights, but if you're hypertensive or have a preexisting heart condition, talk with your doctor before using the Valsalva maneuver. Furthermore, if you use the technique and experience chest pain, dizziness, or other side effects, stop using it and talk with your doctor.

"What type of grip should I use?"

When you're new to deadlifting, a double overhand grip (both palms facing down) works well, but as the weights get heavier, the bar begins to slip out of your hands in the middle of your sets. A common workaround is the "mixed grip," which involves alternating one of your hands so it's palm-up. This allows you to maintain a strong grip on the bar with heavy weights, but it also has downsides:

1. It causes you to turn toward your palm-down hand, and this makes it harder to keep the bar close to your body.
2. It places additional strain on the biceps of your palm-up arm.

I don't know of any scientific data on how the mixed grip affects the safety of the exercise, but I'll say this: While biceps tears are rare, when they do happen, it's often the palm-up biceps during a heavy mixed-grip deadlift. You can make the mixed grip safer by alternating your palm-up hand every workout or even set, but I'd rather you just use a double-overhand grip with lifting straps instead.

Now, I know, many people snub straps as a form of "cheating," but this is silly. When used properly, straps allow you to safely pull maximum weight

without the limitations of the mixed grip. Plus, you don't need to use straps on every set of deadlifts—only the ones that you can't complete without them. For most people, that's their second or third hard set and beyond. Straps can help with barbell and dumbbell rows and lat pulldowns, too, if they're needed (the less you use straps, the faster your grip strength will improve).

The final type of deadlift grip you can use is the hook grip, which is a variation of the double-overhand grip where you wrap your fingers over your thumbs instead of vice versa. This grip is virtually as secure as straps and also avoids the mixed grip's snags, but it's very uncomfortable at first. If you stick with it, however, the pain subsides, and it's likely better for training your grip than using straps.

"Can I bounce the weight off the floor with each rep?"

The "tap-and-go transition" of maintaining tension as you tap the weights to the floor and immediately begin the next rep is fine for warming up, but I recommend the "stop-and-go" method for your hard sets. With this technique, you fully release the weight to the ground and reset your bottom position—including your breath—before starting the next rep. This is tougher than tap-and-go, but that's good, and it's safer as well.

Barbell Romanian Deadlift



Adjust the pins or hooks in a squat rack so the bar is at your mid-thigh, and load it with plates. If you don't have access to a squat rack, load the bar on the floor instead. Grip the bar just like you would during the conventional deadlift, lift it off the rack (or floor), take one step back with each foot, and turn your toes out and bend your knees slightly.

Stand up tall with your chest out, take a deep breath into your stomach, and brace your abs. Squeeze the bar as hard as you can, press your upper arms into your sides and, with a flat back, lower the bar toward the floor in a straight line, allowing your butt to move backward as the bar descends. Keep lowering the bar until you feel a stretch in your hamstrings and your lower back starts to round—just below the knees for most people, and about mid-shin for those who are particularly flexible. Bring the bar upward by driving your hips toward it and return to the starting position.

Barbell Row



Set up the same way you would for the barbell deadlift, with a few differences:

You'll probably want to position your feet slightly wider than you would for the deadlift—6-to-12 inches apart works well for most people.

You'll want to grip the bar slightly wider, too, with your hands about 3-to-6 inches wider than you would have them in the deadlift.

You start with your hips high and legs straight enough to allow your back to be more or less parallel to the floor.

Once you're in the starting position, start the pull by extending your legs (raising your hips) to "pop" the weight off the floor, and then start pulling your elbows toward the ceiling to keep it moving upward. Ensure your shoulders rise with your hips and your back maintains its flat, neutral position as you lift the bar (no "whipping" your torso backward to lift more weight or get more reps).

Once you've got the weight moving, keep pulling until the barbell makes contact with the bottom of your rib cage. The bar should move up your shins, and once it reaches knee height, your legs should be straight enough that the bar can easily pass over your knees. Try to move the bar on as vertically straight of a path as possible because any deviations are just going to slow you down and make it

harder to maintain good form. Reverse the motion to return to the starting position, then begin your next rep.

One-Arm Dumbbell Row



Hold a dumbbell in your right hand. Plant your left knee and arm firmly on a bench, your right foot on the floor a foot or two from the bench, and let your right arm (the one holding the dumbbell) hang straight down toward the floor. Keeping your back straight, pull the dumbbell upward until it touches your torso, and then return the dumbbell to the starting position. Once you've completed the desired number of reps, repeat the process with your left arm.

Pull-Up



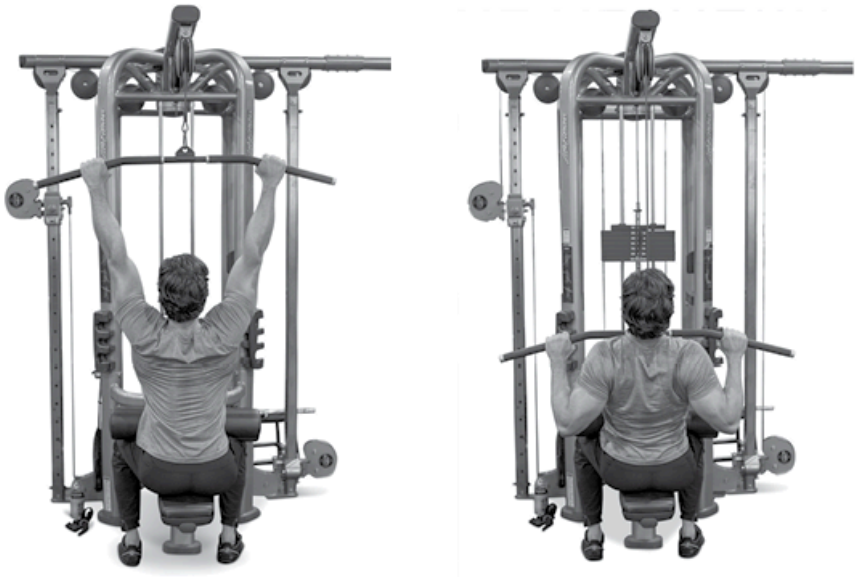
Grab a pull-up bar with your palms facing away from you and your hands just outside of shoulder-width apart. Lift up your feet so that you're hanging with your arms straight. Flex your glutes and hamstrings to keep your lower body from swinging (a good cue to remember is "tighter is lighter"), pull your body upward until your chin rises above your hands, and then lower yourself to the starting position. To add weight to the pull-up, use a dip belt or pinch a dumbbell between your thighs.

Chin-Up



Perform the chin-up just like the pull-up, except begin with your hands facing toward you and about shoulder-width apart.

Lat Pulldown



Adjust the thigh pad so that it locks your legs in place and prevents your butt from rising off the seat when pulling the bar down. Stand in front of the seat, grab the bar with your palms facing away from you and a few inches wider than shoulder-width apart, and sit down, wedging your legs underneath the thigh pad. Relax your arms and shoulders, letting the weight pull them straight. Pull the bar down until it's a few inches from your collarbone (usually an inch or two below your chin), and then return to the starting position.

This is referred to as the pronated, medium-grip lat pulldown, because it involves gripping the bar with your palms facing away from you (a pronated grip) and a little wider than shoulder-width apart (a medium grip). You'll often see people doing wide-grip (hands about twice as wide as shoulders) and supinated (palms facing you) lat pulldowns, but I generally don't recommend these variations for a few reasons.

First, recent research shows that a medium grip better activates both the lats and the biceps, whereas a wide grip primarily activates the lats while reducing biceps activation. Basically, the medium-grip lat pulldown offers a bit more return on investment with no downsides and tends to be more comfortable for most people. Research also shows that a pronated grip is more effective for training the lats than a supinated grip. Although you could argue that a supinated grip activates the biceps a bit more, you'll be doing plenty of biceps training on this program, making this a moot point.

There is one variation of the lat pulldown worth doing, though: The close-grip lat pulldown, which uses a close-grip handle (which has your palms facing each other). It's not better than the regular (pronated-grip) lat pulldown, but it does train your back muscles in a slightly different way. To do this variation, simply attach the close-grip handle to the cable and then follow the same steps you would with the regular lat pulldown.

Seated Cable Row (Wide- and Close-Grip)

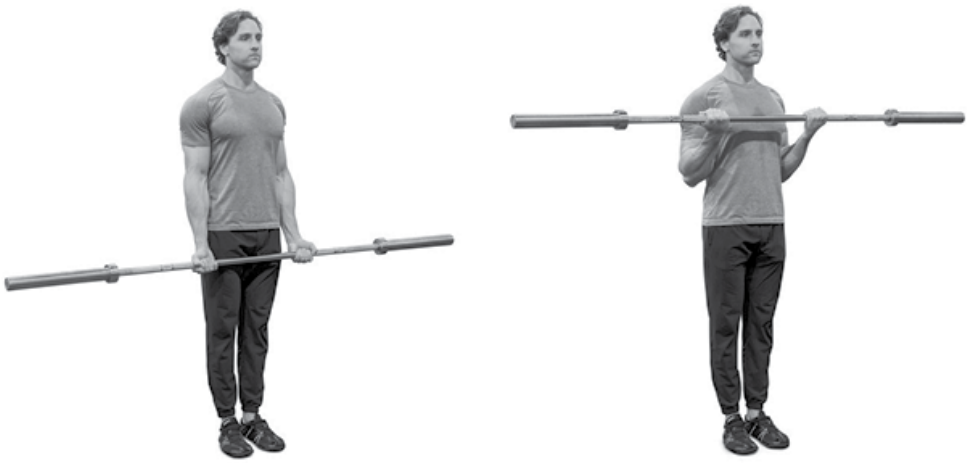


Most gyms have two handles for this exercise: A narrow-grip and a wide-grip. Both are useful, and I like to alternate between them every couple of months.

Attach the handle to the cable row machine, sit on the pad, and place your feet against the footrests. Lean forward, grab the handle, and lean backward slightly so that your back is perpendicular to the floor and your arms are straight out in front of you. Keeping your back straight, pull the handle to your abdomen, allowing your torso to move backward just enough to help you finish the rep, but no more. Once the handle touches your torso, straighten your arms, and return to the starting position.

The *Bigger Leaner Stronger* Accessory Pulling Exercises

Barbell Curl



Place a barbell on the floor or in the hooks of a bench (or even in a squat rack!), and load it with plates. Grab the bar with your palms facing away from you a little wider than shoulder-width apart. Stand up straight, take a deep breath of air, and brace your core. Then, without leaning backward, curl the bar up toward your chest until your forearms are roughly perpendicular to the floor. Allow your elbows to move forward slightly as the bar rises, so that the bar is right in front of your mouth at the end of the rep. Then, lower the bar to the starting position.

Alternating Dumbbell Curl



Hold one dumbbell in each hand and stand up straight with your feet about shoulder-width apart. Curl one hand toward your shoulder with your palm facing upward until your forearm is roughly perpendicular to the floor. Allow your elbow to move forward slightly as the dumbbell rises. Then, lower the dumbbell, return to the starting position, and repeat the exercise with your other hand.

Dumbbell Hammer Curl



Follow the same steps as the alternating dumbbell curl, but instead of holding the dumbbell so your palm faces upward, turn the dumbbell vertically (so it looks like a hammer from the side). You'll probably notice that you can't quite get your forearm perpendicular to the floor with this exercise, which is fine.

EZ-Bar Preacher Curl



Perform the EZ-bar preacher curl the same way as the barbell curl, except use an EZ bar (the squiggly looking one) and rest your upper arms against an angled pad known as a preacher curl bench.

Barbell Rear Delt Row



Load a barbell on the floor in front of you, bend over at the waist, and grab the bar with an overhand (palms facing you) grip a little wider than shoulder-width apart. Position your feet about shoulder-width apart, flatten your back so that it's parallel to the floor, and let your arms hang straight down. Pull the bar straight up until it touches your torso, keeping your spine straight and back parallel to the floor. Then, lower the bar to the starting position.

Dumbbell Rear Lateral Raise (Standing or Seated)



Holding a dumbbell in each hand, stand with your feet about shoulder-width apart. Keeping your back straight, lean forward, hinging at your hips and allowing your knees to bend slightly, just as you would when doing a deadlift. Take a breath, then extend your arms out in an arc toward the ceiling with your palms facing the ground. Keep raising the dumbbells until your arms are roughly parallel to the floor, then lower them and return to the starting position.

This is what's known as the standing dumbbell rear lateral raise, but there's also a seated version. In this case, you sit on the edge of a bench with your feet and knees together. Then, you lean forward, grab a dumbbell in each hand, and follow the same steps as the standing dumbbell rear lateral raise. Which version you use comes down to personal preference, but I like to alternate between the

two roughly every eight weeks or so. You can do the same or just stick with whichever is more comfortable.

Machine Reverse Fly



Adjust the handles and seat so the handles are in line with your shoulders and your hands are as close together as possible. Grip the handles, push your torso into the chest pad, and push your hands away from each other. Once your arms are roughly parallel with your torso, reverse the motion and return to the starting position.

Leg Curl (Seated or Lying)



Sit on the machine with the back of your knees a couple of inches off the pad supporting your thighs and adjust the heel pad so it's across your Achilles tendons. Press your heels toward your butt until your shins are roughly perpendicular to your thighs (or closer), and then raise your heels and return to the starting position.

This is the seated leg curl, but there's also a lying version. In this case, you perform the exercise the same way, except lying on your stomach instead of sitting on your keister. Both exercises are acceptable, but some research shows that seated leg curls are more effective for building your hamstring muscles than lying leg curls. Thus, if you have a choice, I recommend you opt for the seated version. That said, it's fine to use the lying version if your gym doesn't have a seated leg curl machine or if you've been doing the seated version for a while and want to mix up your training.

The *Bigger Leaner Stronger* Primary Squatting Exercises

Barbell Squat



Adjust the hooks in a power rack or squat rack so the bar is at the height of your midchest. This might feel a bit low, but it's better to be on the low side than having to tippy-toe the weight off the rack. If your squat rack has safety arms or bars, set them to about the height of your midthigh. Load the bar with your desired amount of weight (or not, if the bar alone is enough).

Face the bar so you can walk it out backward. Don't ever walk the bar out forward, as trying to rerack it by walking backward is dangerous. Grip the bar with your palms facing forward, thumbs on top of the bar, and your hands about 4-to-8 inches wider than shoulder-width apart. Keeping your palms on the bar, place your feet underneath it a little wider than shoulder-width apart. Then, push your torso under and up against the bottom of the bar, sliding your back up until the bar rests against the back of your shoulders. Pull your shoulder blades

together and down to create a little "shelf" out of the bony protrusions of your shoulder blades, and then maneuver your torso so the bar nestles into this shelf. You may need to narrow or widen your grip to find the correct position. The weight of the bar should be resting almost entirely on your back muscles, not in your hands or on your spine.

Unrack the bar by standing up and taking one step back with each foot (one at a time). Adjust your feet so they're a little wider than shoulder-width apart and point your toes out about 20-to-25 degrees (around one and eleven o'clock). The weight should feel evenly balanced over the middle of your feet, not primarily over your toes or your heels.

Take a deep breath into your stomach, push your chest out, brace your abs, and sit down by pushing your hips backward and bending your knees at the same time. Keep sitting down until your thighs are parallel to the floor or slightly lower, then stand up. Start the ascent by driving your feet into the floor, ensuring your shoulders move upward at the same rate as your hips—a mirror image of what you did during the descent. About halfway up, push your hips forward and underneath the bar to return to the starting position.

When you're on the last rep of a set, finish the rep (legs straight and bar motionless) and then move the bar back to the hooks. Don't try to squat the bar directly back into the hooks because if you miss, the bar might fall off your back.

"Is the barbell squat bad for my knees?"

In some circles, the barbell squat has a reputation for ravaging the knees. The more you barbell squat, so it goes, the more you'll damage the tendons, ligaments, and cartilage in your knees until finally your knees just won't "knee" anymore.

While some people's anatomies don't play well with the barbell squat, studies show that when performed correctly, it's safe for most people and may even

prevent joint pain. In a study published in the *Journal of Biomechanics*, scientists analyzed the forces placed on the knee joints of twelve experienced male powerlifters who squatted between 375 and 650 pounds. The researchers found that during heavy squatting—up to 2.5 times body weight—knee tendons and ligaments were only pushed to about 50 percent of their maximum strength and never came close to their breaking points.

Some people do experience knee pain from barbell squatting, however, and this is almost always for the same reason their shoulders hurt when they bench press or their lower back hurts when they deadlift—they're not squatting correctly. Maybe they round their upper or lower back, maybe they let their knees collapse inward, maybe they use too much weight and compensate by doing half-reps, and so forth. Another way to nettle your knees with squatting is to just do too much of it too often. That's why the *Bigger Leaner Stronger* program doesn't rely solely on the barbell squat to train the lower body and instead uses a combination of squats and other effective exercises.

"Is it bad if my knees move over my toes?"

No. You've probably heard this is bad for your knees, but research shows otherwise—the forces placed on the knees when they extend to and even beyond the toes in a squat are well within safe limits. This is what you'd expect, too, because most people find that their knees naturally move over their toes at the bottom of the squat, and sometimes slightly farther if they have long femurs.

"Should I use the high-bar or low-bar position?"

There are two ways to perform the barbell squat:

1. High-bar
2. Low-bar

The high-bar squat has the bar resting directly on your upper traps, whereas the low-bar squat has the bar resting between the upper traps and rear deltoids (about 2-to-3 inches lower down your back).

Your torso remains more upright in the high-bar squat, and some people claim that this is better for training your quads versus the low-bar squat, which is better for training your glutes and lower-back. Research published in the *Journal of Human Kinetics* shows otherwise, though—both exercises activate the same lower-body muscles in almost identical ways.

Therefore, if you're new to the barbell squat, start with the low-bar position because you'll probably find that you're stronger in it, but if it causes significant shoulder and wrist discomfort, switch to the high-bar position. If the low-bar position feels fine, you can also switch to the high-bar squat from time to time if you'd like.

"Should I breathe in and out throughout each rep?"

No. Use the Valsalva maneuver just as you would during the deadlift (unless it's contraindicated for the reasons given earlier).

"How do I keep my knees from collapsing inward?"

Think about spreading the floor with your feet as you descend and ascend, and make sure you take a deep breath of air into your stomach before every rep (many technique errors stem from a lack of torso stability).

"What should I do if I can't get my thighs parallel to the ground?"

This is often from inadequate mobility or coordination (or a bit of both), and it typically resolves itself with a few weeks of practice. So, try to get a little lower each time you squat, and you should get there fairly quickly. If this doesn't fix the issue, though, do the following mobility exercise several times per day until you

can squat correctly: With no equipment, squat down to the bottom position, and place your elbows against your knees and touch the palms of your hands together. Use your elbows to press your knees out and into the proper position (in line with your toes), and hold this position for 20-to-30 seconds.

"Can I use a Smith machine instead of a barbell?"

I wouldn't forbid this, but I also wouldn't advise it. The Smith machine forces an unnatural and often uncomfortable movement pattern for the squat, and research shows it's less effective than the free weight barbell squat for gaining muscle and strength.

Barbell Front Squat



Position the bar at the same height you'd use for the back squat. Grip the bar with your palms facing the floor about shoulder-width apart. Then, keeping your grip on the bar, squat down and press the top of your shoulders against the bottom of the bar and push your elbows up and out in front of the bar. For most people, this will put the bar right at the base of the throat (it should feel slightly

uncomfortable—like a tight-fitting button-up shirt).

Stand up, take one step back with both feet, and position your feet about shoulder-width apart with your toes pointed outward at about 45-degrees (most people prefer to front squat with a slightly narrower stance and their toes pointed more outward than they do with the back squat). Take a deep breath into your stomach, push your elbows up, brace your abs, and sit down by pushing your hips backward and bending your knees at the same time. Keep sitting down until your thighs are parallel to the floor or slightly lower, then stand up to return to the starting position.

When you're on the last rep of a set, finish the rep (legs straight and bar motionless) and then move the bar back to the hooks. Don't try to squat the bar directly back into the hooks because if you miss, the bar might fall off your shoulders.

"What kind of grip should I use?"

The two most common methods for gripping the bar in the front squat are wrapping all four fingers of each hand around the bar (full grip) or wrapping one, two, or three fingers around the bar (partial grip). Use whichever you prefer. If you have the wrist mobility to make the full grip work, go for it, but the partial grip works just as well. Personally, I like a two-finger grip because it's enough to keep the bar in place and is more comfortable than a full-grip.

"How do I keep the bar from slipping off my shoulders?"

Make sure the bar is resting against (but not crushing) your throat, and keep your elbows as high as possible during the entirety of each rep.

"What should I do if the front squat hurts my wrists or shoulders?"

Rub some dirt on it? Jokes aside, everyone finds their first date with the front

squat at least a little unpleasant, but this tends to disappear within a few workouts.

Leg Press



Load a leg press with plates and adjust the seat to its lowest position (with the backrest closest to the floor, at about a 30-degree angle). Then wedge your butt down into the base of the seat and plant your feet on the footplate just outside of shoulder-width apart.

Bend your knees slightly, use the safety handles to release the weight, and lower the footplate toward your chest until your thighs are 12-to-16 inches from your torso. Make sure your butt remains firmly in place and your lower back doesn't round at the bottom of the rep. Finally, push the footplate upward until your legs are almost but not completely straight (knees slightly bent at the top of the rep).

The *Bigger Leaner Stronger* Accessory Squatting Exercises

Dumbbell Lunge (Walking or In-Place)



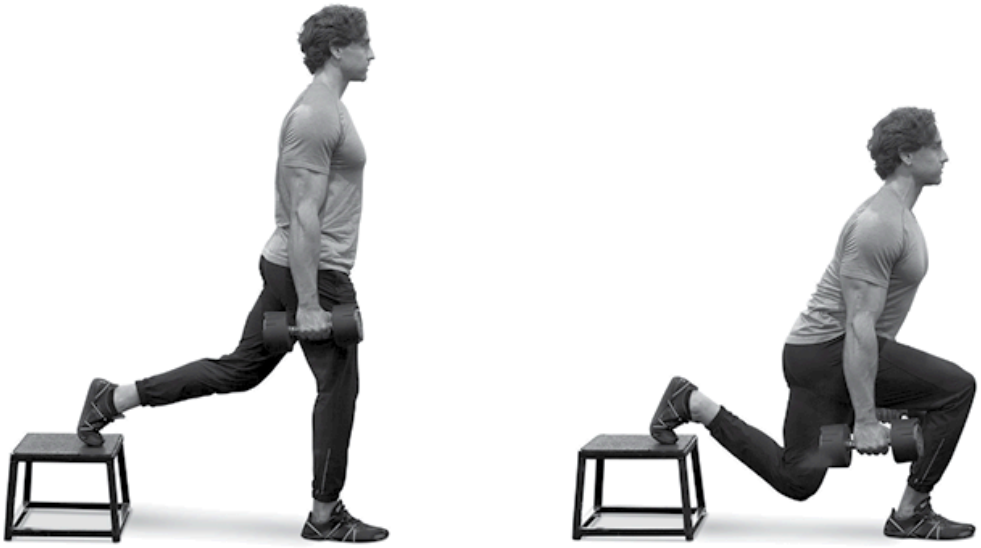
Stand up straight with both feet about shoulder-width apart. Take a long step forward with your right foot—about two-to-three feet—and, with most of your weight on your front foot, lower your body until your left knee touches the floor. Then reverse the motion by pushing off the floor with your front foot and leaning slightly backward, allowing your legs to straighten. Once you're standing, bring your right foot back to the starting position, and then repeat the exercise with your other foot.

This is known as the *in-place* lunge, and it's ideal for when you have limited space. There's also the *walking* lunge, which works much the same way except

instead of pushing off of your right foot and leaning back to return to the starting position, you lean forward, pushing off the floor with your left foot and bringing it forward to rest next to your right foot. Then, you repeat the process with your left foot, continuing to walk forward in a series of steps. Make sure you have plenty of space, as you'll be moving two-to-three feet forward with every stride.

Both exercises are equally effective, so choose whichever version you prefer. Try to stick with one version for at least a whole phase of training before switching to the other, though.

Dumbbell Bulgarian Split Squat



Hold a dumbbell in each hand, and stand two-to-three feet in front of a bench or other surface that's about knee height off the ground. Lean slightly forward, putting most of your weight on your right foot, and extend your left foot out behind you, resting the top of your foot on the bench. Keeping your right foot firmly planted on the floor, lower your butt toward the floor by bending both knees at the same time. Keep descending until your left knee touches the floor or

your right thigh is parallel with the floor (whichever occurs first), and then stand up, returning to the starting position.

You now have all the know-how needed to do your first *Bigger Leaner Stronger* workouts! We've discussed the truth about some of the most common exercise lies and blunders, the anatomy of effective strength training, and the best exercises for building a powerful physique. All that's required now is a plan for integrating and mobilizing all of the components, and that's exactly what I'll give you in the next chapter.

KEY TAKEAWAYS

- As great as primary exercises are, they don't adequately train all of the major muscle groups that we want to develop, especially the more stubborn ones like the shoulders, biceps, and lats.
- Accessory exercises will further stimulate these muscle groups, which will enhance their growth, boost performance on primary exercises, and help prevent muscle imbalances that can slow your progress.
- While there are no *direct* ab exercises in the program, like crunches, planks, and situps, many of the compound exercises in it heavily train your abdominals, obliques, and other core muscles.
- If you'd like to include some ab exercises in your workouts to speed up your core development (or if you just like doing them), simply add 2 or 3 sets of an ab/core exercise of your choice to the end of your legs and pull days.
- Many lower body exercises in the *Bigger Leaner Stronger* program train the calves indirectly, but if you want your calves to develop faster, add 3 sets of calf exercises to the end of your pull and lower body workouts (6 sets per week).

The Big Ideas

1. Choose the right workout routine for you.
2. Use double progression to get bigger and stronger.
3. Do some cardio to optimize your health and fitness.

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The Bigger Leaner Stronger Workout Program

We who cut mere stones must always be envisioning cathedrals.

—QUARRY WORKER'S CREED

In chapter 12, you learned the *Bigger Leaner Stronger* strength training formula:

- Do 3-to-5 strength training workouts per week.
- Do 9-to-12 hard sets per major muscle group per week.
- Do 9-to-12 hard sets per workout.
- Use 75-to-85 percent of one-rep max.
- End most hard sets 1-to-3 reps shy of muscular failure.
- Rest 2-to-4 minutes in between hard sets.
- Take it easy every 8 weeks.

In this chapter, we'll turn that blueprint into three-day, four-day, and five-day (per week) strength training programs for gaining muscle, losing fat, and getting strong. Then we'll discuss cardiovascular exercise, which isn't an essential part of the *Bigger Leaner Stronger* program, but it's certainly valuable enough to

include in your regimen.

You don't have to do the workouts I'll share with you here if you'd rather create your own routine, but if you've never trained this way before, follow mine for at least a few months before going it alone. Workout programming can be tricky. There are several aspects that must come together—phases, routines, workouts, progressions, deloads—and many factors to weigh, including goals, schedule, frequency, intensity, volume, and recovery, among others. Also, if you follow my directions for at least one phase of training, you'll better understand what is and isn't likely to work in actual practice and be able to create better plans going forward.

To make it easy for you to get started with my workouts, you'll find spreadsheets and printable templates in the free bonus material available at www.biggerleanerstronger.com/bonus along with an entire year of programming if you want to continue with my recommendations. You can also pick up a copy of *The Year One Challenge for Men* (www.biggerleanerstronger.com/challenge) if you'd prefer the workouts in the form of a digital or hardcopy book.

So, let's explore exactly how to do *Bigger Leaner Stronger* training, starting with the first of the three layers I just mentioned: the training phase.

The *Bigger Leaner Stronger* Training Phase

A training phase is a period that lasts anywhere from several weeks to months, designed to achieve a specific goal, like more strength, muscle, endurance, or recovery. In *Bigger Leaner Stronger*, each training phase will last nine weeks, will focus on increasing strength and muscularity, and will consist of two parts:

1. Hard training. Each training phase begins with eight weeks of challenging workouts to gain muscle and strength.
2. Deloading. Each training phase ends with one week of deloading to facilitate recovery from the hard training.

Each year on the program, then, can be divided roughly into six training phases (54 weeks).

The *Bigger Leaner Stronger* Workout Routines

A workout routine outlines what you'll do in each training phase to achieve its goals— specifically, how often you'll work out and what type of workouts you'll do every week. In *Bigger Leaner Stronger*, you can choose from a three-, four-, or five-day workout routine. Each are weekly (seven-day) routines, so the most strength training you can do is five workouts per week. Cardio is optional, and how much you do (if any) will depend on how much time you have to give to it and whether you're cutting, lean bulking, or maintaining.

As far as results go, the five-day routine is better than the four- and three-day routines, and the four-day routine is better than the three-day. That doesn't mean you can't do well with the four- or three-day routines, though, because you absolutely can. Pick whatever routine you *know* you can consistently do every week. For example, if you can sometimes train five days per week but usually only four, pick the four-day routine. Ideally, you'd choose one routine and stick with it for the entire phase, but if your schedule changes mid-phase, feel free to "upgrade" or "downgrade" as needed.

Depending on your goals, progress, and other factors, your workout routines can change or not as you move from one training phase to another. For instance, bodybuilders tend to change their routines more often and more significantly than strength athletes. If you plan on following my workouts in the bonus material, though, you'll see that the workout routines change slightly from phase to phase to expose your muscles to new and different types of movement patterns to more fully develop them. For example, horizontal and vertical pulling movements train your back muscles slightly differently, and you can alternate between emphasizing one or the other in different training phases to good effect.

Every one of my workout routines will also include at least three sets of

heavy squatting, deadlifting, and bench pressing per week, and these exercises will always come first in your workouts. No exercises will help you build a lean, muscular, and powerful body more than these, so you want to make sure you're doing them every week, and you want to do them first in your workouts because they require the most physical and mental energy.

The 5-Day *Bigger Leaner Stronger* Workout Routine

Workout 1	Workout 2	Workout 3	Workout 4	Workout 5
Push	Pull	Upper Body A	Legs	Upper Body B

If you have the time and inclination, make this your first workout routine because it'll produce the fastest results. If you don't like it for whatever reason (or just want to spend less time in the gym), you can always try the other routines in later phases.

Most people on this routine train Monday through Friday and take the weekends off, but you can incorporate your rest days however you'd like. The important thing is that you do each of the workouts every seven days and in the order given.

One caveat, however. Include at least one day of rest between the Upper Body B and Push workouts because doing them back-to-back won't give your body enough time to recuperate. Your larger upper-body muscles need more time to recover than smaller muscle groups that can survive daily beatings, like the abs or calves.

So, if you need to train on the weekends due to your schedule or lifestyle, you might train Monday (push), Tuesday (pull and calves), Wednesday (upper body), and rest Thursday, then train Friday (legs), Saturday (upper body), and rest Sunday.

The 4-Day *Bigger Leaner Stronger* Workout Routine

Workout 1	Workout 2	Workout 3	Workout 4
Push	Pull	Upper Body	Legs

The main difference between this and the five-day routine is two upper-body workouts are combined into one. Again, you can do these workouts on any days of the week that you like so long as you do each once per seven days and in the order given.

The 3-Day *Bigger Leaner Stronger* Workout Routine

Workout 1	Workout 2	Workout 3
Push	Pull	Legs

This is your time-proven "push-pull-legs" routine that allows you to squeeze a lot of gains from a minimalist setup. Do the workouts on whatever days suit you and put at least one day of rest in between your Pull and Leg workouts.

The *Bigger Leaner Stronger* Workouts

A workout is an individual training session geared toward a specific goal. In *Bigger Leaner Stronger*, each workout is designed to train a particular movement pattern or muscle group, such as your "push" or "upper body" muscles.

You'll notice the following workouts are for "Phase One" of the program. As there are six phases of training per year and each phase comprises eight weeks of hard training and one week of deloading, the workouts below are for your first eight weeks of hard training (and I'll explain how to deload later in this chapter).

When you've completed Phase One, you can repeat it, modify it based on what you're learning in this book, or follow my "Phase Two" (and Three and beyond) workouts available in the bonus material that comes with this book (www.biggerleanerstronger.com/bonus).

The 5-Day *Bigger Leaner Stronger* Workout Routine

Workout 1 Push	Workout 2 Pull	Workout 3 Upper Body A	Workout 4 Legs	Workout 5 Upper Body B
Barbell Bench Press 3 hard sets 4–6 reps	Barbell Deadlift 3 hard sets 4–6 reps	Seated Dumbbell Press 3 hard sets 4–6 reps	Barbell Squat 3 hard sets 4–6 reps	Close-Grip Bench Press 3 hard sets 4–6 reps
Incline Barbell Bench Press 3 hard sets 4–6 reps	One-Arm Dumbbell Row 3 hard sets 4–6 reps	Dumbbell Side Lateral Raise 3 hard sets 6–8 reps	Leg Curl 3 hard sets 6–8 reps	Chin-up 3 hard sets 4–6 reps

Dumbbell Bench Press 3 hard sets 6–8 reps	Lat Pulldown 3 hard sets 6–8 reps	Dumbbell Rear Lateral Raise (Seated) 3 hard sets 6–8 reps	Leg Press 3 hard sets 6–8 reps	Seated Cable Row (Close Grip) 3 hard sets 6–8 reps
Triceps Pushdown 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps	Seated Triceps Press 3 hard sets 6–8 reps	Dumbbell Lunge (In-Place) 3 hard sets 6–8 reps	Barbell Curl 3 hard sets 6–8 reps

The 4-Day *Bigger Leaner Stronger* Workout Routine

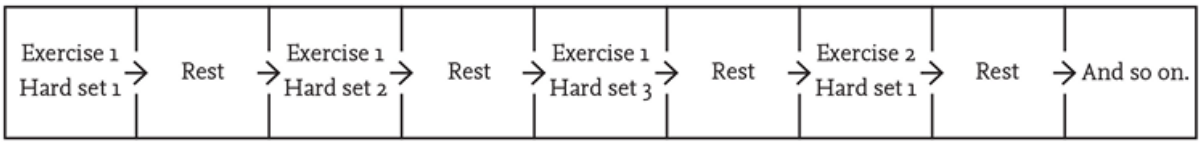
Workout 1 Push	**Workout 2 Pull**	**Workout 3 Upper Body**	**Workout 4 Lower Body**
Barbell Bench Press 3 hard sets 4–6 reps	Barbell Deadlift 3 hard sets 4–6 reps	Seated Dumbbell Press 3 hard sets 4–6 reps	Barbell Squat 3 hard sets 4–6 reps
Incline Barbell Bench Press 3 hard sets 4–6 reps	One-Arm Dumbbell Row 3 hard sets 4–6 reps	Seated Cable Row (Close Grip) 3 hard sets 6–8 reps	Leg Curl 3 hard sets 6–8 reps
Dumbbell Bench Press 3 hard sets 6–8 reps	Lat Pulldown 3 hard sets 6–8 reps	Close-Grip Bench Press 3 hard sets 6–8 reps	Leg Press 3 hard sets 6–8 reps
Triceps Pushdown 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps	Dumbbell Rear Lateral Raise (Seated) 3 hard sets 6–8 reps	Dumbbell Lunge (In-Place) 3 hard sets 6–8 reps

The 3-Day *Bigger Leaner Stronger* Workout Routine

Workout 1 Push	**Workout 2 Pull**	**Workout 3 Lower Body**

Barbell Bench Press 3 hard sets 4–6 reps	Barbell Deadlift 3 hard sets 4–6 reps	Barbell Squat 3 hard sets 4–6 reps
Incline Barbell Bench Press 3 hard sets 4–6 reps	One-Arm Dumbbell Row 3 hard sets 4–6 reps	Leg Curl 3 hard sets 6–8 reps
Dumbbell Bench Press 3 hard sets 6–8 reps	Lat Pulldown 3 hard sets 6–8 reps	Leg Press 3 hard sets 6–8 reps
Triceps Pushdown 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps	Dumbbell Lunge (In-Place) 3 hard sets 6–8 reps

Do the exercises in a workout one at a time and complete all the hard sets for one exercise before moving on to another, like this:



How to Make Exercise Substitutions

Most people can do all of the *Bigger Leaner Stronger* exercises without issue, but your mileage may vary. For example, some elbows are tormented by triceps pushdowns, some shoulders sniff at dips, and some lower-backs don't like Romanian deadlifts. If an exercise gives you grief in any way, swap it out for one that doesn't—just make sure it's an exercise of the same type (a primary exercise for a primary exercise, an accessory for an accessory) and for the same muscle group. This will make your workouts safer, more enjoyable, and more fruitful because you can progress much faster when you aren't in pain.

Here's a chart to help you make the right replacements:

Instead of the . . .	Do the . . .
Barbell Bench Press	Dumbbell Bench Press, Machine Chest Press
Incline Barbell Bench Press	Incline Dumbbell Bench Press, Incline Machine Chest Press
Close-Grip Bench Press	Dip, JM Press
Dumbbell Bench Press	Barbell Bench Press, Machine Chest Press
Incline Dumbbell Bench Press	Incline Barbell Bench Press, Incline Machine Chest Press
Dip	Close-Grip Bench Press, Bench Dip
Standing Barbell Overhead Press	Seated Dumbbell Press, Arnold Dumbbell Press
Seated Barbell Overhead Press	Seated Dumbbell Press, Arnold Dumbbell Press
Seated Dumbbell Press	Arnold Press, Standing or Seated Barbell Overhead Press

Arnold Dumbbell Press	Seated Dumbbell Press, Standing or Seated Barbell Overhead Press
Triceps Pushdown	Seated Triceps Press, Lying Triceps Extension, Triceps Kickback, Dumbbell Skullcrusher
Seated Triceps Press	Triceps Pushdown, Lying Triceps Extension, Triceps Kickback, Dumbbell Skullcrusher
Lying Triceps Extension (“Skullcrusher”)	Dumbbell Skullcrusher, Triceps Pushdown, Triceps Kickback, Seated Triceps Press
Dumbbell Side Lateral Raise	Cable Side Lateral Raise, Machine Side Lateral Raise
Barbell Deadlift	Dumbbell Deadlift, Trap-Bar Deadlift, Sumo Deadlift
Barbell Romanian Deadlift	Dumbbell Romanian Deadlift, Barbell Good Morning, Rack Pulls
Barbell Row	Seated Cable Row, Helms Row, Seal Row, Two- or One-Arm Dumbbell Row, Yates Row
One-Arm Dumbbell Row	Barbell Row, Helms Row, Seal Row, Seated Cable Row, Two-Arm Dumbbell Row, Yates Row
Pull-up	Chin-up, Lat Pulldown
Chin-up	Pull-up, Lat Pulldown
Lat Pulldown	Chin-up, Pull-up, Lat Pushdown
Seated Cable Row (Wide- and Close-Grip)	Barbell Row, Helms Row, Seal Row, One- or Two-Arm Dumbbell Row, Yates Row
Barbell Curl	Alternating Dumbbell Curl, Dumbbell Hammer Curl, EZ-Bar Preacher Curl, Dumbbell Preacher Curl, Cable Curl
Alternating Dumbbell Curl	Barbell Curl, Dumbbell Hammer Curl, EZ-Bar Preacher Curl, Dumbbell Preacher Curl, Cable Curl
Dumbbell Hammer Curl	Barbell Curl, Alternating Dumbbell Curl, EZ-Bar Preacher Curl, Dumbbell Preacher Curl, Cable Curl

EZ-Bar Preacher Curl	Barbell Curl, Alternating Dumbbell Curl, Dumbbell Hammer Curl, EZ-Bar Preacher Curl, Dumbbell Preacher Curl, Cable Curl
Barbell Rear Delt Row	Dumbbell Rear Delt Row, Cable Facepull
Dumbbell Rear Lateral Raise	Machine Reverse Fly, Cable Facepull, Barbell Rear Delt Row, Dumbbell Rear Delt Row
Machine Reverse Fly	Dumbbell Rear Lateral Raise, Cable Facepull, Barbell Rear Delt Row, Dumbbell Rear Delt Row
Leg Curl (Seated or Lying)	Barbell Romanian Deadlift, Barbell Good Morning, Nordic Curl, Single-Leg Dumbbell Deadlift
Barbell Squat	Barbell Front Squat, Safety Bar Squat, Leg Press, Single-Leg Split Squat (Barbell or Dumbbell), Pit Shark, Hack Squat, Zercher Squat, Dumbbell Goblet Squat
Leg Press	Barbell Back Squat, Barbell Front Squat, Single-Leg Split Squat (Barbell or Dumbbell), Safety Bar Squat, Pit Shark, Hack Squat, Zercher Squat, Dumbbell Goblet Squat
Dumbbell Lunge (Walking or In-Place)	Barbell Lunge (In-Place), Dumbbell Bulgarian Split Squat, Single-Leg Split Squat (Barbell or Dumbbell)
Dumbbell Bulgarian Split Squat	Dumbbell Lunge (Walking or In-Place), Barbell Lunge (In-Place), Single-Leg Split Squat (Barbell or Dumbbell)

Another reason for exercise substitutions is lacking access to the right equipment, and again, just find a similar exercise that fits the bill. For example, if your gym doesn't have EZ bars, you can use a straight bar or dumbbells instead, and if you only have access to dumbbells, you can still do the program with the following changes:

Instead of the . . .	Do the . . .
Barbell Bench Press	Dumbbell Bench Press, Dumbbell Fly

Incline Barbell Bench Press	Incline Dumbbell Bench Press
Close-Grip Bench Press	Close-Grip Dumbbell Bench Press
Dip	Bench Dip
Standing or Seated Barbell Overhead Press	Seated Dumbbell Press, Arnold Dumbbell Press
Triceps Pushdown	Seated Triceps Press
Lying Triceps Extension ("Skullcrusher")	Dumbbell Skull Crusher
Barbell Deadlift	Dumbbell Deadlift
Barbell Romanian Deadlift	Dumbbell Romanian Deadlift
Barbell Row	Two-Arm Dumbbell Row
Lat Pulldown	Chin-up, Negative Chip-up, Band-Assisted Chin-up
Seated Cable Row (Wide- and Close-Grip)	Two-Arm Dumbbell Row
Barbell Curl	Alternating Dumbbell Curl
EZ-Bar Preacher Curl	Dumbbell Preacher Curl
Barbell Rear Delt Row	Dumbbell Rear Delt Row
Machine Reverse Fly	Dumbbell Rear Lateral Raise
Leg Curl	Dumbbell Romanian Deadlift
Barbell Squat	Dumbbell Goblet Squat, Dumbbell Front Squat
Leg Press	Dumbbell Goblet Squat, Dumbbell Front Squat, Dumbbell Lunge

And here are a few dumbbell workout routines to get you started:

The 5-Day *Bigger Leaner Stronger* Dumbbell Workout Routine

Workout 1 Push	Workout 2 Pull	Workout 3 Upper Body A	Workout 4 Legs	Workout 5 Upper Body B
Dumbbell Bench Press 3 hard sets 4–6 reps	Dumbbell Deadlift 3 hard sets 4–6 reps	Seated Dumbbell Press 3 hard sets 4–6 reps	Dumbbell Goblet Squat 3 hard sets 4–6 reps	Close-Grip Dumbbell Bench Press 3 hard sets 4–6 reps
Incline Dumbbell Bench Press 3 hard sets 4–6 reps	One-Arm Dumbbell Row 3 hard sets 4–6 reps	Dumbbell Side Lateral Raise 3 hard sets 6–8 reps	Single Leg Split Squat (Dumbbell) 3 hard sets 4–6 reps	Chin-up 3 hard sets 4–6 reps
Dip 3 hard sets 6–8 reps	Chin-up 3 hard sets 4–6 reps	Dumbbell Rear Lateral Raise (Seated) 3 hard sets 6–8 reps	Dumbbell Romanian Deadlift 3 hard sets 6–8 reps	Two-Arm Dumbbell Row 3 hard sets 6–8 reps
Seated Triceps Press 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps	Seated Triceps Press 3 hard sets 6–8 reps	Dumbbell Lunge (In-Place) 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps

The 4-Day *Bigger Leaner Stronger* Dumbbell Workout Routine

Workout 1 Push	Workout 2 Pull	Workout 3 Upper Body	Workout 4 Lower Body
Dumbbell Bench Press 3 hard sets 4–6 reps	Dumbbell Deadlift 3 hard sets 4–6 reps	Seated Dumbbell Press 3 hard sets 4–6 reps	Dumbbell Goblet Squat 3 hard sets 4–6 reps
Incline Dumbbell Bench Press 3 hard sets 4–6 reps	One-Arm Dumbbell Row 3 hard sets 4–6 reps	Two-Arm Dumbbell Row 3 hard sets 6–8 reps	Single Leg Split Squat (Dumbbell) 3 hard sets 4–6 reps

Dip 3 hard sets 6–8 reps	Chin-up 3 hard sets 4–6 reps	Close-Grip Dumbbell Bench Press 3 hard sets 6–8 reps	Dumbbell Romanian Deadlift 3 hard sets 6–8 reps
Seated Triceps Press 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps	Dumbbell Rear Lateral Raise (Seated) 3 hard sets 6–8 reps	Dumbbell Lunge (Walking or In-Place) 3 hard sets 6–8 reps

The 3-Day *Bigger Leaner Stronger* Dumbbell Workout Routine

Workout 1 Push	**Workout 2 Pull**	**Workout 3 Lower Body**
Dumbbell Bench Press 3 hard sets 4–6 reps	Dumbbell Deadlift 3 hard sets 4–6 reps	Dumbbell Goblet Squat 3 hard sets 4–6 reps
Incline Dumbbell Bench Press 3 hard sets 4–6 reps	One-Arm Dumbbell Row 3 hard sets 4–6 reps	Single Leg Split Squat (Dumbbell) 3 hard sets 4–6 reps
Dip 3 hard sets 6–8 reps	Chin-up 3 hard sets 6–8 reps	Dumbbell Romanian Deadlift 3 hard sets 6–8 reps
Seated Triceps Press 3 hard sets 6–8 reps	Alternating Dumbbell Curl 3 hard sets 6–8 reps	Dumbbell Lunge (Walking or In-Place) 3 hard sets 6–8 reps

If you still can't find a suitable substitute for whatever reason (lack of equipment, sufficiently heavy dumbbells, mobility, etc.), simply do 3 more sets of an exercise already in your workout. And finally, if all you have is your body weight and maybe some bands, you can do plenty of different resistance training workouts, but you can't follow most of the training modalities you've learned

because they require weights.

How to Progress in *Bigger Leaner Stronger* Workouts

In chapter 12, you learned about double progression—the system of working with a weight until you can do a number of sets at the top of a rep range, then increasing the weight. Here's how it works in *Bigger Leaner Stronger*:

- When you can do one top-rep set of a primary exercise with at least 1-to-2 good reps left or one top-rep set of an accessory exercise with at least 0-to-1 good reps left, immediately add 10 pounds (total) to your *working weight* (the amount of weight used in hard sets).
- If you can at least hit the bottom of your rep range with the new, heavier weight with at least 0-to-2 good reps left (depending on the type of exercise), work with it until you reach the progression target (one top-rep set) again, increase the weight again, and so forth.
- If you can't reach the bottom of your rep range with sufficient *reps in reserve* (at least 0- to-2 reps left depending on the exercise) with the new, heavier weight, you have three options:

    1. Decrease the weight by 5 pounds so you've only added 5 pounds instead of 10.
    2. If number one doesn't work—you still can't stay in your rep range—or you can't do it because your gym doesn't have 2.5-pound plates or its dumbbells move up in 5-pound increments, reduce the weight to the original, lighter load and work there until you can do 2 top-rep sets in a single workout. Then, increase by 10 pounds, and if that doesn't work, by 5.
    3. If you do that, and you still can't stay in your rep range, go back to the lighter weight and work with it until you can do 3 top-

rep sets in a single workout. At this point, your progress should stick (either at 10 or 5 pounds), and if it doesn't, the culprit is usually diet (calories, macros, nutrition) or lifestyle (especially stress management and sleep habits) or both.

- If you hit your progression target on the last set of an exercise, increase the weight on the first set of that exercise the next time you do that workout. In *Bigger Leaner Stronger*, you'll bench press on Monday, so if you get 6 reps on your third (and final) set, you'd increase the weight on the first set of the bench press on the following Monday.
- If you're doing the same exercise several times per week, progress on each exercise in each workout independently. That is, if you bench press on Mondays and Fridays, and you get 155 x 6 on Friday, you shouldn't increase the weight in your Monday pressing—just your next Friday session. This procedure helps because you'll experience varying levels of fatigue in each workout, depending on your workout programming, lifestyle, and physiology. For instance, if you train Monday through Friday and take the weekend off, you'll feel fresher on Monday (after two days of rest) than on Friday (after four days of training). Thus, you may find that you can progress faster in workouts earlier rather than later in the week. On the other hand, if you do an exercise later in a workout earlier in the week and then earlier in a session later in the week, you may find that your performance is better in the second workout because the "first-workout advantage" isn't enough to offset the "later-in-the-workout" drawback. So, since it's hard to account for these variables in your programming, it's best to progress on each exercise in each workout individually.

To better understand how this works, let's go through some examples. Let's say you're bench pressing in the 4-to-6-rep range for 3 sets, and on your first set, you get 6 reps with 135 pounds with 1 good rep left. The *Bigger Leaner Stronger* program calls for just one top-rep set to progress, so it's time to increase the weight. You add 10 pounds to the bar (145 pounds), rest a few minutes, and get 4 reps on your next two sets of the exercise in that workout. Because you're still within your target rep range (4-to-6 reps) with 145 pounds, you'd keep working with it until you get 6 reps for one set, and then you'd increase the weight to 155 pounds. Here's how this could look over the course of four weeks of training:

	Set 1	Set 2	Set 3
Week 1	135 x 4	135 x 4	135 x 5
Week 2	135 x 6 (top-rep set)	145 x 4 (increase)	145 x 4
Week 3	145 x 4	145 x 5	145 x 5
Week 4	145 x 6 (top-rep set)	155 x 4 (increase)	155 x 4

Now let's say that after a few months of training, you're moving up from 185 to 195 pounds, but you only manage to get 2 or 3 reps with the new, heavier weight before your technique breaks down—something that'll happen at times as you get stronger. What should you do next? Reduce the weight by 5 pounds to 190 pounds, so you've only added 5 pounds instead of 10, because this is usually enough to get you back in your target rep range. If, however, you still can't stay in your target rep range (you only get 3 or fewer reps with 190 pounds), or your gym doesn't have 2.5-pound plates or dumbbells that move up in 5-pound increments, then you'd reduce the weight to 185 pounds and work with that until you can do 2 sets of 6 reps before adding the 10 pounds back to the bar.

Two other scenarios to address involve insufficient reps in reserve to

progress. Say in the above bench press example, you get 6 reps with 185 pounds but have 0 reps in reserve (0 good reps left). In this case, you're not ready to add weight yet—you need to keep bench pressing 185 pounds until you get 6 reps *and* have at least 1 good rep left. Then, when you achieve that and bump up to 195 pounds, you don't just need to get 4 reps—you also need to have at least 1 good rep left, and if you don't, you should drop the weight to 190 pounds and continue from there.

Also, know that missing a rep range after adding weight to an exercise isn't a cause for concern. Although people new to strength training with double progression can typically progress in 10-pound increases with just one top-rep set for their first year or so, at some point, this approach falls flat, and they have to use the techniques I outlined above to keep advancing.

How to Warm Up for Your Workouts

For many people, warming up for a strength training workout is like a little workout unto itself—ten minutes on the treadmill, ten minutes of stretching, ten minutes of foam rolling, and so on. This type of routine is mostly a waste of time. Although raising your core temperature and the temperature of your muscles can improve performance, it doesn't necessarily decrease the risk of injury (as proponents of drawn-out warm-ups often claim). Ironically, "loosening" your muscles up with stretching before a workout can actually *reduce* your strength and athletic performance.

A much better way to warm up for your workouts involves doing warm-up sets of the exercises in those workouts. This method increases body and muscle temperature and enhances blood flow to the muscles you're about to train (which can also boost performance). It also helps you "groove in" proper technique so your form doesn't crumble as you approach failure in your hard sets. This latter point is particularly important as you get stronger, when the weights get heavy and poor form becomes more dangerous.

To warm up for your *Bigger Leaner Stronger* workouts, you'll do several warm-up sets with the first exercises for each of the major muscle groups you'll be training:

Set 1: 6 reps with about 50 percent of your hard-set weight, and rest for a minute.
Set 2: 4 reps with about 70 percent of your hard-set weight, and rest for a minute.

And that's it—you're ready to do your hard sets.

Now, that doesn't mean you'll need to do warm-up sets for *all* of the exercises

in a workout because if you're doing an exercise for a muscle group that's already primed, no warm-up is needed.

For example, let's say you're doing a lower-body workout of squatting, leg curling, and leg pressing, in that order. You'd warm up on the squat and then do your hard sets. Next is the leg curl, but you won't need to warm up first because your hamstrings are plenty warm after squatting. The same goes for the leg press—the major muscle groups involved are the same as in the squat. Therefore, your warm-up on the squat serves as your warm-up for the entire workout.

One more example for the sake of clarity. Say you're starting a pull workout that calls for deadlifting, dumbbell rowing, and biceps curling, in that order. Based on what you now know, how would you do this workout? You've got it—warm up on the deadlift first, followed by your hard sets, followed by the rest of the workout without any further warm-up sets because the next exercises train the same muscles as the deadlift.

How to Find Your Starting Weights

You now know which rep ranges you'll be working in, but how do you figure out how much weight to use on all of the exercises you'll be doing? It's mostly a matter of experimentation. You start light on a new exercise and increase the weight after each set until you've found the right amount. As a rule, you'll lose 2 reps for every 10 pounds you add to an exercise.

If you're completely unsure of where to start, the following chart will help you find your starting weights faster:

Exercise	Starting Weight (lb.)
Barbell Bench Press	95
Incline Barbell Bench Press	65
Dumbbell Bench Press	30 (per dumbbell)
Triceps Pushdown	30
Barbell Deadlift	135
One-Arm Dumbbell Row	30 (per dumbbell)
Lat Pulldown	80
Alternating Dumbbell Curl	15 (per dumbbell)
Seated Dumbbell Press	20 (per dumbbell)
Dumbbell Side Lateral Raises	10 (per dumbbell)
Dumbbell Rear Lateral Raise	10 (per dumbbell)
Seated Triceps Press	30
Barbell Squat	135
Leg Curl	50

Leg Press	185
Dumbbell Lunge	20 (per dumbbell)

How to Deload

As you learned a couple of chapters ago, deloading entails reducing your workout intensity or volume for a period, usually a week. By doing this, you can reduce joint, tendon, and ligament strain; lower the risk of injury; and alleviate psychological stress.

Some people say deloading by reducing workout volume is better than reducing intensity and vice versa. I lean toward deloading volume because studies show that reducing volume instead of intensity is more effective for decreasing fatigue, which is the main goal, as well as for maintaining performance (making it easier to pick up where you left off when you get back to your hard training).

As for frequency, there's no pat answer for how often you should deload—some people's bodies can endure more strain than others' before needing a break—but in *Bigger Leaner Stronger*, here's how it'll work:

Every ninth week, do the following:

1. Do the previous week's workout.
2. Do 2 hard sets for each exercise instead of 3.
3. Warm up and use your normal working weights, but do 2 reps per set for primary exercises and 4 reps per set for accessory exercises.

That's all there is to it.

"Can I do cardio on my deload week?"

Sure. That said, remember the goal is to significantly reduce the amount of stress on your body. So walking or light physical activity is okay, but skip the high-intensity and high-impact cardio (sprinting, for example).

"What should I do with my diet when deloading?"

If you're cutting, you can maintain your current calorie intake while deloading, unless you feel the need for a diet break, in which case you can increase your intake to your approximate total daily energy expenditure.

If you're lean bulking, you can maintain your current calorie intake or reduce it to your approximate TDEE if you'd like a break from all the food.

"Can I just take a week off instead?"

Yes, but deloading is preferable because it'll produce better results. That said, if you're traveling, sick (or on the brink), or ragged, take the week off instead. It's also helpful to plan your deloads or rest weeks to coincide with trips, holidays, vacations, or any other forthcoming disruptions to your routine. This way, you don't have to interrupt your hard training.

How to Include Cardio in Your Regimen

Strength training is more important than cardio for improving and maintaining body composition, health, and vitality, but there are several good reasons to doing some cardio as well, including improved heart and circulatory health (which reduces the risk of high blood pressure, diabetes, cognitive decline, and other undesirable conditions), anti-aging effects, additional energy expenditure, and more.

It also doesn't take much cardio to make a major difference—just a couple of hours per week is plenty—and if you combine it correctly with your strength training, it won't hurt (and may even enhance) your results in the gym. Therefore, if you want to maximize your health and fitness, I recommend making cardio a part of your routine. Here's a winning formula:

1. Do one-to-three hours of cardio per week.
2. Mostly do low- and moderate-intensity cardio.
3. Do cardio that you enjoy.

If you follow those simple instructions, you'll reap many of the benefits I just mentioned without any of the potential handicaps. Let's review each of these points.

1. Do one to three hours of cardio per week.

One hour of cardio per week is a sensible "minimum effective dose" for improving cardiovascular and metabolic health, and three hours per week is a reasonable cutoff for preventing several problems that occur when you do too much cardio:

1. The more cardio you do, the more it can wear you out, making it harder

to give your strength training everything it needs. You won't always notice the systemic fatigue that can slowly set in when you do too much cardio, either.

2. Depending on what you're doing, cardio can cause enough muscle damage and soreness to hinder your performance in the gym and recovery from your strength training.
3. Research shows that some of the cellular adaptations to cardio can interfere with those of strength training. In substance, by doing too much cardio in addition to strength training, we're sending "mixed messages" to our body (and muscles in particular), and they can't optimally respond and adapt to either.

By limiting your cardio to a couple of hours per week, however, you can minimize these disadvantages (and eliminate them altogether if you follow the other two guidelines). If you like cardio and want to do more than I'm recommending here, you can, but just know that it'll likely slow your progress in your strength training.

2. Mostly do low- and moderate-intensity cardio.

Broadly, cardio comes in three varieties: Low-intensity, moderate-intensity, and high-intensity. Low-intensity cardio allows you to talk in full sentences throughout the workout. When you can only talk in a few short sentences before catching your breath, you're doing moderate-intensity cardio. And when you're breathing too heavily to even get out full sentences, that's high-intensity cardio.

You've probably heard that high-intensity cardio is far better for losing fat than less demanding workouts, but this isn't always true. High-intensity cardio can help you lose more fat than low-intensity cardio, but only because it burns more calories per unit of time (a 30-minute high-intensity cardio session will

burn about 2-to-3 times more calories than a leisurely 30-minute walk, for instance). That said, when you compare high-intensity cardio to moderate-intensity cardio, the difference largely disappears. For example, a study conducted by scientists at Arizona State University found that a 30-minute moderate intensity cardio workout burned just as many calories as a 30-minute high-intensity interval workout (about 300 calories).

What's more, high-intensity cardio also causes more muscle damage, soreness, and general fatigue than less intense types, and this can cut into your ability to gain muscle and strength. It requires more moxie than lower-intensity cardio, too, which can be in short supply after a long day of work, strength training, chores, arguing with strangers on the internet, and so forth.

Thus, I recommend you mostly do low- and moderate-intensity cardio while on the *Bigger Leaner Stronger* program, and if you want to include high-intensity training in your regime to further improve your cardiovascular endurance, keep it to one 20-to-45-minute high-intensity cardio workout per week. These don't have to be "workouts" in the formal sense, either—you could play a competitive tennis match instead of hopping on the elliptical, for example.

3. Do cardio that you enjoy.

Exactly what you do for cardio isn't nearly as important as how consistently you do it, so make sure to choose something that you like. This can include fun physical activities as well—going for a hike, walking a golf course, playing tag with your kids, whatever! That said, it's best to prioritize low-impact types of cardio, such as swimming, cycling, walking, rowing, hiking, cross-country skiing, elliptical, and so on. These kinds of exercise cause little muscle damage or soreness, which means they don't directly interfere with strength training like high-impact cardio does.

Higher-impact activities, like running, tennis, or basketball, aren't verboten in *Bigger Leaner Stronger*, however—they just shouldn't make up more than half of your total cardio. If you do two hours of cardio per week, then, no more than one hour should be high-impact activities.

Finally, if you aren't sure what kind of cardio to do, try my favorite: Cycling. It's basically all pros and no cons:

- It burns a lot of calories.
- It's easy on your muscles, tendons, ligaments, and joints.
- Research shows it can help you gain lower-body muscle and strength faster.
- It's something you can do inside when the weather's bad and outside when it's nice.
- When indoors, you can pair it with something else that you enjoy or need to do, such as listening to music, podcasts, or audiobooks, making work or personal calls, reading, or watching TV.

With everything I've given you in this chapter, it's not a matter of *if* you can get bigger, leaner, and stronger but *how quickly*. I'll also bet you a shiny shekel that soon, you'll have a lot more than a new body—you'll have a new lease on life. You'll feel more confident and competent as you gain muscle and strength. Your motivation will soar as you relish the changes you see in the mirror and on the scale. Friends and family will start noticing and asking how the heck you're doing it. You may hardly believe the results yourself. All of that and more is inevitable if you just follow the way I'm lighting for you.

To ensure you get there, we need to add a couple more strings to your bow,

however, so let's continue thrusting ahead.

KEY TAKEAWAYS

- A training phase is a period that lasts anywhere from several weeks to months, designed to achieve a specific goal, like more strength, muscle, endurance, or recovery. In *Bigger Leaner Stronger*, each training phase will last nine weeks, will focus on increasing strength and muscularity, and will consist of two parts: hard training and deloading.
- In *Bigger Leaner Stronger*, you can choose from a three-, four-, or five-day workout routine. Pick whatever routine you *know* you can consistently do every week.
- If an exercise gives you grief in any way, swap it out for one that doesn't—just make sure it's an exercise of the same type (a primary exercise for a primary exercise, an accessory for an accessory) and for the same muscle group.
- To warm up for your *Bigger Leaner Stronger* workouts, do two warm-up sets with the first exercises for each of the major muscle groups you'll be training: 1 set of 6 reps with about 50 percent of your hard-set weight, and rest for a minute, then 1 set of 4 reps with about 70 percent of your hard-set weight, and rest for a minute.
- Every ninth week of training, take a deload.
- Instead of deloading you can take a week off instead, but deloading is preferable because it'll produce better results. That said, if you're traveling, sick (or on the brink), or ragged, take the week off instead.
- One hour of cardio per week is a sensible "minimum effective dose" for improving cardiovascular and metabolic health, and three hours per week is a reasonable cutoff for preventing several problems that occur when you do too much cardio.

- Mostly do low- and moderate-intensity cardio, and if you want to include high-intensity training in your regime to further improve your cardiovascular endurance, keep it to one 20-to-45-minute high-intensity cardio workout per week.
- Exactly what you do for cardio isn't nearly as important as how consistently you do it, so make sure to choose something that you like.
- If you aren't sure what kind of cardio to do, try cycling. It's basically all pros and no cons.

The Big Ideas

1. Track your body composition with weight, pictures, and body measurements.

2. Don’t obsess over fluctuations in your weight.

3. Track your strength training with a journal or app.

15

The Right and Wrong Ways to Track Your Progress

Moderation is a fatal thing. Nothing succeeds like excess.

—OSCAR WILDE

Sir William Thomson (also known as Lord Kelvin) was a nineteenth-century physicist who famously said that "when you can measure what you are speaking about, and express it in numbers, you know something about it." This insight applies to diet and exercise just as much as the scientific method: unless you can measure your progress (or lack thereof) and express it in numbers, you likely won't know whether you're headed in the right direction.

As it happens, one of the least appreciated reasons why so many people fail to achieve their fitness goals is because people fail to consistently track and evaluate how their bodies are responding to their eating and training. When you're new to flexible dieting, for instance, it's important to closely monitor your calories and macros because it's easy to accidentally over- and undereat, but as you become familiar with the routine, your estimations and intuitions become more accurate and reliable. The same is true of training, but in the reverse. As time goes on, tracking workouts becomes *more* important, not less, because without precise

records of past workouts to refer to, you'll lose sight of the trajectory of your training and with it your ability to adjust (or not) accordingly—a vital skill for long-term fitness success.

You can also be misled by watching the wrong things, or the right ones incorrectly. For example, many people use smartphone apps or electronic gizmos to gauge calorie burning and body composition, but research shows that the data provided by such software and gadgets can be wildly erroneous. And while it's useful to watch body weight, daily fluctuations aren't meaningful, so cheering or cursing them is pointless.

So what should you track, exactly, and how do you do it? The answer to the first question is twofold:

1. Your body composition
2. Your training

And to answer the second question, let's address each of these elements separately.

How to Track Your Body Composition

Tracking your body composition properly is crucial because even when you do everything right, transformation takes time. And when your fleshy bits aren't firming up as quickly as you'd like, it's easy to feel deflated. By observing your body composition correctly, however, you'll always know whether you're advancing (and whether you should keep plugging away) or stuck, and change your diet or training or both. This process has three steps:

1. Weigh yourself every three days and calculate the average every two weeks.
2. Take progress pictures every two weeks.
3. Take body measurements every two weeks.

That may sound like a lot of hassle, but don't worry—it'll only require five minutes per week. Plus, as you get bigger, leaner, and stronger, you'll look forward to the next round of measurements for the same reason it's fun to keep score when you're playing a game (especially when you're winning!).

1. Weigh yourself every three days and calculate the average every two weeks.

One of the easiest ways to drive yourself to distraction in your fitness journey is to obsess over daily shifts in your weight, which often have nothing to do with gaining or losing fat or muscle. For instance, even slight swings in fluid retention, glycogen levels, and bowel movements can produce noticeable ups and downs.

A much better way to use body weight is to look at longer-range averages, which are less erratic and better register the stuff that we actually care about (fat and muscle). If, over the course of weeks and months, these averages are moving

down, you're clearly losing weight, and if they're moving up, you're clearly gaining weight. Here's the procedure:

1. Weigh yourself every one-to-three days first thing in the morning, naked, after the bathroom and before eating or drinking anything. Record the numbers somewhere accessible like an Excel file or Google Sheets or the notepad app in your phone. If you want to take your weight-tracking game a step further, you can graph the numbers in a spreadsheet.
2. Every two weeks, add your weigh-ins together and divide the sum by the number of weigh-ins to get your average daily weight for the period. Record this as well.

Simple and clean. Here's an example of how this could look for someone cutting:

Monday: 193 pounds
Thursday: 194 pounds
Sunday: 192 pounds
Wednesday: 191 pounds
Saturday: 191 pounds
Tuesday: 190 pounds

Average daily weight: 1,151 (pounds) / 6 (weigh-ins) = 192 pounds

With this method of tracking your weight, you can focus on the broader picture instead of stewing over minor and meaningless variances.

2. Take progress pictures every two weeks.

In theory, taking progress pictures is better than taking body measurements because ultimately, we don't care about numbers nearly as much as what we see in the mirror. Such pictures can cause guilt and frustration when they're brute reminders of how far we are from where we want to be, however, so many people avoid them. I understand this reaction—it's perfectly normal—but I also think it can be framed more positively. Don't look at these pictures as reflections of *who you are* but merely as another set of data to help you with *who you're becoming*. The following from David Schwartz's classic self-help book *The Magic of Thinking Big* comes to mind:

> "If I looked at myself strictly as I am—old car, low income, cheap apartment, and hamburger diet—I couldn't help but be discouraged. I'd see a nobody and I'd be a nobody for the rest of my life. I've made up my mind to look at myself as the person I'm going to be in a few short years. I see myself not as a rate clerk but as an executive. I don't see a crummy apartment, I see a fine new suburban home. And when I look at myself that way, I feel bigger and think bigger. And I've got plenty of personal experiences to prove it's paying off."

Remember too that nobody else needs to see these pictures, but don't be surprised if, after some time, you're sharing your "before" and "after" pictures with at least a few people (and I hope I'm one of them!).

So, even if you're feeling sheepish, how about we take your "before" pictures now? Then, take another set of pictures every two weeks to document your transformation, and mark my words: before long, you'll be shocked by how much your physique is changing.

Here's how to take progress pictures properly:

- Take front, side, and back shots first thing in the morning, after using the bathroom and before eating or drinking anything.
- The more skin you show, the better (most people do them in their underwear).
- Keep the lighting and setting the same, if possible.
- If you have someone to take pictures of you and you want to better see how your muscles are developing, take both flexed and unflexed pictures.

I also recommend saving all your progress photos in an individual album on your phone or computer so you can easily scroll through them, like a flipbook, and see how your physique is changing.

3. Take body measurements every two weeks.

If you only watch your weight, you won't have a handle on how your body composition is changing because gaining or losing weight can mean gaining or losing fat or muscle or water or glycogen or excreta or some combination thereof. Furthermore, when you're new to strength training, you can expect to gain muscle and lose fat at the same time, rendering weight measurements less important. In fact, your weight may not change as much as you'd expect while you're on the program, especially in the beginning. I've seen some radical physical overhauls where weight only changed by 15-to-20 pounds!

To help you understand how your body is responding to your eating and training, there's one body measurement worth noting every two weeks: Your waist circumference. This is useful because the size of your waist is a reliable indicator of body fatness regardless of what's happening with your weight. Eventually, you won't feel it necessary to measure your waist, even when cutting (the scale and mirror will be enough), but until then, you'll probably find it

beneficial.

To measure your waist properly, do the following in the morning, naked, after using the bathroom and before eating or drinking anything. Wrap a tape measure around your bare stomach, right at your navel, and make sure the tape is straight (parallel to the floor) and snug, but not so tight that it compresses your skin. Breathe in, exhale until your lungs are mostly empty, and with your stomach relaxed (no sucking it in), record the number somewhere handy (workout journal, notepad app, cloud document, etc.).

If you'd like to collect even more data on how your body is developing, here are several more measurements to take every two weeks:

- Your chest circumference. Wrap a measuring tape around your chest at the fullest part (usually across your nipples) by first pulling it across your back, under your armpits, and then around your chest. Adjust the measuring tape so that it's parallel to the floor, stand tall, relax your upper body, don't flare your elbows or flex your lats, and note the number.
- Your flexed arms. Flex one arm and with your other hand, wrap a tape measure around the largest part of the flexed arm (the peak of the biceps and middle of the triceps), and note the number. Then do your other arm.
- Your shoulder circumference. Wrap a measuring tape straight around your torso (over your shoulders and chest) right around the top of your armpits, and make sure the measuring tape is parallel to the floor. Hold the measuring tape with one hand, stand tall, and relax your upper body (again don't flare your elbows or flex your lats), and note the number.

- Your upper-leg circumference. While standing, wrap a tape measure around the widest part of one thigh, straighten and flex your leg, and note the number. Then do the same with your left leg.
- Your flexed calves. While standing, flex one calf by raising your heel off the ground, wrap a measuring tape around the largest part of it, and note the number. Then do your other calf.

How to Track Your Strength Training Workouts

As I mentioned earlier in this chapter, tracking your strength training is important because it's the only way to know if you're getting fitter and stronger (and therefore progressively overloading your muscles). At first, progress is obvious because your strength climbs by leaps and bounds, but once it begins to crawl, the scene gets hazy if you don't have a training journal. You'll forget exactly what you did in previous workouts and thus have no reliable reference points for future workouts—a surefire recipe for stagnation when a successful workout is one that's just a little better than the last, like getting another rep or two on just one exercise, for instance. Strength and muscle grow one rep at a time, so you always want a clear goal in mind for every set of every exercise. This is impossible if you don't know what you did the last time. For example, if you're about to squat, and you know that in your previous workout, you did 3 sets of 4 reps with 185 pounds, in this workout, your overarching objective is always to get 6 reps in one set (likely your first) with at least 1-to-2 good reps left, but failing that, it's to get at least one more rep across all sets than last time—13 reps or more in this case.

This level of focus and intentionality requires a training journal, not grunting and groaning with death metal blaring in your headphones. It's the gateway to great workouts—ones where you're full of energy and concentration, where the weights feel lighter than usual, and where you leave the gym feeling invigorated, not wiped out.

To track your strength training workouts, the three easiest options are a pen and paper, the notepad app in your smartphone, or a workout tracking app. If you prefer pen and paper, you can use a blank notepad or workout journal, or you may like the pre-formatted workout journal I created specifically for *Bigger*

Leaner Stronger called *The Year One Challenge for Men* (www.). There are just as many workout apps to choose from as journals, and I have a free one called Stacked, which you can get at . Some people (like me) also like using Excel or Google Sheets (my preference). To make this easy for you, I've included an entire year's worth of *Bigger Leaner Stronger* training workouts in the free bonus material that comes with this book ().

Regardless of which method of tracking you choose, it needs to show the workouts you'll be doing (and the number of hard sets and rep ranges for each), and it needs to allow you to record how it goes (weight and reps for each set). It's also helpful if you can include relevant notes like feeling strong or weak on a set or exercise, experiencing an ache or pain, not having slept well the night before, etc. These comments give context to your numbers when you come back to them later ("I wonder why my bench press was down 10 pounds that workout?").

Here's a simple way to lay all of this out in an analogue or digital notepad or journal:

Phase 1

Week 1

Workout 1

Monday 8/23/2021

Push

Barbell Bench Press	Incline Barbell Bench Press
4–6 reps	4–6 reps
Set 1:	Set 1:

Set 2:	Set 2:
Set 3:	Set 3:
Notes:	Notes:

Dumbbell Bench Press	Triceps Pushdown
6–8 reps	6–8 reps
Set 1:	Set 1:
Set 2:	Set 2:
Set 3:	Set 3:
Notes:	Notes:

Then, when you do this workout, you can fill it out like this:

Phase 1

Week 1

Workout 1

Monday 8/23/2021

Push

Barbell Bench Press	Incline Barbell Bench Press
4–6 reps	4–6 reps
Set 1: 125 x 5	Set 1: 100 x 6

Set 2: 125 x 5	Set 2: 110 x 4
Set 3: 125 x 4	Set 3: 110 x 4
Notes:	Notes: Felt strong despite adding weight on bench press before this.

Dumbbell Bench Press	Triceps Pushdown
6–8 reps	6–8 reps
Set 1: 40 x 6	Set 1: 40 x 8
Set 2: 40 x 6	Set 2: 40 x 7
Set 3: 40 x 7	Set 3: 40 x 6
Notes:	Notes: Triceps were toast at the end.

This is a major moment for us. You've learned all of the salient diet and exercise principles and strategies, and you're officially ready to stop reading about the *Bigger Leaner Stronger* style of strength training and start doing it! To make your first phase of training as smooth as possible, in the next chapter I'll share with you a quickstart guide that'll quickly get you up to speed.

KEY TAKEAWAYS

- By observing your body composition correctly, you'll always know whether you're advancing (and whether you should keep plugging away) or stuck, and change your diet or training or both.
- Weigh yourself every one-to-three days first thing in the morning, naked, after the bathroom and before eating or drinking anything. Record the numbers somewhere accessible like an Excel file or Google Sheets or the notepad app in your phone. Every two weeks, add your weigh-ins together and divide the sum by the number of weigh-ins to get your average daily weight for the period. Record this as well.
- Take progress pictures every two weeks to document your transformation. Take front, side, and back shots first thing in the morning, after using the bathroom and before eating or drinking anything; show as much skin as possible (most people do them in their underwear); keep the lighting and setting the same, if possible; and if you have someone to take pictures of you and you want to better see how your muscles are developing, take both flexed and unflexed pictures.
- Tracking your waist circumference is useful because the size of your waist is a reliable indicator of body fatness regardless of what's happening with your weight. To measure your waist properly, do the following in the morning, naked, after using the bathroom and before eating or drinking anything: wrap a tape measure around your bare stomach, right at your navel, and make sure the tape is straight (parallel to the floor) and snug, but not so tight that it compresses your skin. Breathe in, exhale until your lungs are mostly empty, and with your stomach relaxed (no sucking it in),

record the number somewhere handy (workout journal, notepad app, cloud document, etc.).

- If you'd like to collect even more data on how your body is developing, here are several more measurements to take every two weeks: chest circumference, flexed arms, shoulder circumference, upper-leg circumference, and flexed calves.
- To track your strength training workouts, the three easiest options are a pen and paper, the notepad app in your smartphone, or a workout tracking app. Regardless of which method of tracking you choose, it needs to show the workouts you'll be doing and the number of hard sets and rep ranges for each), and it needs to allow you to record how it goes (weight and reps for each set). It's also helpful if you can include relevant notes like feeling strong or weak on a set or exercise, experiencing an ache or pain, not having slept well the night before, etc.

The Big Ideas

1. Buy the right gear.
2. Join the right gym or set up the right home gym.
3. Create a training schedule that works for you.

16

The Bigger Leaner Stronger Quickstart Guide

Any airplane is off track much of the time but just keeps coming back to the flight plan.

—STEPHEN R. COVEY

The nutritional component of the *Bigger Leaner Stronger* program is easy enough to enact—make a meal plan, go grocery shopping, and start following the meal plan—but the training portion is harder to embark on because you have to consider gear, gyms, logistical and technical agendas, and more. Therefore, to help grease the skids, I've prepared a five-step checklist that walks you through the whole process:

1. Buy your supplies.
2. Join or set up a gym.
3. Create your workout schedule.
4. Prepare for your first week.
5. Do your first week.

Let's go over each.

1. Buy Your Supplies

You don't need much in the way of gewgaws to do your *Bigger Leaner Stronger* workouts. The only must-haves are a measuring tape and a bathroom scale for tracking your body composition and a notepad or workout journal if you want to record your workouts manually. There are other items that can be useful, however:

1. A pair of weightlifting shoes for stabler squatting and deadlifting.
2. A pair of lifting straps for heavy pulling.
3. A pair of neoprene knee sleeves, shin guards, or a couple pairs of knee-high socks for protecting your shins while deadlifting.
4. A weightlifting belt to boost your squat and deadlift strength. I recommend a powerlifting belt made of rigid leather, 9-to-13 millimeters thick, and 3-to-4 inches wide.
5. A comfortable pair of running shoes if you're going to run or walk for your cardio. Ignore most of the marketing puffery used to sell these products and just pick whatever feels the best on your feet when you're walking or jogging.
6. Equipment for your home gym (more details to follow).

Also, if you want my specific product recommendations, you can find them in the free bonus material that comes with this book (www.biggerleanerstronger.com/bonus).

2. Join or Set Up a Gym

Some people are put off by gyms, and I can't blame them. What with the violent outbursts of groaning and grunting, the gaggles of would-be Instagram celebrities snapping selfies, and the occasional stony-faced bodybuilder laying claim to entire

rows of equipment, getting in a workout can seem about as appealing as skinny dipping in the Ganges River. Fortunately, strength training is more popular than ever before (and its star is still rising), so it's also easier than ever to find a clean and comfortable gym that more resembles a classy community center than a dingy dungeon.

To find the right gym for you, there are a few things to take into account:

1. Does it have all of the equipment you'll need? I explicitly designed the *Bigger Leaner Stronger* workouts to work with the standard gym complement of free weights and machines, so all you need is a spot with at least one bench press and squat rack (but a few is better so you don't end up waiting), a full set of dumbbells, and some basic machines (and the gym has to allow deadlifting).
2. Is it relatively nearby? I've noticed that when a gym commute is more than about thirty minutes of driving (total), the chances of missing or skipping workouts rises markedly. So, if possible, pick a gym that's fairly close to your home or office.
3. Is it affordable? Many entry-level gyms cost $10-to-$50 per month, depending on where you live, and are clean, well-equipped, and have other perks you may want to use, such as showers, towels, a pool, etc. My favorite entry-level franchises are Gold's Gym, 24 Hour Fitness, LA Fitness, and Anytime Fitness. If you have the budget or inclination to spend more, for $100-to-$300 per month, you can get nicer and newer facilities and equipment and more perks like fitness classes, saunas and steam rooms, spa services, and more. My favorite premium gym franchises are Equinox and Life Time Fitness.
4. Do you like it? It may sound silly, but the more you like training at your

gym, the more you'll like working out in general and the better you'll do on the program. So if you're fortunate enough to have several gyms to choose from, go with the one that you like best, even if it's not the fanciest of the lot (I currently train at an awesome Gold's Gym in Ocala, Florida that I wouldn't trade for any of the higher-end gyms I've been to). By the same token, if the gym you really like costs a bit more than something you don't like and you can afford the upgrade, do it—you'll get many times your additional investment back in the more valuable coin of muscle, strength, and health and vitality.

You can also train in a home gym, which has its pros and cons. On the one hand, you can't beat the convenience—you don't have to get in the car and you can train whenever you want and don't need to worry about holiday hours or other restrictions. You also never have to wait for equipment and can ensure everything stays clean and functional, and decorating the space with whatever takes your fancy and blaring your favorite music can make your training more fun. There's quite a bit of time and money to be saved too when you don't have to get to and from the gym and pay monthly dues.

A home gym comes with tradeoffs, however. If you want new equipment, you'll need at least $1,000-to-$1,500 in hardware for your *Bigger Leaner Stronger* workouts. If you want to look for deals on used equipment, Facebook Marketplace, Craigslist, and eBay are good resources for this. You'll also need at least 100-to-200 square feet for a barebones barbell-and-dumbbells setup and double or triple that if you want to include some additional equipment to expand your exercise options (and for reference, the average two-car garage is about 650 square feet). You'll also want a room with a concrete floor that's on the ground floor of your home or in the basement because carpet, tile, and wood will get

damaged, and working out on an upper level can scare others or damage the floor or even the framework. Although most people like being able to walk to their gym instead of driving, working out at home usually means working out alone, and you may find that less enjoyable than at a gym with a friendly training partner. Home workouts are also notoriously susceptible to interruptions by visitors, kids, pets, your spouse or partner, etc. You have to be willing to take the time to clean, maintain, repair, and replace your equipment, too, or it'll slowly go to seed.

All things considered, I'd recommend a commercial gym if you have to compromise too much to train at home. For example, if I had to choose between a meager home gym and a plain but well-appointed local gym, I'd take the latter because it'll mean better workouts and results, and I'd only think about changing to a home gym if it had all of the equipment that I share below.

If you'll be doing *Bigger Leaner Stronger* at home, however, you'll want the following equipment:

- A barbell. Get an 87-inch long, 45-pound (20-kilogram) bar with a 28.5-mm wide shaft (the part you grip).
- A set of weight plates. Start with two 2.5-, 5-, 10-, and 25-pound plates, as well as four-to-six 45-pound plates, depending on how strong you are, and then add plates as you get stronger (more 10- and 45-pound plates should suffice). Also, get round plates and not polygonal ones, which move around when you deadlift and row.
- A set of dumbbells. You have two options here: Regular dumbbells (each is a fixed amount of weight) and adjustable dumbbells (each dumbbell's weight can be adjusted). Regular dumbbells are the easiest to handle, but they also take up a lot of space. If you have the room,

however, I recommend regular dumbbells. Start with a set that ranges from 10-to-80 pounds per dumbbell in 5-pound increments. You can add more dumbbells later, as you get stronger, and if you need to, you can switch to 10-pound increases with fractional plates (small plates ranging from 0.25-to-1.25 pounds that can be attached to dumbbells) for filling in the gaps. In contrast to regular dumbbells, adjustable ones have a small footprint but are more awkward to use. That said, they can get the job done, so if you need to go this route, get ones that go up to at least 50 pounds per dumbbell, and if they don't allow you to add further weight later (Bowflex, for instance), consider a model that goes up to 90-to-100 pounds or choose a product that can be gradually augmented as you get stronger (PowerBlock, for example). This way, you won't have to buy a whole new set of dumbbells after your first six months or so.

- A set of barbell collars. These are clasps that prevent plates from slipping off the barbell.
- An adjustable bench. This is a padded bench with wheels that can be put flat or upright, allowing you to do many seated and lying exercises among others that require a stable surface.
- A set of rubber flooring tiles. These are rubber interlocking tiles for deadlifting and rowing without damaging your floor or plates or making a racket.
- A power rack (also called a *squat rack*). This is a tall metal frame with movable hooks for holding a barbell and safety pins for extra safety when training alone. With a power rack, you can do various types of squats, bench presses, deadlifts, and other barbell exercises. Many power racks also come with a built-in pull-up bar.

- A dip stand (also called a *dip bar* and *dip station*). This is a metal frame for bodyweight dips and rows. Most power rack manufacturers also make dip stations that can be hooked onto their racks, which is convenient.
- A dip belt. This is a synthetic or leather belt with a hook for holding weight plates when doing dips, pull-ups, or chin-ups.
- (Optional) A pull-up bar. This is a bar you install in a doorway for doing pull-ups and chin-ups. If your power rack comes with a pull-up bar, you don't need this.
- (Optional) Power tower. This is a metal frame for pull-ups, chin-ups, dips, and leg raises (abs). If you get this, you don't need a dip stand or pull-up bar, and if your power rack can be fit with a dip station and comes with a pull-up bar, you don't need this.
- (Optional) A trap-bar (also called a *hex bar*). This is a special type of barbell for the trap-bar deadlift.

With this setup, you won't be able to do all of the exercises in the workouts in chapter 14 and in the bonus material, but you'll be able to do most of them. For the exercises you don't have the equipment for, you can swap them for ones you do as described in chapter 13.

Now, if you have funds and room to buy more equipment for your home gym, here's what else I'd recommend:

- A set of kettlebells. These are spherical weights with a handle similar to dumbbells that can be used for a variety of exercises. A good starter set goes from 10-to-50 pounds in 5-pound increments.
- A deadlift bar jack. This is a lever that allows you to lift the barbell a few inches off the ground for loading and unloading plates.

- A set of bumper plates. These are thick, rubberized weight plates that create less noise and vibration when dropped on the floor.
- A hamstring curl machine (ideally, a seated one). Aside from a few deadlift variations, there aren't many hamstring exercises you can do with just a barbell and dumbbells. A hamstring curl machine allows you to effectively target these muscles.
- A cable machine. This machine allows you to do triceps pushdowns, cable curls, cable side raises, and myriad other isolation exercises that train your muscles in slightly different ways than with free weights.
- A leg press. This allows you to keep training your quads after your back is beat from squats.
- A lat pulldown machine. Aside from chin-ups, pull-ups, and a few other exercises, it's difficult to focus on the lats like you can with a lat pulldown machine.
- A horizontal row machine. Unlike most barbell and dumbbell row exercises, a horizontal row machine allows you to train your upper back and lats without taxing your lower back.

There are countless other tools and toys you can buy, but with the lists I've just given you, you'll have a first-rate home gym with everything you need to build the body of your dreams.

3. Create Your Workout Schedule

Life is hectic and it always feels like there's never enough time in the 24 hours we get each day. As I say in my book *The Little Black Book of Workout Motivation*:

> "Who has the time for half of all the stuff we want to do? I'm sorry that life isn't gift wrapping a chunk of your days so you can train in Zen-like

comfort and solitude. Join the club. Face the fact that you're going to die with a long to-do list. Make damn sure 'start training' isn't on that list."

What we're really talking about here is *priorities*. We have to make our fitness a priority and only then will we be able to "find the time" to get our workouts in. Thankfully, we don't need to find *that* much time to make it all work. As you'll recall from chapter 14, you have three *Bigger Leaner Stronger* workout routines to choose from—a five-day, a four-day, and a three-day. And as stated before, the five-day routine beats the four- and three-day routines, and the four-day routine beats the three-day. But you can get great results with all of them, so don't feel that the four- or five-day routine is required.

So to create your workout routine, let's first decide when you'll do your strength training. For the most part, you can set this up however you'd like, but do try to space your strength training workouts evenly throughout the week (Monday, Wednesday, and Friday, instead of Monday, Tuesday, and Wednesday, for instance). Also, many people enjoy strength training during the week and taking the weekends off so they're free to do cardio or other activities.

If you plan on doing cardio, here are some tips for scheduling it properly:

Try to do cardio on days with no strength training, but when that's not possible, try to do your strength training first, and try to wait at least 3-to-6 hours before doing your cardio workout (the longer, the better). This decreases the chances of your cardio workouts interfering with your strength training.

When combining strength training and cardio on the same days, try to do cardio on upper-body days. This also helps minimize the impact of your cardio on your strength training by not exposing the same muscle groups to both on the same days.

Make sure you have at least one rest day per week where you don't do any vigorous physical activities (low-intensity cardio is fine).

Here are some examples of workout routines that meet our criteria:

	Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
Morning	Push	Pull	Upper Body A	Legs	Upper Body B	Cardio	
Afternoon/ Evening	Cardio		Cardio				

	Mon.	**Tue.**	**Wed.**	**Thu.**	**Fri.**	**Sat.**	**Sun.**
Morning	Push	Pull	Cardio	Upper Body	Legs	Cardio	
Afternoon/ Evening	Cardio						

	Mon.	**Tue.**	**Wed.**	**Thu.**	**Fri.**	**Sat.**	**Sun.**
Morning	Push	Cardio	Pull	Cardio	Legs	Cardio	
Afternoon/ Evening							

Next, think about your schedule and what else has to happen for you to follow it. If you want to train first thing in the morning before work, what time will you need to wake up? And what time will you need to be in bed the night before to ensure you're well rested? Or if your plan is to hit the gym after work, what time will you need to leave the office to have an hour or so for your workouts? Create what–when–where statements (from chapter 6) for any important preconditions you can think of and write them down.

Now reflect on your what–when–where statements and search for ways they can go astray. What if you have to miss a workout? What will you do? Do it on another day or skip it? (Either is fine.) What about when you're running late to

the gym? Can you stay longer or do you need to shorten the workout? What will you do when you travel? Will you find a gym and train as usual or do something else like a dumbbell-only or bodyweight routine or something else? Formulate if–then statements (also from chapter 6) for the hurdles you're likely to face now and again, and write them down.

4. Prepare for Your First Week

You now know what you need to start training and where and when to do it, so all that remains now is, well, doing it! So when will you officially start the program? Next week? Hogwash! How about today? Or tomorrow? That's more like it!

A final bit of groundwork before you do your inaugural *Bigger Leaner Stronger* workout: Make certain that you know how to do all of the exercises in it by reviewing the videos linked in the bonus material that comes with this book (www.biggerleanerstronger.com/bonus). Remember too that it also contains neatly organized Excel and Google spreadsheets with a year's worth of workouts that you can simply read and do.

5. Do Your First Week

The main goals of week one are to start getting comfortable with the exercises (remember, the primary compound movements will be at least a little awk for the first month or two) and to learn your working weights, so don't be ashamed if you feel like you're easing into the program. In fact, that's exactly what I want you to do, and to that end, in your first week, you can stop adding weight once you're able to do a top-rep set with 5 or so reps in reserve (easy), instead of 0-to-2 reps in reserve (strenuous).

Why do this? If you're new to strength training and try to tear right into it, you'll likely experience so much muscle soreness that it'll interfere with your daily

living and even the following week's workouts. Research shows that you prevent this by starting with a week of lighter training, however.

So, if you're figuring out your starting weights for the barbell bench press, your sets might look like this:

95 x 6 (easy, 3 out of 10)
105 x 6 (a little harder, 4 out of 10)
115 x 6 (harder, 5 out of 10)

Then you'd do two more sets with 115 x 6, and move on to your next exercise.

Also, don't be dismayed if your first month is a bit of a brute. I remember month one clearly: I felt weak, clumsy, and uneasy, but I knew that even the fittest people in the gym had to run this gauntlet many times to win their spurs. The initial struggle isn't a sign that it's not worth it or you're not worthy—it's the other way around. The strain at the start signals worth and to prevail is to prove you're worthy.

Transforming your body is much more than merely adding and removing tight and squishy tissues, too—you're *sacrificing who you are for who you want to be*. You're wielding iron and steel to forge a new form—a mystical act of heat and fire, hammer and anvil, thunder and lightning. It's not picking your nose or popping a pimple. It's hard, just as it's supposed to be. But it's not *that* hard. I'm not asking you to slap a honey badger or eat a fried tarantula. I'm only asking you to pick stuff up and put it down until your muscles burn and your body aches. And then do it again. And again. I'm not asking you to always like it, either, because you won't—you're not going to enjoy every workout. You'll always enjoy having worked out, though. Remember that.

At any rate, here's a quick refresher on some of the more important points to

remember when you're doing your workouts:

- Do exercises one at a time and finish all of the hard sets for each before doing the next.
- You need to do warm-up sets for some machine and free-weight exercises. When this is the case, do two warm-up sets before your hard sets. On the first set, do 6 reps with about 50 percent of your hard-set weight, and rest for a minute. On the second, do 4 reps with about 70 percent of your hard-set weight, and rest for a minute.
- For primary exercises, rest three-to-four minutes between hard sets, and for accessory exercises, two-to-three minutes.
- For all exercises, use double progression to first increase the reps and then the weight. To do this, increase the weight on a primary exercise when you do one top-rep set with at least 1-to-2 good reps left, and on an accessory exercise when you do one top-rep set with at least 0-to-1 good reps left.
- Generally, you'll lose about 2 reps for every 10 pounds (total) you add to an exercise, and gain 2 reps for every 10 pounds you remove.
- End all hard sets of primary exercises with 1-to-2 good reps left (2-to-3 reps from failure) and all hard sets of accessory exercises with 0-to-1 good reps left (1-to-2 reps from failure).
- Use a "1-0-1" rep tempo for all exercises—about one second for the first part of the exercise, a brief pause, and about one second for the second part.
- To inspect or troubleshoot your form, take a video of yourself and compare your technique to what you see in the videos from the bonus material.

- If an exercise feels bad (you experience a strange or painful sensation in your joints, especially on one side of your body), stop immediately, rest for a couple of minutes, and try the exercise again. If it continues giving you trouble, substitute it with another similar exercise that doesn't.
- Limit your cardio to a couple of hours of mostly low- and moderate-intensity exercise per week

You should also expect a fair amount of muscle soreness for the first month or so, even if you're not new to resistance training. *Bigger Leaner Stronger* will put your body on its mettle, and if you want to boost its ability to adapt, get thirty minutes of extra sleep each night for the first month. (And keep doing this if it makes a big difference inside and outside of the gym!)

Early on in this book, I said it isn't one you just read. It's one that you do. And now it's time to get off the mark and into the gym. You should keep reading as well because I have more to teach you about getting and staying fit and healthy, but everything that follows is meant to supplement (literally, in the case of the next section of this book) your work in the kitchen and gym. The sooner you're following your meal and training plans, the sooner you can implement the rest of the program.

Are you all set? Maybe feeling a little gun-shy? Great! That's precisely how you should feel—the "pre-event nerves" that everyone regularly experiences, even professional athletes. Such feelings aren't a warning that something is wrong or a sign that you need to "settle down" or "stop worrying." Rather than suppressing these sensations, research shows that you can harness them for better performance by simply saying to yourself "I'm excited!" This reframe shifts your

attitude from what scientists call a *threat mindset* (anxious and reluctant) to an *opportunity mindset* (confident and positive). So lean into this state. Do you feel your heart beating a little faster? Is your blood pumping a little harder? Are you breathing a little faster? Perfect! Let's go! Show me what you're made of!

PART 5

The Last Supplement Advice You'll Ever Need

The Big Ideas

1. Supplements are supplementary by definition, not vital.

2. You don’t need supplements to get the body you want.

3. There are four supplements you should consider taking to improve your results on the program.

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The Smart Supplement Buyer's Guide

In this age, which believes that there is a shortcut to everything, the greatest lesson to be learned is that the most difficult way is, in the long run, the easiest.

—HENRY MILLER

"What type of supplements do you sell?" I once asked an acquaintance.

He shrugged. "Pills in a bottle, brother."

I frowned. Was he serious?

"It's pretty slick, right?" Anthony said with a sideways grin.

This character had a simple business. He spent about $2 million per month on pay-per-click advertising to make close to double that in sales. What kind of "pills in a bottle" was he selling, exactly? Anthony didn't know. He couldn't tell me a single ingredient. What he did know, however, is a bottle cost him $3 to make and sold for $39.99. He also knew that, on average, he got close to $100 out of customers before they finally figured out how to stop him from charging their credit cards every month. Oh, and he also knew that his million-dollar renovation of his multimillion-dollar mansion was coming along splendidly.

When I first entered the fitness industry, I thought Anthony was an anomaly. A bad apple in the bushel. Sadly, I was wrong. People like him are the

norm. In fact, the supplement industry is best described by Ben Kenobi's famous words: "a wretched hive of scum and villainy." Seriously. Fake news, fake science, fake products—you can find it all in the supplement racket. It's almost funny … in the not-so-funny kind of way.

Don't believe me? In 2015, the New York State Attorney General's office accused four national retailers of selling dietary supplements that were fraudulent and in many cases contaminated with unlisted ingredients. The authorities said they had run tests on popular store brands of herbal supplements at the retailers —Walmart, Walgreens, Target, and GNC—which showed that roughly four out of five of the products contained none of the herbs listed on their labels. In many cases, the supplements contained little more than cheap fillers like rice and house plants, or substances that could be hazardous to people with food allergies.

According to lab tests obtained by the Michigan law firm Barbat, Mansour and Suciu in 2015, a number of sports supplement brands including Giant Sports, MusclePharm, CVS Health, 4 Dimension Nutrition, NBTY, and Inner Armour were mixing cheap fillers into their protein powders to bring down costs and were falsifying their supplement fact panels to cover their tracks.

Even worse is the supplement company Driven Sports, which was busted in 2013 for putting a methamphetamine-like drug in their popular pre-workout supplement "Craze." Apparently, meth makes for some pretty intense workouts. Oh, and their CEO was also previously busted for selling anabolic steroids and illegal weight loss drugs. Yet another supplement company, USPlabs, was slapped by the FDA in the same year for selling a now-notorious pre-workout supplement ("Jack3d") spiked with a dangerously powerful stimulant called *DMAA*, and then again in 2015 for selling a popular fat loss product ("OxyELITE Pro") laced with fluoxetine (Prozac) and drugs that caused liver damage and failure. A genius plan —if you love jail.

The sad lesson I've learned is most supplement companies, especially sports supplement companies, are first and foremost marketing companies. They're in the game of telling people what they want to hear to sell them what they don't need—a game that's easy to play if you have the capital and sales chops. Anybody can reach out to a contract manufacturer, ask for some off-the-shelf formulations full of useless ingredients, create some sleek "science-based" branding, recruit a mob of Instagram shills, and move a lot of product.

Why all the skulduggery and shenanigans? Filthy lucre, of course. When someone's presented with an unethical opportunity to coin money fast, they learn who they really are. Professed principles can become surprisingly supple when plied with cash, and many pill and powder pushers are an object lesson in moral relativism. This is why many workout supplements can't deliver a fraction of the results they promise. They are, for lack of a better term, worthless crap. And some are even dangerous.

Sports supplements don't have to suck, though. There *are* safe, natural ingredients that can help you gain muscle, lose fat, and get healthy faster. They're not silver bullets or heal-alls; however, high-quality, unbiased research shows they *can* give you an edge. The trick, then, is knowing which supplements are actually worth buying and which aren't, and that's exactly what you're going to learn in this chapter.

Let's start with a universal law of supplementation: Supplements are *supplementary* by definition, not *vital*. You don't need supplements to succeed with *Bigger Leaner Stronger* or any other fitness program. That said, there are supplements that can benefit your body and mind in many ways—supporting faster muscle growth and fat loss, more strength and energy, better heart, brain and gut health, and less inflammation and risk of disease. Such minor improvements accumulate and compound over time and can add up to significant

upswings.

It would take an entire book to cover everything you see on the shelves of your local supplement store, so instead, let's just focus on the four supplements that'll be most helpful for you:

1. Protein powder
2. Multivitamin
3. Fish oil
4. Creatine

That's also their ranking. Protein powder is at the top of the pile because it makes it easy to eat enough protein. A high-quality multivitamin is next because it plugs nutritional "holes" in your diet and boosts your intake of micronutrients. Then there's fish oil because maintaining adequate omega-3 intake is tricky with food alone. And last is creatine because it's a safe and effective supplement for boosting muscle growth, strength, and post-workout recovery.

So let's learn about each of these supplements and how to choose and use them properly. Before we begin, however, full disclosure: the specific products I'll recommend aren't only the ones that I use personally—they're also produced by my sports nutrition company, Legion.

Now, before your eyes roll back so hard that your neck snaps, let me explain. Years ago, I was struggling to find the type of all-natural high-quality sports supplements that I really wanted, so I wondered: Should I "scratch my own itch" and create the products that I wish someone else was making? This wasn't an easy decision. I've made my bones as an author and educator. I've sold nearly two million books and published nearly two million words of free content on my blog and nearly a thousand episodes of my podcast. And most people think that's awesome. Go me. What would happen if I were to sell supplements, though?

I feared that no matter how good my products might be, or how honestly or fairly I might try to offer them, many of my readers and followers would assume the worst, grab their pitchforks and torches, and run me off the Internet. So I was on the horns of a dilemma. On the one hand, I saw an opportunity to do things better and create 100 percent natural, science-based, safe supplements that really work. On the other, doing so would mean entering the shady supplement industry and trying to convince people that I wasn't a shyster like everyone else. After much deliberation and many sacrificial offerings to the gods of commerce and capitalism, I went with my gut, threw my hat into the ring, and started Legion. That was 2014, and I'm glad I made that leap of faith. Legion is now the number-one bestselling brand of all-natural sports supplements in the world, with over 300,000 customers who have left us over 25,000 5-star reviews. My team's mission is to change the supplement industry for the better, and we're proud to be making significant inroads in this direction.

With that out of the way, let's learn how to get the most out of each of the four supplements I recommend.

1. Protein Powder

Whey and casein and soy, oh my! There are scores of popular brands and types of protein powder to choose from. Which is right for you? An animal-based protein powder like whey, casein, or collagen? Or maybe a plant-based product like soy, hemp, or rice? Or something else, like beef or egg protein powder? To help you decide, a top-flight protein powder meets several criteria:

1. It tastes good and mixes well. You'll be eating it every day, so it has to go down easily.
2. It has a lot of protein and little carbohydrate and fat. This keeps liquid calories to a minimum so we can eat more food.
3. It's packed with essential amino acids and is absorbed well by the body. This makes it effective for enhancing body composition.

Ideally, the product you select should also have a reasonable cost per serving (good value), and it's also wise to choose a protein powder that doesn't have artificial sweeteners or food dyes. While these chemicals may not be as dangerous as some people claim, studies suggest that regular consumption of them could be harmful to our health.

The protein powders that best meet our requirements are whey, casein, rice, and pea protein, so let's learn a bit about each, and then talk about a couple of popular powders that don't make the cut.

The Scoop on Whey Protein

Whey protein is the poo-bah of protein powder because it's mostly protein by weight, is affordable (for now, at least—the price is surging these days), is rich in essential amino acids, is digested and absorbed well, and is easy to flavor. What is

it, though? Whey is a liquid leftover of cheesemaking that was once considered waste, but in the late 1800s, scientists discovered it contained a large amount of protein. Businesspeople realized it could have commercial value and began investing in the development of technology to turn it into a food product. They succeeded, and an industry was born. Later, scientists discovered that whey is easily digested and absorbed, and brimming with the essential amino acid leucine, which, we recall, stimulates protein synthesis. These qualities made whey particularly attractive to athletes and bodybuilders, and grabbed the attention of mainstream food companies as well, who were looking for cost-effective ways to add protein to various foodstuffs like breads, baby formula, ice cream, etc. Now, with the benefits of high-protein eating known to many, whey is more popular than ever.

There are three types of whey protein powder to choose from:

1. Whey concentrate. This form of whey protein contains dietary fat and lactose, and depending on the quality, ranges from 25 (bad)-to-80 (good) percent protein by weight.
2. Whey isolate. This form is specially processed to increase its protein content and remove the fat and lactose. It's at least 90 percent protein by weight.
3. Whey hydrolysate. This form is whey concentrate or isolate that has undergone additional processing to improve digestion and absorption.

Supplement marketers will often claim that whey isolate and hydrolysate powders are better for muscle building than whey concentrate, but this isn't always true. Whey isolate has more protein by weight than concentrate and is easier to mix and digest, while whey hydrolysate may be absorbed faster (the scientific evidence is mixed), but your bottom-line results will be about the same

with each for—so long as you use a high-quality whey concentrate that's mostly protein by weight. Therein lies the rub, however—how do you know if a whey concentrate is a corker or clinker? Unfortunately, there's no way to be sure, but checking the ingredients and protein per serving can help.

Ingredients are listed according to predominance by weight (there's more of the first one than the second, more of the second than the third, and so forth), so if a "whey isolate" protein powder lists whey concentrate as its first ingredient followed by whey isolate, it contains more concentrate than isolate (and it could be mostly or even almost entirely concentrate). You can also find cheap "whey concentrate" products that are predominantly milk protein (the cheapest dairy-based protein powder on the market).

You should also look at how the amount of protein per serving compares to the weight of each serving. The two numbers will never match perfectly because unless the powder is unflavored, it'll have sweetener and flavoring and small amounts of other necessary ingredients, but a large difference suggests that something is amiss. For example, if a protein powder has 20 grams of protein and 40 grams of powder per serving, make sure those 20 extra grams are stuff that you want.

Comparing prices per serving can also be illuminating because as with most things in life, when picking a protein powder, you get what you pay for. If a protein powder (any type) is substantially less than other similar products, there's a good chance it contains inferior ingredients. My sports nutrition company, Legion, gets offers every week from Chinese suppliers of shoddy whey concentrate costing less than half of what I pay to source first-class whey isolate from small Irish dairy farms. If I were more mercenary, I could add a handsome sum to my bottom line merely by switching vendors. However, just because low prices generally mean low-grade protein doesn't necessarily mean that high prices

always indicate a premium product. For instance, some supplement companies start with a base of second-rate whey concentrate and then sprinkle in some whey isolate or hydrolysate to create a "whey protein blend" that they promote as mostly (or even wholly) isolate or hydrolysate.

"Amino spiking" is another way that supplement sellers shortchange consumers. Essentially, this involves a testing and labeling loophole that allows manufacturers to falsely claim a protein powder has more protein per serving than it really does by "spiking" it with inexpensive and nitrogen-rich amino acids such as glycine, taurine, and glutamine—molecules that are *not* protein and mostly useless for muscle building to boot.

In summary, here's how to spot a promising whey protein:

1. The first ingredient is whey concentrate, isolate, or hydrolysate. I prefer whey isolate (and sell a 100-percent whey isolate powder called Whey+) because it's highest in protein and lowest in calories per serving and has no lactose (easier on the stomach).
2. The protein per serving is similar to the overall weight of each serving.
3. It doesn't claim to contain added amino acids.

The Scoop on Casein Protein

Casein protein also comes from milk, and like whey, it's also fantastic for enhancing body composition. Where it differs, however, is in mouthfeel (it's thicker and creamier than whey), taste (milder and less sweet), and digestion rate (whereas whey digests quickly, casein is a "slow-burning" protein that results in a more stable, continuous release of amino acids into the blood). Theoretically, this final point could make casein slightly superior for muscle building, but research suggests any such benefit is probably too small to matter.

There are two types of casein protein to choose from:

1. Sodium, calcium, or potassium caseinate. These less-expensive forms of casein protein are processed to improve mixability but are digested faster than micellar casein.
2. Micellar casein. This form of casein protein is specially processed to preserve the natural structure of small bundles of protein (*micelles*), which require more time to digest than the protein in caseinate powders.

Both forms are roughly comparable for general supplementation, although research shows that less processed forms of casein (such as micellar casein) may be slightly superior for muscle growth. Thus, I recommend you choose a micellar casein (mine is called Casein+ if you want to have a look). Also, inspect the ingredients, protein, protein and powder per serving, and added amino acids like you would with a whey protein.

The Scoop on Soy Protein

Studies show that soy is a great plant-based source of protein for gaining muscle and strength, but it's also a controversial one, especially among men. Soybeans contain estrogen-like molecules called *isoflavones*, and according to some research, these molecules can produce feminizing effects in men. Other studies contradict this, however, and demonstrate that eating normal amounts of soy and isoflavones has no effect on male fertility or hormones. What's the deal? Scientists aren't sure yet. Such effects may depend on the presence or absence of certain intestinal bacteria that turn isoflavones into a stronger estrogenic chemical called *equol*, for instance. What's clear, however, is that men have no reason to fear small amounts of soy protein or other soy-derived substances (soy lecithin, for example) found in various foods and supplements.

As for a protein supplement, however, I recommend choosing something

other than soy (rice, pea, or a blend of the two if you want to stick with plant-based powder) in case you're one of the people whose hormones will be negatively impacted by it.

The Scoop on Collagen Protein

Collagen is the primary protein that comprises the connective tissues in animals, and thanks to wily supplement marketers and prominent health and nutrition gurus, collagen protein supplements are having their day. Unfortunately, they don't even deserve a spot in the wings, let alone centerstage.

As you know, the amount of essential amino acids provided by a source of protein is a major factor in its utility for improving body composition. The more essential amino acids a protein contains, the better it can nourish our muscles and support growth and repair. Collagen protein is rotten in this regard because it's short of the three essential amino acids most related to muscle building: leucine, isoleucine, and valine (also known as *branched-chain amino acids*).

Collagen protein does have one thing going for it, however: It's rich in the amino acid *glycine*, and some people believe supplementing with this may improve the look of your hair, skin, and nails. Whereas collagen protein is often priced like a premium product, glycine is cheap as chips and easy to drink in water (it's naturally sweet), so if you want to see if it can make you look prettier, buy a pound or two as a bulk powder and take 2 to 3 grams per day.

The Scoop on Rice Protein

Rice isn't known for its protein, but it makes for quite the protein powder because its amino acid profile is similar to soy's, it's about as good for building muscle as whey, and it has a naturally mild taste and pleasant texture that doesn't need much tinkering to make it appetizing.

Rice protein isolate is the best form of rice protein to buy because it contains

the most protein by weight.

The Scoop on Pea Protein

You've probably never heard a meathead brag about all of the peas he's eating to bulk up, but this wouldn't be as silly as you might think because, like rice protein, pea protein is well digested and absorbed, and thick with essential amino acids, especially leucine. Pea protein can also be mixed with rice protein to yield an amino acid profile that looks a lot like whey's, which is why this blend is often referred to as the "vegan's whey" (and why I used a rice-and-pea blend in my plant-based protein powder Plant+).

There are two forms of pea protein on the market—concentrate and isolate. I prefer isolate because most high-quality products are about 80-to-90 percent protein by weight (more protein as well as fewer carbs and fats per serving).

How to Take Protein Powder

Normally, people will have a scoop or two of protein powder before or after workouts and another scoop or two in between other meals because it makes a good snack. This works well.

You may want to rely more on powder than food to get enough protein every day, but I don't recommend this because it'll likely cause gastrointestinal issues like gassiness, bloating, and stomachaches. Tolerance varies from person to person, but for me, if I eat more than 70-to-80 grams of powdered protein per day, my stomach starts to feel off. Food-derived protein is also more nutritious than protein powder and thus contributes more to the overall quality of your diet. It also tends to be more satiating than powdered protein, which helps you better stick to your meal plan (especially when cutting).

Eating too much powdered protein in one sitting is easy to do and can upset your stomach. Without thinking, you can gulp down a large amount of protein

powder that's digested much quicker than whole food, placing an instant and intense demand on your digestive system. In just a few seconds, you can drink the amount of protein in a pile of chicken, for instance, and our bodies weren't designed to deal well with that scenario.

Here's a practical (and more comfortable) protocol for protein powder supplementation:

1. Don't get more than half of your daily protein from powders.
2. Don't eat more than 40-to-50 grams of powdered protein at a time.

Or, put differently, a scoop or two before or after your strength training workouts and another scoop or two between meals (midafternoon and before bed are popular times).

2. Multivitamin

It's no secret that the average American diet is abysmal. In 2005, research conducted by Colorado State University scientists found at least 50 percent of Americans weren't eating enough vitamin B6, vitamin A, magnesium, calcium, and zinc, and that 33 percent weren't getting enough folate. In 2017, a study conducted by scientists at Tufts University found that over 30 percent of Americans weren't getting enough calcium, magnesium, and vitamins A, C, D, and E. Other research shows that people likely aren't getting enough vitamin K and D as well, which is especially concerning as we get older because of the role these substances play in preventing disease, preserving blood vessel and immune function, boosting bone growth and repair, and protecting joint health.

If you eat according to the advice in this book, your blood work will look a lot better than most, but you can still have nutritional insufficiencies and even deficiencies depending on the exact foods you eat and how often you eat them. For example, avoid red meat, and you may not get enough zinc or B12. What's more, some nutrients (like vitamin K and D) are difficult to get enough of through food alone, even if you try.

Theoretically, then, a high-quality multivitamin can act as a nutritional "insurance policy," if you will, and help you in a few ways:

1. It can plug nutritional "holes" in your diet that are common even among people who eat well.
2. It can give you larger (and beneficial) doses of certain nutrients than you can practically get from food, like B vitamins, zinc, and chromium.
3. It can contain ingredients that you wouldn't or can't eat as food, like grape-seed extract, ashwagandha, and alpha-lipoic acid.

How do you know if a multivitamin is a winner, though? Many aren't. Your average multivitamin is a mess—a lot of stuff most of us don't need to supplement like molybdenum, B vitamins (apart from niacin), and manganese; potentially dangerous megadoses of vitamins like vitamin E and retinol (vitamin A); inexpensive and less effective forms of certain ingredients like B12 (the synthetic form *cyanocobalamin* vs. a natural form like *methylcobalamin*) and folate (the synthetic form *folic acid* vs. the natural form *5-methyltetrahydrofolate*, aka 5-MTHF); and occasionally, needlessly expensive forms of ingredients that only make for better marketing. For example, most people believe that natural vitamins and minerals are best, and while I just mentioned a couple of natural forms of vitamins that outdo synthetic ones, that isn't always true—not all natural forms of vitamins are better than artificial ones, and not all man-made forms are bad. Some natural substances are uniquely desirable, like with vitamins A and E, while some synthetic substances are more useful (folic acid vs. folate). And sometimes, it doesn't matter either way (natural vitamin C vs. ascorbic acid).

Superior multivitamins do exist, however, and here's how to find them:

1. Don't buy one-a-day multivitamins. It takes at least two-to-three pills per serving to produce a decent formulation, and upward of six-to-eight to produce a top-quality one.
2. Don't buy tablets. They're not absorbed as well as capsules.
3. Don't buy a product that includes exactly 100 percent of the *Reference Daily Intake* (the amount needed by most people daily to maintain health) of many nutrients. This may seem positive, but it's actually a sign that it's a lousy, off-the-shelf formulation that doesn't account for actual eating habits and nutritional needs. It's counterintuitive, but in a well-formulated multivitamin, some substances will be at a fraction of

RDI and others a multiple of it.

4. Don't buy a product that contains retinol. This is a form of vitamin A that shouldn't be taken orally because it can damage the liver at higher doses. Instead, pigments found in plants called *carotenoids* should be used because the body converts them into retinol as necessary and eliminates or uses the rest as antioxidants.
5. Buy a product that has a proper dose of vitamin D, 1,500-to-2,000 IU per day. Not long ago, experts believed that vitamin D was essential only for bone health, but more recent research shows otherwise. Vitamin D receptors are present in nearly every type of tissue and cell, including the brain, heart, and even fat cells, and vitamin D influences many major physiological functions, such as immune responses, metabolic activities, and cell development. Getting enough vitamin D is tricky, though, because our bodies make it when exposed to the sun, but most of us aren't in the sun enough to meet our vitamin D needs. Food isn't a good source, either, considering how little is in nutritious, relatively unprocessed foods compared to how much is needed to maintain health. Vitamin D supplementation shines, however, and research shows that 1,500-to- 2,000 IU per day is adequate for ages 19 and up.
6. If a product contains 5-methyltetrahydrofolate (5-MTHF), that's a good sign. I mentioned 5-MTHF earlier as a "natural form" of vitamin B9 (folic acid), but it's more than that—it's the substance produced by our body when we eat vitamin B9 that it then uses for various physiological processes. Studies show that many people have a genetic mutation that impedes this conversion of B9 into 5-MTHF, however, and that this defect can cause a nutritional deficiency even when folate

or folic acid intake is high through food and supplementation. This problem can be avoided by supplementing with 5-MTHF directly, however, which is equally beneficial to those with and without the genetic mutation. Many supplement creators don't know this, though, or don't want to use 5-MTHF because it's expensive, so when you find a multivitamin with it, the formulator may have had a deeper understanding of human biology than others or a larger budget or both. (And yes, I have a multivitamin as well called Triumph, and it has 5-MTHF.)

How to Take a Multivitamin

Multivitamins are best taken with food, ideally meals with a bit of dietary fat because this helps with absorption.

3. Fish Oil

Fish oil is oil from fatty fish like anchovies, herring, mackerel, salmon, and sardines, and it's an excellent source of two important omega-3 fatty acids: *Eicosapentaenoic acid* (EPA) and *docosahexaenoic acid* (DHA). Studies show that here in the West, people on average get one-tenth of the EPA and DHA needed to preserve health—an insufficiency that significantly increases the risk of heart disease, dementia, depression, Alzheimer's, cancer, and other diseases. Correcting this shortage has other benefits, too, including:

- Faster fat loss
- Less muscle and joint soreness
- Less fat gain
- Less anxiety and stress
- Better memory, attention, and reaction times
- Increased immunity
- Possibly faster muscle gain

The best way to increase your intake of EPA and DHA with food is fatty fish (ensuring to avoid those highest in pollutants), but grass-fed meat, free-range eggs, and vegetable oils can also help. Meat and eggs have much lower levels of omega-3 fatty acids than fish, however, and vegetable oils don't have EPA and DHA but *alpha-linolenic acid* (ALA), which is converted into EPA and DHA rather inefficiently.

If you don't want to eat a few servings of "cleaner" fatty fish (herring, mackerel, or salmon, for instance) per week, a fish oil supplement is the ideal alternative. There are three types on the market today, though:

1. Triglyceride. This form of fish oil is in its natural state.
2. Ethyl ester. This form of fish oil is created by replacing the glycerol molecules in natural triglycerides with *ethanol* (alcohol). This method of processing removes contaminants and raises EPA and DHA levels.
3. Re-esterified triglyceride. This form of fish oil is created by using special enzymes to convert ethyl ester oil back into triglyceride oil.

All three are workable, but the gold standard isn't the natural triglyceride oil, as you might expect. Instead, re-esterified triglyceride oil is the champ because it contains very low levels of toxins and is better absorbed than natural triglyceride and ethyl ester oil. That's why I chose re-esterified triglyceride oil for my fish oil supplement, Triton.

How to Take Fish Oil

Research shows that you can reap most fish oil's benefits by taking 500 milligrams to 1.8 grams of EPA and DHA per day (combined). It's also possible that taking 2-to-3 grams of combined EPA and DHA per day may deliver additional benefits related to physical performance and recovery.

Take your fish oil with food to help with absorption. If you bought a high-quality supplement, you shouldn't experience noxious fish-oil burps, but if you do (it can happen even with good products), store the bottle in the freezer.

4. Creatine

Most of the supplements that claim to boost muscle growth and recovery are a letdown, but not creatine—the most-researched molecule in all of sports nutrition. Creatine is composed of several amino acids (L-arginine, glycine, and methionine), exists in most cells, and acts as an energy reserve. Our body produces small amounts of creatine, and foods like meat, eggs, and fish also contain it.

Hundreds of studies have shown that creatine enhances muscle and strength gain, anaerobic endurance, and post-workout recovery, and it does these things safely, too. Research shows that creatine isn't harmful to the kidneys, as is sometimes claimed, even if you have impaired kidney function. As a precaution, however, if you have any kidney issues, speak with your doctor before taking creatine.

Creatine comes in many forms, including creatine monohydrate, buffered creatine, creatine hydrochloride, and others, and while we could spend a few pages discussing them all, let's make the long story short: powdered creatine monohydrate is the most-researched and time-proven form, and if you find the texture of regular creatine monohydrate too grainy, choose a micronized monohydrate powder (which is what you'll find in my post-workout supplement Recharge).

How to Take Creatine

Studies show that 0.02 grams per pound of body weight (3-to-5 grams for most men) of creatine monohydrate once per day is effective. While some people recommend taking upward of 20 grams per day for the first week or so ("loading phase"), this only helps you benefit from creatine faster (but not more) and it can upset your stomach, so I generally don't recommend it.

You can take creatine whenever to good effect. You don't need to take it with carbs or protein or after your workout, as some people recommend.

The *Bigger Leaner Stronger* Supplementation Plan

Here's a quick overview of how to use supplements you've just learned about to get better results on the program:

What	Why	When	How
Protein powder	Helps you eat enough high-quality protein	Whenever (before or after workouts; between lunch and dinner; and before bed are typical times)	20-to-40 grams at once, not more than half of daily protein from powder
Multivitamin	Helps you get enough nutrients that enhance health and well-being	With food	Follow instructions
Fish oil	Helps you get enough essential omega-3 fatty acids	With food	0.5-to-3 grams of EPA and DHA (combined) per day
Creatine monohydrate	Helps you gain muscle and strength and recover from your workouts faster	Whenever	3-to-5 grams per day

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Sports nutrition is a $44 billion market, and if you trust even half of the marketing hype, your shelves can quickly fill with flashy bottles and bags that do little or nothing to help you achieve your health and fitness goals faster. Remember, then, you don't need a single supplement to get the body you want, including those highlighted in this chapter. Vitamin D and fish oil are as close to

necessary as they come, but if you're in the sun enough and eat a fair amount of fatty fish, even those may not be needed. Otherwise, eating a superb diet (like I've taught you) will keep your body well nourished.

If your diet is mediocre or worse, however, or if you just want to enjoy the additional advantages of a smart supplementation plan, then the basic regime shared here will serve you well. If you'll be taking supplements on the program, order them now, and then keep reading because I have more insights and tactics to help you get the most from *Bigger Leaner Stronger*.

KEY TAKEAWAYS

- You don't need supplements to succeed with any fitness program, but there are supplements that can benefit your body and mind in many ways.
- Top-flight protein powder mixes well, contains a lot of protein and little carbohydrate and fat, is packed with essential amino acids, and is absorbed well by the body.
- Whey protein is the poo-bah of protein powder because it's mostly protein by weight, is affordable, is rich in essential amino acids, is digested and absorbed well, and is easy to flavor.
- A high-quality whey protein is easy to spot: the first ingredient is whey concentrate, isolate, or hydrolysate; the protein per serving is similar to the overall weight of each serving; and it doesn't claim to contain added amino acids.
- The best form of casein protein powder is micellar casein because it's specially processed to preserve the natural structure of small bundles of protein (*micelles*), which require more time to digest than the protein in caseinate powders.
- Soybeans contain estrogen-like molecules called isoflavones, which can produce feminizing effects in some men, which is why I recommend choosing something other than soy.
- Collagen protein is short of the three essential amino acids most related to muscle building: leucine, isoleucine, and valine (also known as branched-chain amino acids).
- Rice makes for quite the protein powder because its amino acid profile is similar to soy's, it's about as good for building muscle as whey, and it has a

naturally mild taste and pleasant texture that doesn't need much tinkering to make it appetizing.

- Pea protein can also be mixed with rice protein to yield an amino acid profile that looks a lot like whey's, which is why this blend is often referred to as the "vegan's whey."
- Don't get more than half of your daily protein from powders, and don't eat more than 40-to-50 grams of powdered protein at a time.
- Here's how to find a superior multivitamin:
    - Don't buy one-a-day multivitamins.
    - Don't buy tablets. They're not absorbed as well as capsules.
    - Don't buy a product that includes exactly 100 percent of the *Reference Daily Intake* (the amount needed by most people daily to maintain health) of many nutrients.
    - Don't buy a product that contains retinol.
    - Buy a product that has a proper dose of vitamin D, 1,500-to-2,000 IU per day.
    - If a product contains 5-methyltetrahydrofolate (5-MTHF), that's a good sign.
- Fish oil is oil from fatty fish like anchovies, herring, mackerel, salmon, and sardines, and it's an excellent source of two important omega-3 fatty acids: eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA).
- Re-esterified triglyceride fish oil is the gold standard because it contains very low levels of toxins and is better absorbed than natural triglyceride and ethyl ester oil.
- You can reap most fish oil's benefits by taking 500 milligrams to 1.8 grams of EPA and DHA per day (combined).

- Creatine enhances muscle and strength gain, anaerobic endurance, and post-workout recovery, and it does these things safely, too.
- Taking 0.02 grams per pound of body weight (3-to-5 grams for most men) of creatine monohydrate once per day is effective.

PART 6

The Beginning

18

Frequently Asked Questions

Courage doesn't always roar. Sometimes courage is the quiet voice at the end of the day saying, "I will try again tomorrow."

—MARY ANNE RADMACHER

At this stage, we've covered all of the most important aspects of the *Bigger Leaner Stronger* program, but you may have lingering questions or uncertainties. Let's see if we can put them to rest.

"I'm in my 30s/40s/50s+. Can I do this program?"

Absolutely. Every week, at least a few people ask me if it's too late to build muscle and get fit, and most are pleasantly surprised when I explain that it's definitely *not* too late. How should people in their 40s and beyond go about it, though? Certainly they can't eat and train like the 20-year-olds, right? Maybe not, but there aren't nearly as many changes to make as you might think.

First, let's look at muscle building. In a study conducted by scientists at the University of Oklahoma, college and middle-aged men followed the same weightlifting routine for eight weeks, and both gained about the same amount of muscle and strength. People in their 60s and beyond aren't left in the cold, either. Studies show that they too can gain significant amounts of muscle and strength, and more importantly, that strength training helps fight the "dwindling health spiral" normally associated with aging.

Then, there's fat loss. Many middle-aged folk believe that their metabolism craters as they get older, but this is mostly a myth. Research shows that the average adult's metabolism slows by just one to three percent per decade—primarily from muscle loss, not biological programming. Therefore, if you maintain your muscle as you age, you maintain your metabolism; similarly, if you add muscle to your frame, you can improve it.

Why do so many people gain weight as they get older, then? For most, it's mainly a matter of lifestyle. They were far more active when they were younger, which allowed them to eat far more food without gaining weight, but now they're mostly sedentary, which makes it very easy to overeat. Unless you've lost significant amounts of muscle through practices like extreme starvation dieting or excessive cardio, your metabolism should be just fine. And even if you have made those mistakes, you can correct them now with proper diet and training.

The same goes for your hormones. It was once believed that the sometimes severe hormonal disturbances associated with aging were inevitable, but we now know this isn't true. Studies show that lifestyle factors are as equally responsible for hormonal changes as aging itself, if not more so. For example, here's a short list of the biggest lifestyle factors that can depress your testosterone levels:

- Gaining weight
- Stopping exercise
- Suffering chronic illness
- Using certain medications
- Sleeping too little
- Consuming large amounts of alcohol

All of these factors are under your control. More to the point, however, is that you don't need stellar hormone levels to get fit. It can speed up the process, but so long as you're willing to work at it, you can have an above-average physique with below-average hormones.

That said, there are several key differences between younger and older bodies that make fitness harder as you get older. Research shows that tendons and ligaments become stiffer and recover slower as we age, for example, and that after about age 50, muscles also recover slower from exercise. Therefore, if you're middle-aged, I recommend taking a few precautionary measures.

1. Be a stickler for form.

The older you get, the fewer shenanigans you can get away with in your strength training. Lumbar rounding in your deadlifts … knees bowing in your squats … elbows flaring in your bench pressing … it all increases the risk of injury at any age, but it becomes more dangerous as the years go by. This is why I put a lot of

emphasis on learning and using proper form from day one, regardless of age or fitness level. It's also why I'm not willing to sacrifice form to hit PRs and chest bump with my buddies. I'm not a competitive powerlifter or strength athlete. I like lifting heavy things and being strong, but I like staying healthy and injury-free more. I suspect you're in the same boat.

2. Make sure you take at least one day off strength training each week. Two is better.

Don't underestimate how taxing strength training is on your body. Even the young'uns can't do it every day, week after week, without eventually running themselves into the ground. Train hard six or seven days per week, and physical fatigue will start accumulating. Your sleep will deteriorate. Your workouts will suffer. You'll continue to feel worse and worse until you rein it in and give your body more time to rest. Proper recovery requires taking time off the weights every week and resisting the urge to replace strength training with some other form of intense physical exercise or activity.

Also, don't fall into the trap of using exercise (particularly cardio) as a way to eat more food. That's a one-way street to exhaustion, not to mention life and psychological imbalances. Forcing yourself to do excessive amounts of exercise just so you can gorge on food is a terrible use of time as well as a great way to develop anxiety or dysfunctional eating habits.

3. Rest or deload more frequently.

Even when you're properly managing your workout volume and intensity and taking a couple of days off each week, your body eventually needs a bigger break. And the older you are, the sooner that time comes. While guys and gals in their 20s can go anywhere from twelve-to-fifteen weeks or longer before needing additional recovery time, people in their 40s and 50s need it more frequently,

sometimes as often as every four-to-six weeks.

There are many factors that determine how long you'll be able to go before needing additional rest—age, training programming and history, genetics, sleep hygiene, diet, etc.—but it's pretty easy to discover for yourself. As you exceed your body's ability to fully recover from your training, you'll start noticing symptoms like more difficult workouts, worse sleep, lower energy levels, joint aches and pains, and less interest in training. Many people mistake these signs as mental obstacles to push through. It doesn't go well. Deloading does. Listen to your body, play the long game, and play it intelligently.

"I travel a lot. Can I follow this program?"

Yes, but you'll have to think ahead. Staying in hotels close to good gyms helps a lot (hotel gyms are usually sparse), as does figuring out beforehand when you can train. If your regular workouts aren't going to happen when you're out of town, that's okay. Anything is better than nothing, so do what you can, even if it's just bodyweight training or cardio. You can use the training principles you've learned in this book to make those workouts plenty productive.

As for your diet, you have three options when traveling:

1. Make a meal plan with foods you can get at a local grocery store (or get delivered to you with a service like Instacart, Walmart+, Amazon Fresh, or Whole Foods Delivery) and then keep, prepare, and eat your meals in your hotel room. Keep it simple, like cold cuts, rotisserie chicken, high-protein dairy, protein powder, protein bars, precut vegetables/crudités, fruit, nuts, etc.
2. Track your calories and macros with an app like MyFitnessPal, try to ensure you eat enough protein, and keep your calories in a reasonable range. This can work well if you have to mostly eat out or on the go.
3. Eat according to your appetite, try to eat enough protein, and keep your calories in check. This is fine for the occasional trip, but if you travel frequently, you'll get better results with the other two approaches.

"What should I do if I miss or have to skip a workout?"

If you miss a strength training workout or two in a week, you can either do them on other days of the week, or skip them and carry on as if you'd done them, depending on your circumstances and preferences. For example, let's say your normal schedule looks like this:

Mon.	Tue.	Wed.	Thu.	Fri.	Sat.	Sun.
Push	Pull		Upper Body	Lower Body		

You miss your Pull workout for one reason or another on Tuesday. You could do it the following day (Wednesday); or, if you don't like doing three workouts in a row, you could do it Wednesday, rest on Thursday, and do your Upper Body workout on Friday and Lower Body workout on Saturday. If you can't shuffle your workout schedule to make up for a missed workout, that's all right. Keep your hair on and continue as you normally would.

If you miss an entire week or two of training, you should be able to pick up where you left off because it takes several weeks without working out to lose a noticeable amount of muscle or strength. If you've been off your routine for several weeks or longer, however, you'll need to use lighter weights when you begin again. There is a bright spot, however: Thanks to "muscle memory," you'll quickly gain back whatever muscle and strength you've lost. Scientists are still studying this phenomenon, but research shows that strength training alters muscle cells in a way that allows for rapid regrowth.

When you're coming back to the program after an extended break, plan on reducing your training weights by the following:

- 20 percent if it has been one to two months
- 30 percent if it has been three to four months

- 50 percent if it has been five to six months or longer

"What if I have to cut a workout short?"

If you don't have time for a full workout, do what you can with the time that you have, even if that's just the first two exercises. Something is always better than nothing. Remember too that the most important exercises come first in *Bigger Leaner Stronger* workouts, so an abbreviated session can do more than you might think.

Also, like with missing a workout, shortening them isn't an issue if it happens infrequently, but if it's a regular occurrence, you need to revisit your workout schedule, priorities, or both. Also, if you cut a workout short, don't tack on what you skipped to your next workout—just carry on as normal.

"What can I do to be less hungry while cutting?"

Here are three easy ways to reduce hunger and cravings while cutting:

1. Drink enough water. This isn't just one of the easiest ways to improve general health and performance—research shows that drinking plenty of water (especially with meals) also increases feelings of fullness and reduces hunger levels.
2. Get enough sleep. Most people wouldn't think of this as a weight loss tip, but inadequate sleep can markedly increase hunger levels. Scientists at the European Center for Taste Sciences found that when people slept four hours per night, they ate almost 600 calories more the next day and were more hungry before meals than people who got eight hours of sleep.
3. Eat slowly and thoughtfully. Instead of scarfing your meals while scrolling on social media, take smaller bites, deliberately chew and enjoy each mouthful, turn off the screens, and focus on your food and company instead. You'll probably find that you get fuller on fewer calories and feel satisfied for longer afterward.

“I’m not getting very sore. Is that a problem?”

No. Muscle soreness is a byproduct of muscle damage, and while this may contribute to muscle growth, it’s not essential. While you can expect at least some soreness after workouts (if you never experience any soreness whatsoever, you’re probably not training hard enough), it doesn’t need to be (and shouldn’t be) exaggerated.

“Can I train muscles that are still sore from a previous workout?”

Yes. If a muscle is still sore, that doesn’t necessarily mean it isn’t recovered and ready for more training. In fact, some research suggests that much of the soreness we feel after training isn’t from the muscle fibers themselves but the connective tissues holding them together. A much better measure of muscle recovery is performance. Say you can normally do 6 reps of dumbbell curls with 35 pounds, and your biceps are sore from yesterday’s workout. Today, if you can only get 4 reps with 35 pounds, your biceps probably haven’t fully recovered yet. If you can get 6 reps or more, though, they probably have, regardless of how they feel.

"How do I switch from cutting to maintaining or lean gaining?"

To go from cutting to maintaining, simply increase your calories to 100 percent of your newly calculated maintenance calories (there's no need for a gradual increase of calories, sometimes referred to as "reverse dieting"). When you do this, you can expect to gain weight for the first week or two mostly from the jump in carbs (which increases water retention), but after that, your weight should stabilize.

To go from cutting to lean gaining, first increase your calories to 100 percent of your newly calculated maintenance calories, and stay there for two or three weeks. Then, increase your calories again by 10 percent (110 percent of maintenance calories).

"I'm having trouble eating enough to gain weight. What should I do?"

You could double-fist cheeseburgers and pizza every day, but instead, I'd recommend figuring out how to eat more of "healthier" high-calorie foods. This is the opposite of how you approach meal planning when cutting. Instead of filling your diet with lower-calorie foods with a lot of fiber and water (fruit and vegetables, for instance), you load up on foods that pack a lot of calories into every square inch. Here are some of my favorites:

- White rice, bread, and pasta
- Dried fruit
- Whole eggs
- Fatty meat and seafood (ground beef, steak, fish, etc.)
- Oats
- "Healthier" breakfast cereal (minimal ingredients and added sugar, such as granola or muesli)
- Full-fat dairy or dairy substitutes (cheese, yogurt, milk, etc.)
- Seeds, nuts, and nut butters
- High-calorie protein smoothies (protein powder plus milk or a substitute, nuts, oats, etc.)
- No-sugar-added fruit juice
- Sauces like pesto, mole, aioli, chimichurri, teriyaki, and gravy
- Higher calorie/carb veggies and beans (lima beans, corn, chickpeas, etc.)

"How often should I change my workout routine?"

If you follow the training phases in the bonus material that comes with this book (www.biggerleanerstronger.com/bonus), changes are baked into the cake—you'll switch up exercises every phase. If you plan on creating your own workouts and routines, however, keep these points in mind:

1. Don't change your workouts (exercises, rep ranges, rest times, etc.) more than once every six to eight weeks. If you change your workouts too often, it'll get in the way of your progress because you'll waste time and energy adjusting to new exercises and rep ranges, finding new working weights, etc.
2. Do your hardest exercises first. You always start a workout with more energy than you finish with, so it makes sense to do the most demanding training first, followed by easier exercise. For example, if you're going to do an upper-body workout that includes the barbell bench press, barbell row, barbell biceps curl, and dumbbell side lateral raise, do it in that order.
3. Change primary exercises less frequently than accessory exercises. As a general rule, you should try to stick with a primary exercise for at least three to four months, as they take longer to master and produce steadier and greater whole-body gains than accessory exercises. Also, when you do swap a primary exercise out, replace it with a similar variation. For example, after several months of the barbell bench press, the incline bench press would be a better alternative than push-ups or machine flyes.

"Should I exercise when I'm sick?"

No. At least not seriously. As much as you may want to shrug illness off and stick with your routine (a desire I completely sympathize with), force yourself to rest because training will only depress immune function and make things worse. That said, research shows that twenty minutes or so of light activity, like walking or jogging, can boost immunity and recovery, so that's an option if you feel up to it. Personally, I take one or two 15-to-20-minute walks and one 30-to-45-minute nap per day when sick, and I notice immediate benefits from each.

"What do people who achieve their fitness goals all have in common?"

I've seen tens of thousands of transformations over the years, and invariably, the people who lose the most fat, gain the most muscle, and then stay fit (or even continue getting fitter) aren't abnormally disciplined, determined, or durable—they're just the most consistent. They make the fewest nutritional mistakes and miss the fewest workouts. They're far from perfect, too. They're just good enough most of the time.

I'm confident you're capable of the same. Why? Because you've made it to the end of this book, which tells me you have the will and wherewithal to win with fitness. I've given you the way. Follow it, and I'll be waiting for you on the other side.

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From Here, Your Body Will Change

To be yourself in a world that is constantly trying to make you something else is the greatest accomplishment.

—RALPH WALDO EMERSON

So ... I guess this is it, right? We've reached the end. No way. You're in a process now—and yes, it has already begun—of transforming your body faster than you ever thought possible. From here, all you have to do is walk the path I've traced for you, and soon, you'll be able to look in the mirror and think, "I'm glad I did," not "I wish I had."

You won't just be bigger, leaner, and stronger, either—you'll also be more confident, capable, and cheerful. This newfound positivity and pride will shine into other areas of your life as well, inspiring you to reach for other goals and improve yourself in other ways. As I say in my book *The Little Black Book of Workout Motivation*: if you have the power to change your body, you have the power to change your life.

My goal is to help you reach your goals, and if we can work together as a team, I know that we can and will succeed. That's why I'd love to connect with you, follow your progress, and address any questions or concerns you may have.

Also, once you're thrilled with your transformation, I'd love to feature your story on my website! The best way to reach me is e-mail: mikem@legionsupplements.com. I get a lot of messages every day, so it may take a week or two to get to yours, but you will get a reply.

I'd also like to invite you to join my Facebook group, which is a community of thousands of like-minded people who can answer questions, celebrate victories, and soothe setbacks. You can find it at www.muscleforlife.group. All you have to do is click the "+ Join Group" button, and one of my administrators will let you in.

Also, speaking of social media, here's where to find me on the big ones:

- Instagram: www.instagram.com/muscleforlifefitness
- Facebook: www.facebook.com/muscleforlifefitness
- YouTube: www.youtube.com/muscleforlifefitness
- Twitter: www.twitter.com/muscleforlife

If you want to publicly announce that you're starting the program or share progress updates, definitely tag me and add the hashtag #BiggerLeanerStronger so other people can find you and follow your journey.

Also, if you've enjoyed this book, please pass it on to someone you care about. Lend them your copy, or, better yet, get them their own as a gift and say, "This book has helped me in my fitness journey, and I think it'll do the same for you, so I got you a copy. Read it." My mission is to get the information in this book into as many hands as possible, and nothing helps me do that more than plain word of mouth. So please spread the word however you can.

Thank you again for reading my book, and I hope to hear from you soon.

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Would You Do Me a Favor?

The only easy day was yesterday.

—U.S. NAVY SEALS MOTTO

I have a small favor to ask.

Would you mind taking a minute to write a blurb on Kobo about this book? I check all my reviews and love to get honest feedback because that's the real pay for my work—knowing that I'm helping people.

To leave me a review, you can:

1. Go to www.biggerleanerstronger.com/kobo, scroll down and click the "Write your review" button, and leave your rating and review.
2. Or go to Kobo in your web browser (or open the app), search for "bigger leaner stronger," click on the book, scroll down and click the "Write your review" button, and leave your rating and review.

Thanks again, and I look forward to reading your review!

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Free Bonus Material (Workouts, Meal Plans, and More!)

My dear fellow, who will let you? That's not the point. The point is, who will stop me?

—AYN RAND

Thank you again for reading this book. I hope you've found it insightful, inspiring, and practical, and I hope it helps you build the fit, strong, and healthy body you really desire.

I want to make sure that you get as much value from this book as possible, so I've put together a number of additional free resources to help you, including:

- A savable, shareable, printable reference guide with all of this book's key takeaways, checklists, and action items.
- Links to form demonstration videos for all *Bigger Leaner Stronger* exercises.
- An entire year's worth of *Bigger Leaner Stronger* workouts neatly laid out and provided in several formats, including PDF, Excel, and Google Sheets. If you'd prefer the workouts in a digital or hard-copy book, check out *The Year One Challenge for Men* (www.).

- Twenty *Bigger Leaner Stronger* meal plans that make losing fat and gaining lean muscle as simple as possible.
- My favorite tools for getting and staying motivated and on track inside and outside of the gym.
- My all-time favorite fitness books.
- And more.

To get instant access to all of those free bonuses (plus a few additional surprise gifts), go here now:

www.biggerleanerstronger.com/bonus

Remember too that if you have any questions or run into any difficulties, you can shoot me an email at mikem@legionsupplements.com, and I'll do my best to help!

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Do You Want One-on-One Coaching?

Have you ever wished you could hop into a DeLorean, go back in time, and shake some sense into your younger self? I know, right?

I mean, I spent a good seven years spinning my hamster wheel in the gym without really knowing what I was doing. So many unproductive "pump" workouts, so many hours on the cardio machines, so many stinky plastic containers of chicken, broccoli, and rice … and so little progress to show for it.

Oh, the wine of youth …

And here's a wild thought: With what I know now, I could've reached the pinnacle of my personal fitness in three-to-five years instead of ten+. And I could've done it spending a fraction of the time in the gym, doing workouts I enjoyed, and eating foods I liked. That's what a "clinically effective dose" of the right kind of knowledge (wisdom) can do for you.

Think of it this way: Imagine a plane taking off from Los Angeles en route to New York. If, during takeoff, the pilot decided to adjust course 3.5 degrees to the south, the plane's nose would move just a few feet. Outside of the cockpit, no one on board would notice the small movement. But over the course of a journey across the country, the impact of the change would be considerable, causing the confused passengers to alight from their plane in Washington, DC, not New

York—a small change, but a major result.

For instance, if I had to start over, here are a few things I wouldn't do again:

- I'd stop paying to be lied to by the muscle mags.
- I'd stop trying to copy the workouts of the biggest Joe-bros online.
- I'd stop wasting hundreds of dollars per month on sham pills, powders, and potions.

And there are, oh, a couple dozen or so things I'd do differently in round two, like ...

- Lift heavier weights
- Focus on progressive overload
- Do the right exercises
- Learn how to make effective meal plans
- Break up with the treadmill ("I'm so miserable without you, it's almost as if you're here!")

But if I could choose just ONE thing to force my younger self to do, it'd be this: I'd hire a coach.

Why?

A good coach would've saved me Cthulu-knows-how-much time, energy, and frustration lost trying to figure everything out through study and trial and error. In fact, even if I had the game taped like I do now, I'd still hire a coach if I were starting over. The upside is just too great to ignore. A coach would save me time by building my training and eating plans, and might even do a better job than I would as an objective expert. A coach would help keep me accountable, track my progress, and bring to light "blind spots" that we all have. And frankly, a

coach would scowl me straight on those days when I just don't feel like picking up the heavy things and putting them down (Yes, I have those too—I blame inflation and climate change).

All this is why even the greatest athletes of all time, like Tiger Woods, Michael Phelps, and Tom Brady, still have a coach—and why you should too if you feel like you've been putting in the effort but not getting the results.

If that's the case, then I'd like to invite you to start fresh with my VIP coaching program, which you can learn more about at www.biggerleanerstronger.com/coaching.

We've worked with thousands of men and women of all ages and circumstances and helped them lose tens of thousands of pounds of fat and gain thousands of pounds of muscle. And you know what? They're often missing just *one* crucial piece of the puzzle. Maybe it's their calories or macros ... maybe it's their exercise selection ... maybe it's their food choices ... or maybe they're not progressively overloading their muscles.

And it's probably the same with you. You're likely doing a lot of things right, but sure as a goose goes barefoot, there's something you're *not* doing correctly that's giving you most of the grief.

Whatever it is, here's what's important: Once you identify what that one thing you're missing is—once you "figure it out"—that's when everything finally clicks. *That's* when you start making serious progress.

This is exactly what we did for Andrew (40 years old), who dropped 10 pounds of fat while also adding 100 pounds to his key lifts:



It's also how we helped Ron (63 years old), who lost 17 pounds, 10% body fat, and 5.5 inches off his waist:



And we did the same for Nick (23 years old), who lost 18 pounds and 7% body fat and went from fit to absolutely ripped:



And get this:

Andrew, Ron, and Nick all made these radical transformations in as little as 90 days.

So, if you want someone to take you by the hand and help you build the body you really want ...

Someone with a proven track record working with people just like you ...

And someone who can make it paint-by-numbers simple and more or less guarantee your success ...

Then head over to www. now, schedule your free consultation call, and let's see if my coaching service is right for you.

This is NOT a high-pressure sales call, by the way. It's a friendly chat that lets us learn about you, your goals, and your lifestyle, and then determine whether the

program is right for you. (And yes, sometimes we do speak with people who aren't a good fit, but we almost always have other experts and resources to refer them to.)

If you feel even slightly interested, you should act now and schedule your free call. You have absolutely nothing to lose and everything to gain because in a few months, YOU could be one of our next success stories!

In fact, if you qualify for the program, show up every day, and do the work, I guarantee it or you'll get your money back. So don't put this off for later and forget. Take the first step: schedule your call now at www..

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Going Beyond Bigger Leaner Stronger

If you obey all the rules, you miss all the fun.

—KATHARINE HEPBURN

If your goal is to gain much of the muscle and strength genetically available to you (at least 20-to-35 pounds of muscle and solid intermediate-level strength for most guys), *Bigger Leaner Stronger* is the last fitness program you'll need. Follow it diligently, and your success is as good as guaranteed. If you want to realize your *full* fitness potential, however, this book likely won't be able to get you there. For that, you'll probably need to make some changes to how you eat and train at some point in the future.

For example, after one-to-two years of *Bigger Leaner Stronger*, many people stop gaining muscle and strength mostly because the program no longer provides enough volume for an intermediate weightlifter to continue making progress. That said, some find they can follow *Bigger Leaner Stronger* for three, four, or even five years with consistent muscle and strength gains. If you're one of them, you don't have to change anything so long as you're still making steady progress. But if you aren't, you might get even better results with my book (and program) for intermediate and advanced weightlifters, *Beyond Bigger Leaner Stronger*.

Another way of determining when to transition from *Bigger* to *Beyond Bigger Leaner Stronger* is strength levels. If you haven't hit at least two of the following targets, chances are you can still profit from *Bigger Leaner Stronger* and don't need to switch just yet.

- Squat one-rep max: 1.5 x body weight
- Bench press one-rep max: 1.2 x body weight
- Deadlift one-rep max: 2 x body weight
- Overhead press one-rep max: 0.8 x body weight
- Chin-up or Pull-up: 8 reps with bodyweight or at least 1 rep with bodyweight plus 20 percent

Why two out of five targets for moving to *Beyond Bigger Leaner Stronger*? Because it's normal for people to be above average in a couple of key lifts and below average in the others. For instance, I have long legs and arms, so I've always struggled with the bench press, overhead press, and squat, but less so on the deadlift, because the advantage provided by my arms negates the disadvantage of my legs.

To get a gist of the *Beyond Bigger Leaner Stronger* program, download the free bonus material that comes with this book, and in it, you'll find a crucial chapter from *Beyond Bigger Leaner Stronger* called "The 'More for Less' Method of Maximum Muscle Gain." If you like it, go ahead and read the entire book even if you're not ready for the program because you'll immediately benefit from the new diet tools and tips as well as the deeper understanding of the mechanics and methods of effective muscle building.

PART 7

Appendix

The Hidden Barrier to Effective Eating & Exercising

The beginning of wisdom is the definition of terms.

—SOCRATES

There once was a Harvard graduate named Johnson O'Connor who spent his early years studying astronomical mathematics under the famous astronomer Percival Lowell. In the 1920s, General Electric hired O'Connor to observe and analyze its successful employees and discover which traits they had in common that made them good at their jobs. The company then wanted to be able to test new hires and, based on the results, assign them to jobs that best fit their personalities and skill sets. This project was the beginning of what would become O'Connor's life's work: the study of human talents and learning.

To expand his efforts, in 1930, he created the Human Engineering Laboratory at Stevens Institute of Technology and worked diligently to gather data on skills specific to various professions as well as data regarding learning and ability in general. He then launched a research project to determine which traits or talents were more important than others in becoming successful and advancing in one's career.

In this investigation, he made an unexpected discovery. O'Connor found a person's vocabulary level was the best single predictor of long-term success in all

disciplines and endeavors that he analyzed. In other words, an understanding of not only general language but of the words specific to the activity was the most important factor that separated the unsuccessful from the successful. This discovery sparked in O'Connor a fascination with language and its connection with skill and success. In his later writings, he concluded the understanding of words was a major key to unlocking human potential.

Why is this so? His hypothesis was that since words are the tools with which we think and interpret ideas and reality, the more words we understand, the more perceptive, subtle, and versatile our minds are; and the more robust our minds, the better we can reason and make constructive decisions that result in ever higher levels of achievement.

Interestingly, O'Connor also fiercely opposed educators who believed only the usage of words mattered and that standard, precise definitions, such as those found in a dictionary, were irrelevant. "We can't let the ignorant define our words for us," he argued.

For my part, I wholeheartedly agree with O'Connor's conclusions, and one of the primary reasons I read every day is to expand my vocabulary. I've found that the more words I learn, the better I can command and connect ideas in my mind and understand the people and world around me. Accordingly, I spend a fair amount of time in the dictionary when I read, checking and clarifying the meaning of words. In fact, I'd estimate that about 30 percent of the time I spend reading is invested in the dictionary, learning definitions, making sample sentences, and reviewing etymologies. And yes, I do this regardless of what I'm reading, whether an article, book, or tweet, or what I'm watching or listening to, including movies, TV shows, podcasts, etc. The frequent interruptions to consult the dictionary can be annoying at times—I'd prefer to just blaze through the material and move on to something else—but I believe enough in the dividends

of the more studied approach to continue with it.

Paying close attention to your understanding of words has another practical benefit: It forces you to slow down and absorb and analyze what you're reading or hearing word by word, as opposed to sailing through sentences and paragraphs believing you're understanding and retaining more than you actually are. Incidentally, research shows that this is probably one of the reasons that people tend to remember more of what they read in print than on a screen (most read digital content faster). If you want to upgrade not just your ability to comprehend and remember information but the foundation of your entire intellectual "operating system," so to speak, get a dictionary and start using it liberally.

Now, what does any of this have to do with getting fit? Well, have you ever wondered why so many people find diet and exercise so confusing? Why are these topics so rife with unworkable, conflicting, and illogical advice and ideas? You know, nonsense like:

- "Counting calories doesn't work."
- "Exercise doesn't help you lose weight."
- "Broccoli has more protein than chicken."
- "Weightlifting is a waste of time."
- "Foods that spike insulin levels make you fat."
- "Hormones drive fat loss."
- "Eating a lot of 'healthy' dietary fat keeps you lean."

These examples are but a few of the many false fitness mantras that continue to maintain currency, despite decades of scientific and anecdotal evidence to the contrary. Why are so many people so susceptible to howlers like these? Would you believe that at least a large part of it can be chalked up to nothing more than

misunderstood words?

You can experience this firsthand. The next time you hear someone declare that "calories in versus calories out is obsolete," ask them this simple question: What is a calorie? Dollars to doughnuts your question will be met with silence or gabble because they don't have a clue what the word means, let alone what "calories in versus calories out" means or how the related metabolic processes work. Chances are your experience with such a misinformed individual will be something like when Jimmy Kimmel asked random people in Los Angeles to define *gluten*. The best response any of them could muster was, "It's like a grain, right?" and my personal favorite, "It's a flour derivative of wheat . . ." (In case you're wondering, gluten is a mixture of two proteins present in many grains, which is responsible for the elastic texture of dough.)

"Calorie" and "gluten" are only the very top of the tip of the proverbial iceberg, however. What is protein? What is carbohydrate, sugar, body fat, muscle, metabolism, insulin? What is a hormone? Very few people can answer these questions simply and accurately, so of course they spin around in circles, believing nearly anything they're told. How can they possibly gain a proper understanding of how to get healthy and fit when they don't even understand the words used to discuss many of the most important concepts?

When you're learning something new, the first major hurdle to clear must be to learn the precise meanings of key words if you're to have any hope of gaining a deep and practical understanding of it. For example, if I were to tell you, "The children have to leave in the gloaming," you might wonder what I mean because you don't understand the word *gloaming*. The sentence doesn't give you any hints, as it could mean early, midday, late, or something else altogether. In school, most of us were taught to guess at word meanings by looking at the surrounding context or by comparing them to other words in our vocabulary. That doesn't

help much in my example, though, because the context only reveals that gloaming may be a time of day. You can then analyze the word itself, but that doesn't offer any clues either. "Well, *gloaming* sounds like *glowing*," you may surmise, "and the sun glows, so I guess it could mean 'in the morning'?" This is a hinky way to communicate and learn. Without a precise, standard, and universal lexicon, we can never really know if we're understanding others correctly or vice versa. It's like trying to play a game without first agreeing on the rules.

So, what is *gloaming*? It means the time of day when the sun is just below the horizon, especially the period between sunset and dark. In other words, twilight. The sentence I shared earlier is now crystal clear, isn't it?

This is why the humble dictionary is an unsung hero of culture and civilization. It provides the intellectual bedrock upon which all ideas are formed and disseminated. (It's also why, in Orwell's *1984*, the authoritarian government made every effort to control the meaning of words—to "make thoughtcrime literally impossible, because there will be no words in which to express it"— but that's another discussion.)

In this chapter, we'll review the dictionary definitions of a number of the most important words related to diet, nutrition, and exercise. We'll also flatten the learning curve by starting with the most elementary terms and progressing gradually to the more complex, in the same way we'd learn to play notes before attempting chords, let alone songs. Also, in case you're wondering, every definition in this chapter comes directly from, or is a combination of definitions from, one of the following dictionaries:

- New Oxford American Dictionary, Third Edition (my personal favorite dictionary)
- *Webster's Third New International Dictionary, Unabridged*

- *Random House Unabridged Dictionary*

I highly recommend that you purchase one of these dictionaries, or all three (great for cross-checking definitions), and start using them to clarify the meanings of words you don't understand.

At any rate, don't underestimate the destructive power of misunderstanding the jargon of something you're trying to learn and apply. It alone can be the difference between success and failure. All it takes is a few major misconceptions to make entire categories of knowledge seemingly incomprehensible.

All right, let's get to the key words, starting with those related to diet and nutrition.

Diet Keywords

Acid

An acid is a chemical substance that can react with and sometimes dissolve other materials.

Amino Acid

An amino acid is a naturally occurring compound found in proteins.

Blood Sugar

1. Blood sugar is glucose in your blood.
2. Blood sugar refers to the concentration of glucose in your blood, measured in milligrams of glucose per 100 milliliters of blood.

Calorie

When referring to food, a calorie is the energy needed to raise the temperature of 1 kilogram of water by 1 degree Celsius. This is also called a *kilocalorie* or *large calorie*.

Carbohydrate

A carbohydrate is a molecule composed of carbon, oxygen, and hydrogen that can be broken down in the body to create energy.

Celsius

Celsius is a scale of temperature on which water freezes at 0 degrees and boils at 100 degrees. In the Fahrenheit scale used in the United States, water freezes at 32 degrees and boils at 212 degrees.

Chemical

1. Chemical (adj) means having to do with chemistry or the interactions of substances as studied in chemistry.
2. A chemical (n) is any substance that can undergo a chemical process or change. When people refer to chemicals, they're usually talking about manmade substances, but the definition isn't limited to just this meaning.

Chemistry

Chemistry is the branch of science concerned with the substances of which matter is composed, the investigation of their properties and reactions, and the use of such reactions to form new substances.

Cholesterol

Cholesterol is a soft, waxy substance found in most body tissues that's an important part of the structure of cells and used to create different hormones.

Complex Carbohydrate

A complex carbohydrate is a form of carbohydrate consisting of a chain of simple carbohydrates linked together. Because of this structure, a complex carbohydrate takes longer to break down into glucose in the body.

The sugars found in whole grains, beans, and vegetables are complex carbohydrates.

Compound

A compound is a substance made up of two or more different elements.

Diet

1. A diet is the food and drink that a person usually consumes.
2. A diet is a special course of controlled or restricted intake of food or drink for a particular purpose, such as weight loss, exercise support, or maintenance therapy (a treatment designed to help another primary treatment succeed).

Element

An element (also called a *chemical element*) is a substance that can't be broken down into smaller parts by a chemical reaction. There are more than 100 elements, and they are the primary building blocks of matter.

Energy

1. Energy is the power received from electricity, fuel, food, and other sources to do work or produce motion.
2. Energy is the physical or mental strength of a person that can be directed toward some activity.

Essential Amino Acid

An essential amino acid is an amino acid needed by the body to maintain growth and health that must be obtained from food.

Essential Fatty Acid

An essential fatty acid is a fatty acid vital for proper bodily function that must be obtained from food.

Fat

1. Fat is an oily or greasy substance found in animals, deposited as a layer under the skin or around certain organs.
2. Fat is an oily or greasy substance derived from animals and plants that's solid or liquid in form and often used in cooking.

Fatty Acid

A fatty acid is an acid found in the fats and oils of animals and plants.

Fiber

Fiber is a mostly indigestible type of carbohydrate found in many types of foods, including fruits, vegetables, legumes, and grains.

Fructose

Fructose is a sweet sugar found in many fruits and honey as well as sucrose and high-fructose corn syrup, both of which are about 50 percent fructose and 50 percent glucose. Fructose is converted into glucose by the liver and then released into the blood for use.

Galactose

Galactose is a type of sugar found in dairy products that's processed similarly to fructose.

Glucose

Glucose is a sugar that occurs widely in nature as a component of many carbohydrates and is an important energy source in organisms.

Glycemic Index

The glycemic index (GI) is a numeric system that ranks how quickly a food is

converted by the body into glucose. Foods are ranked on a scale of 0-to-100 depending on how they affect blood sugar levels once eaten. A GI rating of 55 and under is considered low on the index, while a rating of 56-to-69 is medium, and a rating of 70 or above is high.

Simple carbohydrates are converted into glucose quickly and thus have high GI ratings. For example, sucrose's rating is 65, white bread's is 71, white rice's is 89, and white potato's is 82. Complex carbohydrates are converted into glucose more slowly and thus have lower GI ratings. For example, apples' rating is 39, black beans' is 30, peanuts' is 7, and whole-grain pasta's is 42.

Glycogen

Glycogen is a form of carbohydrate found primarily in the liver and muscle tissue that's readily converted to glucose to satisfy the body's energy needs, especially during exercise.

Gram

A gram is a unit of weight in the metric system. One pound is about 454 grams.

Healthy

1. If a body is healthy, it has good strength, high energy levels, and is free from pain, illness, damage, and dysfunction.
2. If something is healthy, it's beneficial to one's physical, mental, or emotional state.

Hormone

A hormone is a chemical transported by the blood or other bodily fluids to cells and organs, where it causes some action or has some specific effect.

Insulin

Insulin is a hormone made in the pancreas and released into the blood when you eat food. Insulin causes muscles, organs, and fat tissue to absorb and use or store nutrients in food.

Kilogram

A kilogram is a unit of weight in the metric system equal to 1,000 grams or 2.2 pounds.

Lactose

Lactose is a type of sugar present in milk that contains glucose and galactose.

Macronutrient

A macronutrient is any of the nutritional components of the diet required in relatively large amounts. Specifically, these are protein, carbohydrate, fat, and certain minerals, such as calcium, zinc, iron, magnesium, and phosphorus.

Metabolism

Metabolism is the series of physical and chemical processes that occur in an organism in order to maintain life. Metabolism involves the production of energy as well as the creation, maintenance, and destruction of cells and tissues.

Milligram

A milligram is a unit of weight in the metric system equal to one-thousandth of a gram.

Milliliter

A milliliter is a unit of capacity in the metric system equal to one-thousandth of a liter, which is equal to about 4.2 cups in the United States system.

Mineral

A mineral is a carbonless substance that forms naturally in the earth. Humans need various minerals, such as sodium, potassium, calcium, and zinc, for many different physiological functions, including building bones, making hormones, and regulating the heartbeat.

Molecule

A molecule is the smallest particle of any compound that still exists as that substance. If you were to break a molecule down any further, it would separate into the elements that make it up (and so would no longer exist as that original substance).

Nourish

To nourish is to provide with the food or other substances needed for growth, health, and good condition.

Nutrient

A nutrient is a substance an organism needs to live and grow.

Nutrition

Nutrition is the process of getting nourishment, especially the process of getting food and nutrients and using them to grow bigger, build and replace tissues, and stay healthy.

Protein

A protein is a naturally occurring compound that's composed of one or more long chains of amino acids. Proteins are an essential part of all organisms and are used to create body tissues such as muscle, hair, and skin, as well as various chemicals vital to life.

Saturated Fat

Saturated fat is a type of fat that's solid at room temperature and found in many animal and some plant sources, including meat, cream, cheese, butter, lard, coconut oil, cottonseed oil, and palm kernel oil.

Simple Carbohydrate

A simple carbohydrate is a form of carbohydrate that breaks down quickly into glucose in the body. Curose, fructose, galactose, and lactose are simple carbohydrates.

Sucrose

Sucrose (commonly known as *table sugar*) is a sugar that occurs naturally in most plants and is obtained commercially from sugarcane or sugar beets. It's composed of equal parts fructose and glucose.

Sugar

Sugar is a class of sweet-tasting carbohydrate that comes from various plants, fruits, grains, and other sources.

Tissue

Tissue is a group of cells in animals and plants that forms a definite kind of structural material with a specific function.

Trans Fatty Acid

A trans fatty acid is a type of unsaturated fatty acid that's uncommon in nature and usually created artificially. "Trans fats" are often found in highly processed foods like cereals, baked goods, fast food, ice cream, and frozen dinners as "partially hydrogenated oil."

Unsaturated Fat

Unsaturated fat is a type of fat that's liquid at room temperature and found in many plant and some animal sources, including avocado, nuts, vegetable oils, and fish.

Vitamin

A vitamin is a substance that an organism needs for cells to function, grow, and develop correctly.

And that's it for the diet and nutrition keywords. When I first learned what many of these words meant, I was surprised to discover how many I had wrong and how clarifying them began to demystify the details of effective dieting. I hope you've experienced something similar. Let's continue by looking at the lexical elements of exercise.

Exercise Keywords

Anabolism

Anabolism is a metabolic process in an organism by which more complex substances (such as tissue) are made from simpler ones (such as proteins).

Body Fat Percentage

Body fat percentage is an estimate of the percentage of a person's body weight that's fat. For example, a 200-pound man with 15% body fat has 30 pounds of body fat.

Catabolism

Catabolism is the metabolic process by which more complex substances (such as proteins) are broken down into simpler ones (such as amino acids).

Compound Exercise

A compound exercise is an exercise that emphasizes the use of two or more joints and muscle groups, typically larger ones. For example, the squat emphasizes the hips, knees, ankles, and hip flexors, spinal erectors, quadriceps, and glutes (although like all compound exercises, it also trains a variety of other smaller muscle groups like the calves, abs, and traps).

Failure

Failure is the point during a strength training set when you can no longer move the weight despite giving your maximal effort.

Fitness

1. The condition of being physically fit and healthy.

2. The quality of being suitable to fulfill a particular role or task. In sports, this refers to your ability to perform the physical demands of a particular sport. For example, “fitness” for powerlifting looks very different from “fitness” for triathlon or golf.

Force

Force is strength or energy expressed as physical action or movement.

Form

Form refers to how well you perform an exercise. Proper form involves using the correct technique for an exercise through a full range of motion.

Frequency

In strength training, frequency refers to how often you train a particular muscle group or exercise (typically per week).

Hypertrophy

Hypertrophy is the enlargement of an organ or tissue from the increase in size of its cells. *Muscle hypertrophy* refers to an increase in the size of muscle cells.

Intensity

Intensity refers to how much energy is used during exercise, and it can be expressed in two ways:

1. Absolute intensity, which is the absolute amount of effort being exerted.
2. Relative intensity, which is the amount of effort being exerted relative to your maximum capabilities.

In strength training, both absolute and relative intensity is generally measured in

terms of how much weight you're lifting. For example, although bench pressing 200 pounds may seem like a high *absolute* intensity, it might be a low *relative* intensity if the person can bench press 350 pounds. Likewise, bench pressing 95 pounds may seem like a low *absolute* intensity, but it could be a high *relative* intensity for someone who can only bench press 115 pounds.

Isolation Exercise

An isolation exercise is an exercise that emphasizes the use of one joint and muscle group, typically a smaller one. For example, the barbell curl emphasizes the elbows and biceps (although like all isolation exercises, it also slightly trains surrounding muscle groups such as the forearms and shoulders).

Lean Mass

Lean mass is all of the mass in the body that isn't fat, including water, protein, minerals, and other substances. Changes in lean mass are often used as an index of changes in muscle mass.

Muscle

Muscle is a tissue in the body, often attached to bones, that can tighten and relax to produce motion.

Muscle Fiber

A muscle fiber, also called a *muscle cell* or *myocyte*, is a collection of long, threadlike strands called *myofibrils* that contract.

One-Repetition Max

A one-repetition max, also known as a one-rep max or 1RM, is the maximum amount of weight you can lift for a single repetition of a given exercise through a full range of motion with proper technique.

Overreaching

Overreaching is a temporary decrease in performance caused by an imbalance between training and recovery (it's also sometimes referred to as "underrecovery"). Symptoms of overreaching include fatigue, lack of motivation to train, joint and muscle soreness, and irritability. Overreaching resolves after one-to-three weeks of less and lighter training and more rest.

Overtraining

Overtraining is a long-term decrease in performance caused by an imbalance between training and recovery that *doesn't* resolve after one-to-three weeks of easier training and more rest. Symptoms of overtraining are the same as those of overreaching but can also include insomnia, loss of libido and appetite, weight loss (not in the good way), and depression. True overtraining is extremely rare and only occurs in athletes doing an exceptionally high volume of high-intensity training for long periods of time (usually months or years). It basically doesn't happen to regular folks in the gym (and definitely won't happen to someone following the *Bigger Leaner Stronger* program).

Power

In sports, power refers to how quickly you can generate force.

Range of Motion

Range of motion refers to the full movement potential of a joint.

Repetition

A repetition, also known as a "rep," is a single raising and lowering of a weight. For example, if you're doing dumbbell biceps curls and curl the weights up from your sides and then lower them back to their starting positions, you've done one rep.

Repetition Range

A repetition range represents the minimum and maximum number of reps you should do in a set for an exercise. For example, if your workout calls for a rep range of 4–6 reps for squats, you should use a weight that allows you to do at least four reps but not more than six.

Reps In Reserve

Reps in reserve (RIR) refers to how many more reps can be done before reaching failure. For example, if you finish a set of squats with two reps in reserve, that means you could have done two more reps before reaching failure.

Resistance Training

Resistance training refers to contracting your muscles against an external resistance with the goal of increasing strength, power, and muscle mass.

Rest Period

A rest period is the amount of time you rest between sets of an exercise.

Set

A set is a fixed number of repetitions of a particular exercise. For instance, if you do six reps of biceps curls and stop, you've done one set (of six reps).

Skeletal Muscle

Skeletal muscle is muscle tissue connected to the skeleton to form part of the system that moves the limbs and other parts of the body.

Strength

Strength is the ability to overcome resistance. In strength training, it's typically expressed as a person's one-rep max.

Strength Training

Strength training refers to resistance training with the goal of becoming as strong as possible on a particular exercise or exercises.

Tempo

Tempo refers to how quickly (in seconds) you raise and lower a weight during a strength training exercise. For example, a 1-0-1 tempo for biceps curls would mean raising the weight in one second, not pausing at all at the top of the rep, and lowering the weight in one second before raising the weight again.

Volume

Volume refers to how much total work is done in an exercise or other athletic activity over a period of time. In strength training, volume is typically expressed in one of three ways: as the total number of reps (reps x weight x sets), also called *volume load*, or the total number of hard sets for a muscle group or exercise (usually per workout or week).

Believe it or not, you've just learned at least a few things that most weightlifters—even experienced ones—don't fully understand. And you may be surprised just how much trouble these "minor" misconceptions can cause. They can block the ability to observe, comprehend, and apply, give rise to dotty (and tenacious) ideas and theories, and lay us open to lies and misinformation.

So, I hope I've convinced you to get a dictionary and start using it. This habit is one of the easiest ways to upgrade not just your ability to comprehend and retain information but your entire intellectual cosmos.

Meal Plans for Cutting

Cutting Meal Plan for a 170-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Egg whites	155 g (5 large whites)	86	18 g	1 g	0 g
	Whole eggs	100 g (2 large)	143	13 g	1 g	10 g
	Grapes	180 g	124	1 g	33 g	0 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Hot sauce, dry seasonings	(if desired)	4	0 g	1 g	0 g
Lunch (12 p.m.)	Chum, pink, or sockeye salmon (farm-raised may be higher in fat/calories)	190 g	241	39 g	0 g	8 g
	White rice (measured dry)	45 g	164	3 g	36 g	0 g
	Broccoli	125 g	40	3 g	8 g	0 g
	Bell pepper	125 g	33	1 g	8 g	0 g
	Carrots	125 g	51	1 g	12 g	0 g
	0-calorie cooking spray	(if desired	15	0 g	0 g	2 g
	Soy sauce	32 g	16	3 g	2 g	0 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	2 scoops (60 grams)	240	44 g	8 g	4 g
	2% milk	120 mL (½ cup)	65	4 g	6 g	3 g
	Rice cakes	18 g	70	2 g	15 g	1 g
	PB®fit Original	16 g	70	8 g	5 g	2 g

Dinner (6 p.m.)	Chicken breast, trimmed of fat	220 g	238	45 g	0 g	7 g
	White potato	170 g	131	4 g	30 g	0 g
	Brussels sprouts	200 g	86	7 g	18 g	1 g
	BBQ sauce	17 g	29	0 g	7 g	0 g
	Dark chocolate (70–85% cocoa solids)	15 g	90	1 g	7 g	6 g
Total			1951	196 g	196 g	46 g
Target			1950	195 g	195 g	45 g

Cutting Meal Plan for a 190-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr or Greek yogurt, 2% fat, plain	250 g	176	25 g	7 g	5 g
	Whey protein isolate (recommend Legion Whey+)	1 scoop (30 grams)	120	22 g	4 g	2 g
	Mixed berries (strawberries, raspberries, blueberries, blackberries, etc.)	85 g	120	2 g	301 g	1 g
	Granola	60 g	273	6 g	44 g	9 g
	Cinnamon, nutmeg, vanilla extract, stevia	(if desired)	12	0 g	2 g	1 g
	Half and half (in coffee)	2 tablespoons (30 mL)	39	1 g	1 g	4 g
Lunch (12 p.m.)	Whole grain bread	2 slices	217	11 g	36 g	4 g
	Oven roasted chicken breast or turkey breast	220 g	240	48 g	4 g	0 g
	Low-fat cheddar cheese	15 g	47	4 g	0 g	3 g
	Tomato, lettuce	100 g	18	1 g	4 g	0 g
	Light mayonnaise	15 g (1 tablespoon)	36	0 g	1 g	3 g
	1 tablespoon mustard, salt, pepper	15 g (1 tablespoon)	9	1 g	1 g	1 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	1 scoop (30 grams)	120	22 g	4 g	2 g
	Almond milk	240 mL (1 cup)	37	1 g	3 g	2 g
	Rice cakes	36 g	140	3 g	29 g	1 g
	Peanut butter	15 g	88	4 g	3 g	8 g
Dinner (6 p.m.)	99% fat free ground turkey	220 g	235	51 g	0 g	3 g

	Pasta (measured dry)	60 g	223	8 g	45 g	1 g
	Tomato pasta sauce	120 mL (½ cup)	66	2 g	10 g	2 g
	Mushrooms	100 g	26	1 g	6 g	0 g
	Zucchini	150 g	26	2 g	5 g	1 g
	Hot sauce, dry seasonings	(if desired)	6	0 g	2 g	0 g
Total			2156	211 g	214 g	49 g
Target			2150	210 g	210 g	45 g

Cutting Meal Plan for a 210-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr or Greek yogurt, 2% fat, plain	250 g	176	25 g	7 g	5 g
	Whey protein isolate (recommend Legion Whey+)	1 scoop (30 grams)	120	22 g	4 g	2 g
	Mixed berries (strawberries, raspberries, blueberries, blackberries)	85 g	120	2 g	31 g	1 g
	Granola	60 g	273	6 g	44 g	9 g
	Cinnamon, nutmeg, vanilla extract, stevia	(if desired)	12	0 g	2 g	1 g
	Half and half (in coffee)	2 tablespoons (30 mL)	39	1 g	1 g	4 g
	Sugar (in coffee)	10 g (2 teaspoons)	38	0 g	10 g	0 g
Lunch (12 p.m.)	Shrimp	260 g	222	52 g	0 g	1 g
	White rice (measured dry)	35 g	128	3 g	28 g	0 g
	Sweet yellow corn (kernels cut off cob)	100 g	86	3 g	19 g	1 g
	Bell pepper	150 g	39	2 g	9 g	1 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Teriyaki sauce	18 g	32	2 g	6 g	0 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	2 scoops (60 grams)	240	44 g	8 g	4 g
	Skim milk	240 mL (1 cup)	83	8 g	12 g	0 g
	Rice cakes	36 g	140	3 g	29 g	1 g
	PB®fit Original	18 g	80	9 g	6 g	2 g

Dinner (6 p.m.)	Wonder® Jumbo Seeded Hamburger Bun (or similar)	74 g (1 bun)	200	6 g	39 g	3 g
	95% lean ground beef	200 g	274	43 g	0 g	10 g
	Low fat cheddar cheese	15 g	47	4 g	0 g	3 g
	Tomato, lettuce	100 g	18	1 g	4 g	0 g
	Ketchup	34 g (2 tablespoons)	34	0 g	9 g	0 g
	Hot sauce, lemon juice, dry seasonings	(if desired)	6	0 g	2 g	0 g
	Milk chocolate	25 g	134	2 g	15 g	7 g
Total			2304	233 g	236 g	52 g
Target			2300	230 g	230 g	50 g

Cutting Meal Plan for a 230-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Egg whites	184 g (6 large whites)	103	22 g	2 g	0 g
	Whole eggs	100 g (2 large)	143	13 g	1 g	10 g
	Canadian bacon	114 g	126	23 g	2 g	3 g
	Whole grain bread	2 slices	217	11 g	36 g	4 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Hot sauce, dry seasonings	(if desired)	4	0 g	1 g	0 g
Lunch (12 p.m.)	Flour tortilla	71 g (1 large)	217	6 g	35 g	6 g
	Turkey breast, trimmed of fat	230 g	262	54 g	0 g	4 g
	Cucumber	50 g	8	0 g	2 g	0 g
	Romaine lettuce	50 g	9	1 g	2 g	0 g
	Tomato	50 g	9	1 g	2 g	0 g
	Avocado	30 g	48	1 g	3 g	4 g
	Bolthouse Farms® organic dressing (or similar)	30 g (2 tablespoons)	45	1 g	1 g	4 g
	Hot sauce, lemon juice, dry seasonings	(if desired)	6	0 g	2 g	0 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	2 scoops (60 grams)	240	44 g	8 g	4 g
	Unsweetened almond milk	240 mL (1 cup)	37	1 g	3 g	2 g
	Apple	165 g (1 small)	86	0 g	23 g	0 g
	PB®fit Original	16 g	70	8 g	5 g	2 g
Dinner (6 p.m.)	Sirloin, trimmed of all fat	240 g	324	53 g	0 g	11 g

	Sweet potato	420 g	361	7 g	85 g	0 g
	Green beans	150 g	46	3 g	11 g	0 g
	Carrots	150 g	62	1 g	14 g	0 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	BBQ sauce	34 g (2 tablespoons)	58	0 g	14 g	0 g
Total			2511	249 g	249 g	58 g
Target			2500	250 g	250 g	55 g

Cutting Meal Plan for a 250-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Whey protein isolate (recommend Legion Whey+) mixed with water or in oatmeal	1½ scoops (45 grams)	180	33 g	6 g	3 g
	Old-fashioned oats (measured dry)	75 g	284	10 g	51 g	5 g
	Skim milk	240 mL (1 cup)	83	8 g	12 g	0 g
	Blueberries	50 g	28	0 g	7 g	0 g
	Chia seeds	5 g	27	1 g	1 g	2 g
	Cinnamon, nutmeg, vanilla extract, stevia	(if desired)	12	0 g	2 g	1 g
	Half and half (in coffee)	2 tablespoons (30 mL)	39	1 g	1 g	4 g
	Sugar (in coffee)	10 g (2 teaspoons)	38	0 g	10 g	0 g
Lunch (12 p.m.)	Chicken breast, trimmed of fat	270 g	292	55 g	0 g	8 g
	Dried (sweetened) cranberries	15 g	45	0 g	12 g	0 g
	Sweet yellow corn (kernels cut off cob)	150 g	129	5 g	28 g	2 g
	Spinach	75 g	17	2 g	3 g	0 g
	Cucumber	50 g	8	0 g	2 g	0 g
	Carrots	30 g	12	0 g	3 g	0 g
	Bolthouse Farms® organic dressing (or similar)	30 g (2 tablespoons)	45	1 g	1 g	4 g
	Hot sauce, lemon juice, dry seasonings	(if desired)	6	0 g	2 g	0 g
Snack (3 p.m.)	2% cottage cheese	280 g	227	28 g	13 g	7 g
	Cinnamon, nutmeg, vanilla extract,	(if desired)	12	0 g	2 g	1 g

	stevia					
	Peach	225 g (1 extra large)	88	2 g	22 g	1 g
	Cashews	25 g	138	5 g	8 g	11 g
Dinner (6 p.m.)	Mahi-mahi (or similar, like tilapia)	310 g	273	58 g	0 g	3 g
	Quinoa (measured dry)	55 g	203	8 g	35 g	3 g
	Green beans	100 g	31	2 g	7 g	0 g
	Cauliflower	150 g	38	3 g	8 g	0 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Teriyaki sauce	18 g (1 tablespoon)	32	2 g	6 g	0 g
	Oreo cookie (or similar)	23 g (2 cookies)	110	1 g	17 g	5 g
Pre Bed Shake (9 p.m.)	Casein protein (recommend Legion Casein+) mixed with water	1½ scoops (45 grams)	210	39 g	8 g	3 g
	Pear	150 g (1 small)	85	1 g	23 g	0 g
Total			2707	265 g	288 g	65 g
Target			2700	265 g	265 g	60 g

Meal Plans for Lean Gaining

Lean Gaining Meal Plan for a 130-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Kodiak Power Cakes Buttermilk Flapjack & Waffle Mix (or similar)	2 servings (106 g)	380	28 g	60 g	4 g
	Skim milk	120 mL (½ cup)	42	4 g	6 g	0 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Maple syrup	25 g	65	0 g	17 g	0 g
	Half and half (in coffee)	2 tablespoons (30 mL)	39	1 g	1 g	4 g
	Sugar (in coffee)	10 g (2 teaspoons)	38	0 g	10 g	0 g
Lunch (12 p.m.)	Whole grain bread	2 slices	217	11 g	36 g	4 g
	Oven roasted chicken breast or turkey breast	130 g	142	28 g	2 g	0 g
	Tomato, lettuce	100 g	18	1 g	4 g	0 g
	Light mayonnaise	15 g (1 tablespoon)	36	0 g	1 g	3 g
	Mustard, salt, pepper	15 grams (1 tablespoon)	9	0 g	2 g	0 g
Snack (3 p.m.)	Legion Protein Bar (or similar)	1 bar	250	20 g	24 g	12 g
	Banana	150 g (1 extra large)	133	2 g	34 g	1 g
	Peanut butter	15 g	88	4 g	3 g	8 g
Dinner (6 p.m.)	Chicken breast, trimmed of fat	110 g	119	22 g	0 g	3 g

	Flour tortilla	71 g (1 large)	217	6 g	35 g	6 g
	Traditional refried beans (canned)	150 g	135	8 g	20 g	3 g
	Sweet yellow corn (kernels cut off cob)	100 g	86	3 g	19 g	1 g
	Onion	50 g	20	1 g	5 g	0 g
	Tomato	150 g	27	1 g	6 g	0 g
	Sour cream	20 g	40	1 g	1 g	4 g
	Salsa, salt pepper	45 g	17	1 g	3 g	0 g
	Gummy bears	50 g	168	3 g	23 g	0 g
Total			2302	145 g	312 g	54 g
Target			2300	145 g	315 g	50 g

Lean Gaining Meal Plan for a 150-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Skyr or Greek yogurt, 2% fat, plain	250 g	176	25 g	7 g	5 g
	Mixed berries (strawberries, raspberries, blueberries, blackberries)	220 g	120	2 g	31 g	1 g
	Granola	60 g	273	6 g	44 g	9 g
	Cinnamon, nutmeg, vanilla extract, stevia	(if desired)	12	0 g	2 g	1 g
	Half and half (in coffee)	2 tablespoons (30 mL)	39	1 g	1 g	4 g
	Sugar (in coffee)	10 g (2 teaspoons)	38	0 g	10 g	0 g
Lunch (12 p.m.)	Pork Tenderloin, trimmed of all fat	150 g	164	32 g	0 g	3 g
	Quinoa	75 g	276	11 g	48 g	5 g
	Carrots	100 g	41	1 g	10 g	0 g
	Cauliflower	125 g	31	2 g	10 g	0 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Teriyaki sauce	18 g (1 tablespoon)	32	2 g	6 g	0 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	1 scoop (30 g)	120	22 g	4 g	2 g
	Rice Milk	240 mL (1 cup)	114	1 g	22 g	2 g
	Rice cakes		140	3 g	29 g	1 g
	Peanut butter		88	4 g	3 g	
Dinner (6 p.m.)	95% lean ground beef	120 g	164	26 g	0 g	6 g
	Pasta (measured dry)	90 g	334	12 g	67 g	1 g

	Tomato pasta sauce	180 mL (¾ cup)	99	3 g	15 g	3 g
	Grated parmesan cheese	10 g (1 tablespoon)	42	3 g	1 g	3 g
	Bell pepper	100 g	26	1 g	6 g	0 g
	Broccoli	150 g	48	4 g	10 g	1 g
	Hot sauce, dry seasonings	(if desired)	6	0 g	2 g	0 g
	Oreo cookie (or similar)	34 g (3 cookies)	160	1 g	25 g	7 g
Total			2558	159 g	349 g	63 g
Target			2550	160 g	350 g	60 g

Lean Gaining Meal Plan for a 170-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Egg white	62 g (2 large whites)	34	7 g	1 g	0 g
	Whole egg	100 g (2 large)	143	13 g	1 g	10 g
	Whole grain bread	3 slices	327	17 g	53 g	5 g
	Jam	30 g (2 tablespoons)	83	0 g	21 g	0 g
	0-calorie cooking spray	(if desired)	15	0 g	0 g	2 g
	Hot sauce, dry seasonings	(if desired)	4	0 g	1 g	0 g
Lunch (12 p.m.)	Sirloin, trimmed of all fat	150 g	203	33 g	0 g	7 g
	Sweet potato	410 g	352	6 g	83 g	0 g
	Green beans	150 g	46	3 g	11 g	0 g
	Carrots	150 g	62	1 g	14 g	0 g
	Butter	5 g	36	0 g	0 g	4 g
	BBQ sauce	34 g (2 tablespoons)	60	0 g	14 g	0 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	1 scoop (30 g)	120	22 g	4 g	2 g
	Rice Milk	240 mL (1 cup)	114	1 g	22 g	2 g
	Banana	150 g (1 extra large)	133	2 g	34 g	1 g
	Mixed berries (strawberries, raspberries, blueberries, blackberries)	180 g	103	1 g	22 g	1 g
Dinner (6 p.m.)	Chicken breast, trimmed of fat	140 g	168	32 g	0 g	4 g
	Dried (sweetened) cranberries	30 g	92	0 g	25 g	0 g

	Sweet yellow corn (kernels cut off cob)	200 g	172	7 g	37 g	3 g
	Romaine lettuce	100 g	17	2 g	3 g	0 g
	Cucumber	50 g	8	0 g	2 g	0 g
	Carrots	30 g	12	0 g	3 g	0 g
	Bolthouse® Farms Organic dressing (or similar)	45 g (3 tablespoons)	68	2 g	2 g	6 g
	Hot sauce, lemon juice, dry seasonings	(if desired)	6	0 g	2 g	0 g
	Chocolate or vanilla ice cream	140 g	302	5 g	40 g	15 g
Total			2800	175 g	397 g	65 g
Target			2800	175 g	390 g	65 g

Lean Gaining Meal Plan for a 190-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Whey protein isolate (recommend Legion Whey+)	1½ scoops (45 g)	180	33 g	6 g	3 g
	Rice Milk	240 mL (1 cup)	114	1 g	22 g	2 g
	Old-fashioned oats (measured dry)		588	21 g	105 g	10 g
	Blueberries		28	0 g	7 g	0 g
	Chia seeds	155 g	27	1 g	1 g	2 g
	Cinnamon, nutmeg, vanilla extract, stevia	50 g	12	0 g	2 g	1 g
Lunch (12 p.m.)	Turkey breast, trimmed of fat	5 g	182	38 g	0 g	2 g
	Brown rice (measured dry)	(if desired)	367	8 g	76 g	3 g
	A1 steak sauce	160 g	15	0 g	3 g	0 g
	Spinach	100 g	17	2 g	3 g	0 g
	Cucumber	17 g (1 tablespoon)	8	0 g	2 g	0 g
	Carrots	75 g	12	0 g	3 g	0 g
	Bolthouse Farms® Organic dressing (or similar)	50 g	45	1 g	1 g	4 g
	Hot sauce, lemon juice, dry seasonings	30 g	6	0 g	2 g	0 g
Snack (3 p.m.)	Skyr or Greek yogurt, 2% fat, plain	250 g	176	25 g	7 g	5 g
	Honey	30 g (2 tablespoons)	91	0 g	25 g	0 g
	Cinnamon, nutmeg, vanilla extract, stevia	(if desired)	12	0 g	2 g	1 g

	Apple	242 g (1 large)	126	1 g	33 g	0 g
	Almonds	25 g	145	5 g	5 g	13 g
Dinner (6 p.m.)	Shrimp	180 g	153	36 g	0 g	1 g
	Pasta (measured dry)	90 g	334	12 g	67 g	1 g
	Tomato pasta sauce	180 mL (¾ cup)	99	3 g	15 g	3 g
	Broccoli	150 g	48	4 g	10 g	1 g
	Mushrooms	100 g	26	1 g	6 g	0 g
	Hot sauce, dry seasonings	(if desired)	6	0 g	2 g	0 g
	Dark chocolate (70–85% cocoa solids)	50 g	300	4 g	23 g	21 g
Total			3112	196 g	429 g	76 g
Target			3100	195 g	425 g	70 g

Lean Gaining Meal Plan for a 210-lb. Man

	Food	Amount	Calories	Protein	Carbs	Fat
Breakfast (9 a.m.)	Smoked salmon (lox)	110 g	129	20 g	0 g	5 g
	Bagel	131 g (1 large, ~5-inch diameter)	346	14 g	69 g	2 g
	Low-fat cream cheese	30 g (2 tablespoons)	62	2 g	2 g	5 g
	Tomato	100 g	18	1 g	4 g	0 g
	Hot sauce, lemon juice, dry seasonings	(if desired)	6	0 g	2 g	0 g
	Orange juice	240 mL (1 cup)	112	2 g	26 g	1 g
	Half and half (in coffee)	2 tablespoons (30 mL)	39	1 g	1 g	4 g
	Sugar (in coffee)	10 g (2 teaspoons)	38	0 g	10 g	0 g
Lunch (12 p.m.)	Chicken breast, trimmed of fat	110 g	119	22 g	0 g	3 g
	Lentils (measured dry)	120 g	422	30 g	76 g	1 g
	Onion	150 g	60	2 g	14 g	0 g
	Tomato	200 g	36	2 g	8 g	0 g
	Chicken broth or stock	480 mL (2 cups)	20	2 g	2 g	1 g
	Dry seasonings (garlic powder, cumin, paprika, oregano, salt)	(if desired)	4	0 g	1 g	0 g
Snack (3 p.m.)	Whey protein isolate (recommend Legion Whey+)	1½ scoops (45 g)	180	33 g	6 g	3 g
	Rice Milk	240 mL (1 cup)	114	1 g	22 g	2 g
	Banana	252 g (2 medium)	224	3 g	58 g	1 g
	Almond butter	25 g	153	5 g	5 g	14 g

Dinner (6 p.m.)	Wonder® Jumbo Seeded Hamburger Buns (or similar)	74 g (1 bun)	200	6 g	39 g	3 g
	95% lean ground beef	100 g	137	21 g	0 g	5 g
	Low-fat cheddar cheese	30 g	95	8 g	1 g	6 g
	Tomato, lettuce	100 g	18	1 g	4 g	0 g
	Ketchup	68 g (4 tablespoons)	68	1 g	19 g	0 g
	Hot sauce, lemon juice, dry seasonings	(if desired)	6	0 g	2 g	0 g
	Ice cream (chocolate or vanilla)	150 g	324	6 g	42 g	17 g
Pre Bed Snack (9 p.m.)	2% cottage cheese	280 g	227	28 g	13 g	7 g
	Pineapple	385 g	193	2 g	51 g	1 g
	Salt, pepper, stevia (if desired)	(if desired)	0	0 g	0 g	0 g
Total			3350	213 g	475 g	80 g
Target			3550	210 g	460 g	75 g

Also by Michael Matthews

The Year One Challenge for Men

The Shredded Chef

Beyond Bigger Leaner Stronger

Muscle for Life

Thinner Leaner Stronger

The Year One Challenge for Women

The Little Black Book of Workout Motivation

Fitness Science Explained

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Chapter 4
Have You Found Your Real Fitness Whys?

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Chapter 5
The Art and Science of Upgrading Your Willpower

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Chapter 9
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Chapter 12
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Chapter 13
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Chapter 18
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